

DevOrSysAdminScripts

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Abstract

Some “small” useful scripts for enhancing development or system-administrators tasks

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Code lines: 12450 total lines, 11460 not empty lines, 990 empty lines.

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1 Files trees

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.lib.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.lib.sh
│   ├── check_shell_scripts_indentation.lib.sh
│   ├── check_URLs.lib.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.lib.sh
│   ├── comparisons.lib.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.lib.sh
│   ├── get_common_text_glob_patterns.lib.sh
│   ├── grammar_and_spelling_check.lib.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.lib.sh
│   ├── lines_filters.lib.sh
│   ├── lines_maps.lib.sh
│   ├── overwrite_if_not_equal.lib.sh
│   ├── python_black_complement.lib.sh
│   ├── python_isort_complement.lib.sh
│   ├── repository_data.lib.sh
│   ├── shell_checks_complement.lib.sh
│   ├── split_score.exec.php
│   ├── split_score.lib.php
│   ├── strings_functions.lib.sh
│   ├── too_long_code_lines.lib.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   └── DevOrSysAdminScripts.html.tpl
```



```

| an_see_that_it_works.txt
|— funny.sh
|— spampty_data.txt
|— string_escaping.html
|— string_escaping.xhtml
|
| yet_another_long_long_long_long_long_long_long_long_filename_so_
| that_you_can_see_that_it_works.txt
|— README.html
|— README.md
|— README.md.tpl
|— string_escaping.lib.php
|— unicode.lib.php
|— upgrade_build_and_checks.exec.sh
|— wget_sha512.lib.sh

```

6 directories, 83 files

```

[ 12K May 12 19:40] ./
|— [4.0K May 12 19:39] build_and_checks_dependencies/
|   |— [4.4K May 12 19:33] build_and_checks.exec.sh.tpl*
|   |— [1.2K May 12 19:31] build_dependencies_notes.exec.php*
|   |— [8.3K May 12 19:31] build_dependencies_notes.lib.php
|   |— [4.2K May 12 19:34] build_md_from_printable_md.exec.sh*
|   |— [3.1K May 12 19:34] check_shell_scripts_beginnings.lib.sh
|   |— [5.3K May 12 19:37] check_shell_scripts_indentation.lib.sh
|   |— [3.9K May 12 19:34] check_URLs.lib.sh
|   |— [ 38K May 12 19:38] common_build_and_checks.exec.sh*
|   |— [ 33K May 12 19:37] common_build_and_checks.exec.sh.tpl
|   |— [ 19K May 12 19:37] common_options.lib.sh
|   |— [6.2K May 12 19:37] comparisons.lib.sh
|   |— [ 22K May 12 19:37] create_PDF.exec.sh*
|   |— [3.7K May 12 19:32] generate_from_template.lib.sh
|   |— [2.0K May 12 19:37] get_common_text_glob_patterns.lib.sh
|   |— [3.3K May 12 19:37] grammar_and_spelling_check.lib.sh
|   |— [4.0K May 12 19:37] licenses_templates/
|       |— [ 15K May 12 19:37] build_licenses_templates.exec.sh*
|       |— [ 720 May 12 19:16] license_file_header_AGPLv3+.tpl
|       |— [ 699 May 12 19:17] license_file_header_GPLv3+.tpl
|       |— [ 720 May 12 19:18] license_file_header_LGPLv3+.tpl
|   |— [6.0K May 12 19:37] lines_counts.lib.sh
|   |— [ 15K May 12 19:37] lines_filters.lib.sh
|   |— [3.0K May 12 19:37] lines_maps.lib.sh
|   |— [3.3K May 12 19:37] overwrite_if_not_equal.lib.sh
|   |— [2.4K May 12 19:37] python_black_complement.lib.sh
|   |— [3.1K May 12 19:37] python_isort_complement.lib.sh

```

```

|— [2.0K May 12 19:37] repository_data.lib.sh
|— [3.9K May 12 19:37] shell_checks_complement.lib.sh
|— [2.0K May 12 19:31] split_score.exec.php*
|— [3.2K May 12 19:31] split_score.lib.php
|— [ 47K May 12 19:37] strings_functions.lib.sh
|— [4.6K May 12 19:37] too_long_code_lines.lib.sh
|— [4.4K May 12 19:37] update_common_build_and_checks.exec.sh*
|— [6.0K May 12 19:37] update_or_check_files_names_listing.exec.sh*
|— [4.6K May 12 19:38] build_and_checks.exec.sh*
|— [4.0K May 12 19:40] build_and_checks_variables/
|— [ 15 Jan 5 20:00] bandit_baseline.json
|— [ 25 Jan 5 18:49] bandit.ini
|— [ 30 Feb 28 20:30] black.toml
|— [5.1K Apr 17 16:50] dependencies_data.json
|— [4.3K Apr 17 16:57] dependencies_notes.txt
|— [5.7K May 12 19:31] DevOrSysAdminScripts.html.tpl
|— [ 35K May 12 19:31] DevOrSysAdminScripts.tex
|— [4.2K May 12 19:38] DevOrSysAdminScripts.tex.tpl
|— [5.5K May 12 19:31] files_names_listing.txt
|— [ 23 Mar 24 19:33] .gitignore
|— [1.4K Apr 17 16:49] licenses_notes.txt
|— [ 13K Apr 17 09:38] phpcs_ruleset.xml
|— [2.2K Apr 17 14:32] phpmd_baseline.xml
|— [ 678 Feb 24 22:38] pylintrc.toml
|— [4.6K Apr 17 17:09] repository_data.txt
|— [ 841 Apr 17 17:08] repository_data.txt.tpl
|— [1.4K May 12 12:26] ruff.toml
|— [ 166 Mar 24 19:33] shellcheck.ini
|— [3.6K Apr 2 09:16] shellcheck_notes.txt
|— [ 34K May 6 2024] COPYING
|— [7.5K Dec 27 21:48] COPYING.LESSER
|— [485K May 12 19:31] DevOrSysAdminScripts.html
|— [417K May 12 19:28] DevOrSysAdminScripts.pdf
|— [4.0K May 12 19:37] git_helpers/
|— [3.8K May 12 19:37] git_checks.lib.sh
|— [2.1K May 12 19:31] post-commit*
|— [ 15K May 12 19:37] post_commit_renaming_comment.lib.sh
|— [1.8K May 12 19:31] pre-commit*
|— [ 8 Mar 26 10:37] .gitignore
|— [8.4K May 12 19:31] IMAP_transfert.exec.py
|— [1.1K Mar 31 19:59] LiensUtiles.txt
|— [4.0K May 12 19:37] miscellaneous/
|— [ 0 Mar 4 10:37]
|— 12345678901234567890123456789012345678901234567890123456789
|— 0123\
|— [ 0 Mar 4 10:03]

```

```

12345678901234567890123456789012345678901234567890123456789
01234\
— [ 0 Mar 4 10:01]
12345678901234567890123456789012345678901234567890123456789
012345
— [ 0 Mar 4 10:37]
12345678901234567890123456789012345678901234567890123456789
012345\
— [ 0 Mar 4 10:37]
12345678901234567890123456789012345678901234567890123456789
0123456
— [ 0 Mar 4 10:37]
12345678901234567890123456789012345678901234567890123456789
0123456\
— [ 0 Mar 4 09:13]
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
an_see_that_it_works.txt
— [1.3K May 12 19:37] funny.sh
— [ 94 Mar 3 19:56] spampty_data.txt
— [ 442 Mar 18 16:50] string_escaping.html
— [ 653 May 12 14:38] string_escaping.xhtml
— [ 0 Mar 4 09:13]
yet_another_long_long_long_long_long_long_long_long_long_filename_so_
that_you_can_see_that_it_works.txt
— [8.1K May 12 12:17] README.html
— [4.3K Mar 22 21:29] README.md
— [4.3K Mar 22 21:29] README.md.tpl
— [7.0K May 12 19:31] string_escaping.lib.php
— [ 17K May 12 19:31] unicode.lib.php
— [5.5K May 12 19:37] upgrade_build_and_checks.exec.sh*
— [2.5K May 12 19:37] wget_sha512.lib.sh

```

6 directories, 83 files

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: <https://www.gnu.org/licenses/>. The git repository of this source code is also available at: <https://github.com/LLyaudet/DevOrSysAdminScripts/>. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:

basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmid;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, utils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:

<https://www.gnu.org/software/bash/>,
<https://tiswww.case.edu/php/chet/bash/bashtop.html>,
<https://savannah.gnu.org/projects/bash/>.

GNU coreutils can be found at the following URLs:

<https://www.gnu.org/software/coreutils/>,
<https://savannah.gnu.org/projects/coreutils/>.

utils coreutils can be found at the following URLs:

<https://uutils.github.io/coreutils/>,
<https://github.com/uutils/coreutils>.

diff (GNU diffutils) can be found at the following URLs:

<https://www.gnu.org/software/diffutils/>,
<https://savannah.gnu.org/projects/diffutils/>.

find, xargs (GNU findutils) can be found at the following URLs:
<https://www.gnu.org/software/findutils/>,
<https://savannah.gnu.org/projects/findutils/>.

grep (GNU grep) can be found at the following URLs:
<https://www.gnu.org/software/grep/>,
<https://savannah.gnu.org/projects/grep/>.

sed (GNU sed) can be found at the following URLs:
<https://www.gnu.org/software/sed/>,
<https://savannah.gnu.org/projects/sed/>.

ShellCheck can be found at the following URLs:
<https://www.shellcheck.net/>,
<https://github.com/koalaman/shellcheck>.

GNU Wget can be found at the following URLs:
<https://www.gnu.org/software/wget/>,
<https://savannah.gnu.org/projects/wget/>.

more of util-linux can be found at the following URLs:
<https://www.kernel.org/pub/linux/utils/util-linux/>,
<https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/>,
<https://github.com/util-linux/util-linux> official?.

pandoc can be found at the following URLs:
<https://pandoc.org/>,
<https://github.com/jgm/pandoc>.

pdftex/pdflatex can be found at the following URLs:
<https://www.tug.org/applications/pdftex/>,
<https://tug.org/texlive/svn/>.

tree can be found at the following URLs:
<https://oldmanprogrammer.net/source.php?dir=projects/tree>,
<https://gitlab.com/OldManProgrammer/unix-tree>,
<https://github.com/Old-Man-Programmer/tree>.

git can be found at the following URLs:
<https://git-scm.com/about>,
<https://github.com/git/git>.

pylint can be found at the following URLs:
<https://www.pylint.org/>,
<https://pylint.readthedocs.io/en/stable/>,
<https://github.com/pylint-dev/pylint>,
<https://pypi.org/project/pylint/>.

notify-send can be found at the following URLs:
<https://gitlab.gnome.org/GNOME/libnotify/>,
<https://gnome.pages.gitlab.gnome.org/libnotify/>.

black can be found at the following URLs:
<https://github.com/psf/black>,
<https://black.readthedocs.io/en/stable/>,
<https://pypi.org/project/black/>.

isort can be found at the following URLs:
<https://pycqa.github.io/isort/>,

<https://github.com/pycqa/isort/>,
<https://pypi.org/project/isort/>.
mypy can be found at the following URLs:
<https://mypy-lang.org/>,
<https://mypy.readthedocs.io/en/stable/>,
<https://github.com/python/mypy>,
<https://pypi.org/project/mypy/>.
phpdoc (phpDocumentor) can be found at the following URLs:
<https://phpdoc.org/>,
<https://docs.phpdoc.org/>,
<https://github.com/phpdocumentor/phpdocumentor>.
pip can be found at the following URLs:
<https://pip.pypa.io/en/stable/>,
<https://github.com/pypa/pip>,
<https://pypi.org/project/pip/> (egg and chicken ;)).
ruff can be found at the following URLs:
<https://docs.astral.sh/ruff/>,
<https://github.com/astral-sh/ruff>,
<https://pypi.org/project/ruff/>.
xmllint (libxml2) can be found at the following URLs:
<https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home>,
<https://gitlab.gnome.org/GNOME/libxml2>.
bandit can be found at the following URLs:
<https://bandit.readthedocs.io/en/latest/>,
<https://github.com/PyCQA/bandit>,
<https://pypi.org/project/bandit/>.
phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.
phpmd can be found at the following URLs:
<https://phpmd.org/>,
<https://github.com/phpmd/phpmd>,
<https://packagist.org/packages/phpmd/phpmd>.
php can be found at the following URLs:
<https://www.php.net/>,
<https://github.com/php/php-src>.
tex can be found at the following URL:
<https://ctan.org/tex-archive/systems/knuth/dist/tex>.
pcgrep, pcre2grep can be found at the following URLs:
<https://www.pcre.org/>,
<https://pcre2project.github.io/pcre2/>,
<https://github.com/PCRE2Project/pcre1>,
<https://github.com/PCRE2Project/pcre2>,
<https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c>.
python3 can be found at the following URLs:
<https://www.python.org/>,

<https://docs.python.org/3/>.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "build_and_checks.exec.sh.tpl" was renamed from
25 # "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
26 # The file "build_and_checks.exec.sh" generated from the file
27 # "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
28 # was renamed from
29 # "build_and_checks.sh" to "build_and_checks.exec.sh".
30
31 build_and_checks() {
32     # $1=work_directory
33     # Options:
34     #   --check-download=y|keep (=keep to keep the downloaded files)
35     #   --verbose
36     local LFBFL_work_directory="."
37     if [[ -n "$1" ]]; then
```

```

38     LFBFL_work_directory="$1"
39 fi
40 readonly LFBFL_work_directory
41
42 local LFBFL_verbose=""
43 if [[ "$*" == *--verbose* ]]; then
44     printf "%s %s\n" "$0" "$*"
45     LFBFL_verbose="--verbose"
46 fi
47 readonly LFBFL_verbose
48
49 local LFBFL_check_download=""
50 if [[ "$*" == *--check-download=keep* ]]; then
51     LFBFL_check_download="--check-download=keep"
52 elif [[ "$*" == *--check-download=y* ]]; then
53     LFBFL_check_download="--check-download=y"
54 fi
55 readonly LFBFL_check_download
56
57 source ./wget_sha512.lib.sh
58
59 LFBFL_subdir="build_and_checks_dependencies"
60 mkdir --parents "${LFBFL_subdir}/licenses_templates"
61 local LFBFL_variables_directory
62 LFBFL_variables_directory="${LFBFL_work_directory}/"
63 LFBFL_variables_directory+="build_and_checks_variables"
64 readonly LFBFL_variables_directory
65 mkdir --parents -- "${LFBFL_variables_directory}/temp"
66 if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
67     printf "temp/\nupgrade_venvs_ts\n" \
68         > "${LFBFL_variables_directory}/.gitignore"
69 fi
70
71 local LFBFL_dependencies_URL
72 LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73 LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74 readonly LFBFL_dependencies_URL
75
76 declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77 local LFBFL_script_download_URL
78 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79 LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80 readonly LFBFL_script_download_URL
81 declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
82 @sha512_common_build_and_checks.exec.sh@
83 wget_sha512 "${LFBFL_file_path}" \

```

```

84     "${LFBFL_script_download_URL}"\
85     "${LFBFL_correct_sha512}"\
86     "${LFBFL_verbose}"\
87     "${LFBFL_check_download}"
88     chmod +x "${LFBFL_file_path}"
89
90     "${LFBFL_file_path}" "${LFBFL_work_directory}"\
91     "${LFBFL_dependencies_URL}"\
92     "${LFBFL_verbose}"\
93     "${LFBFL_check_download}"
94 }
95
96 if [[ "$*" == *--fixed-point-build* ]]; then
97     printf -- "--fixed-point-build\n"
98     source "./build_and_checks_dependencies/lines_filters.lib.sh"
99     LFBFL_work_directory="."
100    if [[ -n "$1" ]]; then
101        LFBFL_work_directory="$1"
102    fi
103    LFBFL_variables_directory="${LFBFL_work_directory}/"
104    LFBFL_variables_directory+="build_and_checks_variables"
105    LFBFL_data_file_name="${LFBFL_variables_directory}/"
106    LFBFL_data_file_name+="repository_data.txt"
107    LFBFL_repository_name=""
108    grep_variable "${LFBFL_data_file_name}" repository_name\
109        --result-variable-prefix="LFBFL_"\
110        --replace-line-returns-by=""
111    # Touching the 3 following files first let us iterate
112    # upgrade_build_and_checks 2 times instead of 3 to reach
113    # a fixed point.
114    touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
115    touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
116    touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
117    # The two iterations of build_and_checks needs to complete
118    # during the same minute to have a fixed point.
119    build_and_checks "$@"
120    # build_and_checks "$@" second iteration below
121    # Then calling this script without --fixed-point-build should not
122    # yield any new superficial modifications.
123 fi
124
125 build_and_checks "$@"

```

3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
1  #!/usr/bin/php
2  <?php
3
4  /**
5   This file is part of DevOrSysAdminScripts.
6
7   DevOrSysAdminScripts is free software:
8   you can redistribute it and/or modify it
9   under the terms of the GNU Lesser General Public License
10  as published by the Free Software Foundation,
11  either version 3 of the License,
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25
26  ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
27
28  @category Executable
29  @package DevOrSysAdminScripts
30  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31  @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33  */
34
35  declare(strict_types=1);
36  declare(encoding='UTF-8');
37
38  require_once 'string_escaping.lib.php';
39  require_once 'build_dependencies_notes.lib.php';
40
41  build_dependencies_notes($argv[1]);
42  ?>
```

3.3 build_and_checks_dependencies/build_dependencies_notes.lib.php

```
1  <?php
2
3  /**
4   This file is part of DevOrSysAdminScripts.
5
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23  If not, see <https://www.gnu.org/licenses/>.
24
25  ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26  2026-03-26T11:59+01:00: This file was renamed from
27  "build_dependencies_notes.exec.php"
28  to
29  "build_dependencies_notes.lib.php".
30
31  @category Library
32  @package DevOrSysAdminScripts
33  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
34  @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
35  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
36  */
37
38  declare(strict_types=1);
39  declare(encoding='UTF-8');
40
41  // require_once 'string_escaping.lib.php';
42
43
44
```



```

45  /**
46  This function looks at the dependencies data found in a work directory,
47  and generate three files: a txt file, a TeX file (.sub),
48  and an HTML file (.sub) with the data in a nicer format.
49
50  @param string $s_work_directory The directory where the work must be done.
51
52  @return void
53  */
54  function build_dependencies_notes(string $s_work_directory) : void {
55      $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
56      $s_subdir2 = $s_subdir1.'/temp';
57      $s_data_file_path = $s_subdir1.'/dependencies_data.json';
58      $s_data = file_get_contents($s_data_file_path);
59      $arr_data = json_decode($s_data, true);
60      $s_data_file_path2 = $s_subdir1.'/repository_data.txt';
61      $s_data2 = file_get_contents($s_data_file_path2);
62      $i_max_line_length = 70;
63      foreach(explode("\n", $s_data2) as $s_line){
64          if(str_starts_with($s_line, 'max_line_length=')){
65              $i_max_line_length = (
66                  (int) substr($s_line, strlen('max_line_length='))
67              );
68              break;
69          }
70      }
71
72      $s_result_content1 = '';
73      $s_result_content2 = '';
74      $s_result_content3 = '';
75
76      foreach($arr_data as $arr_dependency){
77          $s_URL_word = 'URL';
78          if(count($arr_dependency['URLs']) > 1){
79              $s_URL_word .= 's';
80          }
81
82          $s_result_content3 .= "<p>\n";
83
84          $s_dependency1 = $arr_dependency['dependency_name'];
85          $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
86          $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);
87
88          $s_sentence_end = ' can be found at the following '.$s_URL_word.' :';
89          $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
90          $s_result_content2 .= $s_dependency2.$s_sentence_end."\n\n";

```

```

91 $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";
92
93 for(
94     $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
95     $i <= $i_max;
96     ++$i
97 ) {
98     $mixed_URL = $arr_dependency['URLs'][$i];
99     $s_comment = '';
100    $s_URL = '';
101    if(is_array($mixed_URL)) {
102        $s_URL = $mixed_URL['URL'];
103        $s_comment = $mixed_URL['comment'];
104    }
105    else {
106        $s_URL = $mixed_URL;
107    }
108    $i_URL_length = strlen($s_URL);
109    $i_comment_length = strlen($s_comment);
110
111    $s_result_content1 .= $s_URL;
112    $i_previous_line_length = $i_URL_length;
113    if($i_comment_length > 0) {
114        if(
115            $i_previous_line_length + $i_comment_length + 1
116            > $i_max_line_length
117        ) {
118            $s_result_content1 .= "\n";
119        }
120        else {
121            $s_result_content1 .= ' ';
122        }
123        $s_result_content1 .= $s_comment;
124    }
125    $s_result_content1 .= "\n";
126
127    if($i_URL_length > 60) {
128        $s_result_content2 .= (
129            "{\\small\\n"
130            . '  \url{'. $s_URL ."}\n"
131            . '}'
132        );
133        $i_previous_line_length = 1;
134    }
135    else {
136        $s_result_content2 .= '\url{'. $s_URL .'}';

```

```

137     $i_previous_line_length = $i_URL_length + 6;
138 }
139 if($i === $i_max){
140     ++$i_previous_line_length;
141 }
142 else{
143     $i_previous_line_length += 9;
144 }
145 $s_comment2 = string_escaping\TeX\escape_text($s_comment);
146 $i_comment_length2 = strlen($s_comment2);
147 if($i_comment_length2 > 0){
148     if(
149         $i_previous_line_length + $i_comment_length2 + 1
150         > $i_max_line_length
151     ){
152         $s_result_content2 .= "\n";
153     }
154     else{
155         $s_result_content2 .= ' ';
156     }
157     $s_result_content2 .= $s_comment2;
158 }
159 if($i === $i_max){
160     $s_result_content2 .= ".\n";
161 }
162 else{
163     $s_result_content2 .= ",\\newline\n";
164 }
165
166 $s_result_content3 .= (
167     "<a\n"
168     . 'href="' . string_escaping\HTML\escape_attribute($s_URL) . "\"\n"
169     . "><span style=\"font-family:monospace\"\n"
170     . '>' . string_escaping\HTML\escape_text($s_URL) . "</span\n"
171     . '></a>'
172 );
173 if($s_comment !== ''){
174     $s_result_content3 .= (
175         ' ' . string_escaping\HTML\escape_text($s_comment)
176     );
177 }
178 if($i === $i_max){
179     $s_result_content3 .= ".\n";
180 }
181 else{
182     $s_result_content3 .= "<br>\n";

```

```

183     }
184 }//end for urls
185
186 $s_result_content1 .= "\n";
187 $s_result_content2 .= "\n";
188 $s_result_content3 .= "</p>\n\n";
189 }//end foreach($arr_data as $arr_dependency)
190
191 // Add a "footer".
192 $s_closing_sentence = (
193     "If you see something that I missed regarding my dependencies,\n"
194     ".please tell me/email me :).\n"
195 );
196 $s_result_content1 .= $s_closing_sentence;
197 $s_result_content2 .= $s_closing_sentence;
198 $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";
199
200 // We overwrite results if needed.
201 $s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
202 $s_existing_result_content1 = '';
203 if(file_exists($s_result_path1)){
204     $s_existing_result_content1 = file_get_contents($s_result_path1);
205 }
206 if($s_result_content1 != $s_existing_result_content1){
207     file_put_contents($s_result_path1, $s_result_content1);
208 }
209 /*
210 Example of generated content of dependencies_notes.txt:
211 -----
212 more of util-linux can be found at the following URLs:
213 https://www.kernel.org/pub/linux/utils/util-linux/
214 https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
215 https://github.com/util-linux/util-linux official?
216 -----
217 */
218 $s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
219 $s_existing_result_content2 = '';
220 if(file_exists($s_result_path2)){
221     $s_existing_result_content2 = file_get_contents($s_result_path2);
222 }
223 if($s_result_content2 != $s_existing_result_content2){
224     file_put_contents($s_result_path2, $s_result_content2);
225 }
226 /*
227 Example of generated content of dependencies_notes.tex.sub:
228 -----

```

```

229 more of util-linux can be found at the following URLs:\newline
230 \url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
231 {\small
232   \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
233 },\newline
234 \url{https://github.com/util-linux/util-linux} official?.
235 -----
236 */
237 $$result_path3 = $$subdir2.'/dependencies_notes.html.sub';
238 $$existing_result_content3 = '';
239 if (file_exists($$result_path3)) {
240   $$existing_result_content3 = file_get_contents($$result_path3);
241 }
242 if ($$result_content3 != $$existing_result_content3) {
243   file_put_contents($$result_path3, $$result_content3);
244 }
245 /*
246 Example of generated content of dependencies_notes.html.sub:
247 <p>
248 more of util-linux can be found at the following URLs:<br>
249 <a
250 href="https://www.kernel.org/pub/linux/utils/util-linux/"
251 ><span style="font-family:monospace"
252 >https://www.kernel.org/pub/linux/utils/util-linux/</span
253 ></a>,<br>
254 <a
255 href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
256 ><span style="font-family:monospace"
257 >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
258 ></a>,<br>
259 <a
260 href="https://github.com/util-linux/util-linux"
261 ><span style="font-family:monospace"
262 >https://github.com/util-linux/util-linux</span
263 ></a> official?.
264 </p>
265 */
266 } //end build_dependencies_notes()
267 ?>

```

3.4 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```

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2 # This file is part of DevOrSysAdminScripts.
3 #
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```

```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "build_md_from_printable_md.sh"
25 # to "build_md_from_printable_md.exec.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=overwrite_if_not_equal.lib.sh
31 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
32
33 build_md_from_printable_md() {
34     # Options:
35     #   --verbose
36     #   --work-directory=""
37     #   --base-name="README"
38     declare -i LFBFL_i_verbose=0
39     get_verbose_option "$@"
40     local LFBFL_work_directory=""
41     get_work_directory_option "$@"
42     declare -a LFBFL_return_traps_stack
43     init_return_trap
44     pushd_to_work_directory --trap-popd
45     can_continue_after_enhanced_pushd || return 1
46
47     local LFBFL_base_name
48     get_some_option LFBFL_base_name --base-name README README 1 "$@"
49     if [[ LFBFL_i_verbose -eq 1 && "${LFBFL_base_name}" != "README" ]]; then
50         printf "Searching md file: %s.\n" "${LFBFL_base_name}"

```

```

51 fi
52
53 # Remove line returns here to keep lines short.
54 local LFBFL_sed_expression
55 LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]): [^\n\\]*\\n/\\1/Mg'
56 LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*\\n/\\1/Mg'
57 LFBFL_sed_expression+=';s/(\n[ ]*-<http[^\n\\]*\\n/\\1/Mg'
58 readonly LFBFL_sed_expression
59 # Since Markdown can contain code, it is limited to Markdown URLs.
60 # Hence, it is way more complicated than 's/\\n/Mg' expression
61 # in other files.
62 # Another downside is that more than one pass is needed.
63
64 if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
65     sed --regexp-extended --null-data\
66         --expression="${LFBFL_sed_expression}"\
67         --expression="${LFBFL_sed_expression}"\
68         --expression="${LFBFL_sed_expression}"\
69         --expression="${LFBFL_sed_expression}"\
70         -- "${LFBFL_base_name}.md.tpl"\
71         > "${LFBFL_base_name}.md.temp"
72     overwrite_if_not_equal "${LFBFL_base_name}.md"\
73         "${LFBFL_base_name}.md.temp"
74 else
75     printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
76     # Unfortunately, we cannot return early, since we want to propose
77     # HTML file generation with pandoc from .md file,
78     # if the file .md.tpl doesn't exist.
79 fi
80
81 if [[ ! -f "${LFBFL_base_name}.md" ]]; then
82     printf "No file %s.md\n" "${LFBFL_base_name}"
83     return 1
84 fi
85
86 pandoc --from=markdown --to=html --standalone\
87     --shift-heading-level-by=-1 --output="${LFBFL_base_name}.html.temp"\
88     -- "${LFBFL_base_name}.md"
89 # Correcting pandoc HTML "by hand".
90 # https://github.com/jgm/pandoc/issues/10957
91 # With the most open-minded answer found on Internet ;) XD.
92 # This second sed expression is what I need,
93 # but it may be unsuitable for you.
94 # Other problems:
95 # https://github.com/jgm/pandoc/issues/11484
96 local LFBFL_prewrap="s~(code \{(\n|[^}])*) ( \}) "

```

```

97 LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
98 readonly LFBFL_prewrap
99 sed --in-place --regexp-extended\
100     --expression='s~( <meta .*) />~\1>~g'\
101     --expression='s~<html .*>~<html lang="en">~'\
102     --expression='/      code\{white-space: pre-wrap;\}/d'\
103     -- "${LFBFL_base_name}.html.temp"
104 sed --in-place --regexp-extended --null-data\
105     --expression='s~(<img(\n|[^a-z0-9]) (\n|[^>]) *) />~\1>~Mg'\
106     --expression="${LFBFL_prewrap}"\
107     -- "${LFBFL_base_name}.html.temp"
108 overwrite_if_not_equal "${LFBFL_base_name}.html"\
109     "${LFBFL_base_name}.html.temp"
110
111 # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
112 #     --verbose "${LFBFL_base_name}.html"
113 }
114
115 build_md_from_printable_md "$@"

```

3.5 build_and_checks_dependencies/check_shell_scripts_beginnings.lib.sh

```

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23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "check_shell_scripts_beginnings.sh"

```



```

25 # to "check_shell_scripts_beginnings.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=lines_filters.lib.sh
31 source "${LFBFL_subdir}/lines_filters.lib.sh"
32
33 declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"
34
35 check_one_shell_script_beginning() {
36     # $1=file_path
37     #
38     # declare -r LFBFL_file_path=$(basename -- "$1")
39     # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
40     #     return 0
41     # fi
42     # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
43     # -----
44     # is of course slower than
45     # -----
46     # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
47     # -----
48     # is of course slower than
49     # -----
50     declare -r LFBFL_head=$(head --lines=1 -- "$1")
51     [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
52     || printf "%s:File has wrong shell script beginning.\n" "$1"
53 }
54
55 check_shell_scripts_beginnings() {
56     # Options:
57     # --verbose
58     # --work-directory=""
59     declare -i LFBFL_i_verbose=0
60     get_verbose_option "$@"
61     local LFBFL_work_directory=""
62     get_work_directory_option "$@"
63     declare -a LFBFL_return_traps_stack
64     init_return_trap
65     pushd_to_work_directory --trap-popd
66     can_continue_after_enhanced_pushd || return 1
67
68     # shopt -s globstar
69     # for LFBFL_file_path in **/*.sh; do
70     local LFBFL_file_path

```

```

71 local LFBFL_head
72 declare -r LFBFL_s_files_paths=$(
73     find . -type f -name "*.sh" -printf '%P\n'\
74     | relevant_find
75 )
76 if [[ -z "${LFBFL_s_files_paths}" ]]; then
77     [[ LFBFL_i_verbose -eq 1 ]]\
78     && printf "check_shell_scripts_beginnings: No file found.\n"
79     return
80 fi
81 declare -a LFBFL_arr_files_paths
82 mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
83 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
84     [[ LFBFL_i_verbose -eq 1 ]]\
85     && printf "%s:Checking shell script beginning.\n"\
86         "${LFBFL_file_path}"
87     check_one_shell_script_beginning "${LFBFL_file_path}"
88 done
89 }

```

3.6 build_and_checks_dependencies/check_shell_scripts_indentation.lib.sh

```

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24

```

```

25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.lib.sh
27 source "${LFBFL_subdir}/common_options.lib.sh"
28 # shellcheck source=lines_filters.lib.sh
29 source "${LFBFL_subdir}/lines_filters.lib.sh"
30
31 check_one_shell_script_indentation() {
32     # $1=file_path
33     declare -r LFBFL_NULL_PREVIOUS_LINE="--"
34     declare -i LFBFL_i_file_with_error=0
35     local LFBFL_some_line
36     local LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
37     local LFBFL_line_end
38     local LFBFL_previous_line_end
39     declare -i LFBFL_i_line_length
40     declare -i LFBFL_i_line_end_length
41     declare -i LFBFL_i_line_start_length
42     local LFBFL_spacing
43     declare -i LFBFL_i_offset
44     local LFBFL_substring1
45     local LFBFL_substring2
46     declare -r LFBFL_pattern='^ *-e '
47
48     # This grep would have been enough without some sed commands.
49     declare -r LFBFL_s_lines=$(
50         grep --before-context=1 --extended-regexp --line-number\
51             --regexp='^ ( )* ([^ ]|$)' --with-filename -- "$1"
52     )
53     if [[ -z "${LFBFL_s_lines}" ]]; then
54         return
55     fi
56
57     declare -a LFBFL_arr_lines
58     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
59     readonly LFBFL_arr_lines
60
61     for LFBFL_some_line in "${LFBFL_arr_lines[@]"; do
62         if [[ "${LFBFL_some_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]]; then
63             # When switching between code fragments, we reinit.
64             # It cannot happen for other grep output lines with line-number.
65             LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
66             continue
67         fi
68         # Note how it's funny that this loop always work because we check
69         # shell scripts beginnings in another script and thus we always
70         # have a line before.

```

```

71     if [[ "${LFBFL_previous_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]];
72     then
73         LFBFL_previous_line="${LFBFL_some_line}"
74         continue
75     fi
76     # Remove the name of the file and then the line number.
77     LFBFL_line_end=${LFBFL_some_line#*:}
78     LFBFL_line_end=${LFBFL_line_end#*:}
79     # But for the previous line '-' or ':' is used instead of ':'
80     # and filenames with '-' are much more frequent than with ':'...
81     LFBFL_i_line_length=${#LFBFL_some_line}
82     LFBFL_i_line_end_length=${#LFBFL_line_end}
83     LFBFL_i_line_start_length=$((
84         LFBFL_i_line_length - LFBFL_i_line_end_length
85     ))
86     LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
87     # Checking if increasing number of digits in the line numbers
88     # (99 -> 100).
89     LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
90     # Nice puzzle below :) Look for comments above :)
91     case ${LFBFL_substring1} in
92         -) ((LFBFL_i_offset += 1));;
93         :) ((LFBFL_i_offset += 1));;
94         [0-9]) ((LFBFL_i_offset += 2));;
95         *) printf ":-(\n";;
96     esac
97     LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
98     if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
99         # Line starts with spacing and -e
100         # Keep max spaces
101         LFBFL_spacing=${LFBFL_line_end%%-*}
102         LFBFL_i_offset=${#LFBFL_spacing}
103         LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
104         LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
105         if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
106             printf "%s\n" "${LFBFL_some_line}"
107             LFBFL_i_file_with_error=1
108         fi
109     else
110         printf "%s\n" "${LFBFL_some_line}"
111         LFBFL_i_file_with_error=1
112     fi
113     LFBFL_previous_line="${LFBFL_some_line}"
114 done
115
116 if [[ LFBFL_i_file_with_error -eq 1 ]]; then

```

```

117     printf "%s:File has some lines with odd number of spaces.\n" "$1"
118 fi
119 }
120
121 check_shell_scripts_indentation(){
122     # Options:
123     #   --verbose
124     #   --work-directory=""
125     declare -i LFBFL_i_verbose=0
126     get_verbose_option "$@"
127     local LFBFL_work_directory=""
128     get_work_directory_option "$@"
129     declare -a LFBFL_return_traps_stack
130     init_return_trap
131     pushd_to_work_directory --trap-popd
132     can_continue_after_enhanced_pushd || return 1
133
134     # shopt -s globstar
135     # for LFBFL_file_path in **/*.sh; do
136     local LFBFL_file_path
137     declare -r LFBFL_s_files_paths=$(
138         find . -type f -name "*.sh" -printf '%P\n'\
139         | relevant_find
140     )
141     if [[ -z "${LFBFL_s_files_paths}" ]]; then
142         [[ LFBFL_i_verbose -eq 1 ]]\
143         && printf "check_shell_scripts_indentation: No file found.\n"
144         return
145     fi
146     declare -a LFBFL_arr_files_paths
147     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
148     readonly LFBFL_arr_files_paths
149     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
150         [[ LFBFL_i_verbose -eq 1 ]]\
151         && printf "%s:Checking shell script indentation.\n"\
152             "${LFBFL_file_path}"
153         check_one_shell_script_indentation "${LFBFL_file_path}"
154     done
155 }

```

3.7 build_and_checks_dependencies/check_URLs.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:

```

```

5 # you can redistribute it and/or modify it
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18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "check_URLs.sh" to "check_URLs.lib.sh".
25
26 LFBFL_subdir="build_and_checks_dependencies"
27 # shellcheck source=common_options.lib.sh
28 source "${LFBFL_subdir}/common_options.lib.sh"
29 # shellcheck source=get_common_text_glob_patterns.lib.sh
30 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
31 # shellcheck source=lines_filters.lib.sh
32 source "${LFBFL_subdir}/lines_filters.lib.sh"
33
34 check_URLs() {
35     # Options:
36     #   --verbose
37     #   --work-directory=""
38     declare -i LFBFL_i_verbose=0
39     get_verbose_option "$@"
40     local LFBFL_work_directory=""
41     get_work_directory_option "$@"
42     declare -a LFBFL_return_traps_stack
43     init_return_trap
44     pushd_to_work_directory --trap-popd
45     can_continue_after_enhanced_pushd || return 1
46
47     enhanced_set_shell_option pipefail --trap-unset
48
49     get_COMMON_TEXT_FILES_GLOB_PATTERNS
50

```

```

51 declare -Ar LFBFL_substitutions=(
52     ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
53 )
54
55 local LFBFL_pattern
56 local LFBFL_base_file_name
57 local LFBFL_substitution
58 local LFBFL_substitution2
59 local LFBFL_file_path
60 local LFBFL_s_files_paths
61 declare -a LFBFL_arr_files_paths
62 for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
63     if [[ LFBFL_i_verbose -eq 1 ]]; then
64         printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
65     fi
66     find . -type f -name "${LFBFL_pattern}" -printf '%P\n' \
67         | xargs grep --with-filename -- 'http:' \
68         | grep --invert-match "|" xargs grep --with-filename -- 'htt'" \
69         | grep --invert-match --perl-regexp \
70             "(['\"])\http(:[^'\"]*)\\1.*(['\"])\https\\2\\3\" \
71         | grep --invert-match 'http://www.w3.org/1999/xhtml' \
72         | grep --invert-match \
73             --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd' \
74             --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd' \
75             --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd' \
76             --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd' \
77         # Second grep after xargs grep just above will remove false positives
78         # from substitutions that fit on one line.
79
80     LFBFL_s_files_paths=$(
81         find . -type f -name "${LFBFL_pattern}" -printf '%P\n' \
82         | relevant_find \
83         | not_main_html_find
84     )
85     if [[ -z "${LFBFL_s_files_paths}" ]]; then
86         [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
87         continue
88     fi
89     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
90     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
91         LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
92         [[ "${LFBFL_base_file_name}" != "check_URLs.lib.sh" ]] \
93         || continue
94         if [[ LFBFL_i_verbose -eq 1 ]]; then
95             printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
96         fi

```

```

97     for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
98         LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
99         if grep --quiet "${LFBFL_substitution}" -- "${LFBFL_file_path}";
100         then
101             sed --in-place\
102                 --expression="s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
103                 -- "${LFBFL_file_path}"
104         fi
105     done
106 done
107 done
108 }

```

3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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11 # DevOrSysAdminScripts is distributed
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18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "common_build_and_checks.exec.sh.tpl" was renamed from
25 # "common_build_and_checks.sh.tpl" to
26 # "common_build_and_checks.exec.sh.tpl".
27 # The file "common_build_and_checks.exec.sh" generated from the file
28 # "common_build_and_checks.sh.tpl"
29 # or "common_build_and_checks.exec.sh.tpl"
30 # was renamed from
31 # "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

```



```

32
33 common_build_and_checks() {
34     # $1 LFBFL_work_directory
35     # $2 LFBFL_dependencies_URL
36     # Options:
37     #   --check-download=y|keep (=keep to keep the downloaded files)
38     #   --verbose
39     local LFBFL_work_directory="${1:-.}"
40     LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
41     declare -r LFBFL_dependencies_URL="$2"
42     local LFBFL_verbose=""
43     declare -i LFBFL_i_verbose=0
44     if [[ "$*" == *--verbose* ]]; then
45         printf "%s %s\n" "$0" "$*"
46         LFBFL_verbose="--verbose"
47         LFBFL_i_verbose=1
48     fi
49     readonly LFBFL_verbose
50     readonly LFBFL_i_verbose
51     local LFBFL_check_download=""
52     if [[ "$*" == *--check-download=keep* ]]; then
53         LFBFL_check_download="--check-download=keep"
54     elif [[ "$*" == *--check-download=y* ]]; then
55         LFBFL_check_download="--check-download=y"
56     fi
57     readonly LFBFL_check_download
58
59     declare -a LFBFL_some_common_options=(
60         "${LFBFL_verbose}"
61         "--work-directory=${LFBFL_work_directory}"
62     )
63
64     source ./wget_sha512.lib.sh
65
66     local LFBFL_subdir="build_and_checks_dependencies"
67     declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
68     declare -r LFBFL_subdir3="build_and_checks_variables"
69     local LFBFL_file_name
70     local LFBFL_script_download_URL
71     local LFBFL_file_path
72     local LFBFL_correct_sha512
73
74     wrapped_wget_sha512() {
75         wget_sha512 "${LFBFL_file_path}" \
76             "${LFBFL_script_download_URL}" \
77             "${LFBFL_correct_sha512}" \

```

```

78     "${LFBFL_verbose}"\
79     "${LFBFL_check_download}"
80 }
81
82 LFBFL_file_name="build_dependencies_notes.exec.php"
83 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
84 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
85 LFBFL_correct_sha512='7fa2566b939e765b70bc48b8b15e0dab50823b50bae40'
86 LFBFL_correct_sha512+='97eeaac9f573139f60427f1d2c02fcf6a1bb4f89e21f'
87 LFBFL_correct_sha512+='ca39fd98859a5f4714b6359588e10ef96ec623d'
88 wrapped_wget_sha512
89 chmod +x "${LFBFL_file_path}"
90
91 LFBFL_file_name="build_dependencies_notes.lib.php"
92 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
93 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
94 LFBFL_correct_sha512='3136857d20aafaabdbcd2468012a55991518430a39a07'
95 LFBFL_correct_sha512+='e2f80572c39e51bda8599ac830cdb9ce83cbb606d034'
96 LFBFL_correct_sha512+='f971f8734b103feae78d48867a29d4d566f2201'
97 wrapped_wget_sha512
98
99 LFBFL_file_name="build_md_from_printable_md.exec.sh"
100 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
101 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
102 LFBFL_correct_sha512='500e10ac1b95a41b4d8c045bc7347a7473e7392168d3d'
103 LFBFL_correct_sha512+='7f40f3d8a08e8fcb1f5229e548729684e33687afa759'
104 LFBFL_correct_sha512+='34330727bf8b1884d45fda03a5407c9a6d798ef'
105 wrapped_wget_sha512
106 chmod +x "${LFBFL_file_path}"
107
108 LFBFL_file_name="check_shell_scripts_beginnings.lib.sh"
109 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
110 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
111 LFBFL_correct_sha512='0b3e861820e09148a141142c8bb5549e7fa29d906dda9'
112 LFBFL_correct_sha512+='5e0fe44132610910687e6f767f1050646abelea262b8'
113 LFBFL_correct_sha512+='37b3a5d5e79497d0cdc42e43d3c74967f266abb'
114 wrapped_wget_sha512
115
116 LFBFL_file_name="check_shell_scripts_indentation.lib.sh"
117 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
118 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
119 LFBFL_correct_sha512='3596fddc4b20ce353facac3e355e38fadb5593caee3c8'
120 LFBFL_correct_sha512+='05464aedd513d34860a79b296efa648dae18717ae4c0'
121 LFBFL_correct_sha512+='24a6df11eb0549c4797643427c49c4f80947840'
122 wrapped_wget_sha512
123

```

```

124 LFBFL_file_name="check_URLs.lib.sh"
125 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
126 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
127 LFBFL_correct_sha512='08561321cb3dee406f492dc29bf3a622036fb7094708d'
128 LFBFL_correct_sha512+='cc25ead2327eb183b4cf6fe2979494e21eb1d05b0e58'
129 LFBFL_correct_sha512+='9f8ce6ba08e190e81f935dc820bd2f2d3041a36'
130 wrapped_wget_sha512
131
132 LFBFL_file_name="common_options.lib.sh"
133 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
134 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
135 LFBFL_correct_sha512='3cfaf6cea99f67ade931cf21e9f49658c421e182dc4ac'
136 LFBFL_correct_sha512+='211be526f9d99b2e57a09844991cf0d2e6d5e49c19e2'
137 LFBFL_correct_sha512+='a63bf9e8f1e3fe999542e9a4e45e5196b375fa9'
138 wrapped_wget_sha512
139
140 LFBFL_file_name="comparisons.lib.sh"
141 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
142 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
143 LFBFL_correct_sha512='f146beca0d7b4046c140f3b2292cec4c31515cf8c7dbe'
144 LFBFL_correct_sha512+='7e9b07fc6dae32513aff6692c6286fa31055caff1a20'
145 LFBFL_correct_sha512+='9a6a7384568fea0b50b3cbb2608a3ed9e6e5125'
146 wrapped_wget_sha512
147
148 LFBFL_file_name="create_PDF.exec.sh"
149 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
150 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
151 LFBFL_correct_sha512='af85eaa0bedae5519bf02e44cf31d825d18f4c6248b9f'
152 LFBFL_correct_sha512+='b256832edacbee60deff90b74ea5935104b246e526de'
153 LFBFL_correct_sha512+='8fd5628649e464dd730b5923d35b53008d16487'
154 wrapped_wget_sha512
155 chmod +x "${LFBFL_file_path}"
156
157 LFBFL_file_name="generate_from_template.lib.sh"
158 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
159 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
160 LFBFL_correct_sha512='5f322f27bd20d7042c729f4cb5d8f14f587b5b415c77e'
161 LFBFL_correct_sha512+='4a224902d8d4cdeb1483639ae85d81873c9f20c175bb'
162 LFBFL_correct_sha512+='4de76a6eaa3dbcbf8a300368e65bbd5303afd3d'
163 wrapped_wget_sha512
164
165 LFBFL_file_name="get_common_text_glob_patterns.lib.sh"
166 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
167 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
168 LFBFL_correct_sha512='6c2f4b4ddc270c3d2ce9c88327efcb17510f10cac9b0f'
169 LFBFL_correct_sha512+='88ca3b585dcba2d7997bed299a97102bff194fccae8c'

```

```

170 LFBFL_correct_sha512+= '771db6e735412ec6d0f9b3ccbe8416f95e79dfa'
171 wrapped_wget_sha512
172
173 LFBFL_file_name="grammar_and_spelling_check.lib.sh"
174 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
175 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
176 LFBFL_correct_sha512='88cc4f031c05d576a74ea7b14e79077f06c53b3b8a294'
177 LFBFL_correct_sha512+= 'e9b8ddcfe884db5e2cb5cebde09a8d9aea7980fb595c'
178 LFBFL_correct_sha512+= '85c562462fb67e10503a471ee46da89c8f73b47'
179 wrapped_wget_sha512
180
181 # /licenses_templates/ -----
182 local LFBFL_dependencies_URL2
183 LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
184 readonly LFBFL_dependencies_URL2
185 LFBFL_file_name="build_licenses_templates.exec.sh"
186 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
187 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
188 LFBFL_correct_sha512='12c9f0f5ceb7bf1baa36b9e8ca7ff09d0c469c5d50fef'
189 LFBFL_correct_sha512+= '4da112efcd2848346eda6c547a053574ec5762e56f8b'
190 LFBFL_correct_sha512+= '4e644932de02948bc3178bce92199ed52aeb7a0'
191 wrapped_wget_sha512
192 chmod +x "${LFBFL_file_path}"
193
194 LFBFL_file_name="license_file_header_AGPLv3+.tpl"
195 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
196 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
197 LFBFL_correct_sha512='5dc0dc20914c57feee2e26c2d329f020f221357b11b0d'
198 LFBFL_correct_sha512+= '0ff6add69941ad953f8356b68be32cd60df8c56e91a2'
199 LFBFL_correct_sha512+= '716c9cbf70ab8698e74e6aacc9c5439a93cc2bc'
200 wrapped_wget_sha512
201
202 LFBFL_file_name="license_file_header_GPLv3+.tpl"
203 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
204 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
205 LFBFL_correct_sha512='2d4b31697e78ba36ee0e3bd2b5393a57e15b5f5e62b2e'
206 LFBFL_correct_sha512+= '802b812dcb087588f63cb92a851b21180be24bf5dee4'
207 LFBFL_correct_sha512+= 'eff7927244cde817e964512eaa1fb51d9064c01'
208 wrapped_wget_sha512
209
210 LFBFL_file_name="license_file_header_LGPLv3+.tpl"
211 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
212 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
213 LFBFL_correct_sha512='43976b6dc9a54fcea0440cfaf87530dade73efe22b5ae'
214 LFBFL_correct_sha512+= 'a968453c9c5ac1fdc4d3798fdd445fb38a53f997da82'
215 LFBFL_correct_sha512+= '7c30be6b7ad42d5e56ae90aac5095b9526b203d'

```

```

216 wrapped_wget_sha512
217 # -----
218
219 LFBFL_file_name="lines_counts.lib.sh"
220 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
221 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
222 LFBFL_correct_sha512='46366ca30eb8333cbab85576746579f34d1b53bc4583c'
223 LFBFL_correct_sha512+='0536373f3aa20f3483b274fa580a792af325337580f7'
224 LFBFL_correct_sha512+='0cf76c48fa6c1ff4f463f8fd918e39ec3d818a3'
225 wrapped_wget_sha512
226
227 LFBFL_file_name="lines_filters.lib.sh"
228 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
229 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
230 LFBFL_correct_sha512='854f138dc0d5a66d85169a1a079aae8592659e92a0f76'
231 LFBFL_correct_sha512+='f7ca52faec634cf0454957c81b4a9d4206bec5f4c441'
232 LFBFL_correct_sha512+='73f1934647a9c5ba55344cb82a617092f88ea8e'
233 wrapped_wget_sha512
234
235 LFBFL_file_name="lines_maps.lib.sh"
236 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
237 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
238 LFBFL_correct_sha512='ccb9115be582a5b39790725dc96a628beb21b0c9951e7'
239 LFBFL_correct_sha512+='03a7a7bc8ab2736e3940ed3df6a2fcb04e75350c10d7'
240 LFBFL_correct_sha512+='3e92bcfada424b4049cc6141d41bf7901c0525b'
241 wrapped_wget_sha512
242
243 LFBFL_file_name="overwrite_if_not_equal.lib.sh"
244 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
245 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
246 LFBFL_correct_sha512='378f03b8ba77430e05db4ef5d5926803b466a87e1a2ec'
247 LFBFL_correct_sha512+='eef0b51036f25460a93a018bd86d3f90146b4dfec9ab'
248 LFBFL_correct_sha512+='15c5d1d57a526296bbdc4a0c03650088f3e9653'
249 wrapped_wget_sha512
250
251 LFBFL_file_name="python_black_complement.lib.sh"
252 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
253 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
254 LFBFL_correct_sha512='4f08461329f1b78b58e54f675fb433aec363de2c96a91'
255 LFBFL_correct_sha512+='34ba5e96f3c77b93c9d7d6e28771a0b886501c286457'
256 LFBFL_correct_sha512+='a4a410389556f180f2dc8f66d7956bf5dd5acf2'
257 wrapped_wget_sha512
258
259 LFBFL_file_name="python_isort_complement.lib.sh"
260 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
261 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"

```

```

262 LFBFL_correct_sha512='570fd623c3d1db0caleb66f612aef77774b19652a8c6f'
263 LFBFL_correct_sha512+='8a579c3d3d7cb7412d6c60cc702d32bd25eafde4aa01'
264 LFBFL_correct_sha512+='b446a186ef4900c6bcc7773ffd59e539991c952'
265 wrapped_wget_sha512
266
267 LFBFL_file_name="repository_data.lib.sh"
268 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
269 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
270 LFBFL_correct_sha512='447d561896b60b6685389661f635ac726af8a2af83f42'
271 LFBFL_correct_sha512+='1d5dea89893dec90f0bba0d93903408056ef56a4ac12'
272 LFBFL_correct_sha512+='c985d5fa931bce90764dfc20253176f3b00dc54'
273 wrapped_wget_sha512
274
275 LFBFL_file_name="shell_checks_complement.lib.sh"
276 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
277 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
278 LFBFL_correct_sha512='143575d86b5fc0fd78e575b6e46054b0c8facaf2c6df6'
279 LFBFL_correct_sha512+='07dc596b6db0494c2b88211d5e0e011dee9abe70a2fc'
280 LFBFL_correct_sha512+='b8f2bc04eb5cd50b248b0afe423103ee1cd568b'
281 wrapped_wget_sha512
282
283 LFBFL_file_name="split_score.exec.php"
284 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
285 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
286 LFBFL_correct_sha512='53830be68f0ef9f1a0b9b16ald98d0a744a38b8fd0ee6'
287 LFBFL_correct_sha512+='68f592aba2df7ec6c856eb1ced9ced7f3d21251699f8'
288 LFBFL_correct_sha512+='7a352c8624d03ce5d5503692488c9e14a9ffb62'
289 wrapped_wget_sha512
290 chmod +x "${LFBFL_file_path}"
291
292 LFBFL_file_name="split_score.lib.php"
293 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
294 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
295 LFBFL_correct_sha512='e8333a215a02dd7250b01a5a8e8ec862b735735a52a95'
296 LFBFL_correct_sha512+='2c9339cba94ffc9c1605fa4d84487be100fdff644c2e'
297 LFBFL_correct_sha512+='e68cae3569fbc2b0a5525817b33e21425cd2e2f'
298 wrapped_wget_sha512
299
300 LFBFL_file_name="strings_functions.lib.sh"
301 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
302 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
303 LFBFL_correct_sha512='73ac813d69d07a53fdcf16938125b7a83c2fd20518809'
304 LFBFL_correct_sha512+='0d8b067ddc89a4a07783f6c0acb0a0aec802a42838b2d'
305 LFBFL_correct_sha512+='8b172f40390d53283103771f8d3b732f1c442a8'
306 wrapped_wget_sha512
307

```

```

308 LFBFL_file_name="too_long_code_lines.lib.sh"
309 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
310 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
311 LFBFL_correct_sha512='ded23acf2574ed706d0cle7e34e866e93e001f3bdfd71'
312 LFBFL_correct_sha512+=' 048a32cd99a197034283cbb55de3d2ble76e5e32529c'
313 LFBFL_correct_sha512+=' a9d4c9ea2bfd1924611d01f8b31b647b5078c4a'
314 wrapped_wget_sha512
315
316 LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
317 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
318 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
319 LFBFL_correct_sha512='188fa5dbcab63be8e0f5b3c8997e099c7259f7c804433'
320 LFBFL_correct_sha512+=' 581c2c05a7d9c7bd3d82bc424e2021afa90cb9e9d6a5'
321 LFBFL_correct_sha512+=' f84f549ed692c18b6351983c31b875846d7e1af'
322 wrapped_wget_sha512
323 chmod +x "${LFBFL_file_path}"
324
325 # shellcheck source=check_shell_scripts_beginnings.lib.sh
326 source "${LFBFL_subdir}/check_shell_scripts_beginnings.lib.sh"
327 # shellcheck source=check_shell_scripts_indentation.lib.sh
328 source "${LFBFL_subdir}/check_shell_scripts_indentation.lib.sh"
329 # shellcheck source=check_URLs.lib.sh
330 source "${LFBFL_subdir}/check_URLs.lib.sh"
331 # shellcheck source=common_options.lib.sh
332 source "${LFBFL_subdir}/common_options.lib.sh"
333 # shellcheck source=comparisons.lib.sh
334 source "${LFBFL_subdir}/comparisons.lib.sh"
335 # shellcheck source=generate_from_template.lib.sh
336 source "${LFBFL_subdir}/generate_from_template.lib.sh"
337 # shellcheck source=get_common_text_glob_patterns.lib.sh
338 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
339 # shellcheck source=grammar_and_spelling_check.lib.sh
340 source "${LFBFL_subdir}/grammar_and_spelling_check.lib.sh"
341 # shellcheck source=lines_counts.lib.sh
342 source "${LFBFL_subdir}/lines_counts.lib.sh"
343 # shellcheck source=lines_filters.lib.sh
344 source "${LFBFL_subdir}/lines_filters.lib.sh"
345 # shellcheck source=lines_maps.lib.sh
346 source "${LFBFL_subdir}/lines_maps.lib.sh"
347 # shellcheck source=overwrite_if_not_equal.lib.sh
348 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
349 # shellcheck source=python_black_complement.lib.sh
350 source "${LFBFL_subdir}/python_black_complement.lib.sh"
351 # shellcheck source=python_isort_complement.lib.sh
352 source "${LFBFL_subdir}/python_isort_complement.lib.sh"
353 # shellcheck source=shell_checks_complement.lib.sh

```

```

354 source "${LFBFL_subdir}/shell_checks_complement.lib.sh"
355 # shellcheck source=strings_functions.lib.sh
356 source "${LFBFL_subdir}/strings_functions.lib.sh"
357 # shellcheck source=too_long_code_lines.lib.sh
358 source "${LFBFL_subdir}/too_long_code_lines.lib.sh"
359
360 enhanced_set_bash_option extglob
361 # shellcheck source=repository_data.lib.sh
362 source "${LFBFL_subdir}/repository_data.lib.sh"
363 # shellcheck disable=SC2154,SC2309
364 [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
365     && enhanced_unset_bash_option extglob
366
367 declare -a LFBFL_return_traps_stack
368 init_return_trap
369
370 enhanced_set_shell_option pipefail --trap-unset
371
372 declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
373 keep_options LFBFL_some_common_options2 --verbose
374
375 printf "Building license headers\n"
376 "${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
377     "${LFBFL_some_common_options[@]}"
378
379 printf "Building README.md\n"
380 "${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
381     --work-directory="${LFBFL_work_directory}"\
382     --base-name=README\
383     "${LFBFL_verbose}"
384
385 printf "Building other Markdown files\n"
386 local LFBFL_some_directory
387 declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
388 local LFBFL_s_files_paths
389 LFBFL_s_files_paths=$(
390     find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
391     | relevant_find
392 )
393 declare -a LFBFL_arr_files_paths
394 if [[ -n "${LFBFL_s_files_paths}" ]]; then
395     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
396     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
397         [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
398         printf "Found template %s.\n" "${LFBFL_file_path}"
399         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")

```



```

400     LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
401     LFBFL_file_name=${LFBFL_file_name%.md.tpl}
402     "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
403         --work-directory="${LFBFL_some_directory}"\
404         --base-name="${LFBFL_file_name}"\
405         "${LFBFL_verbose}"
406     done
407 fi
408
409 printf "Building dependencies_notes\n"
410 # https://github.com/php/php-src/issues/21538
411 local LFBFL_annoying_warning
412 LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
413 LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
414 LFBFL_annoying_warning=" by settings in"
415 readonly LFBFL_annoying_warning
416 php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
417     -- "${LFBFL_work_directory}"\
418     2>&1\
419     | grep --invert-match "${LFBFL_annoying_warning}"
420
421 declare -i LFBFL_i_directory_changed
422 pushd_to_work_directory
423 LFBFL_i_directory_changed=$?
424 can_continue_after_enhanced_pushd || return 1
425
426 local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
427
428 declare -i LFBFL_i_max_line_length
429 grep_variable "${LFBFL_data_file_path}" max_line_length\
430     --result-variable-prefix=LFBFL_i_
431
432 LFBFL_some_common_options+=(
433     "--max-line-length=${LFBFL_i_max_line_length}"
434 )
435 LFBFL_some_common_options2+=(
436     "--max-line-length=${LFBFL_i_max_line_length}"
437 )
438
439 declare -i LFBFL_i_upgrade_venvs=0
440 local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
441 readonly LFBFL_upgrade_venvs_ts_file
442 declare -i LFBFL_i_upgrade_venvs_ts
443 declare -i LFBFL_i_current_ts
444 local LFBFL_upgrade_venvs_answer
445

```

```

446 local LFBFL_upgrade_venvs_time_interval_in_seconds=""
447 get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
448 "${LFBFL_some_common_options2[@]}"
449
450 if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
451     LFBFL_i_upgrade_venvs_ts=$(
452         stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
453     )
454     LFBFL_i_current_ts=$(date +%s)
455     ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
456     ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
457     if [[ LFBFL_i_current_ts -gt 0 ]]; then
458         LFBFL_i_upgrade_venvs=1
459     fi
460 else
461     LFBFL_i_upgrade_venvs=1
462 fi
463 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
464     local LFBFL_upgrade_venvs=""
465     grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
466         --result-variable-prefix=LFBFL_
467     if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
468         read -r -n 1 -t 10\
469             -p "Upgrade venvs and composer global? [Y/n]"\
470             LFBFL_upgrade_venvs_answer
471         if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
472             LFBFL_i_upgrade_venvs=0
473         fi
474     fi
475 fi
476
477 printf "Analyzing too long lines\n"
478 too_long_code_lines "${LFBFL_some_common_options2[@]}"
479
480 printf "Analyzing URLs\n"
481 check_URLs "${LFBFL_some_common_options2[@]}" \
482     | relevant_grep
483
484 printf "Analyzing strange characters: hover over in doubt\n"
485 # shellcheck disable=SC1111
486 LFBFL_usual_characters="\x00-\x7Fàâéèëîïôûç@\"`|«»..."
487 grep --binary-files=without-match --color=always --exclude-dir .git\
488     --line-number --perl-regexp --recursive --\
489     "[^${LFBFL_usual_characters}]" .\
490     | relevant_grep
491

```

```

492 printf "Grammar and spelling check\n"
493 grammar_and_spelling_check "${LFBFL_data_file_path}"\
494     "${LFBFL_some_common_options2[@]}"
495
496 printf "Running shellcheck\n"
497 LFBFL_s_files_paths=$(
498     find . -type f -name "*.sh"\
499     | relevant_find
500 )
501 if [[ -n "${LFBFL_s_files_paths}" ]]; then
502     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
503     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
504         shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
505             -- "${LFBFL_file_path}"
506     done
507 fi
508 printf "Running shell_checks_complement\n"
509 shell_checks_complement "${LFBFL_some_common_options2[@]}"
510
511 printf "Analyzing shell scripts beginnings\n"
512 check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
513     | relevant_grep
514
515 printf "Analyzing shell scripts indentation\n"
516 check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
517     | relevant_grep
518
519 printf -- "---Python---\n"
520
521 maybe_deactivate(){
522     if command -v deactivate
523     then
524         deactivate
525     fi
526 }
527
528 source_without_return_trap(){
529     declare -i LFBFL_i_no_return_traps=1
530     # shellcheck disable=SC1090,SC1091
531     source -- "$1"
532 }
533
534 printf "Running isort\n"
535 local LFBFL_isort_venv=""
536 grep_variable "${LFBFL_data_file_path}" isort_venv\
537     --result-variable-prefix=LFBFL_\

```

```

538     --replace-line-returns-by=""
539 if [[ -n "${LFBFL_isort_venv}" ]]; then
540     maybe_deactivate
541     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
542 fi
543 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
544     pip install --upgrade isort
545 fi
546 isort --profile=black .
547 if [[ -n "${LFBFL_isort_venv}" ]]; then
548     deactivate
549 fi
550 printf "Running python_isort_complement\n"
551 python_isort_complement "${LFBFL_some_common_options2[@]}"
552
553 printf "Running black\n"
554 # First, we update the configuration file with max_line_length.
555 local LFBFL_update_max_length="s/^line-length = [0-9]*$/\
556 LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/\
557 sed --regextp-extended --expression="${LFBFL_update_max_length}"\
558     "${LFBFL_subdir3}/black.toml"\
559     > "${LFBFL_subdir3}/black.toml.temp"
560 overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
561     "${LFBFL_subdir3}/black.toml.temp"
562
563 local LFBFL_black_venv=""
564 grep_variable "${LFBFL_data_file_path}" black_venv\
565     --result-variable-prefix=LFBFL_\
566     --replace-line-returns-by=""
567 if [[ -n "${LFBFL_black_venv}" ]]; then
568     maybe_deactivate
569     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
570 fi
571 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
572     pip install --upgrade black
573 fi
574 black --config="${LFBFL_subdir3}/black.toml" .
575 if [[ -n "${LFBFL_black_venv}" ]]; then
576     deactivate
577 fi
578 printf "Running python_black_complement\n"
579 python_black_complement "${LFBFL_some_common_options2[@]}"
580
581 printf "Probing if mypy should be runned\n"
582 local LFBFL_mypy_venv=""
583 grep_variable "${LFBFL_data_file_path}" mypy_venv\

```

```

584     --result-variable-prefix=LFBFL\
585     --replace-line-returns-by=""
586 if [[ -n "${LFBFL_mypy_venv}" ]]; then
587     maybe_deactivate
588     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
589 fi
590 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
591     pip install --upgrade mypy
592 fi
593 local LFBFL_directory_path
594 declare -i LFBFL_i_no_toml=1
595 LFBFL_s_files_paths=$(
596     find . -type f -name "pyproject.toml" \
597     | relevant_find
598 )
599 if [[ -n "${LFBFL_s_files_paths}" ]]; then
600     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
601     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"}; do
602         if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
603             printf "Running mypy\n"
604             LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
605             mypy -- "${LFBFL_directory_path}"
606         fi
607         LFBFL_i_no_toml=0
608     done
609 fi
610 if [[ LFBFL_i_no_toml -eq 1 ]]; then
611     printf "Running mypy\n"
612     mypy .
613 fi
614 if [[ -n "${LFBFL_mypy_venv}" ]]; then
615     deactivate
616 fi
617
618 printf "Running bandit\n"
619 LFBFL_bandit_venv=""
620 grep_variable "${LFBFL_data_file_path}" bandit_venv\
621     --result-variable-prefix=LFBFL\
622     --replace-line-returns-by=""
623 if [[ -n "${LFBFL_bandit_venv}" ]]; then
624     maybe_deactivate
625     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
626 fi
627 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
628     pip install --upgrade bandit
629 fi

```

```

630 bandit --ini="${LFBFL_subdir3}/bandit.ini"\
631       --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
632       --recursive .
633 # Saving new baseline in temp if necessary.
634 bandit --ini="${LFBFL_subdir3}/bandit.ini"\
635       --format=json\
636       --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
637       --recursive .
638 if [[ -n "${LFBFL_bandit_venv}" ]]; then
639     deactivate
640 fi
641
642 printf "Running pylint\n"
643 LFBFL_pylint_venv=""
644 grep_variable "${LFBFL_data_file_path}" pylint_venv\
645     --result-variable-prefix=LFBFL_\
646     --replace-line-returns-by=""
647 if [[ -n "${LFBFL_pylint_venv}" ]]; then
648     maybe_deactivate
649     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
650 fi
651 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
652     pip install --upgrade pylint
653 fi
654 pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
655 if [[ -n "${LFBFL_pylint_venv}" ]]; then
656     deactivate
657 fi
658
659 printf "Running ruff\n"
660 LFBFL_ruff_venv=""
661 grep_variable "${LFBFL_data_file_path}" ruff_venv\
662     --result-variable-prefix=LFBFL_\
663     --replace-line-returns-by=""
664 if [[ -n "${LFBFL_ruff_venv}" ]]; then
665     maybe_deactivate
666     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
667 fi
668 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
669     pip install --upgrade ruff
670 fi
671 ruff check --config="${LFBFL_subdir3}/ruff.toml"
672 if [[ -n "${LFBFL_ruff_venv}" ]]; then
673     deactivate
674 fi
675 printf -- "----Python end---\n"

```

```

676
677 printf -- "---PHP---\n"
678 printf "Running PHP native linter\n"
679 LFBFL_s_files_paths=$(
680     find . -type f -name "*.php"\
681     | relevant_find
682 )
683 if [[ -n "${LFBFL_s_files_paths}" ]]; then
684     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
685     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"}; do
686         php --syntax-check -f="${LFBFL_file_path}"
687     done
688 fi
689
690 printf "Running PHP_CodeSniffer\n"
691 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
692     composer global require squizlabs/php_codesniffer
693 fi
694 phpcs --report=code\
695     --standard=build_and_checks_variables/phpcs_ruleset.xml
696
697 printf "Running PHPMD\n"
698 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
699     composer global require phpmd/phpmd
700 fi
701
702 declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
703 local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
704 LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
705 readonly LFBFL_temp_phpmd_baseline
706
707 local LFBFL_phpmd_rulesets="cleancode, codesize, controversial, "
708 LFBFL_phpmd_rulesets+="design, naming, unusedcode"
709 readonly LFBFL_phpmd_rulesets
710
711 phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
712     . text "${LFBFL_phpmd_rulesets}"
713 # Saving new baseline in temp if necessary.
714 rm "${LFBFL_temp_phpmd_baseline}"
715 phpmd --color --generate-baseline\
716     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
717     . text "${LFBFL_phpmd_rulesets}"
718 sed --in-place --expression='s/" file="/"\n    file="/g'\
719     "${LFBFL_temp_phpmd_baseline}"
720 sed --in-place --regextp-extended\
721     --expression='s~(file=".*)/>~\1\n    />~g'\

```

```

722     "${LFBFL_temp_phpmd_baseline}"
723 diff --suppress-common-lines\
724     "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
725     | grep --invert-match --regexp='<!-- ' --regexp='^[0-9]\+d[0-9]\+$'
726
727 printf "Running phpDocumentor\n"
728 # shellcheck disable=SC2312
729 docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
730 printf -- "----PHP end---\n"
731
732 printf -- "----JS---\n"
733 local LFBFL_npm_lint_directories=""
734 grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
735     --result-variable-prefix=LFBFL_
736 if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
737     printf "Running ESLint\n"
738     local LFBFL_JS_directory
739     declare -a LFBFL_arr_paths
740     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
741     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
742         (cd -- "${LFBFL_JS_directory}" && npm run lint)
743     done
744 fi
745 printf -- "----JS end---\n"
746
747 [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
748
749 printf "Checking listed files\n"
750 "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
751     "${LFBFL_some_common_options[@]}"
752
753 printf "Creating the PDF file of the listing of the source code\n"
754 "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
755
756 pushd_to_work_directory --trap-popd
757 can_continue_after_enhanced_pushd || return 1
758
759 if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
760     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
761 fi
762
763 printf "Running xmllint\n"
764 # Unfortunately, we need to handle a cache of common DTDs by hand.
765 # XHTML DTDs validation yields errors currently:
766 # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
767 LFBFL_file_name="xhtml1-strict.dtd"

```



```

768 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
769 LFBFL_script_download_URL="{LFBFL_a}"
770 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
771 LFBFL_correct_sha512='fe11db06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
772 LFBFL_correct_sha512+='d8352bf19947168c9bcf1ae212eca035824293abb1f9'
773 LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
774 wrapped_wget_sha512
775
776 LFBFL_file_name="xhtml1-transitional.dtd"
777 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
778 LFBFL_script_download_URL="{LFBFL_a}"
779 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
780 LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
781 LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
782 LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
783 wrapped_wget_sha512
784
785 LFBFL_file_name="xhtml1-frameset.dtd"
786 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
787 LFBFL_script_download_URL="{LFBFL_a}"
788 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
789 LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
790 LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
791 LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
792 wrapped_wget_sha512
793
794 LFBFL_file_name="xhtml11.dtd"
795 LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
796 LFBFL_script_download_URL="{LFBFL_a}"
797 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
798 LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
799 LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
800 LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
801 wrapped_wget_sha512
802
803 LFBFL_file_name="xhtml-lat1.ent"
804 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
805 LFBFL_script_download_URL="{LFBFL_a}"
806 LFBFL_file_path=".{LFBFL_subdir3}/temp/{LFBFL_file_name}"
807 LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
808 LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
809 LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
810 wrapped_wget_sha512
811
812 LFBFL_file_name="xhtml-special.ent"
813 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"

```

```

814 LFBFL_script_download_URL="${LFBFL_a}"
815 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
816 LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
817 LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
818 LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
819 wrapped_wget_sha512
820
821 LFBFL_file_name="xhtml-symbol.ent"
822 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
823 LFBFL_script_download_URL="${LFBFL_a}"
824 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
825 LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
826 LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
827 LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
828 wrapped_wget_sha512
829
830 declare -r LFBFL_s_files_for_xmllint=$(
831     find . -type f -iregex ".*\\.\\(xml\\|xhtml\\)" \\
832     | relevant_find\\
833     | sort
834 )
835 if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
836     declare -a LFBFL_arr_files_for_xmllint
837     mapfile -t LFBFL_arr_files_for_xmllint\\
838     <<< "${LFBFL_s_files_for_xmllint}"
839     readonly LFBFL_arr_files_for_xmllint
840     local LFBFL_xmllint_files_without_DTD=""
841     grep_variable "${LFBFL_data_file_path}" xmllint_files_without_DTD\\
842     --result-variable-prefix=LFBFL_
843     declare -a LFBFL_arr_xmllint_files_without_DTD
844     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
845         mapfile -t LFBFL_arr_xmllint_files_without_DTD\\
846         <<< "${LFBFL_xmllint_files_without_DTD}"
847     fi
848     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
849         [[ LFBFL_i_verbose -eq 1 ]]\\
850         && printf "xmllint on %s\\n" "${LFBFL_file_path}"
851         # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
852         # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
853         # xmllint --pedantic --noout -- "${LFBFL_file_path}"
854         xmllint --pedantic --noout "${LFBFL_file_path}"
855     done
856
857     local LFBFL_file_path2
858     for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
859         for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";

```

```

860     do
861         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
862             LFBFL_file_path=""
863         fi
864     done
865     if [[ "${LFBFL_file_path}" == "" ]]; then
866         continue
867     fi
868     [[ LFBFL_i_verbose -eq 1 ]]\
869     && printf "xmllint on %s\n" "${LFBFL_file_path}"
870     if [[ "${LFBFL_file_path}" == *.html ]]; then
871         if grep --fixed-strings\
872             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
873             "${LFBFL_file_path}"
874         then
875             xmllint --pedantic --noout\
876                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
877                 "${LFBFL_file_path}"
878         elif grep --fixed-strings\
879             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
880             "${LFBFL_file_path}"
881         then
882             xmllint --pedantic --noout\
883                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
884                 "${LFBFL_file_path}"
885         elif grep --fixed-strings\
886             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
887             "${LFBFL_file_path}"
888         then
889             xmllint --pedantic --noout\
890                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
891                 "${LFBFL_file_path}"
892         elif grep --fixed-strings\
893             "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
894             "${LFBFL_file_path}"
895         then
896             xmllint --pedantic --noout\
897                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
898                 "${LFBFL_file_path}"
899         else
900             printf "Unknown XHTML DTD.\n"
901         fi
902     else
903         xmllint --pedantic --noout --valid "${LFBFL_file_path}"
904     fi
905 done

```

```

906 fi
907 # -----
908
909 # -----
910 # It is at the end that all the HTML files are generated and that
911 # we can check there are no mistakes.
912 printf "Running Nu W3C HTML CSS and SVG validator\n"
913 local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
914 grep_variable "${LFBFL_data_file_path}" \
915     upgrade_vnu_jar_time_interval_in_seconds \
916     --result-variable-prefix=LFBFL_
917
918 if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
919 then
920     LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
921 fi
922
923 declare -i LFBFL_i_vnu_jar_ts
924 local LFBFL_vnu_jar_path=""
925 grep_variable "${LFBFL_data_file_path}" vnu_jar_path \
926     --result-variable-prefix=LFBFL_ \
927     --replace-line-returns-by=""
928 if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
929     LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
930     LFBFL_i_current_ts=$(date +%s)
931     ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
932     ((
933         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
934     ))
935     if [[ LFBFL_i_current_ts -gt 0 ]]; then
936         printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\n'
937         printf "Go get the latest version here:\n"
938         printf "https://github.com/validator/validator/releases\n"
939     fi
940     # vnu has false positives for xhtml.
941     local LFBFL_use_vnu_for_xhtml=""
942     grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml \
943         --result-variable-prefix=LFBFL_
944     local LFBFL_find_regexp
945     if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
946         LFBFL_find_regexp=".*\.(css|html|svg|xhtml)"
947     else
948         LFBFL_find_regexp=".*\.(css|html|svg)"
949     fi
950     declare -r LFBFL_s_files_for_Nu=$(
951         find . -type f -iregex "${LFBFL_find_regexp}" \

```

```

952         | relevant_find
953     )
954     if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
955         declare -a LFBFL_arr_files_for_Nu
956         mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
957         readonly LFBFL_arr_files_for_Nu
958         java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
959             --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
960     fi
961 fi
962 # -----
963
964 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
965     touch -- "${LFBFL_upgrade_venvs_ts_file}"
966 fi
967 }
968
969 common_build_and_checks "$@"

```

3.9 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "common_build_and_checks.exec.sh.tpl" was renamed from
25 # "common_build_and_checks.sh.tpl" to

```

```

26 # "common_build_and_checks.exec.sh.tpl".
27 # The file "common_build_and_checks.exec.sh" generated from the file
28 # "common_build_and_checks.sh.tpl"
29 # or "common_build_and_checks.exec.sh.tpl"
30 # was renamed from
31 # "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".
32
33 common_build_and_checks() {
34     # $1 LFBFL_work_directory
35     # $2 LFBFL_dependencies_URL
36     # Options:
37     #   --check-download=y|keep (=keep to keep the downloaded files)
38     #   --verbose
39     local LFBFL_work_directory="${1:-.}"
40     LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
41     declare -r LFBFL_dependencies_URL="$2"
42     local LFBFL_verbose=""
43     declare -i LFBFL_i_verbose=0
44     if [[ "$*" == *--verbose* ]]; then
45         printf "%s %s\n" "$0" "$*"
46         LFBFL_verbose="--verbose"
47         LFBFL_i_verbose=1
48     fi
49     readonly LFBFL_verbose
50     readonly LFBFL_i_verbose
51     local LFBFL_check_download=""
52     if [[ "$*" == *--check-download=keep* ]]; then
53         LFBFL_check_download="--check-download=keep"
54     elif [[ "$*" == *--check-download=y* ]]; then
55         LFBFL_check_download="--check-download=y"
56     fi
57     readonly LFBFL_check_download
58
59     declare -a LFBFL_some_common_options=(
60         "${LFBFL_verbose}"
61         "--work-directory=${LFBFL_work_directory}"
62     )
63
64     source ./wget_sha512.lib.sh
65
66     local LFBFL_subdir="build_and_checks_dependencies"
67     declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
68     declare -r LFBFL_subdir3="build_and_checks_variables"
69     local LFBFL_file_name
70     local LFBFL_script_download_URL
71     local LFBFL_file_path

```

```

72     local LFBFL_correct_sha512
73
74     wrapped_wget_sha512() {
75         wget_sha512 "${LFBFL_file_path}" \
76             "${LFBFL_script_download_URL}" \
77             "${LFBFL_correct_sha512}" \
78             "${LFBFL_verbose}" \
79             "${LFBFL_check_download}"
80     }
81
82     LFBFL_file_name="build_dependencies_notes.exec.php"
83     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
84     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
85     @sha512_build_dependencies_notes.exec.php@
86     wrapped_wget_sha512
87     chmod +x "${LFBFL_file_path}"
88
89     LFBFL_file_name="build_dependencies_notes.lib.php"
90     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
91     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
92     @sha512_build_dependencies_notes.lib.php@
93     wrapped_wget_sha512
94
95     LFBFL_file_name="build_md_from_printable_md.exec.sh"
96     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
97     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
98     @sha512_build_md_from_printable_md.exec.sh@
99     wrapped_wget_sha512
100    chmod +x "${LFBFL_file_path}"
101
102    LFBFL_file_name="check_shell_scripts_beginnings.lib.sh"
103    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
104    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
105    @sha512_check_shell_scripts_beginnings.lib.sh@
106    wrapped_wget_sha512
107
108    LFBFL_file_name="check_shell_scripts_indentation.lib.sh"
109    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
110    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
111    @sha512_check_shell_scripts_indentation.lib.sh@
112    wrapped_wget_sha512
113
114    LFBFL_file_name="check_URLs.lib.sh"
115    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
116    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
117    @sha512_check_URLs.lib.sh@

```

```

118 wrapped_wget_sha512
119
120 LFBFL_file_name="common_options.lib.sh"
121 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
122 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
123 @sha512_common_options.lib.sh@
124 wrapped_wget_sha512
125
126 LFBFL_file_name="comparisons.lib.sh"
127 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
128 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
129 @sha512_comparisons.lib.sh@
130 wrapped_wget_sha512
131
132 LFBFL_file_name="create_PDF.exec.sh"
133 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
134 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
135 @sha512_create_PDF.exec.sh@
136 wrapped_wget_sha512
137 chmod +x "${LFBFL_file_path}"
138
139 LFBFL_file_name="generate_from_template.lib.sh"
140 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
141 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
142 @sha512_generate_from_template.lib.sh@
143 wrapped_wget_sha512
144
145 LFBFL_file_name="get_common_text_glob_patterns.lib.sh"
146 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
147 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
148 @sha512_get_common_text_glob_patterns.lib.sh@
149 wrapped_wget_sha512
150
151 LFBFL_file_name="grammar_and_spelling_check.lib.sh"
152 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
153 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
154 @sha512_grammar_and_spelling_check.lib.sh@
155 wrapped_wget_sha512
156
157 # /licenses_templates/ -----
158 local LFBFL_dependencies_URL2
159 LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
160 readonly LFBFL_dependencies_URL2
161 LFBFL_file_name="build_licenses_templates.exec.sh"
162 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
163 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"

```



```

164 @sha512_build_licenses_templates.exec.sh@
165 wrapped_wget_sha512
166 chmod +x "${LFBFL_file_path}"
167
168 LFBFL_file_name="license_file_header_AGPLv3+.tpl"
169 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
170 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
171 @sha512_license_file_header_AGPLv3+.tpl@
172 wrapped_wget_sha512
173
174 LFBFL_file_name="license_file_header_GPLv3+.tpl"
175 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
176 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
177 @sha512_license_file_header_GPLv3+.tpl@
178 wrapped_wget_sha512
179
180 LFBFL_file_name="license_file_header_LGPLv3+.tpl"
181 LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
182 LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
183 @sha512_license_file_header_LGPLv3+.tpl@
184 wrapped_wget_sha512
185 # -----
186
187 LFBFL_file_name="lines_counts.lib.sh"
188 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
189 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
190 @sha512_lines_counts.lib.sh@
191 wrapped_wget_sha512
192
193 LFBFL_file_name="lines_filters.lib.sh"
194 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
195 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
196 @sha512_lines_filters.lib.sh@
197 wrapped_wget_sha512
198
199 LFBFL_file_name="lines_maps.lib.sh"
200 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
201 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
202 @sha512_lines_maps.lib.sh@
203 wrapped_wget_sha512
204
205 LFBFL_file_name="overwrite_if_not_equal.lib.sh"
206 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
207 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
208 @sha512_overwrite_if_not_equal.lib.sh@
209 wrapped_wget_sha512

```

```

210
211 LFBFL_file_name="python_black_complement.lib.sh"
212 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
213 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
214 @sha512_python_black_complement.lib.sh@
215 wrapped_wget_sha512
216
217 LFBFL_file_name="python_isort_complement.lib.sh"
218 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
219 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
220 @sha512_python_isort_complement.lib.sh@
221 wrapped_wget_sha512
222
223 LFBFL_file_name="repository_data.lib.sh"
224 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
225 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
226 @sha512_repository_data.lib.sh@
227 wrapped_wget_sha512
228
229 LFBFL_file_name="shell_checks_complement.lib.sh"
230 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
231 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
232 @sha512_shell_checks_complement.lib.sh@
233 wrapped_wget_sha512
234
235 LFBFL_file_name="split_score.exec.php"
236 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
237 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
238 @sha512_split_score.exec.php@
239 wrapped_wget_sha512
240 chmod +x "${LFBFL_file_path}"
241
242 LFBFL_file_name="split_score.lib.php"
243 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
244 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
245 @sha512_split_score.lib.php@
246 wrapped_wget_sha512
247
248 LFBFL_file_name="strings_functions.lib.sh"
249 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
250 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
251 @sha512_strings_functions.lib.sh@
252 wrapped_wget_sha512
253
254 LFBFL_file_name="too_long_code_lines.lib.sh"
255 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"

```

```

256 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
257 @sha512_too_long_code_lines.lib.sh@
258 wrapped_wget_sha512
259
260 LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
261 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
262 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
263 @sha512_update_or_check_files_names_listing.exec.sh@
264 wrapped_wget_sha512
265 chmod +x "${LFBFL_file_path}"
266
267 # shellcheck source=check_shell_scripts_beginnings.lib.sh
268 source "${LFBFL_subdir}/check_shell_scripts_beginnings.lib.sh"
269 # shellcheck source=check_shell_scripts_indentation.lib.sh
270 source "${LFBFL_subdir}/check_shell_scripts_indentation.lib.sh"
271 # shellcheck source=check_URLs.lib.sh
272 source "${LFBFL_subdir}/check_URLs.lib.sh"
273 # shellcheck source=common_options.lib.sh
274 source "${LFBFL_subdir}/common_options.lib.sh"
275 # shellcheck source=comparisons.lib.sh
276 source "${LFBFL_subdir}/comparisons.lib.sh"
277 # shellcheck source=generate_from_template.lib.sh
278 source "${LFBFL_subdir}/generate_from_template.lib.sh"
279 # shellcheck source=get_common_text_glob_patterns.lib.sh
280 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
281 # shellcheck source=grammar_and_spelling_check.lib.sh
282 source "${LFBFL_subdir}/grammar_and_spelling_check.lib.sh"
283 # shellcheck source=lines_counts.lib.sh
284 source "${LFBFL_subdir}/lines_counts.lib.sh"
285 # shellcheck source=lines_filters.lib.sh
286 source "${LFBFL_subdir}/lines_filters.lib.sh"
287 # shellcheck source=lines_maps.lib.sh
288 source "${LFBFL_subdir}/lines_maps.lib.sh"
289 # shellcheck source=overwrite_if_not_equal.lib.sh
290 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
291 # shellcheck source=python_black_complement.lib.sh
292 source "${LFBFL_subdir}/python_black_complement.lib.sh"
293 # shellcheck source=python_isort_complement.lib.sh
294 source "${LFBFL_subdir}/python_isort_complement.lib.sh"
295 # shellcheck source=shell_checks_complement.lib.sh
296 source "${LFBFL_subdir}/shell_checks_complement.lib.sh"
297 # shellcheck source=strings_functions.lib.sh
298 source "${LFBFL_subdir}/strings_functions.lib.sh"
299 # shellcheck source=too_long_code_lines.lib.sh
300 source "${LFBFL_subdir}/too_long_code_lines.lib.sh"
301

```

```

302 enhanced_set_bash_option extglob
303 # shellcheck source=repository_data.lib.sh
304 source "${LFBFL_subdir}/repository_data.lib.sh"
305 # shellcheck disable=SC2154,SC2309
306 [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
307     && enhanced_unset_bash_option extglob
308
309 declare -a LFBFL_return_traps_stack
310 init_return_trap
311
312 enhanced_set_shell_option pipefail --trap-unset
313
314 declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
315 keep_options LFBFL_some_common_options2 --verbose
316
317 printf "Building license headers\n"
318 "${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
319     "${LFBFL_some_common_options[@]}"
320
321 printf "Building README.md\n"
322 "${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
323     --work-directory="${LFBFL_work_directory}"\
324     --base-name=README\
325     "${LFBFL_verbose}"
326
327 printf "Building other MarkDown files\n"
328 local LFBFL_some_directory
329 declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
330 local LFBFL_s_files_paths
331 LFBFL_s_files_paths=(
332     find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
333     | relevant_find
334 )
335 declare -a LFBFL_arr_files_paths
336 if [[ -n "${LFBFL_s_files_paths}" ]]; then
337     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
338     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
339         [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
340         printf "Found template %s.\n" "${LFBFL_file_path}"
341         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
342         LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
343         LFBFL_file_name=${LFBFL_file_name%.md.tpl}
344         "${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
345             --work-directory="${LFBFL_some_directory}"\
346             --base-name="${LFBFL_file_name}"\
347             "${LFBFL_verbose}"

```

```

348     done
349 fi
350
351 printf "Building dependencies_notes\n"
352 # https://github.com/php/php-src/issues/21538
353 local LFBFL_annoying_warning
354 LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
355 LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
356 LFBFL_annoying_warning=" by settings in"
357 readonly LFBFL_annoying_warning
358 php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
359     -- "${LFBFL_work_directory}" "\
360     2>&1\
361     | grep --invert-match "${LFBFL_annoying_warning}"
362
363 declare -i LFBFL_i_directory_changed
364 pushd_to_work_directory
365 LFBFL_i_directory_changed=$?
366 can_continue_after_enhanced_pushd || return 1
367
368 local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
369
370 declare -i LFBFL_i_max_line_length
371 grep_variable "${LFBFL_data_file_path}" max_line_length\
372     --result-variable-prefix=LFBFL_i_
373
374 LFBFL_some_common_options+=(
375     "--max-line-length=${LFBFL_i_max_line_length}"
376 )
377 LFBFL_some_common_options2+=(
378     "--max-line-length=${LFBFL_i_max_line_length}"
379 )
380
381 declare -i LFBFL_i_upgrade_venvs=0
382 local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
383 readonly LFBFL_upgrade_venvs_ts_file
384 declare -i LFBFL_i_upgrade_venvs_ts
385 declare -i LFBFL_i_current_ts
386 local LFBFL_upgrade_venvs_answer
387
388 local LFBFL_upgrade_venvs_time_interval_in_seconds=""
389 get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
390     "${LFBFL_some_common_options2[@]}"
391
392 if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
393     LFBFL_i_upgrade_venvs_ts=$(

```

```

394     stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
395 )
396 LFBFL_i_current_ts=$(date +%s)
397 ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
398 ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
399 if [[ LFBFL_i_current_ts -gt 0 ]]; then
400     LFBFL_i_upgrade_venvs=1
401 fi
402 else
403     LFBFL_i_upgrade_venvs=1
404 fi
405 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
406     local LFBFL_upgrade_venvs=""
407     grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
408         --result-variable-prefix=LFBFL_
409     if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
410         read -r -n 1 -t 10\
411             -p "Upgrade venvs and composer global? [Y/n]" \
412             LFBFL_upgrade_venvs_answer
413         if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
414             LFBFL_i_upgrade_venvs=0
415         fi
416     fi
417 fi
418
419 printf "Analyzing too long lines\n"
420 too_long_code_lines "${LFBFL_some_common_options2[@]}"
421
422 printf "Analyzing URLs\n"
423 check_URLs "${LFBFL_some_common_options2[@]}" \
424     | relevant_grep
425
426 printf "Analyzing strange characters: hover over in doubt\n"
427 # shellcheck disable=SC1111
428 LFBFL_usual_characters="\x00-\x7Fàâéèëîïôûç@'\"`|«»..."
429 grep --binary-files=without-match --color=always --exclude-dir .git\
430     --line-number --perl-regexp --recursive --\
431     "[^${LFBFL_usual_characters}]" .\
432     | relevant_grep
433
434 printf "Grammar and spelling check\n"
435 grammar_and_spelling_check "${LFBFL_data_file_path}" \
436     "${LFBFL_some_common_options2[@]}"
437
438 printf "Running shellcheck\n"
439 LFBFL_s_files_paths=$(

```

```

440     find . -type f -name "*.sh"\
441     | relevant_find
442 )
443 if [[ -n "${LFBFL_s_files_paths}" ]]; then
444     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
445     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
446         shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
447             -- "${LFBFL_file_path}"
448     done
449 fi
450 printf "Running shell_checks_complement\n"
451 shell_checks_complement "${LFBFL_some_common_options2[@]}"
452
453 printf "Analyzing shell scripts beginnings\n"
454 check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
455     | relevant_grep
456
457 printf "Analyzing shell scripts indentation\n"
458 check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
459     | relevant_grep
460
461 printf -- "---Python---\n"
462
463 maybe_deactivate(){
464     if command -v deactivate
465     then
466         deactivate
467     fi
468 }
469
470 source_without_return_trap(){
471     declare -i LFBFL_i_no_return_traps=1
472     # shellcheck disable=SC1090,SC1091
473     source -- "$1"
474 }
475
476 printf "Running isort\n"
477 local LFBFL_isort_venv=""
478 grep_variable "${LFBFL_data_file_path}" isort_venv\
479     --result-variable-prefix=LFBFL_\
480     --replace-line-returns-by=""
481 if [[ -n "${LFBFL_isort_venv}" ]]; then
482     maybe_deactivate
483     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
484 fi
485 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then

```

```

486     pip install --upgrade isort
487 fi
488 isort --profile=black .
489 if [[ -n "${LFBFL_isort_venv}" ]]; then
490     deactivate
491 fi
492 printf "Running python_isort_complement\n"
493 python_isort_complement "${LFBFL_some_common_options2[@]}"
494
495 printf "Running black\n"
496 # First, we update the configuration file with max_line_length.
497 local LFBFL_update_max_length="s/^line-length = [0-9]*$"
498 LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
499 sed --regextype=extended --expression="${LFBFL_update_max_length}"\
500     "${LFBFL_subdir3}/black.toml"\
501     > "${LFBFL_subdir3}/black.toml.temp"
502 overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
503     "${LFBFL_subdir3}/black.toml.temp"
504
505 local LFBFL_black_venv=""
506 grep_variable "${LFBFL_data_file_path}" black_venv\
507     --result-variable-prefix=LFBFL_\
508     --replace-line-returns-by=""
509 if [[ -n "${LFBFL_black_venv}" ]]; then
510     maybe_deactivate
511     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
512 fi
513 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
514     pip install --upgrade black
515 fi
516 black --config="${LFBFL_subdir3}/black.toml" .
517 if [[ -n "${LFBFL_black_venv}" ]]; then
518     deactivate
519 fi
520 printf "Running python_black_complement\n"
521 python_black_complement "${LFBFL_some_common_options2[@]}"
522
523 printf "Probing if mypy should be runned\n"
524 local LFBFL_mypy_venv=""
525 grep_variable "${LFBFL_data_file_path}" mypy_venv\
526     --result-variable-prefix=LFBFL_\
527     --replace-line-returns-by=""
528 if [[ -n "${LFBFL_mypy_venv}" ]]; then
529     maybe_deactivate
530     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
531 fi

```



```

532 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
533     pip install --upgrade mypy
534 fi
535 local LFBFL_directory_path
536 declare -i LFBFL_i_no_toml=1
537 LFBFL_s_files_paths=$(
538     find . -type f -name "pyproject.toml" \
539     | relevant_find
540 )
541 if [[ -n "${LFBFL_s_files_paths}" ]]; then
542     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
543     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"}; do
544         if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
545             printf "Running mypy\n"
546             LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
547             mypy -- "${LFBFL_directory_path}"
548         fi
549         LFBFL_i_no_toml=0
550     done
551 fi
552 if [[ LFBFL_i_no_toml -eq 1 ]]; then
553     printf "Running mypy\n"
554     mypy .
555 fi
556 if [[ -n "${LFBFL_mypy_venv}" ]]; then
557     deactivate
558 fi
559
560 printf "Running bandit\n"
561 LFBFL_bandit_venv=""
562 grep_variable "${LFBFL_data_file_path}" bandit_venv \
563     --result-variable-prefix=LFBFL \
564     --replace-line-returns-by=""
565 if [[ -n "${LFBFL_bandit_venv}" ]]; then
566     maybe_deactivate
567     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
568 fi
569 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
570     pip install --upgrade bandit
571 fi
572 bandit --ini="${LFBFL_subdir3}/bandit.ini" \
573     --baseline="${LFBFL_subdir3}/bandit_baseline.json" \
574     --recursive .
575 # Saving new baseline in temp if necessary.
576 bandit --ini="${LFBFL_subdir3}/bandit.ini" \
577     --format=json \

```

```

578     --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
579     --recursive .
580 if [[ -n "${LFBFL_bandit_venv}" ]]; then
581     deactivate
582 fi
583
584 printf "Running pylint\n"
585 LFBFL_pylint_venv=""
586 grep_variable "${LFBFL_data_file_path}" pylint_venv\
587     --result-variable-prefix=LFBFL_\
588     --replace-line-returns-by=""
589 if [[ -n "${LFBFL_pylint_venv}" ]]; then
590     maybe_deactivate
591     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
592 fi
593 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
594     pip install --upgrade pylint
595 fi
596 pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
597 if [[ -n "${LFBFL_pylint_venv}" ]]; then
598     deactivate
599 fi
600
601 printf "Running ruff\n"
602 LFBFL_ruff_venv=""
603 grep_variable "${LFBFL_data_file_path}" ruff_venv\
604     --result-variable-prefix=LFBFL_\
605     --replace-line-returns-by=""
606 if [[ -n "${LFBFL_ruff_venv}" ]]; then
607     maybe_deactivate
608     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
609 fi
610 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
611     pip install --upgrade ruff
612 fi
613 ruff check --config="${LFBFL_subdir3}/ruff.toml"
614 if [[ -n "${LFBFL_ruff_venv}" ]]; then
615     deactivate
616 fi
617 printf -- "---Python end---\n"
618
619 printf -- "---PHP---\n"
620 printf "Running PHP native linter\n"
621 LFBFL_s_files_paths=$(
622     find . -type f -name "*.php"\
623     | relevant_find

```

```

624 )
625 if [[ -n "${LFBFL_s_files_paths}" ]]; then
626     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
627     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
628         php --syntax-check -f="${LFBFL_file_path}"
629     done
630 fi
631
632 printf "Running PHP_CodeSniffer\n"
633 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
634     composer global require squizlabs/php_codesniffer
635 fi
636 phpcs --report=code\
637     --standard=build_and_checks_variables/phpcs_ruleset.xml
638
639 printf "Running PHPMD\n"
640 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
641     composer global require phpmd/phpmd
642 fi
643
644 declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
645 local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
646 LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
647 readonly LFBFL_temp_phpmd_baseline
648
649 local LFBFL_phpmd_rulesets="cleancode, codesize, controversial, "
650 LFBFL_phpmd_rulesets+="design, naming, unusedcode"
651 readonly LFBFL_phpmd_rulesets
652
653 phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
654     . text "${LFBFL_phpmd_rulesets}"
655 # Saving new baseline in temp if necessary.
656 rm "${LFBFL_temp_phpmd_baseline}"
657 phpmd --color --generate-baseline\
658     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
659     . text "${LFBFL_phpmd_rulesets}"
660 sed --in-place --expression='s/" file="/"\n    file="/g'\
661     "${LFBFL_temp_phpmd_baseline}"
662 sed --in-place --regexp-extended\
663     --expression='s~(file=".*)/>~\1\n    />~g'\
664     "${LFBFL_temp_phpmd_baseline}"
665 diff --suppress-common-lines\
666     "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
667     | grep --invert-match --regexp='<!-- ' --regexp='^[0-9]\+d[0-9]\+${'
668
669 printf "Running phpDocumentor\n"

```

```

670 # shellcheck disable=SC2312
671 docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
672 printf -- "---PHP end---\n"
673
674 printf -- "---JS---\n"
675 local LFBFL_npm_lint_directories=""
676 grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
677 --result-variable-prefix=LFBFL_
678 if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
679     printf "Running ESLint\n"
680     local LFBFL_JS_directory
681     declare -a LFBFL_arr_paths
682     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
683     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]"; do
684         (cd -- "${LFBFL_JS_directory}" && npm run lint)
685     done
686 fi
687 printf -- "---JS end---\n"
688
689 [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
690
691 printf "Checking listed files\n"
692 "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
693 "${LFBFL_some_common_options[@]}"
694
695 printf "Creating the PDF file of the listing of the source code\n"
696 "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
697
698 pushd_to_work_directory --trap-popd
699 can_continue_after_enhanced_pushd || return 1
700
701 if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
702     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
703 fi
704
705 printf "Running xmllint\n"
706 # Unfortunately, we need to handle a cache of common DTDs by hand.
707 # XHTML DTDs validation yields errors currently:
708 # see https://gitlab.gnome.org/GNOME/libxml2/-/work\_items/1119
709 LFBFL_file_name="xhtml1-strict.dtd"
710 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
711 LFBFL_script_download_URL="${LFBFL_a}"
712 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
713 LFBFL_correct_sha512='felldb06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
714 LFBFL_correct_sha512+='d8352bf19947168c9bcflae212eca035824293abb1f9'
715 LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'

```

```

716 wrapped_wget_sha512
717
718 LFBFL_file_name="xhtml1-transitional.dtd"
719 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
720 LFBFL_script_download_URL="${LFBFL_a}"
721 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
722 LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
723 LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
724 LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
725 wrapped_wget_sha512
726
727 LFBFL_file_name="xhtml1-frameset.dtd"
728 LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
729 LFBFL_script_download_URL="${LFBFL_a}"
730 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
731 LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
732 LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
733 LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
734 wrapped_wget_sha512
735
736 LFBFL_file_name="xhtml11.dtd"
737 LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
738 LFBFL_script_download_URL="${LFBFL_a}"
739 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
740 LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
741 LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
742 LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
743 wrapped_wget_sha512
744
745 LFBFL_file_name="xhtml-lat1.ent"
746 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
747 LFBFL_script_download_URL="${LFBFL_a}"
748 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
749 LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
750 LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
751 LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
752 wrapped_wget_sha512
753
754 LFBFL_file_name="xhtml-special.ent"
755 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
756 LFBFL_script_download_URL="${LFBFL_a}"
757 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
758 LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
759 LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
760 LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
761 wrapped_wget_sha512

```

```

762
763 LFBFL_file_name="xhtml-symbol.ent"
764 LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
765 LFBFL_script_download_URL="${LFBFL_a}"
766 LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
767 LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
768 LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
769 LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
770 wrapped_wget_sha512
771
772 declare -r LFBFL_s_files_for_xmllint=$(
773     find . -type f -iregex ".*\\.\\(xml\\|xhtml\\)" "\
774     | relevant_find\
775     | sort
776 )
777 if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
778     declare -a LFBFL_arr_files_for_xmllint
779     mapfile -t LFBFL_arr_files_for_xmllint\
780     <<< "${LFBFL_s_files_for_xmllint}"
781     readonly LFBFL_arr_files_for_xmllint
782     local LFBFL_xmllint_files_without_DTD=""
783     grep_variable "${LFBFL_data_file_path}" xmllint_files_without_DTD\
784     --result-variable-prefix=LFBFL_
785     declare -a LFBFL_arr_xmllint_files_without_DTD
786     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
787         mapfile -t LFBFL_arr_xmllint_files_without_DTD\
788         <<< "${LFBFL_xmllint_files_without_DTD}"
789     fi
790     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
791         [[ LFBFL_i_verbose -eq 1 ]]\
792         && printf "xmllint on %s\n" "${LFBFL_file_path}"
793         # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
794         # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
795         # xmllint --pedantic --noout -- "${LFBFL_file_path}"
796         xmllint --pedantic --noout "${LFBFL_file_path}"
797     done
798
799     local LFBFL_file_path2
800     for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
801         for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
802         do
803             if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
804                 LFBFL_file_path=""
805             fi
806         done
807         if [[ "${LFBFL_file_path}" == "" ]]; then

```

```

808         continue
809     fi
810     [[ LFBFL_i_verbose -eq 1 ]]\
811     && printf "xmllint on %s\n" "${LFBFL_file_path}"
812     if [[ "${LFBFL_file_path}" == *.html ]]; then
813         if grep --fixed-strings\
814             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
815             "${LFBFL_file_path}"
816         then
817             xmllint --pedantic --noout\
818                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
819                 "${LFBFL_file_path}"
820         elif grep --fixed-strings\
821             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
822             "${LFBFL_file_path}"
823         then
824             xmllint --pedantic --noout\
825                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
826                 "${LFBFL_file_path}"
827         elif grep --fixed-strings\
828             "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
829             "${LFBFL_file_path}"
830         then
831             xmllint --pedantic --noout\
832                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
833                 "${LFBFL_file_path}"
834         elif grep --fixed-strings\
835             "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
836             "${LFBFL_file_path}"
837         then
838             xmllint --pedantic --noout\
839                 --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
840                 "${LFBFL_file_path}"
841         else
842             printf "Unknown XHTML DTD.\n"
843         fi
844     else
845         xmllint --pedantic --noout --valid "${LFBFL_file_path}"
846     fi
847 done
848 fi
849 # -----
850
851 # -----
852 # It is at the end that all the HTML files are generated and that
853 # we can check there are no mistakes.

```

```

854 printf "Running Nu W3C HTML CSS and SVG validator\n"
855 local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
856 grep_variable "${LFBFL_data_file_path}" \
857     upgrade_vnu_jar_time_interval_in_seconds \
858     --result-variable-prefix=LFBFL_
859
860 if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
861 then
862     LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
863 fi
864
865 declare -i LFBFL_i_vnu_jar_ts
866 local LFBFL_vnu_jar_path=""
867 grep_variable "${LFBFL_data_file_path}" vnu_jar_path \
868     --result-variable-prefix=LFBFL_ \
869     --replace-line-returns-by=""
870 if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
871     LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
872     LFBFL_i_current_ts=$(date +%s)
873     ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
874     ((
875         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
876     ))
877     if [[ LFBFL_i_current_ts -gt 0 ]]; then
878         printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\n'
879         printf "Go get the latest version here:\n"
880         printf "https://github.com/validator/validator/releases\n"
881     fi
882     # vnu has false positives for xhtml.
883     local LFBFL_use_vnu_for_xhtml=""
884     grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml \
885         --result-variable-prefix=LFBFL_
886     local LFBFL_find_regexp
887     if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
888         LFBFL_find_regexp=".*\\.\\(css\\|html\\|svg\\|xhtml\\)"
889     else
890         LFBFL_find_regexp=".*\\.\\(css\\|html\\|svg\\)"
891     fi
892     declare -r LFBFL_s_files_for_Nu=$(
893         find . -type f -iregex "${LFBFL_find_regexp}" \
894         | relevant_find
895     )
896     if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
897         declare -a LFBFL_arr_files_for_Nu
898         mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
899         readonly LFBFL_arr_files_for_Nu

```



```

900         java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
901             --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
902     fi
903 fi
904 # -----
905
906 if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
907     touch -- "${LFBFL_upgrade_venvs_ts_file}"
908 fi
909 }
910
911 common_build_and_checks "$@"

```

3.10 build_and_checks_dependencies/common_options.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24
25 get_where_was_i(){
26     # Auxiliary function needed below.
27     # See usages below if you want to reuse it.
28     # $1=offset
29     LFBFL_where_was_i="$0"
30     if [[ "${BASH_SOURCE[1]}" != "" && "${BASH_SOURCE[1]}" != "$0" ]]; then
31         LFBFL_where_was_i+=" ${BASH_SOURCE[1]}"

```

```

32     fi
33     if [[ "${FUNCNAME[$1]}" != "" ]]; then
34         LFBFL_where_was_i+=" ${FUNCNAME[$1]}() "
35     fi
36     readonly LFBFL_where_was_i
37     # Comment out line above in case someone would use distinct
38     # offsets in the same function.
39 }
40
41 # RETURN traps handling
42 # Add the following lines at the start of a function.
43 # declare -a LFBFL_return_traps_stack
44 # init_return_trap
45
46 init_return_trap() {
47     local LFBFL_previous_return_trap
48     local LFBFL_previous_return_trap2
49     LFBFL_previous_return_trap=$(trap -p RETURN)
50     # shellcheck disable=SC2154,SC2309
51     if [[ LFBFL_i_verbose -eq 1 ]]; then
52         local LFBFL_where_was_i
53         get_where_was_i 2
54         printf "%s previous_return_trap is: %s.\n"\
55             "${LFBFL_where_was_i}"\
56             "${LFBFL_previous_return_trap}"
57     fi
58     if [[ -z "${LFBFL_previous_return_trap}" ]]; then
59         LFBFL_previous_return_trap="trap - RETURN"
60     fi
61     readonly LFBFL_previous_return_trap
62     declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ) )
63     LFBFL_previous_return_trap2="${LFBFL_function_depth}:"
64     LFBFL_previous_return_trap2+="${LFBFL_previous_return_trap}"
65     readonly LFBFL_previous_return_trap2
66     LFBFL_return_traps_stack+=("${LFBFL_previous_return_trap2}")
67     trap 'execute_return_traps' RETURN
68 }
69
70 execute_return_traps() {
71     if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
72         return
73     fi
74     # -----
75     # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
76     if [[ "${FUNCNAME[1]}" == "source" ]]; then
77         # Not Working

```

```

78     return
79 fi
80 if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
81     return
82 fi
83 # -----
84 local LFBFL_some_return_trap
85 local LFBFL_where_was_i
86 get_where_was_i 2
87 declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
88 while [[ ${#LFBFL_return_traps_stack[@]} -gt 0 ]]; do
89     LFBFL_some_return_trap=${LFBFL_return_traps_stack[-1]}
90     if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}* ]]; then
91         # We only treat the top of the stack that should correspond to the
92         # current function.
93         break
94     fi
95     LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
96     # shellcheck disable=SC2154,SC2309
97     if [[ LFBFL_i_verbose -eq 1 ]]; then
98         if [[ ${#LFBFL_return_traps_stack[@]} -gt 1 ]]; then
99             printf "%s executing some return trap: %s.\n" \
100                 "${LFBFL_where_was_i}" \
101                 "${LFBFL_some_return_trap}"
102         else
103             printf "%s setting back to previous_return_trap: %s.\n" \
104                 "${LFBFL_where_was_i}" \
105                 "${LFBFL_some_return_trap}"
106         fi
107     fi
108     eval -- "${LFBFL_some_return_trap}"
109     unset 'LFBFL_return_traps_stack[-1]'
110 done
111 }
112
113
114 is_top_dirstack_directory() {
115     # Although "${DIRSTACK[0]}" may seem to be already a realpath,
116     # possible use of symbolic links makes this function necessary.
117     # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
118     # $1=to_directory needed to be "realpathed" already
119     # but I removed this micro-optimization.
120     # A compiled language could have a "realpathed property" on
121     # "path object/string" to optimize when possible.
122     if [[ -z "$1" ]]; then
123         return 0

```

```

124     fi
125     local LFBFL_real_path
126     LFBFL_real_path=$(realpath -- "${DIRSTACK[0]}")
127     readonly LFBFL_real_path
128     local LFBFL_real_path2
129     LFBFL_real_path2=$(realpath -- "$1")
130     readonly LFBFL_real_path2
131     [[ "${LFBFL_real_path}" == "${LFBFL_real_path2}" ]]
132 }
133
134 enhanced_pushd() {
135     # Auxiliary function needed below.
136     # See usages below if you want to reuse it.
137     # $1=to_directory (needed to be sanitized with realpath before call)
138     # $2=offset_for_where_was_i
139     # $3=error_message_intermediate_complement
140     # returns an error code whenever no pushd happened
141     # $1 can be any path now, see my comment on the micro-optimization in
142     # is_top_dirstack_directory() just above.
143     # shellcheck disable=SC2154,SC2309
144     [[ LFBFL_i_verbose -eq 1 ]]\
145         && printf "enhanced_pushd requested: %s.\n" "$1"
146     # Value 0 will always be overwritten, it is superfluous.
147     declare -gi i_enhanced_pushd_result=0
148     if [[ -z "$1" ]]; then
149         i_enhanced_pushd_result=110
150         # shellcheck disable=SC2248
151         return ${i_enhanced_pushd_result}
152     fi
153     local LFBFL_real_path
154     LFBFL_real_path=$(realpath -- "$1")
155     readonly LFBFL_real_path
156     # shellcheck disable=SC2154,SC2309
157     [[ LFBFL_i_verbose -eq 1 ]]\
158         && printf "enhanced_pushd passed validation 1.\n"
159     is_top_dirstack_directory "${LFBFL_real_path}" && {
160         i_enhanced_pushd_result=111
161         # shellcheck disable=SC2248
162         return ${i_enhanced_pushd_result}
163     }
164     # shellcheck disable=SC2154,SC2309
165     [[ LFBFL_i_verbose -eq 1 ]]\
166         && printf "enhanced_pushd passed validation 2.\n"
167
168     pushd -- "${LFBFL_real_path}" || {
169         i_enhanced_pushd_result=$?

```

```

170     local LFBFL_where_was_i
171     get_where_was_i "$2"
172     printf "%s %s\n" "${LFBFL_where_was_i}" "$3" \
173         "${LFBFL_real_path}"
174     # shellcheck disable=SC2248
175     return ${i_enhanced_pushd_result}
176 }
177 # shellcheck disable=SC2154,SC2309
178 [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed.\n"
179 # shellcheck disable=SC2248
180 return ${i_enhanced_pushd_result}
181 }
182
183 enhanced_popd() {
184     # Auxiliary function needed below.
185     # See usages below if you want to reuse it.
186     # $1=to_directory
187     # $2=offset_for_where_was_i
188     # shellcheck disable=SC2154,SC2309
189     [[ LFBFL_i_verbose -eq 1 ]] \
190         && printf "enhanced_popd requested: %s.\n" "$1"
191     if [[ -z "$1" ]]; then
192         return 0
193     fi
194
195     declare -i LFBFL_i_popd_result
196     popd || {
197         LFBFL_i_popd_result=$?
198         readonly LFBFL_i_popd_result
199         local LFBFL_where_was_i
200         get_where_was_i "$2"
201         printf "%s popd failed.\n" "${LFBFL_where_was_i}"
202         # shellcheck disable=SC2248
203         return ${LFBFL_i_popd_result}
204     }
205     # shellcheck disable=SC2154,SC2309
206     [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed.\n"
207     return 0
208 }
209
210 can_continue_after_enhanced_pushd() {
211     [[
212         i_enhanced_pushd_result -eq 0
213         || i_enhanced_pushd_result -eq 110
214         || i_enhanced_pushd_result -eq 111
215     ]]

```

```

216 }
217
218 # Main usage for the next functions relative to verbose mode:
219 # Add the two following lines at the start of a function.
220 # declare -i LFBFL_i_verbose=0
221 # get_verbose_option "$@"
222 # /\ You really need all of the two lines. /\
223 # Otherwise, the call to get_verbose_option may impact a function above;
224 # in case you didn't copy all of "$@" in nested function calls...
225 # /\ The same warning applies for other functions below. /\
226 # /\ I will not repeat it below. /\
227 # Use get_verbose_options_array, if needed.
228
229 get_verbose_options_array(){
230     # Use this constant with code like:
231     # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
232     # It could be extended if there is more than one level of verboseness.
233     if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
234         return
235     fi
236
237     declare -agr LFBFL_VERBOSE_OPTIONS=(
238         ""
239         "--verbose"
240     )
241 }
242
243 get_some_flag(){
244     # $1=variable_name
245     # $2=some_flag_long_name
246     # $3=read_only 0 or 1 to make the variable readonly
247     # This command is to be called in another one with same arguments on top
248     # of the three previous.
249     # This test:
250     # if [[ "$*" == *--verbose* ]]; then
251     # is faster than below.
252     # But it is not bug free in a calling function,
253     # and even incompatible with the current function where
254     # the flag_long_name is a mandatory argument.
255     printf -v "$1" "%d" 0
256     declare -i LFBFL_i
257     local LFBFL_argument
258     for ((LFBFL_i = 4; LFBFL_i <= $#; ++LFBFL_i)); do
259         LFBFL_argument="${!LFBFL_i}"
260         if [[ "${LFBFL_argument}" == "$2" ]]; then
261             printf -v "$1" "%d" 1

```

```

262         break
263     fi
264 done
265 if [[ $3 -eq 1 ]]; then
266     readonly "$1"
267 fi
268 }
269
270 get_verbose_option() {
271     # This command is to be called in another one with same arguments.
272     # Options:
273     #   --verbose
274     get_some_flag LFBFL_i_verbose --verbose 1 "$@"
275     # shellcheck disable=SC2154,SC2309
276     if [[ LFBFL_i_verbose -eq 1 ]]; then
277         local LFBFL_where_was_i
278         get_where_was_i 2
279         printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
280     fi
281 }
282
283
284 get_some_option() {
285     # $1=variable_name
286     # $2=some_option_long_name
287     # $3=option_default_value_if_flag_without_option_value_or_empty_value
288     # $4=option_default_value_if_nothing
289     # $5=read_only 0 or 1 to make the variable readonly
290     # This command is to be called in another one with same arguments on top
291     # of the four previous.
292     # If you truly have a need to distinguish:
293     # option_default_value_if_flag_without_option_value --thing
294     # option_default_value_if_flag_with_empty_option_value --thing= or ='
295     # then think twice before modifying this function.
296     printf -v "$1" "%s" "$4"
297     declare -i LFBFL_i
298     local LFBFL_argument
299     for ((LFBFL_i = 6; LFBFL_i <= $#; ++LFBFL_i)); do
300         LFBFL_argument="${!LFBFL_i}"
301         if [[ "${LFBFL_argument}" == "$2" ]]; then
302             printf -v "$1" "%s" "$3"
303             break
304         fi
305         if [[ "${LFBFL_argument}" == "$2"*=* ]]; then
306             printf -v "$1" "%s" "${LFBFL_argument#"${2}"="}"
307             printf -v "$1" "%s" "${!1:-$3}"

```

```

308         break
309     fi
310 done
311 if [[ $5 -eq 1 ]]; then
312     readonly "$1"
313 fi
314 }
315
316 keep_options() {
317     # $1=variable_name of array of options
318     # Other arguments: the options to keep.
319     declare -a LFBFL_array_of_filtered_options
320     declare -n LFBFL_array_of_options=$1
321     local LFBFL_some_argument
322     local LFBFL_some_argument2
323     for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
324         for LFBFL_some_argument2 in "$@"; do
325             if [[
326                 "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
327                 || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"*=*
328             ]]; then
329                 LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
330             fi
331         done
332     done
333     LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
334 }
335
336
337 # Main usage for the next functions relative to work_directory:
338 # Add the five following lines at the start of a function.
339 #   local LFBFL_work_directory=""
340 #   get_work_directory_option "$@"
341 #   pushd_to_work_directory\
342 #   && trap 'popd_from_work_directory' RETURN
343 #   can_continue_after_enhanced_pushd || return 1
344 # Or:
345 #   local LFBFL_work_directory=""
346 #   get_work_directory_option "$@"
347 #   pushd_to_work_directory --trap-popd
348 #   can_continue_after_enhanced_pushd || return 1
349
350 get_work_directory_option() {
351     # This command is to be called in another one with same arguments.
352     # Options:
353     #   --work-directory=""

```



```

354     get_some_option LFBFL_work_directory --work-directory "." "." 0 "$@"
355     LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
356     readonly LFBFL_work_directory
357 }
358
359 pushd_to_work_directory() {
360     # Options:
361     #   --trap-popd
362     enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
363     if [[ i_enhanced_pushd_result -eq 0 ]]; then
364         declare -i LFBFL_i_trap_popd
365         get_some_flag LFBFL_i_trap_popd --trap-popd 1 "$@"
366         if [[ LFBFL_i_trap_popd -eq 1 ]]; then
367             declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
368             local LFBFL_some_return_trap
369             LFBFL_some_return_trap="${LFBFL_function_depth}:"
370             LFBFL_some_return_trap+="popd_from_work_directory 4"
371             readonly LFBFL_some_return_trap
372             LFBFL_return_traps_stack+="${LFBFL_some_return_trap}"
373         fi
374     fi
375     # shellcheck disable=SC2248
376     return ${i_enhanced_pushd_result}
377 }
378
379 popd_from_work_directory() {
380     # $1=offset for where_was_i
381     enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
382 }
383
384 work_directory_is_top_dirstack_directory() {
385     is_top_dirstack_directory "${LFBFL_work_directory}"
386 }
387
388
389 get_offset_option() {
390     # This command is to be called in another one with same arguments on top
391     # of the default value.
392     # $1=option_default_value
393     # Options:
394     #   --offset=2
395     get_some_option LFBFL_i_offset --offset "$1" "$1" 1 "$@"
396 }
397
398 get_max_line_length_option() {
399     # This command is to be called in another one with same arguments.

```

```

400 # Its most useful feature is that it factorizes the default maximum line
401 # length of 70 characters.
402 # Options:
403 # --max-line-length
404 get_some_option LFBFL_i_max_line_length --max-line-length 70 70 1 "$@"
405 }
406
407
408 # Functions relative to shell and bash options
409 # (See SHELL_OPTS and BASH_OPTS, set and shopt):
410 # - shell options usage, example with pipefail:
411 #     enhanced_set_shell_option pipefail\
412 #         && trap 'enhanced_unset_shell_option pipefail' RETURN
413 # or maybe:
414 #     enhanced_unset_shell_option pipefail\
415 #         && trap 'enhanced_set_shell_option pipefail' RETURN
416 # if you need to unset it in some function.
417 # or better if you use my RETURN traps handler:
418 #     enhanced_set_shell_option pipefail --trap-unset
419 #     enhanced_unset_shell_option pipefail --trap-set
420 # - bash options usage, example with globstar:
421 #     enhanced_set_bash_option globstar\
422 #         && trap 'enhanced_unset_bash_option globstar' RETURN
423 # or maybe:
424 #     enhanced_unset_bash_option globstar\
425 #         && trap 'enhanced_set_bash_option globstar' RETURN
426 # or better if you use my RETURN traps handler:
427 #     enhanced_set_bash_option globstar --trap-unset
428 #     enhanced_unset_bash_option globstar --trap-set
429
430 enhanced_set_shell_option() {
431     # $1=optname some shell option for set
432     # Options:
433     # --offset=2
434     # --trap-unset
435     declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
436     declare -gi "${LFBFL_result_name}"=0
437     if [[ -o "$1" ]]; then
438         printf -v "${LFBFL_result_name}" "%d" 1
439         # shellcheck disable=SC2086
440         return ${!LFBFL_result_name}
441     fi
442     set -o "$1"
443     declare -i LFBFL_i_offset=2
444     get_offset_option 2 "$@"
445     # shellcheck disable=SC2154,SC2309

```

```

446 if [[ LFBFL_i_verbose -eq 1 ]]; then
447     local LFBFL_where_was_i
448     # shellcheck disable=SC2248
449     get_where_was_i ${LFBFL_i_offset}
450     printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
451 fi
452 declare -i LFBFL_i_trap_unset
453 get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
454 if [[ LFBFL_i_trap_unset -eq 1 ]]; then
455     declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
456     local LFBFL_some_return_trap
457     LFBFL_some_return_trap="${LFBFL_function_depth}:"
458     LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
459     readonly LFBFL_some_return_trap
460     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
461 fi
462 # shellcheck disable=SC2086
463 return ${!LFBFL_result_name}
464 }
465
466 enhanced_unset_shell_option() {
467     # $1=optname some shell option for set
468     # Options:
469     #   --offset=2
470     #   --trap-set
471     declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
472     declare -gi "${LFBFL_result_name}"=0
473     if [[ ! -o "$1" ]]; then
474         printf -v "${LFBFL_result_name}" "%d" 1
475         # shellcheck disable=SC2086
476         return ${!LFBFL_result_name}
477     fi
478     set +o "$1"
479     declare -i LFBFL_i_offset=2
480     get_offset_option 2 "$@"
481     # shellcheck disable=SC2154,SC2309
482     if [[ LFBFL_i_verbose -eq 1 ]]; then
483         local LFBFL_where_was_i
484         # shellcheck disable=SC2248
485         get_where_was_i ${LFBFL_i_offset}
486         printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
487     fi
488     declare -i LFBFL_i_trap_set
489     get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
490     if [[ LFBFL_i_trap_set -eq 1 ]]; then
491         declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))

```

```

492     local LFBFL_some_return_trap
493     LFBFL_some_return_trap="${LFBFL_function_depth}:"
494     LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
495     readonly LFBFL_some_return_trap
496     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
497 fi
498 # shellcheck disable=SC2086
499 return ${!LFBFL_result_name}
500 }
501
502 enhanced_set_bash_option() {
503     # $1=optname some bash option for shopt
504     # Options:
505     #     --offset=2
506     #     --trap-unset
507     declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
508     declare -gi "${LFBFL_result_name}"=0
509     # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
510     # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
511     if shopt -q "$1"; then
512         printf -v "${LFBFL_result_name}" "%d" 1
513         # shellcheck disable=SC2086
514         return ${!LFBFL_result_name}
515     fi
516     shopt -s "$1"
517     declare -i LFBFL_i_offset=2
518     get_offset_option 2 "$@"
519     # shellcheck disable=SC2154,SC2309
520     if [[ LFBFL_i_verbose -eq 1 ]]; then
521         local LFBFL_where_was_i
522         # shellcheck disable=SC2248
523         get_where_was_i ${LFBFL_i_offset}
524         printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
525     fi
526     declare -i LFBFL_i_trap_unset
527     get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
528     if [[ LFBFL_i_trap_unset -eq 1 ]]; then
529         declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
530         local LFBFL_some_return_trap
531         LFBFL_some_return_trap="${LFBFL_function_depth}:"
532         LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
533         readonly LFBFL_some_return_trap
534         LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
535     fi
536     # shellcheck disable=SC2086
537     return ${!LFBFL_result_name}

```

```

538 }
539
540 enhanced_unset_bash_option() {
541     # $1=optname some bash option for shopt
542     # Options:
543     #     --offset=2
544     #     --trap-set
545     declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
546     declare -gi "${LFBFL_result_name}"=0
547     if ! shopt -q "$1"; then
548         printf -v "${LFBFL_result_name}" "%d" 1
549         # shellcheck disable=SC2086
550         return ${!LFBFL_result_name}
551     fi
552     shopt -u "$1"
553     declare -i LFBFL_i_offset=2
554     get_offset_option 2 "$@"
555     # shellcheck disable=SC2154,SC2309
556     if [[ LFBFL_i_verbose -eq 1 ]]; then
557         local LFBFL_where_was_i
558         # shellcheck disable=SC2248
559         get_where_was_i ${LFBFL_i_offset}
560         printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
561     fi
562     declare -i LFBFL_i_trap_set
563     get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
564     if [[ LFBFL_i_trap_set -eq 1 ]]; then
565         declare -ir LFBFL_function_depth=$(( ${#FUNCNAME[@]} - 1 ))
566         local LFBFL_some_return_trap
567         LFBFL_some_return_trap="${LFBFL_function_depth}:"
568         LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
569         readonly LFBFL_some_return_trap
570         LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
571     fi
572     # shellcheck disable=SC2086
573     return ${!LFBFL_result_name}
574 }

```

3.11 build_and_checks_dependencies/comparisons.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License

```

```

7 # as published by the Free Software Foundation,
8 # either version 3 of the License,
9 # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "comparisons.sh" was renamed to "comparisons.lib.sh".
25
26 # shellcheck source=common_options.lib.sh
27 source "./build_and_checks_dependencies/common_options.lib.sh"
28
29 equal() {
30     # Returns 1 if all the arguments strings are equal.
31     # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
32     # J'ai rerepensé au 3-way-merge
33     # et à mon article sur le principe de première différence
34     # appliqué à la largeur modulaire, de clique ou de rang avec
35     # l'idée d'un principe de première différence ternaire.
36     declare -i LFBFL_i_first=1
37     local LFBFL_current
38     local LFBFL_argument
39     for LFBFL_argument in "$@"; do
40         if [[ LFBFL_i_first -eq 1 ]]; then
41             LFBFL_current="${LFBFL_argument}"
42             LFBFL_i_first=0
43             continue
44         fi
45         if [[ "${LFBFL_current}" != "${LFBFL_argument}" ]]; then
46             return 0
47         fi
48     done
49     return 1
50 }
51
52 some_distinct() {

```

```

53     ! equal "$@"
54 }
55
56 all_distinct() {
57     # Returns 1 if all the arguments strings are distinct.
58     declare -ir LFBFL_ij_max=$#
59     declare -i LFBFL_i
60     declare -i LFBFL_j
61     for ((LFBFL_i = 1; LFBFL_i < LFBFL_ij_max; ++LFBFL_i)); do
62         for ((LFBFL_j = LFBFL_i + 1; LFBFL_j <= LFBFL_ij_max; ++LFBFL_j)); do
63             [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
64         done
65     done
66     return 1
67 }
68
69 all_distinct2() {
70     # Returns 1 if all the arguments strings are distinct.
71     local LFBFL_element
72     local LFBFL_previous_element
73     if [[ $# -le 1 ]]; then
74         return 1
75     fi
76     declare -r LFBFL_s_sorted_arguments=$(
77         printf "%s\n" "$@" \
78         | sort
79     )
80     declare -a LFBFL_arr_sorted_arguments
81     mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
82     declare -i LFBFL_i
83     declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
84     LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
85     for ((LFBFL_i = 1; LFBFL_i < LFBFL_i_max; ++LFBFL_i)); do
86         LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
87         if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
88             return 0
89         fi
90         LFBFL_previous_element="${LFBFL_element}"
91     done
92     return 1
93 }
94
95 all_distinct3() {
96     # Returns 1 if all the arguments strings are distinct.
97     # all_distinct3 is slower than old all_distinct2 since a call to sort is
98     # needed:

```

```

99 # the manual of uniq says it doesn't detect non-adjacent repeated lines.
100 # A few weeks ago, when testing without sort, I had no bug...
101 # I may have tested all_distinct3 a a b and not all_distinct3 a b a
102 # There is a sort also in all_distinct2, but uniq could have an option to
103 # sort before "applying -d", or sort could have an option for printing
104 # duplicates only. It would complement -u option of sort.
105 # Both solutions would avoid one pipe and one more shell command call.
106 # Note that old all_distinct2 was faster with pipes and read -r commands,
107 # but read -r has small drawbacks. Using mapfile is a cleaner way to do
108 # it, since it handles arguments with spaces from scratch.
109 # Arguments with line-returns are still not handled.
110 # Use C, PHP or Python for example for that.
111 # Somehow Bash was made to push people to stop being lazy and do actual C
112 # code, even if it needs compiling something, adding a makefile, etc. XD
113 # Note that C code would be faster with all_distinct2 blueprint,
114 # that's the reason I kept all 3 versions, see the (asymptotic)
115 # complexity and simple algorithms ideas on this toy problem.
116 if [[ $# -le 1 ]]; then
117     return 1
118 fi
119 declare -r LFBFL_s_duplicated_arguments=$(
120     printf "%s\n" "$@" \
121     | sort\
122     | uniq --repeated
123 )
124 [[ -n "${LFBFL_s_duplicated_arguments}" ]]
125 }
126
127 some_equal() {
128     ! all_distinct "$@"
129 }
130
131 some_equal2() {
132     ! all_distinct2 "$@"
133 }
134
135 some_equal3() {
136     ! all_distinct3 "$@"
137 }
138
139 max() {
140     # Returns the maximum string among the arguments.
141     # Renvoie la chaîne la plus grande parmi celles proposées.
142     # $1=sort_command with the flags you want to be applied.
143     # $ max sort a b c
144     # c

```



```

145     if [[ $# -eq 0 ]]; then
146         return 1
147     fi
148     enhanced_set_shell_option pipefail\
149     && trap 'enhanced_unset_shell_option pipefail' RETURN
150     printf "%s\n" "${@:2}" \
151         | eval -- "$1 --reverse"\
152         | head --lines=1
153 }
154
155 min(){
156     # Returns the minimum string among the arguments.
157     # Renvoie la chaîne la plus petite parmi celles proposées.
158     # $1=sort_command with the flags you want to be applied.
159     # $ min sort a b c
160     # a
161     if [[ $# -eq 0 ]]; then
162         return 1
163     fi
164     enhanced_set_shell_option pipefail\
165     && trap 'enhanced_unset_shell_option pipefail' RETURN
166     printf "%s\n" "${@:2}" \
167         | eval -- "$1"\
168         | head --lines=1
169 }
170
171 is_substring(){
172     # $1=string
173     # $2=substring
174     declare -r LFBFL_var_1=$(
175         printf "%s" "$1"\
176         | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
177     )
178     declare -r LFBFL_var_2=$(
179         printf "%s" "$2"\
180         | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
181     )
182     local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
183     LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
184     LFBFL_var_3+=" \$c = '/'.addslashes(\$b, '/').'/';"
185     LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
186     readonly LFBFL_var_3
187     php --run="${LFBFL_var_3}"
188     return $?
189 }
190

```

```

191 is_subfile() {
192     # $1=file_path
193     # $2=sub_file_path
194     local LFBFL_var_1="\$a = file_get_contents('$1');"
195     LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2')));"
196     LFBFL_var_1+=" \$c = '/' .addslashes(\$b, '/') .'/';"
197     LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
198     readonly LFBFL_var_1
199     php --run="\${LFBFL_var_1}"
200     return $?
201 }
202
203 alias min_from_int_sort="min 'sort --numeric-sort'"
204 alias max_from_int_sort="max 'sort --numeric-sort'"

```

3.12 build_and_checks_dependencies/create_PDF.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".
25
26 LFBFL_subdir="build_and_checks_dependencies"
27 # shellcheck source=common_options.lib.sh
28 source "${LFBFL_subdir}/common_options.lib.sh"
29 # shellcheck source=generate_from_template.lib.sh

```

```

30 source "${LFBFL_subdir}/generate_from_template.lib.sh"
31 # shellcheck source=lines_counts.lib.sh
32 source "${LFBFL_subdir}/lines_counts.lib.sh"
33 # shellcheck source=lines_filters.lib.sh
34 source "${LFBFL_subdir}/lines_filters.lib.sh"
35 # shellcheck source=lines_maps.lib.sh
36 source "${LFBFL_subdir}/lines_maps.lib.sh"
37 # shellcheck source=overwrite_if_not_equal.lib.sh
38 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
39 # shellcheck source=strings_functions.lib.sh
40 source "${LFBFL_subdir}/strings_functions.lib.sh"
41
42 create_PDF() {
43     # Options:
44     #   --verbose
45     #   --work-directory=""
46     declare -i LFBFL_i_verbose=0
47     get_verbose_option "$@"
48     local LFBFL_work_directory=""
49     get_work_directory_option "$@"
50     declare -a LFBFL_return_traps_stack
51     init_return_trap
52     pushd_to_work_directory --trap-popd
53     can_continue_after_enhanced_pushd || return 1
54
55     enhanced_set_shell_option pipefail --trap-unset
56
57     declare -a LFBFL_common_options=("$@")
58     keep_options LFBFL_common_options --verbose
59
60     local LFBFL_subdir2="build_and_checks_variables"
61     local LFBFL_data_file_path="${LFBFL_subdir2}/repository_data.txt"
62     local LFBFL_repository_name=""
63     grep_variable "${LFBFL_data_file_path}" repository_name\
64         --result-variable-prefix="LFBFL_"\
65         --replace-line-returns-by=""
66
67     local LFBFL_repository_web_url=""
68     grep_variable "${LFBFL_data_file_path}" repository_web_url\
69         --result-variable-prefix="LFBFL_"\
70         --replace-line-returns-by=""
71
72     local LFBFL_repository_git_url=""
73     grep_variable "${LFBFL_data_file_path}" repository_git_url\
74         --result-variable-prefix="LFBFL_"\
75         --replace-line-returns-by=""

```

```

76
77 local LFBFL_abstract=""
78 grep_variable "${LFBFL_data_file_path}" abstract\
79     --result-variable-prefix="LFBFL_"
80
81 local LFBFL_acknowledgments=""
82 grep_variable "${LFBFL_data_file_path}" acknowledgments\
83     --result-variable-prefix="LFBFL_"
84
85 local LFBFL_author_email=""
86 grep_variable "${LFBFL_data_file_path}" author_email\
87     --result-variable-prefix="LFBFL_"\
88     --replace-line-returns-by=""
89
90 local LFBFL_author_full_name=""
91 grep_variable "${LFBFL_data_file_path}" author_full_name\
92     --result-variable-prefix="LFBFL_"\
93     --replace-line-returns-by=""
94
95 local LFBFL_author_website=""
96 grep_variable "${LFBFL_data_file_path}" author_website\
97     --result-variable-prefix="LFBFL_"\
98     --replace-line-returns-by=""
99
100 declare -i LFBFL_i_max_line_length
101 grep_variable "${LFBFL_data_file_path}" max_line_length\
102     --result-variable-prefix="LFBFL_i_"
103
104 declare -r LFBFL_current_date=$(date --iso-8601="date")
105
106 declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
107
108 declare -ir LFBFL_i_number_of_commits=$(
109     git shortlog\
110     | space_starting_lines\
111     | wc --lines
112 )
113
114 local LFBFL_redirect="/dev/stdout"
115 if [[ LFBFL_i_verbose -eq 0 ]]; then
116     LFBFL_redirect="/dev/null"
117 fi
118 {
119     code_lines_count_all "${LFBFL_common_options[@]}"
120     code_lines_count_not_empty "${LFBFL_common_options[@]}"
121     code_lines_count_empty "${LFBFL_common_options[@]}"

```

```

122 } &> "${LFBFL_redirect}"
123
124 local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
125 LFBFL_number_of_lines+=" total lines,"
126 LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
127 LFBFL_number_of_lines+=" not empty lines,"
128 LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
129 LFBFL_number_of_lines+=" empty lines."
130 readonly LFBFL_number_of_lines
131
132 declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"
133
134 # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
135 # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
136 # ZXIvdW5peC10cmVlIiwiaWQiOiJE4NTE0NTgzOH0%3D
137 tree -a --gitignore\
138     -I "node_modules/"\
139     -I "__pycache__/"\
140     -I ".mypy_cache/"\
141     -I ".ruff_cache/"\
142     -I ".git/"\
143     | replace_non_ascii_spaces\
144     > "${LFBFL_temp_path}/current_tree_light.txt"
145
146 tree -a -DFh --gitignore\
147     -I "node_modules/"\
148     -I "__pycache__/"\
149     -I ".mypy_cache/"\
150     -I ".ruff_cache/"\
151     -I ".git/"\
152     | replace_non_ascii_spaces\
153     > "${LFBFL_temp_path}/current_tree.txt.temp"
154
155 local LFBFL_s_files_paths
156 # Remove line returns that are here to keep lines short.
157 LFBFL_s_files_paths=$(
158     sed --regexp-extended --null-data\
159         --expression='s/(.)\\n\\1/Mg'\
160         -- "${LFBFL_subdir2}/files_names_listing.txt"\
161         | grep --invert-match '^/'
162 )
163 readonly LFBFL_s_files_paths
164 declare -a LFBFL_arr_files_paths
165 if [[ -n "${LFBFL_s_files_paths}" ]]; then
166     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
167 fi

```

```

168     readonly LFBFL_arr_files_paths
169
170     local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
171     LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
172     readonly LFBFL_temp_files_listing
173     : > "${LFBFL_temp_files_listing}"
174     local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
175     LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
176     readonly LFBFL_temp_files_listing2
177     : > "${LFBFL_temp_files_listing2}"
178     # HTML <li> elements, hence "lis".
179     local LFBFL_temp_files_lis
180     LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
181     readonly LFBFL_temp_files_lis
182     : > "${LFBFL_temp_files_lis}"
183     # shellcheck disable=SC2248
184     get_split_score_simple 1 ${LFBFL_i_max_line_length} /
185     declare -r LFBFL_score_command="${get_split_score_result}"
186     declare -i LFBFL_i_score_command_properties
187     LFBFL_i_score_command_properties=${i_get_split_score_result2}
188     readonly LFBFL_i_score_command_properties
189     # shellcheck disable=SC2248
190     get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
191     declare -r LFBFL_score_command2="${get_split_score_result}"
192     declare -i LFBFL_i_score_command_properties2
193     LFBFL_i_score_command_properties2=${i_get_split_score_result2}
194     readonly LFBFL_i_score_command_properties2
195     declare -r LFBFL_suffix='%'
196     declare -i LFBFL_i=0
197     local LFBFL_cleaned_path1
198     local LFBFL_cleaned_path2
199     local LFBFL_new_lines
200     local LFBFL_new_lines2
201     local LFBFL_new_lines3
202     local LFBFL_file_path
203     local LFBFL_s_format
204     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
205         printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
206         # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
207         # LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
208         LFBFL_cleaned_path1="${LFBFL_file_path//_/\_}"
209         LFBFL_cleaned_path2=$(
210             printf "%s" "${LFBFL_file_path}"\
211             | sed --expression='s/\//:/g;s/\././g'
212         )
213         LFBFL_new_lines="    ${LFBFL_cleaned_path1}"

```

```

214 if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
215     # shellcheck disable=SC2248
216     repeated_split_last_line "${LFBFL_new_lines}"\
217         "\
218         ${LFBFL_i_max_line_length}\
219         "${LFBFL_suffix}"\
220         "${LFBFL_score_command}"\
221         ${LFBFL_i_score_command_properties}\
222         "\
223     LFBFL_new_lines=${repeated_split_last_line_result}
224 fi
225 LFBFL_new_lines2="    ${LFBFL_cleaned_path2}"
226 if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
227     # shellcheck disable=SC2248
228     repeated_split_last_line "${LFBFL_new_lines2}"\
229         "\
230         ${LFBFL_i_max_line_length}\
231         "${LFBFL_suffix}"\
232         "${LFBFL_score_command2}"\
233         ${LFBFL_i_score_command_properties2}
234     LFBFL_new_lines2=${repeated_split_last_line_result}
235 fi
236 LFBFL_new_lines3="${LFBFL_file_path}"
237 if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
238     # shellcheck disable=SC2248
239     repeated_split_last_line "${LFBFL_new_lines3}"\
240         "\
241         ${LFBFL_i_max_line_length}\
242         "${LFBFL_suffix}"\
243         "${LFBFL_score_command}"\
244         ${LFBFL_i_score_command_properties}
245     LFBFL_new_lines3=${repeated_split_last_line_result}
246 fi
247 LFBFL_s_format="\subsection{\n%\n}\n\n\label{\n%\n}\n\n"
248 LFBFL_s_format+="\VerbatimInput[numbers=left,xleftmargin=-5mm] "
249 LFBFL_s_format+="{\n%\n}\n\n\n"
250 # shellcheck disable=SC2059
251 printf "${LFBFL_s_format}"\
252     "${LFBFL_new_lines}"\
253     "${LFBFL_new_lines2}"\
254     "${LFBFL_new_lines3}"\
255     >> "${LFBFL_temp_files_listing}"
256
257 # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
258 ((++LFBFL_i))
259 LFBFL_new_lines="${LFBFL_file_path}"

```

```

260 if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
261     # shellcheck disable=SC2248
262     repeated_split_last_line "${LFBFL_new_lines}"\
263         "-->"\
264         ${LFBFL_i_max_line_length}\
265         "<!--"\
266         "${LFBFL_score_command}"\
267         "${LFBFL_i_score_command_properties}"
268     LFBFL_new_lines=${repeated_split_last_line_result}
269 fi
270 {
271     LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
272     LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
273     # shellcheck disable=SC2059
274     printf "${LFBFL_s_format}"\
275         "${LFBFL_i}"\
276         "${LFBFL_i}"\
277         "${LFBFL_new_lines}"
278     sed --expression='s/&/\&/g;s/</\&lt;/g'\
279         < "${LFBFL_file_path}"
280     printf "</pre>\n\n\n"
281 } >> "${LFBFL_temp_files_listing2}"
282 LFBFL_s_format="<li><a href=\"\#subsection3.%s\">\n%s\n"
283 LFBFL_s_format+="</a></li>\n"
284 # shellcheck disable=SC2059
285 printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
286 >> "${LFBFL_temp_files_lis}"
287 done
288 overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
289     "${LFBFL_temp_files_listing}"
290 # Example of generated content of files_listing.tex.sub:
291 # -----
292 # \subsection{
293 #   build\_and\_checks\_dependencies/licenses\_templates/%
294 # build\_licenses\_templates.exec.sh
295 # }
296 # \label{
297 #   build_and_checks_dependencies:licenses_templates:%
298 # build_licenses_templatesexecsh
299 # }
300 #
301 # \VerbatimInput[numbers=left,xleftmargin=-5mm]{
302 # build_and_checks_dependencies/licenses_templates/%
303 # build_licenses_templates.exec.sh
304 # }
305 # -----

```



```

306 sed --in-place --expression="s|\\n</pre>|</pre>|g"\  

307 -- "${LFBFL_temp_files_listing2}"\  

308 overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\  

309 "${LFBFL_temp_files_listing2}"\  

310 # Example of generated content of files_listing.html.sub:  

311 # -----  

312 # <h3 id="subsection3.15">3.15  

313 # build_and_checks_dependencies/licenses_templates/<!--  

314 # -->build_licenses_templates.exec.sh  

315 # </h3>  

316 # ...  

317 # <h3 id="subsection3.33">3.33  

318 # build_and_checks_variables/.gitignore  

319 # </h3>  

320 # <pre class="numbered_lines">  

321 # temp/  

322 # upgrade_venvs_ts  

323 # </pre>  

324 # -----  

325 overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\  

326 "${LFBFL_temp_files_lis}"\  

327 # Example of generated content of files_lis.html.sub:  

328 # -----  

329 # <li><a href="#subsection3.15">  

330 # build_and_checks_dependencies/licenses_templates/<!--  

331 # -->build_licenses_templates.exec.sh  

332 # </a></li>  

333 # -----  

334  

335 # shellcheck disable=SC2248  

336 get_overlength_regexp ${LFBFL_i_max_line_length}  

337  

338 # We verify if some lines are beyond max_line_length characters  

339 # in current_tree_light.txt and current_tree.txt.  

340 declare -ar LFBFL_trees=(  

341     "current_tree_light.txt"  

342     "current_tree.txt.temp"  

343 )  

344 local LFBFL_tree  

345 local LFBFL_tree_path  

346 local LFBFL_line  

347 local LFBFL_prefix  

348 declare -i LFBFL_i_prefix_last_position  

349 local LFBFL_char  

350 local LFBFL_line_start  

351 local LFBFL_file_name

```

```

352 local LFBFL_s_lines
353 declare -a LFBFL_arr_lines
354 declare -i LFBFL_i_keep_length
355 for LFBFL_tree in "${LFBFL_trees[@]}"; do
356     LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
357     # shellcheck disable=SC2248
358     LFBFL_s_lines=$(
359         grep --regexp="${get_overlength_regexp_result}" --\
360             "${LFBFL_tree_path}"
361     )
362     if [[ -z "${LFBFL_s_lines}" ]]; then
363         continue
364     fi
365     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
366     for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
367         # printf "LFBFL_line: %s\n" "${LFBFL_line}"
368         LFBFL_prefix=$(
369             printf "%s" "${LFBFL_line}"\
370             | sed --regexp-extended --expression='s/(.*)—[^-]+$/\1/g'
371         )
372         # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
373         LFBFL_i_prefix_last_position=$(( ${#LFBFL_prefix} - 1 ) )
374         LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
375         # printf "LFBFL_char: %s\n" "${LFBFL_char}"
376         LFBFL_prefix=$(
377             printf "%s" "${LFBFL_prefix}"\
378             | sed --regexp-extended --expression='s/[^ ]+$/g'
379         )
380         if [[ "${LFBFL_char}" == "L" ]]; then
381             LFBFL_prefix+=" "
382         else
383             LFBFL_prefix+="|"
384         fi
385         # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
386         LFBFL_file_name=$(
387             printf "%s" "${LFBFL_line}"\
388             | sed --regexp-extended --expression='s|.* ([^ ]+)$|\1|g'
389         )
390         # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
391         LFBFL_i_keep_length=$(( ${#LFBFL_line} - ${#LFBFL_file_name} ))
392         LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
393         LFBFL_line_start=$(
394             printf "%s" "${LFBFL_line_start}"\
395             | sed --expression='s/ */g'
396         )
397         # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"

```

```

398     LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
399     if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
400         # shellcheck disable=SC2248
401         repeated_split_last_line "${LFBFL_new_lines}"\
402             "${LFBFL_prefix}"\
403             "${LFBFL_i_max_line_length}"\
404             ""\
405             ""\
406             ""
407         LFBFL_new_lines=${repeated_split_last_line_result}
408     fi
409     printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
410         > "${LFBFL_tree_path}.some_line.temp"
411     insert_file_at_token "${LFBFL_tree_path}"\
412         "${LFBFL_line}"\
413         "${LFBFL_tree_path}.some_line.temp"\
414         ""\
415         --quiet
416     done
417     rm --force -- "${LFBFL_tree_path}.some_line.temp"
418 done
419
420 overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
421     "${LFBFL_temp_path}/current_tree.txt.temp" --keep-temp --tree-mode
422
423 local LFBFL_tex_path_start
424 LFBFL_tex_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
425 readonly LFBFL_tex_path_start
426 local LFBFL_html_path_start
427 LFBFL_html_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.html"
428 readonly LFBFL_html_path_start
429
430 if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
431     # If there is a template, we init the process from it.
432     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
433         "${LFBFL_tex_path_start}.1"
434 elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
435     # Otherwise if there is a tex file, we init the process from it.
436     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
437         "${LFBFL_tex_path_start}.1"
438 else
439     printf "Neither .tex.tpl, nor .tex in ./s/.\n" "${LFBFL_subdir2}"
440 fi
441
442 # Same logic with repository HTML file.
443 if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then

```

```

444     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
445         "${LFBFL_html_path_start}.1"
446 elif [[ -f "${LFBFL_repository_name}.html" ]]; then
447     cp -- "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
448 else
449     printf "Neither .html.tpl in ./%/ , nor .html in ./.\n"\
450         "${LFBFL_subdir2}"
451 fi
452
453 # shellcheck disable=SC2001
454 printf "%s\n" "${LFBFL_abstract}"\
455     | sed --expression='s/\\n/\\n/g'\
456     > "${LFBFL_temp_path}/abstract_temp"
457
458 # shellcheck disable=SC2001
459 printf "%s\n" "${LFBFL_acknowledgments}"\
460     | sed --expression='s/\\n/\\n/g'\
461     > "${LFBFL_temp_path}/acknowledgments_temp"
462
463 # But the filling still occurs, in case the dev want to refill
464 # part of the tex/html file that he modified with @token@
465 # using some computed results.
466 # Tex filling:
467 if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
468     sed --in-place\
469         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
470         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
471         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
472         --expression="s|@author_email@|${LFBFL_author_email}|g"\
473         --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
474         --expression="s|@author_website@|${LFBFL_author_website}|g"\
475         --expression="s|@current_date@|${LFBFL_current_date}|g"\
476         --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
477         --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
478         --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
479         -- "${LFBFL_tex_path_start}.1"
480
481     insert_file_at_token "${LFBFL_tex_path_start}.1"\
482         @abstract@\
483         "${LFBFL_temp_path}/abstract_temp"\
484         "${LFBFL_tex_path_start}.2"
485
486     insert_file_at_token "${LFBFL_tex_path_start}.2"\
487         @acknowledgments@\
488         "${LFBFL_temp_path}/acknowledgments_temp"\
489         "${LFBFL_tex_path_start}.3"

```

```

490
491 insert_file_at_token "${LFBFL_tex_path_start}.3"\
492 @current_tree_light@\
493 "${LFBFL_temp_path}/current_tree_light.txt"\
494 "${LFBFL_tex_path_start}.4"
495
496 insert_file_at_token "${LFBFL_tex_path_start}.4"\
497 @current_tree@\
498 "${LFBFL_temp_path}/current_tree.txt"\
499 "${LFBFL_tex_path_start}.5"
500
501 insert_file_at_token "${LFBFL_tex_path_start}.5"\
502 @dependencies_notes@\
503 "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
504 "${LFBFL_tex_path_start}.6"
505
506 insert_file_at_token "${LFBFL_tex_path_start}.6"\
507 @files_listing_VerbatimInput@\
508 "${LFBFL_temp_path}/files_listing.tex.sub"\
509 "${LFBFL_tex_path_start}.7"
510
511 overwrite_if_not_equal\
512 "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
513 "${LFBFL_tex_path_start}.7" --keep-temp
514 fi
515
516 # HTML filling:
517 declare -i LFBFL_i_HTML_updated=0
518 if [[ -f "${LFBFL_html_path_start}.1" ]]; then
519     sed --in-place\
520         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
521         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
522         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
523         --expression="s|@author_email@|${LFBFL_author_email}|g"\
524         --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
525         --expression="s|@author_website@|${LFBFL_author_website}|g"\
526         --expression="s|@current_date@|${LFBFL_current_date}|g"\
527         --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
528         --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
529         --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
530         -- "${LFBFL_html_path_start}.1"
531
532 insert_file_at_token "${LFBFL_html_path_start}.1"\
533 @abstract@\
534 "${LFBFL_temp_path}/abstract_temp"\
535 "${LFBFL_html_path_start}.2"

```

```

536
537 insert_file_at_token "${LFBFL_html_path_start}.2"\
538     @acknowledgments@\
539     "${LFBFL_temp_path}/acknowledgments_temp"\
540     "${LFBFL_html_path_start}.3"
541
542 insert_file_at_token "${LFBFL_html_path_start}.3"\
543     @files_lis@\
544     "${LFBFL_temp_path}/files_lis.html.sub"\
545     "${LFBFL_html_path_start}.4"
546
547 insert_file_at_token "${LFBFL_html_path_start}.4"\
548     @current_tree_light@\
549     "${LFBFL_temp_path}/current_tree_light.txt"\
550     "${LFBFL_html_path_start}.5"
551
552 insert_file_at_token "${LFBFL_html_path_start}.5"\
553     @current_tree@\
554     "${LFBFL_temp_path}/current_tree.txt"\
555     "${LFBFL_html_path_start}.6"
556
557 insert_file_at_token "${LFBFL_html_path_start}.6"\
558     @dependencies_notes@\
559     "${LFBFL_temp_path}/dependencies_notes.html.sub"\
560     "${LFBFL_html_path_start}.7"
561
562 insert_file_at_token "${LFBFL_html_path_start}.7"\
563     @files_listing_HTMLPreInput@\
564     "${LFBFL_temp_path}/files_listing.html.sub"\
565     "${LFBFL_html_path_start}.8"
566
567 overwrite_if_not_equal "${LFBFL_repository_name}.html"\
568     "${LFBFL_html_path_start}.8" --keep-temp
569 LFBFL_i_HTML_updated=$?
570 fi
571
572 # The HTML contains everything including listings with source code.
573 # If it doesn't get updated,
574 # then it is useless to recompute the PDF.
575 if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
576     return
577 fi
578
579 local LFBFL_latex_command=""
580 grep_variable "${LFBFL_data_file_path}" latex_command\
581     --result-variable-prefix="LFBFL_"\

```

```

582     --replace-line-returns-by=""
583
584     case ${LFBFL_latex_command} in
585         pdflatex) : ;;
586         lualatex) : ;;
587         *) printf "Unsupported latex command" ; return 1 ;;
588     esac
589
590     local LFBFL_command
591     LFBFL_command="${LFBFL_latex_command}"
592     LFBFL_command+=" ' ${LFBFL_subdir2}/${LFBFL_repository_name}.tex' "
593     if [[ LFBFL_i_verbose -eq 0 ]]; then
594         LFBFL_command+=" > /dev/null"
595     fi
596     readonly LFBFL_command
597
598     for ((LFBFL_i = 0; LFBFL_i < 3; ++LFBFL_i)); do
599         eval -- "${LFBFL_command}"
600     done
601
602     LFBFL_files_to_temp=(\
603         "${LFBFL_repository_name}.aux"\
604         "${LFBFL_repository_name}.log"\
605         "${LFBFL_repository_name}.out"\
606         "${LFBFL_repository_name}.toc"
607     )
608     for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
609         mv -- "${LFBFL_file_name}" "${LFBFL_temp_path}/${LFBFL_file_name}"
610     done
611 }
612
613 create_PDF "$@"

```

3.13 build_and_checks_dependencies/generate_from_template.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed

```

```

12 # in the hope that it will be useful,
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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "generate_from_template.sh" to
25 # "generate_from_template.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=lines_filters.lib.sh
29 source "${LFBFL_subdir}/lines_filters.lib.sh"
30 # shellcheck source=overwrite_if_not_equal.lib.sh
31 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
32
33 generate_from_template_with_block_comments() {
34     # $1=base_file_path
35     # $2=target_file_path
36     # $3=enter_block_comment
37     # $4=exit_block_comment
38     local LFBFL_temp
39     LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
40     readonly LFBFL_temp
41     {
42         printf "%s\n" "$3"
43         cat -- "$1"
44         printf "%s\n" "$4"
45     } > "${LFBFL_temp}"
46     overwrite_if_not_equal "$2" "${LFBFL_temp}"
47 }
48
49 generate_from_template_with_line_comments() {
50     # $1=base_file_path
51     # $2=target_file_path
52     # $3=line_comment_prefix
53     # $4=optional_post_processing
54     # $5=file_prefix
55     local LFBFL_temp
56     LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57     readonly LFBFL_temp

```



```

58     if [[ -n "$5" ]]; then
59         printf "%s\n" "$5"\
60             > "${LFBFL_temp}"
61     fi
62     sed --expression="s~^~$3~g" -- "$1"\
63         >> "${LFBFL_temp}"
64     if [[ -n "$4" ]]; then
65         eval -- "$4"
66     fi
67     overwrite_if_not_equal "$2" "${LFBFL_temp}"
68 }
69
70 split_file_in_two() {
71     # $1=file_path
72     # $2=token assume that token is the only thing on his line,
73     #     except for a possible indentation.
74     # $3=path_of_part1_of_file
75     # $4=path_of_part2_of_file
76     # We cannot add --line-regexp with --fixed-strings because of the
77     # possible indentation in file,
78     # instead we use grep_fixed_string_with_anchor.
79     declare -ir LFBFL_i_line_number=$(
80         grep_fixed_string_with_anchor "$1" "$2"\
81             --enforce-line-ends-with-fixed-string --line-number\
82             | head --lines=1\
83             | cut --delimiter=':' --fields=1
84     )
85     declare -ir LFBFL_i_line_count=$((ll_wc -l -n "$1"))
86     declare -ir LFBFL_i_lines_after=$((
87         LFBFL_i_line_count - LFBFL_i_line_number
88     ))
89     head --lines="$((LFBFL_i_line_number - 1))" -- "$1"\
90         > "$3"
91     tail --lines="${LFBFL_i_lines_after}" -- "$1"\
92         > "$4"
93 }
94
95 insert_file_at_token() {
96     # $1=file_path
97     # $2=token assume that token is the only thing on his line,
98     #     except for a possible indentation.
99     # $3=path_of_file_to_insert
100    # $4=path_of_result_file
101    # Options:
102    # $5=--quiet to suppress diff output in overwrite_if_not_equal
103    declare -r LFBFL_start_file_path="$1.insert_file_at_token1.temp"

```

```

104 declare -r LFBFL_end_file_path="$1.insert_file_at_token2.temp"
105 local LFBFL_result_file_path
106 if [[ -n "$4" ]]; then
107     LFBFL_result_file_path="$4"
108 else
109     LFBFL_result_file_path="$1.insert_file_at_token3.temp"
110 fi
111 readonly LFBFL_result_file_path
112 split_file_in_two "$1" "$2" "${LFBFL_start_file_path}"\
113     "${LFBFL_end_file_path}"
114 cat -- "${LFBFL_start_file_path}" "$3" "${LFBFL_end_file_path}"\
115     > "${LFBFL_result_file_path}"
116 rm -- "${LFBFL_start_file_path}" "${LFBFL_end_file_path}"
117 if [[ -z "$4" ]]; then
118     overwrite_if_not_equal "$1" "${LFBFL_result_file_path}" "$5"
119 fi
120 }

```

3.14 build_and_checks_dependencies/get_common_text_glob_patterns.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
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23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "get_common_text_glob_patterns.sh"
25 # to "get_common_text_glob_patterns.lib.sh".
26

```

```

27 get_COMMON_TEXT_PATHS_GLOB_PATTERNS() {
28     if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
29         return
30     fi
31
32     declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
33         '**/*.c'
34         '**/*.css'
35         '**/*.h'
36         '**/*.htm'
37         '**/*.html'
38         '**/*.ini'
39         '**/*.js'
40         '**/*.json'
41         '**/*.md'
42         '**/*.php'
43         '**/*.py'
44         '**/*.rs'
45         '**/*.rlib'
46         '**/*.sh'
47         '**/*.sql'
48         '**/*.sub'
49         '**/*.tex'
50         '**/*.toml'
51         '**/*.tpl'
52         '**/*.ts'
53         '**/*.txt'
54         '**/*.xml'
55         '**/*.yaml'
56         '**/COPYING'
57         '**/COPYING.LESSER'
58         '**/pre-commit'
59         '**/post-commit'
60         '**/.gitignore'
61         '**/py.typed'
62     )
63 }
64
65
66 get_COMMON_TEXT_FILES_GLOB_PATTERNS() {
67     if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
68         return
69     fi
70
71     declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
72         '*.c'

```

```

73     '*.css'
74     '*.h'
75     '*.htm'
76     '*.html'
77     '*.ini'
78     '*.js'
79     '*.json'
80     '*.md'
81     '*.php'
82     '*.py'
83     '*.rs'
84     '*.rlib'
85     '*.sh'
86     '*.sql'
87     '*.sub'
88     '*.tex'
89     '*.toml'
90     '*.tpl'
91     '*.ts'
92     '*.txt'
93     '*.xml'
94     '*.yaml'
95     'COPYING'
96     'COPYING.LESSER'
97     'pre-commit'
98     'post-commit'
99     '.gitignore'
100    'py.typed'
101  )
102  }

```

3.15 build_and_checks_dependencies/grammar_and_spelling_check.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
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```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # 2026-03-22T15:30+01:00: This file was renamed from
25 # "grammar_and_spell_check.lib.sh"
26 # to
27 # "grammar_and_spelling_check.lib.sh".
28 # This file was renamed from "grammar_and_spell_check.sh"
29 # to "grammar_and_spell_check.lib.sh".
30
31 LFBFL_subdir="build_and_checks_dependencies"
32 # shellcheck source=common_options.lib.sh
33 source "${LFBFL_subdir}/common_options.lib.sh"
34 # shellcheck source=get_common_text_glob_patterns.lib.sh
35 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
36 # shellcheck source=lines_filters.lib.sh
37 source "${LFBFL_subdir}/lines_filters.lib.sh"
38
39 grammar_and_spelling_check() {
40     # $1=configuration_path build_and_checks_variables/repository_data.txt
41     # Options:
42     #   --verbose
43     #   --work-directory=""
44     declare -i LFBFL_i_verbose=0
45     get_verbose_option "$@"
46     local LFBFL_work_directory=""
47     get_work_directory_option "$@"
48     declare -a LFBFL_return_traps_stack
49     init_return_trap
50     pushd_to_work_directory --trap-popd
51     can_continue_after_enhanced_pushd || return 1
52
53     enhanced_set_shell_option pipefail --trap-unset
54
55     get_COMMON_TEXT_FILES_GLOB_PATTERNS
56     local LFBFL_grammar_and_spelling_checker_command=""
57     grep_variable "$1" grammar_and_spelling_checker_command\
58         --result-variable-prefix=LFBFL_\
59         --replace-line-returns-by=""

```

```

60 declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
61 if [[ -z "${LFBFL_command}" ]]; then
62     if [[ LFBFL_i_verbose -eq 1 ]]; then
63         printf "grammar_and_spelling_check: No command in configuration.\n"
64     fi
65     return
66 fi
67 if [[ LFBFL_i_verbose -eq 1 ]]; then
68     printf "grammar_and_spelling_check: Command found: %s.\n"\
69         "${LFBFL_command}"
70 fi
71 local LFBFL_file_path
72 local LFBFL_s_files_paths
73 declare -a LFBFL_arr_files_paths
74 local LFBFL_pattern
75 local LFBFL_eval_string
76 for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
77     if [[ LFBFL_i_verbose -eq 1 ]]; then
78         printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
79             "${LFBFL_pattern}"
80     fi
81     LFBFL_s_files_paths=$(
82         find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
83         | relevant_find
84     )
85     if [[ -z "${LFBFL_s_files_paths}" ]]; then
86         [[ LFBFL_i_verbose -eq 1 ]]\
87         && printf "grammar_and_spelling_check: No file found.\n"
88         continue
89     fi
90     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
91     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
92         LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
93         eval -- "${LFBFL_eval_string}"
94     done
95 done
96 }

```

3.16 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```

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23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "build_licenses_templates.sh"
25 # to "build_licenses_templates.exec.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=check_shell_scripts_beginnings.lib.sh
29 source "${LFBFL_subdir}/check_shell_scripts_beginnings.lib.sh"
30 # shellcheck source=common_options.lib.sh
31 source "${LFBFL_subdir}/common_options.lib.sh"
32 # shellcheck source=comparisons.lib.sh
33 source "${LFBFL_subdir}/comparisons.lib.sh"
34 # shellcheck source=generate_from_template.lib.sh
35 source "${LFBFL_subdir}/generate_from_template.lib.sh"
36 # shellcheck source=lines_filters.lib.sh
37 source "${LFBFL_subdir}/lines_filters.lib.sh"
38 # shellcheck source=overwrite_if_not_equal.lib.sh
39 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
40
41 build_licenses_templates() {
42     # Options:
43     #   --verbose
44     #   --work-directory=""
45     declare -i LFBFL_i_verbose=0
46     get_verbose_option "$@"
47     local LFBFL_work_directory=""
48     get_work_directory_option "$@"
49     declare -a LFBFL_return_traps_stack
50     init_return_trap
51
52     enhanced_set_shell_option pipefail --trap-unset

```

```

53
54     local LFBFL_s_format
55
56     # Source
57     local LFBFL_license_source_subdir
58     LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
59     readonly LFBFL_license_source_subdir
60     local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
61     LFBFL_source_license_prefix+="license_file_header_"
62     readonly LFBFL_source_license_prefix
63     # Target
64     local LFBFL_license_target_subdir
65     LFBFL_license_target_subdir="${LFBFL_work_directory}/"
66     LFBFL_license_target_subdir="build_and_checks_variables/temp/"
67     readonly LFBFL_license_target_subdir
68     local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
69     LFBFL_target_license_prefix+="license_file_header_"
70     readonly LFBFL_target_license_prefix
71
72     declare -ar LFBFL_licenses=(
73         "AGPLv3+"
74         "GPLv3+"
75         "LGPLv3+"
76     )
77
78     # -----
79     # Various language definitions for text file formats
80     declare -ar LFBFL_block_comment_languages=(
81         "c"
82         "py"
83         "xml"
84     )
85     declare -ar LFBFL_block_comment_enters=(
86         '/ * '
87         ' " " '
88         ' <!-- '
89     )
90     declare -ar LFBFL_block_comment_exits=(
91         ' * / '
92         ' " " '
93         ' --> '
94     )
95     equal "${#LFBFL_block_comment_languages[@]}" "\
96         "${#LFBFL_block_comment_enters[@]}" "\
97         "${#LFBFL_block_comment_exits[@]}" "
98     declare -i LFBFL_i_equal_result=$?

```



```

99     if [[ LFBFL_i_equal_result -eq 0 ]]; then
100         printf $'/!\Problem with the definitions of languages arrays 1/!\n'
101         return 1
102     fi
103
104     declare -ar LFBFL_line_comment_languages=(
105         "rs"
106         "sh"
107         "sql"
108         "tex"
109     )
110     declare -ar LFBFL_line_comment_prefixes=(
111         '// '
112         '# '
113         '-- '
114         '% '
115     )
116     equal "${#LFBFL_line_comment_languages[@]}" "\
117         "${#LFBFL_line_comment_prefixes[@]}"
118     LFBFL_i_equal_result=$?
119     if [[ LFBFL_i_equal_result -eq 0 ]]; then
120         printf $'/!\Problem with the definitions of languages arrays 2/!\n'
121         return 1
122     fi
123     # -----
124
125     # -----
126     # We generate the license header templates adapted to various text
127     # files formats.
128     # Kamoulox do endfor, Jacques Beauregard XD
129     # (C'est toujours mieux que l'inverse :) !)
130     # for do endfor -> for do done (Fort Doux Donne).
131     local LFBFL_some_license
132     local LFBFL_license_prefix
133     local LFBFL_license_prefix2
134     local LFBFL_extension
135     local LFBFL_license_file_path
136     local LFBFL_enter_string
137     local LFBFL_exit_string
138     local LFBFL_temp2
139     local LFBFL_prefix_string
140     local LFBFL_intermediate_file_path
141     local LFBFL_file_prefix
142     declare -i LFBFL_i_generate_from_template_result
143     declare -i LFBFL_i
144     for LFBFL_some_license in "${LFBFL_licenses[@]}; do

```

```

145 LFBFL_license_prefix="${LFBFL_source_license_prefix}"
146 LFBFL_license_prefix+="${LFBFL_some_license}"
147 LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
148 LFBFL_license_prefix2+="${LFBFL_some_license}"
149 for ((
150     LFBFL_i = 0;
151     LFBFL_i < ${#LFBFL_block_comment_languages[@]};
152     ++LFBFL_i
153 )); do
154     LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
155     LFBFL_license_file_path="${LFBFL_license_prefix2}"
156     LFBFL_license_file_path+=".${LFBFL_extension}"
157     LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
158     LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
159     generate_from_template_with_block_comments\
160         "${LFBFL_license_prefix}.tpl"\
161         "${LFBFL_license_file_path}.tpl"\
162         "${LFBFL_enter_string}" "${LFBFL_exit_string}"
163     LFBFL_i_generate_from_template_result=$?
164     if [[ LFBFL_i_verbose -eq 1 ]]; then
165         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
166             printf "License template %s.tpl updated.\n"\
167                 "${LFBFL_license_file_path}"
168         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
169             printf "License template %s.tpl created.\n"\
170                 "${LFBFL_license_file_path}"
171         fi
172     fi
173 done
174 LFBFL_temp2=".generate_from_template_with_line_comments.temp"
175 for ((
176     LFBFL_i = 0;
177     LFBFL_i < ${#LFBFL_line_comment_languages[@]};
178     ++LFBFL_i
179 )); do
180     LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
181     LFBFL_license_file_path="${LFBFL_license_prefix2}"
182     LFBFL_license_file_path+=".${LFBFL_extension}"
183     LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
184     LFBFL_intermediate_file_path="${LFBFL_license_file_path}.tpl"
185     LFBFL_intermediate_file_path+="${LFBFL_temp2}"
186     LFBFL_file_prefix=""
187     if [[ "${LFBFL_extension}" == "sh" ]]; then
188         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
189     fi
190     # The sed command is here to remove trailing spaces of previously

```

```

191     # empty lines after comment prefix is added to the line.
192     generate_from_template_with_line_comments\
193         "${LFBFL_license_prefix}.tpl"\
194         "${LFBFL_license_file_path}.tpl"\
195         "${LFBFL_prefix_string}"\
196         "sed -Ei -e 's/\s*$/g' '${LFBFL_intermediate_file_path}'"\
197         "${LFBFL_file_prefix}"
198     LFBFL_i_generate_from_template_result=$?
199     if [[ LFBFL_i_verbose -eq 1 ]]; then
200         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
201             printf "License template %s.tpl updated.\n"\
202                 "${LFBFL_license_file_path}"
203         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
204             printf "License template %s.tpl created.\n"\
205                 "${LFBFL_license_file_path}"
206         fi
207     fi
208 done
209 done
210 # -----
211
212 # Now that base license templates generated languages license
213 # templates in work directory, we can go there.
214 pushd_to_work_directory --trap-popd
215 can_continue_after_enhanced_pushd || return 1
216
217 local LFBFL_data_file_path="build_and_checks_variables/"
218 LFBFL_data_file_path+="repository_data.txt"
219 readonly LFBFL_data_file_path
220 local LFBFL_repository_name=""
221 grep_variable "${LFBFL_data_file_path}" repository_name\
222     --result-variable-prefix="LFBFL_"\
223     --replace-line-returns-by=""
224 local LFBFL_license=""
225 grep_variable "${LFBFL_data_file_path}" license\
226     --result-variable-prefix="LFBFL_"
227 local LFBFL_license2=""
228 grep_variable "${LFBFL_data_file_path}" license2\
229     --result-variable-prefix="LFBFL_"
230 local LFBFL_author_full_name=""
231 grep_variable "${LFBFL_data_file_path}" author_full_name\
232     --result-variable-prefix="LFBFL_"\
233     --replace-line-returns-by=""
234
235 LFBFL_license_prefix="build_and_checks_variables/temp/"
236 LFBFL_license_prefix+="license_file_header_"

```

```

237     readonly LFBFL_license_prefix
238     LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
239     readonly LFBFL_license_prefix2
240     local LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
241     readonly LFBFL_license_prefix3
242
243     # First year according to current state of git repository.
244     declare -ir LFBFL_i_first_year=$(
245         git log\
246         | grep 'Date:'\
247         | tail --lines=1\
248         | cut --delimiter=' ' --fields=8
249     )
250     # Last year according to current state of git repository.
251     declare -ir LFBFL_i_last_year=$(
252         git log\
253         | grep 'Date:'\
254         | head --lines=1\
255         | cut --delimiter=' ' --fields=8
256     )
257     local LFBFL_copyright_string
258     LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
259     LFBFL_copyright_string+="${LFBFL_author_full_name}"
260     readonly LFBFL_copyright_string
261
262     prepare_filled_license_file_block() {
263         # $1=license_file_path
264         sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
265             --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
266             -- "$1.tpl"\
267             > "$1.temp"
268         overwrite_if_not_equal "$1" "$1.temp"
269         declare -ir LFBFL_i_result=$?
270         if [[ LFBFL_i_verbose -eq 1 ]]; then
271             if [[ LFBFL_i_result -eq 1 ]]; then
272                 printf "License template %s updated.\n" "$1"
273             elif [[ LFBFL_i_result -eq 2 ]]; then
274                 printf "License template %s created.\n" "$1"
275             fi
276         fi
277         head --lines=-1 -- "$1"\
278         | tail --lines=+2\
279         > "$1.temp"
280         # Strange world: tail --lines=-1 was working
281         # to exclude first line.
282         # I tested it months ago, and saw with my eyes the start and end

```

```

283     # block comments removed with:
284     # head --lines=-1 "$1" | tail --lines=-1.
285     # Now, man page says to use --lines=+2 instead.
286     # Welcome in the world where AI change your environment and your
287     # documentation on the fly...
288 }
289
290 declare -Ar LFBFL_all_block_comment_languages=(
291     ["c"]="c"
292     ["css"]="c"
293     ["html"]="xml"
294     ["php"]="c"
295     ["py"]="py"
296     ["js"]="c"
297     ["ts"]="c"
298     ["xml"]="xml"
299 )
300
301 local LFBFL_key
302 local LFBFL_dest
303 local LFBFL_license_file_path2
304 declare -i LFBFL_i_not_subfile
305 declare -i LFBFL_i_not_subfile2
306 local LFBFL_file_path
307 local LFBFL_s_files_paths
308 declare -a LFBFL_arr_files_paths
309 for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]"; do
310     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
311     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
312     prepare_filled_license_file_block "${LFBFL_license_file_path}"
313     if [[ -n "${LFBFL_license2}" ]]; then
314         LFBFL_license_file_path2="${LFBFL_license_prefix3}"
315         LFBFL_license_file_path2+=".${LFBFL_dest}"
316         prepare_filled_license_file_block "${LFBFL_license_file_path2}"
317     fi
318
319     LFBFL_s_files_paths=(
320         find . -type f -iregex ".*\\.\\(${LFBFL_key}\\|\\${LFBFL_key}\\\\.tpl\\)" \
321             -printf '%P\n' \
322             | relevant_find
323     )
324     if [[ -z "${LFBFL_s_files_paths}" ]]; then
325         if [[ LFBFL_i_verbose -eq 1 ]]; then
326             LFBFL_s_format="build_licenses_templates: "
327             LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
328             # shellcheck disable=SC2059

```

```

329         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
330     fi
331     continue
332 fi
333 mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
334 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]"; do
335     is_subfile "${LFBFL_file_path}"\
336         "${LFBFL_license_file_path}.temp"
337     LFBFL_i_not_subfile=$?
338     LFBFL_i_not_subfile2=1
339     if [[ -n "${LFBFL_license2}" ]]; then
340         is_subfile "${LFBFL_file_path}"\
341             "${LFBFL_license_file_path2}.temp"
342         LFBFL_i_not_subfile2=$?
343     fi
344     if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
345     then
346         printf "File %s has no/wrong license header?\n"\
347             "${LFBFL_file_path}"
348     fi
349 done
350 rm -- "${LFBFL_license_file_path}.temp"
351 if [[ -n "${LFBFL_license2}" ]]; then
352     rm -- "${LFBFL_license_file_path2}.temp"
353 fi
354 done
355
356 prepare_filled_license_file_line() {
357     # $1=LFBFL_license_file_path
358     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
359         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
360         -- "$1.tpl"\
361         > "$1.temp"
362     overwrite_if_not_equal "$1" "$1.temp"
363     declare -ir LFBFL_i_result=$?
364     if [[ LFBFL_i_verbose -eq 1 ]]; then
365         if [[ LFBFL_i_result -eq 1 ]]; then
366             printf "License template %s updated.\n" "$1"
367         elif [[ LFBFL_i_result -eq 2 ]]; then
368             printf "License template %s created.\n" "$1"
369         fi
370     fi
371 }
372
373 declare -Ar LFBFL_all_line_comment_languages=(
374     ["rlib"]="rs"

```

```

375     ["rs"]="rs"
376     ["sh"]="sh"
377     ["sql"]="sql"
378     ["tex"]="tex"
379     ["toml"]="sh"
380     ["yaml"]="sh"
381 )
382
383 for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}; do
384     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
385     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
386     prepare_filled_license_file_line "${LFBFL_license_file_path}"
387     if [[ -n "${LFBFL_license2}" ]]; then
388         LFBFL_license_file_path2="${LFBFL_license_prefix3}"
389         LFBFL_license_file_path2+=".${LFBFL_dest}"
390         prepare_filled_license_file_line "${LFBFL_license_file_path2}"
391     fi
392
393     LFBFL_s_files_paths=(
394         find . -type f -iregex ".*\\.\\(${LFBFL_key}\\|${LFBFL_key}\\.tpl\\)" \
395             -printf '%P\n' \
396             | relevant_find
397     )
398     if [[ -z "${LFBFL_s_files_paths}" ]]; then
399         if [[ LFBFL_i_verbose -eq 1 ]]; then
400             LFBFL_s_format="build_licenses_templates: "
401             LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
402             # shellcheck disable=SC2059
403             printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
404         fi
405         continue
406     fi
407     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
408     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}; do
409         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}"
410         LFBFL_i_not_subfile=$?
411         LFBFL_i_not_subfile2=1
412         if [[ -n "${LFBFL_license2}" ]]; then
413             is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}"
414             LFBFL_i_not_subfile2=$?
415         fi
416         if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
417         then
418             printf "File %s has no/wrong license header?\n" \
419                 "${LFBFL_file_path}"
420         fi

```

```

421     done
422 done
423 }
424
425 build_licenses_templates "$@"

```

3.17 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```

1 This file is part of @repository_name@.
2
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6 as published by the Free Software Foundation,
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8 or (at your option) any later version.
9
10 @repository_name@ is distributed
11 in the hope that it will be useful,
12 but WITHOUT ANY WARRANTY;
13 without even the implied warranty of
14 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 See the GNU Affero General Public License for more details.
16
17 You should have received a copy of
18 the GNU Affero General Public License
19 along with @repository_name@.
20 If not, see <https://www.gnu.org/licenses/>.
21
22 ©Copyright @copyright_string@

```

3.18 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

```

1 This file is part of @repository_name@.
2
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5 under the terms of the GNU General Public License
6 as published by the Free Software Foundation,
7 either version 3 of the License,
8 or (at your option) any later version.
9
10 @repository_name@ is distributed
11 in the hope that it will be useful,
12 but WITHOUT ANY WARRANTY;
13 without even the implied warranty of

```



```

14  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15  See the GNU General Public License for more details.
16
17  You should have received a copy of
18  the GNU General Public License
19  along with @repository_name@.
20  If not, see <https://www.gnu.org/licenses/>.
21
22  ©Copyright @copyright_string@

```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```

1  This file is part of @repository_name@.
2
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6  as published by the Free Software Foundation,
7  either version 3 of the License,
8  or (at your option) any later version.
9
10 @repository_name@ is distributed
11 in the hope that it will be useful,
12 but WITHOUT ANY WARRANTY;
13 without even the implied warranty of
14 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 See the GNU Lesser General Public License for more details.
16
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18 the GNU Lesser General Public License
19 along with @repository_name@.
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21
22 ©Copyright @copyright_string@

```

3.20 build_and_checks_dependencies/lines_counts.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.

```

```

10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "lines_counts.sh" to
25 # "lines_counts.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=get_common_text_glob_patterns.lib.sh
31 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
32 # shellcheck source=lines_filters.lib.sh
33 source "${LFBFL_subdir}/lines_filters.lib.sh"
34
35 all_code_lines() {
36     # Options:
37     #   --verbose
38     #   --work-directory=""
39     declare -i LFBFL_i_verbose=0
40     get_verbose_option "$@"
41     local LFBFL_work_directory=""
42     get_work_directory_option "$@"
43     declare -a LFBFL_return_traps_stack
44     init_return_trap
45     pushd_to_work_directory --trap-popd
46     can_continue_after_enhanced_pushd || return 1
47
48     enhanced_set_shell_option pipefail --trap-unset
49
50     get_COMMON_TEXT_FILES_GLOB_PATTERNS
51     local LFBFL_pattern
52     for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
53         if [[ LFBFL_i_verbose -eq 1 ]]; then
54             printf "all_code_lines: Iterating on pattern: %s.\n\"
55                 "${LFBFL_pattern}"

```

```

56     fi
57     find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
58         | xargs grep --invert-match --perl-regexp --with-filename\
59         -- 'a(?!a)a'
60 done
61 }
62
63 all_self_code_lines(){
64     # Options:
65     #   --verbose
66     #   --work-directory=""
67     declare -i LFBFL_i_verbose=0
68     get_verbose_option "$@"
69     local LFBFL_work_directory=""
70     get_work_directory_option "$@"
71     declare -a LFBFL_return_traps_stack
72     init_return_trap
73     pushd_to_work_directory --trap-popd
74     can_continue_after_enhanced_pushd || return 1
75
76     enhanced_set_shell_option pipefail --trap-unset
77
78     declare -a LFBFL_common_options=("$@")
79     keep_options LFBFL_common_options --verbose
80
81     # below and another lines, see SC2145 false positive
82     all_code_lines "${LFBFL_common_options[@]}" \
83         | relevant_grep\
84         | not_license_grep\
85         | not_main_tex_grep\
86         | not_main_html_grep
87 }
88
89 # shellcheck disable=SC2120
90 all_self_empty_code_lines(){
91     # Options:
92     #   --verbose
93     #   --work-directory=""
94     declare -i LFBFL_i_verbose=0
95     get_verbose_option "$@"
96     local LFBFL_work_directory=""
97     get_work_directory_option "$@"
98     declare -a LFBFL_return_traps_stack
99     init_return_trap
100    pushd_to_work_directory --trap-popd
101    can_continue_after_enhanced_pushd || return 1

```

```

102
103     enhanced_set_shell_option pipefail --trap-unset
104
105     declare -a LFBFL_common_options=("$@")
106     keep_options LFBFL_common_options --verbose
107
108     all_self_code_lines "${LFBFL_common_options[@]}" \
109         | empty_lines_after_file_name
110 }
111
112 # shellcheck disable=SC2120
113 all_self_not_empty_code_lines() {
114     # Options:
115     #   --verbose
116     #   --work-directory=""
117     declare -i LFBFL_i_verbose=0
118     get_verbose_option "$@"
119     local LFBFL_work_directory=""
120     get_work_directory_option "$@"
121     declare -a LFBFL_return_traps_stack
122     init_return_trap
123     pushd_to_work_directory --trap-popd
124     can_continue_after_enhanced_pushd || return 1
125
126     enhanced_set_shell_option pipefail --trap-unset
127
128     declare -a LFBFL_common_options=("$@")
129     keep_options LFBFL_common_options --verbose
130
131     all_self_code_lines "${LFBFL_common_options[@]}" \
132         | not_empty_lines_after_file_name
133 }
134
135 code_lines_count_all() {
136     # Options:
137     #   --verbose
138     #   --work-directory=""
139     declare -i LFBFL_i_verbose=0
140     get_verbose_option "$@"
141     local LFBFL_work_directory=""
142     get_work_directory_option "$@"
143     declare -a LFBFL_return_traps_stack
144     init_return_trap
145     pushd_to_work_directory --trap-popd
146     can_continue_after_enhanced_pushd || return 1
147

```

```

148     enhanced_set_shell_option pipefail --trap-unset
149
150     declare -gi i_code_lines_count_all_result
151     i_code_lines_count_all_result=$(
152         all_self_code_lines\
153         | wc --lines
154     )
155     # The printf is for command line use.
156     # A script should use the result value instead.
157     # shellcheck disable=SC2248
158     printf "%s\n" "${i_code_lines_count_all_result}"
159 }
160
161 code_lines_count_empty() {
162     # Options:
163     #   --verbose
164     #   --work-directory=""
165     declare -i LFBFL_i_verbose=0
166     get_verbose_option "$@"
167     local LFBFL_work_directory=""
168     get_work_directory_option "$@"
169     declare -a LFBFL_return_traps_stack
170     init_return_trap
171     pushd_to_work_directory --trap-popd
172     can_continue_after_enhanced_pushd || return 1
173
174     enhanced_set_shell_option pipefail --trap-unset
175
176     declare -gi i_code_lines_count_empty_result
177     # shellcheck disable=SC2119
178     i_code_lines_count_empty_result=$(
179         all_self_empty_code_lines\
180         | wc --lines
181     )
182     # The printf is for command line use.
183     # A script should use the result value instead.
184     # shellcheck disable=SC2248
185     printf "%s\n" "${i_code_lines_count_empty_result}"
186 }
187
188 code_lines_count_not_empty() {
189     # Options:
190     #   --verbose
191     #   --work-directory=""
192     declare -i LFBFL_i_verbose=0
193     get_verbose_option "$@"

```

```

194 local LFBFL_work_directory=""
195 get_work_directory_option "$@"
196 declare -a LFBFL_return_traps_stack
197 init_return_trap
198 pushd_to_work_directory --trap-popd
199 can_continue_after_enhanced_pushd || return 1
200
201 enhanced_set_shell_option pipefail --trap-unset
202
203 declare -gi i_code_lines_count_not_empty_result
204 # shellcheck disable=SC2119
205 i_code_lines_count_not_empty_result=$(
206     all_self_not_empty_code_lines\
207     | wc --lines
208 )
209 # The printf is for command line use.
210 # A script should use the result value instead.
211 # shellcheck disable=SC2248
212 printf "%s\n" "${i_code_lines_count_not_empty_result}
213 }

```

3.21 build_and_checks_dependencies/lines_filters.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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20 # along with DevOrSysAdminScripts.
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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet

```

```

24 # This file was renamed from "lines_filters.sh" to
25 # "lines_filters.lib.sh".
26
27 # The goal of this file is to have some common line filters
28 # for bash processing and pipes.
29 # As is the case for other .sh files in this repository,
30 # it can be copied somewhere in your home directory
31 # and sourced in your .bashrc.
32
33 # shellcheck source=common_options.lib.sh
34 source "../build_and_checks_dependencies/common_options.lib.sh"
35
36 ll_wc() {
37     # current options to wc:
38     # -c --bytes
39     # -m --chars
40     # -l --lines
41     # --files0-from # Incompatible with cat,
42     #                 # another thing intentional by ?
43     #                 # I don't see the meaning of "files0".
44     # -L --max-line-length
45     # -w --words
46     # --help
47     # --version
48     # New option by Laurent Lyaudet (and probably other before):
49     # -n --no-filenames display only number(s)
50     # I think I did that the first time many years ago (Teliae ?).
51     # Since then, they added it as ignored... (-n)
52     enhanced_set_shell_option pipefail\
53     && trap 'enhanced_unset_shell_option pipefail' RETURN
54     declare -a ll_wc_var_args=()
55     declare -i ll_wc_var_i_number_only=0
56     declare -i ll_wc_var_i=0
57     for ll_wc_var_arg in "$@"; do
58         # printf "%s\n" "${ll_wc_var_arg}"
59         if [[ "${ll_wc_var_arg}" == "-n" ]]; then
60             ll_wc_var_i_number_only=1
61         elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
62             ll_wc_var_i_number_only=1
63         elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
64             ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
65             ll_wc_var_i=$((ll_wc_var_i + 1))
66         elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
67             ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
68             ll_wc_var_i=$((ll_wc_var_i + 1))
69         fi

```

```

70 done
71 # typeset -p ll_wc_var_args
72 # printf "%s\n" "${ll_wc_var_i_number_only}"
73 if [[ ll_wc_var_i_number_only -gt 0 ]]; then
74     # Too simple code, only the base use case is handled now.
75     # Convenient but incomplete.
76     # With more time,
77     # the simpler would be to submit a patch to GNU coreutils,
78     # unless blockade from assholes.
79     # Otherwise, it may be more simply handled in PHP or Python
80     # (argparse, etc.).
81     # You'll need to start by improving cat first,
82     # with --files0-from handling.
83     # wc does not display file_name when stream...
84     # hence cat of last arg "${!#}"
85     cat -- "${!#}" \
86         | wc "${ll_wc_var_args[@]}"
87 else
88     # printf "normal\n"
89     wc "${ll_wc_var_args[@]}"
90 fi
91 }
92
93 # I submitted a patch for wc :).
94 # printf "https://domain/list-path/2024-05/msg00013.html" | \
95 #   sed --expression='s|domain|lists.gnu.org|' \
96 #     --expression='s|list-path|archive/html/bug-coreutils|'
97 # I found another way to have a long URL fit on 70 characters.
98 # If it is merged, the following would be better:
99 # If --no-filenames option available,
100 # we redefine ll_wc command as an alias.
101 wc_help_text=$(wc --help)
102 if [[ "${wc_help_text}" == *" --no-filenames "*" ]]; then
103     alias ll_wc='wc'
104 fi
105
106 # Trace avant le (2e ?) renommage
107 # de wget_sha512.sh à wget_sha512.lib.sh.
108 # $ ll_wc wget_sha512.sh
109 # 41 189 1224 wget_sha512.sh
110 # $ ll_wc -n wget_sha512.sh
111 #      41      189      1224
112 # $ ll_wc -l wget_sha512.sh
113 # 41 wget_sha512.sh
114 # $ ll_wc -l -n wget_sha512.sh
115 # 41

```



```

116
117 in_place_grep() {
118     # Same arguments and options as grep.
119     declare -r LFBFL_temp="${!#}.in_place_grep.temp"
120     grep "$@" \
121         > "${LFBFL_temp}"
122     # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
123     # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
124     # printf "%s\n" "${LFBFL_i_lines_before}"
125     # printf "%s\n" "${LFBFL_i_lines_after}"
126     # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
127     # Almost as fast (on large files) and correct in all cases :).
128     # Thanks Paul Eggert for the suggestion :).
129     # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
130     if cmp --silent -- "${LFBFL_temp}" "${!#}"; then
131         rm -- "${LFBFL_temp}"
132         return
133     fi
134     mv -- "${LFBFL_temp}" "${!#}"
135 }
136
137 grep_fixed_string_with_anchor() {
138     # $1=file_path
139     # $2=some_fixed_string
140     # Options:
141     #   --enforce-line-starts-with-fixed-string
142     #   --enforce-line-ends-with-fixed-string
143     #   Apart from $1 and $2 and the options above,
144     #   all other arguments are considered grep_options like --quiet
145     #   for example.
146     #   But note that --fixed-strings will be added below,
147     #   you don't need to supply it.
148     declare -a LFBFL_grep_options=()
149     declare -a LFBFL_grep_options2=()
150     declare -i LFBFL_i=0
151     declare -i LFBFL_i_enforce_line_starts=0
152     declare -i LFBFL_i_enforce_line_ends=0
153     declare -i LFBFL_i_quiet=0
154     local LFBFL_argument
155     for LFBFL_argument in "$@"; do
156         ((++LFBFL_i))
157         if [[ LFBFL_i -lt 3 ]]; then
158             continue
159         fi
160         if [[
161             "${LFBFL_argument}" == "--enforce-line-starts-with-fixed-string"

```

```

162     ]]; then
163         LFBFL_i_enforce_line_starts=1
164         continue
165     fi
166     if [[
167         "${LFBFL_argument}" == "--enforce-line-ends-with-fixed-string"
168     ]]; then
169         LFBFL_i_enforce_line_ends=1
170         continue
171     fi
172     if [[
173         "${LFBFL_argument}" == "-q"
174         || "${LFBFL_argument}" == "--quiet"
175     ]]; then
176         LFBFL_i_quiet=1
177         LFBFL_grep_options+=("${LFBFL_argument}")
178         continue
179     fi
180     LFBFL_grep_options2+=("${LFBFL_argument}")
181 done
182
183 if [[
184     LFBFL_i_enforce_line_starts -eq 0
185     && LFBFL_i_enforce_line_ends -eq 0
186 ]]; then
187     grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
188     return
189 fi
190 declare -r LFBFL_s_lines=$(
191     grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
192         "$2" "$1"
193 )
194 declare -i LFBFL_i_result=1
195 if [[ -z "${LFBFL_s_lines}" ]]; then
196     # shellcheck disable=SC2248
197     return ${LFBFL_i_result}
198 fi
199 declare -a LFBFL_arr_lines
200 mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
201 readonly LFBFL_arr_lines
202
203 declare -i LFBFL_i_fixed_string_length=${#2}
204 local LFBFL_some_line
205 local LFBFL_line_part
206 for LFBFL_some_line in "${LFBFL_arr_lines[@]"; do
207     if [[ "${LFBFL_some_line}" == "$2" ]]; then

```

```

208     LFBFL_i_result=0
209     if [[ LFBFL_i_quiet -eq 1 ]]; then
210         break
211     else
212         printf "%s\n" "${LFBFL_some_line}"
213     fi
214     continue
215 fi
216 if [[
217     LFBFL_i_enforce_line_starts -eq 1
218     && LFBFL_i_enforce_line_ends -eq 1
219 ]]; then
220     continue
221 fi
222 if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
223     LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
224 fi
225 if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
226     LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
227 fi
228 if [[ "${LFBFL_line_part}" == "$2" ]]; then
229     LFBFL_i_result=0
230     if [[ LFBFL_i_quiet -eq 1 ]]; then
231         break
232     else
233         printf "%s\n" "${LFBFL_some_line}"
234     fi
235 fi
236 done
237 # shellcheck disable=SC2248
238 return ${LFBFL_i_result}
239 }
240
241 grep_variable() {
242     # $1=file_path
243     # $2=variable_name
244     # Options :
245     # --replace-line-returns-by=""
246     # --result-variable-prefix="LFBFL_" for example
247     declare -r LFBFL_regexp="(?!<^$2=).*$"
248     # printf "%s\n" "${LFBFL_regexp}"
249     local LFBFL_variable_value
250     local LFBFL_s_replace
251     get_some_option LFBFL_s_replace --replace-line-returns-by ' ' $'\n' 1 "$@"
252     if [[ "${LFBFL_s_replace}" != $'\n' ]]; then
253         # sed -Ez -e...

```

```

254     LFBFL_variable_value=$(
255         grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"\
256         | sed --null-data --regexp-extended\
257             --expression='s/\n/' "${LFBFL_s_replace}"'/Mg'
258     )
259     # We remove the trailing replace string,
260     # since the intent is only to replace the intermediate \n.
261     if [[ -n "${LFBFL_s_replace}" ]]; then
262         LFBFL_variable_value=${LFBFL_variable_value% "${LFBFL_s_replace}"}
263     fi
264     else
265         LFBFL_variable_value=$(
266             grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"
267         )
268     fi
269     local LFBFL_prefix=""
270     get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
271     declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
272     # printf "%s : %s\n" "${LFBFL_variable_name}" "${LFBFL_variable_value}"
273     # The declare line may be dropped by a local or declare
274     # in outer scope.
275     # declare -g "${LFBFL_variable_name}"
276     # So the assignment must be done on another line.
277     printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
278     # printf "%s\n" "${!LFBFL_variable_name}"
279 }
280
281 empty_lines() {
282     grep '^$'
283 }
284
285 not_empty_lines() {
286     grep --invert-match '^$'
287 }
288
289 space_starting_lines() {
290     grep '^ '
291 }
292
293 not_space_starting_lines() {
294     grep --invert-match '^ '
295 }
296
297 # Space starting or empty lines = spampty
298 spampty_lines() {
299     grep --extended-regexp '^( |$)'

```

```

300 }
301
302 not_spamty_lines() {
303     grep '^[^ ]'
304     # grep --extended-regexp --invert-match '^( |$)'
305 }
306
307 all_space_lines() {
308     grep '^[ ]\+$'
309 }
310
311 not_all_space_lines() {
312     grep --invert-match '^[ ]\+$'
313 }
314
315 # All space or empty lines = aspmpty
316 aspmpty_lines() {
317     grep '^[ ]*$'
318 }
319
320 not_aspmpty_lines() {
321     grep '[^ ]'
322     # grep --invert-match '^[ ]*$'
323 }
324
325 empty_lines_after_file_name() {
326     grep '^[^:]\+:$'
327 }
328
329 not_empty_lines_after_file_name() {
330     grep --invert-match '^[^:]\+:$'
331 }
332
333 space_starting_lines_after_file_name() {
334     grep '^[^:]\+:'
335 }
336
337 not_space_starting_lines_after_file_name() {
338     grep --invert-match '^[^:]\+:'
339 }
340
341 spamty_lines_after_file_name() {
342     grep --extended-regexp '^[^:]+\:( |$)'
343 }
344
345 not_spamty_lines_after_file_name() {

```

```

346 # The two variants results differ on lines without filename before.
347 # Since the intent of the function is only on lines "blabla.sh:whatever",
348 # use the fastest one. I didn't do extensive testing and just assumed
349 # that the "positive" one is faster as usual when querying database.
350 # But there is no index here ;) XD.
351 grep '^[:]\+:[^ ]'
352 # grep --extended-regexp --invert-match '^[:]\+:( |$)'
353 }
354
355 all_space_lines_after_file_name() {
356     grep '^[:]\+:[ ]+$'
357 }
358
359 not_all_space_lines_after_file_name() {
360     grep --invert-match '^[:]\+:[ ]+$'
361 }
362
363 asampthy_lines_after_file_name() {
364     grep '^[:]\+:[ ]*$'
365 }
366
367 not_asampthy_lines_after_file_name() {
368     grep --invert-match '^[:]\+:[ ]*$'
369 }
370
371 # If you need to test the previous 20 functions,
372 # look at file spampty_data.txt in miscellaneous.
373 # And give a try at something like:
374 # cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
375 # -E = --show-ends :). Nice command :).
376
377 not_JS_dependencies_find() {
378     grep --extended-regexp --invert-match\
379         --regexp="(^|/)node_modules/"\
380         --regexp="(^|/)package-lock\.json$"
381 }
382
383 not_JS_dependencies_grep() {
384     grep --extended-regexp --invert-match\
385         --regexp="^([:]+)/?node_modules/"\
386         --regexp="^([:]+)/?package-lock\.json:"
387 }
388
389 not_dependencies_find() {
390     not_JS_dependencies_find
391 }

```

```

392
393 not_dependencies_grep() {
394     not_JS_dependencies_grep
395 }
396
397 not_php_cache_find() {
398     grep --extended-regexp --invert-match\
399         --regexp="(^|/)\.phpdoc/cache/"
400 }
401
402 not_php_cache_grep() {
403     grep --extended-regexp --invert-match\
404         --regexp="^( [^:]+/)?\.phpdoc/cache/"
405 }
406
407 not_python_cache_find() {
408     grep --extended-regexp --invert-match\
409         --regexp="(^|/)\__pycache__/"\
410         --regexp="(^|/)\.mypy_cache/"\
411         --regexp="(^|/)\.ruff_cache/"
412 }
413
414 not_python_cache_grep() {
415     grep --extended-regexp --invert-match\
416         --regexp="^( [^:]+/)?__pycache__/"\
417         --regexp="^( [^:]+/)?\.mypy_cache/"\
418         --regexp="^( [^:]+/)?\.ruff_cache/"
419 }
420
421 not_cache_find() {
422     not_php_cache_find\
423     | not_python_cache_find
424 }
425
426 not_cache_grep() {
427     not_php_cache_grep\
428     | not_python_cache_grep
429 }
430
431 not_git_find() {
432     grep --extended-regexp --invert-match "(^|/)\.git/"
433 }
434
435 not_git_grep() {
436     grep --extended-regexp --invert-match "^( [^:]+/)?\.git/"
437 }

```

```

438
439 not_archive_find() {
440     grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
441 }
442
443 not_archive_grep() {
444     grep --extended-regexp --invert-match\
445         --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
446 }
447
448 not_license_find() {
449     grep --extended-regexp --invert-match\
450         --regexp="(^|/)COPYING$\\"
451         --regexp="(^|/)COPYING\.LESSER$"
452 }
453
454 not_license_grep() {
455     grep --extended-regexp --invert-match\
456         --regexp="^([^:]+)/?COPYING:$\\"
457         --regexp="^([^:]+)/?COPYING\.LESSER:"
458 }
459
460 not_main_tex_find() {
461     local LFBFL_repository_name=""
462     grep_variable build_and_checks_variables/repository_data.txt\
463         repository_name\
464         --result-variable-prefix="LFBFL_"\
465         --replace-line-returns-by=""
466     grep --extended-regexp --invert-match\
467         --regexp="(^|/){LFBFL_repository_name}\.tex$"
468 }
469
470 not_main_tex_grep() {
471     local LFBFL_repository_name=""
472     grep_variable build_and_checks_variables/repository_data.txt\
473         repository_name\
474         --result-variable-prefix="LFBFL_"\
475         --replace-line-returns-by=""
476     grep --extended-regexp --invert-match\
477         --regexp="^([^:]+)/?${LFBFL_repository_name}\.tex:"
478 }
479
480 not_main_html_find() {
481     local LFBFL_repository_name=""
482     grep_variable build_and_checks_variables/repository_data.txt\
483         repository_name\

```



```

484     --result-variable-prefix="LFBFL_"\
485     --replace-line-returns-by=""
486     grep --extended-regexp --invert-match\
487         --regexp="(^|/){LFBFL_repository_name}\.html$"
488 }
489
490 not_main_html_grep() {
491     local LFBFL_repository_name=""
492     grep_variable build_and_checks_variables/repository_data.txt\
493         repository_name\
494         --result-variable-prefix="LFBFL_"\
495         --replace-line-returns-by=""
496     grep --extended-regexp --invert-match\
497         --regexp="^( [^:]+/)?${LFBFL_repository_name}\.html:"
498 }
499
500 not_temp_file_find() {
501     grep --extended-regexp --invert-match\
502         --regexp="(^|/)build_and_checks_variables/temp/"
503 }
504
505 not_temp_file_grep() {
506     grep --extended-regexp --invert-match\
507         --regexp="^( [^:]+/)?build_and_checks_variables/temp/[^:]*:"
508 }
509
510 not_unchecked_builds_find() {
511     grep --extended-regexp --invert-match\
512         --regexp="(^|/)\.phpdoc/build/"
513 }
514
515 not_unchecked_builds_grep() {
516     grep --extended-regexp --invert-match\
517         --regexp="^( [^:]+/)?\.phpdoc/build/"
518 }
519
520 relevant_find() {
521     enhanced_set_shell_option pipefail\
522         && trap 'enhanced_unset_shell_option pipefail' RETURN
523     not_dependencies_find\
524         | not_cache_find\
525         | not_git_find\
526         | not_archive_find\
527         | not_temp_file_find\
528         | not_unchecked_builds_find
529 }

```

```

530
531 relevant_grep() {
532     enhanced_set_shell_option pipefail\
533     && trap 'enhanced_unset_shell_option pipefail' RETURN
534     not_dependencies_grep\
535     | not_cache_grep\
536     | not_git_grep\
537     | not_archive_grep\
538     | not_temp_file_grep\
539     | not_unchecked_builds_grep
540 }

```

3.22 build_and_checks_dependencies/lines_maps.lib.sh

```

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2  # This file is part of DevOrSysAdminScripts.
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20 # along with DevOrSysAdminScripts.
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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "lines_maps.sh" to "lines_maps.lib.sh".
25
26 # https://en.wikibooks.org/wiki/Unicode/Character\_reference/2000–2FFF
27 # Some invisible unicode characters that are a problem:
28 # 2000–200F 2028–202F 205F 2060–206F
29 # [
30 #   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
31 #   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
32 #   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',

```

```

33 # '205F',
34 # '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
35 # '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
36 # ]
37 replace_non_ascii_spaces() {
38     sed\
39         --expression='s/\xc2\xa0/ /g'\
40         --expression='s/\xe2\x80\x80/ /g'\
41         --expression='s/\xe2\x80\x81/ /g'\
42         --expression='s/\xe2\x80\x82/ /g'\
43         --expression='s/\xe2\x80\x83/ /g'\
44         --expression='s/\xe2\x80\x84/ /g'\
45         --expression='s/\xe2\x80\x85/ /g'\
46         --expression='s/\xe2\x80\x86/ /g'\
47         --expression='s/\xe2\x80\x87/ /g'\
48         --expression='s/\xe2\x80\x88/ /g'\
49         --expression='s/\xe2\x80\x89/ /g'\
50         --expression='s/\xe2\x80\x8a/ /g'\
51         --expression='s/\xe2\x80\x8b/ /g'\
52         --expression='s/\xe2\x80\x8c/ /g'\
53         --expression='s/\xe2\x80\x8d/ /g'\
54         --expression='s/\xe2\x80\x8e/ /g'\
55         --expression='s/\xe2\x80\x8f/ /g'\
56         --expression='s/\xe2\x80\xa8/ /g'\
57         --expression='s/\xe2\x80\xa9/ /g'\
58         --expression='s/\xe2\x80\xaa/ /g'\
59         --expression='s/\xe2\x80\xab/ /g'\
60         --expression='s/\xe2\x80\xac/ /g'\
61         --expression='s/\xe2\x80\xad/ /g'\
62         --expression='s/\xe2\x80\xae/ /g'\
63         --expression='s/\xe2\x80\xaf/ /g'\
64         --expression='s/\xe2\x81\x9f/ /g'\
65         --expression='s/\xe2\x81\xa0/ /g'\
66         --expression='s/\xe2\x81\xa1/ /g'\
67         --expression='s/\xe2\x81\xa2/ /g'\
68         --expression='s/\xe2\x81\xa3/ /g'\
69         --expression='s/\xe2\x81\xa4/ /g'\
70         --expression='s/\xe2\x81\xa5/ /g'\
71         --expression='s/\xe2\x81\xa6/ /g'\
72         --expression='s/\xe2\x81\xa7/ /g'\
73         --expression='s/\xe2\x81\xa8/ /g'\
74         --expression='s/\xe2\x81\xa9/ /g'\
75         --expression='s/\xe2\x81\xaa/ /g'\
76         --expression='s/\xe2\x81\xab/ /g'\
77         --expression='s/\xe2\x81\xac/ /g'\
78         --expression='s/\xe2\x81\xad/ /g'\

```

```

79     --expression='s/\xe2\x81\xae/ /g'\
80     --expression='s/\xe2\x81\xaf/ /g'
81 }

```

3.23 build_and_checks_dependencies/overwrite_if_not_equal.lib.sh

```

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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "overwrite_if_not_equal.sh"
25 # to "overwrite_if_not_equal.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30
31 overwrite_if_not_equal() {
32     # $1=target_file_path
33     # $2=temp_file_path
34     # Options:
35     #   --keep-temp : keep the temporary file
36     #   --quiet : we even don't want the message of diff saying that
37     #             files are distinct
38     #   --tree-mode : when the two files are the output of tree command,
39     #   we ignore if all the differences are only on directories timestamps.
40     # Return 0 target was the same

```

```

41 #           1 target was different
42 #           2 target did not already exist
43 declare -i LFBFL_i_keep_temp
44 get_some_flag LFBFL_i_keep_temp --keep-temp 1 "$@"
45 declare -i LFBFL_i_quiet
46 get_some_flag LFBFL_i_quiet --quiet 1 "$@"
47 declare -i LFBFL_i_tree_mode
48 get_some_flag LFBFL_i_tree_mode --tree-mode 1 "$@"
49
50 if [[ ! -f "$1" ]]; then
51     if [[ LFBFL_i_keep_temp -eq 1 ]]; then
52         cp --preserve=mode,ownership,timestamps -- "$2" "$1"
53     else
54         mv -- "$2" "$1"
55     fi
56     return 2
57 fi
58 declare -i LFBFL_i_is_equal
59 if [[ LFBFL_i_tree_mode -eq 1 ]]; then
60     # Not setting pipefail since the result would still be incorrect.
61     # shellcheck disable=SC2312
62     diff -- "$1" "$2" \
63         | grep --extended-regexp "^(>|<)" \
64         | grep --invert-match "/"$
65     LFBFL_i_is_equal=${PIPESTATUS[2]}
66     if [[ LFBFL_i_is_equal -eq 1 ]]; then
67         grep --only-matching --regexp="^[^ ]*/$" -- "$1" \
68             > "overwrite_if_not_equal.file1.temp"
69         grep --only-matching --regexp="^[^ ]*/$" -- "$2" \
70             > "overwrite_if_not_equal.file2.temp"
71         if [[ LFBFL_i_quiet -eq 1 ]]; then
72             diff overwrite_if_not_equal.file1.temp \
73                 overwrite_if_not_equal.file2.temp \
74                 > /dev/null
75         else
76             diff overwrite_if_not_equal.file1.temp \
77                 overwrite_if_not_equal.file2.temp
78         fi
79         LFBFL_i_is_equal=1-$?
80         rm overwrite_if_not_equal.file1.temp \
81             overwrite_if_not_equal.file2.temp
82     fi
83 else
84     if [[ LFBFL_i_quiet -eq 1 ]]; then
85         diff --brief -- "$1" "$2" > /dev/null
86     else

```

```

87         diff --brief -- "$1" "$2"
88     fi
89     LFBFL_i_is_equal=1-$?
90 fi
91 if [[ LFBFL_i_is_equal -eq 1 ]]; then
92     if [[ LFBFL_i_keep_temp -eq 0 ]]; then
93         rm -- "$2"
94     fi
95     return 0
96 fi
97 if [[ LFBFL_i_keep_temp -eq 1 ]]; then
98     cp --preserve=mode,ownership,timestamps -- "$2" "$1"
99 else
100     mv -- "$2" "$1"
101 fi
102 return 1
103 }

```

3.24 build_and_checks_dependencies/python_black_complement.lib.sh

```

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23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "python_black_complement.sh"
25 # to "python_black_complement.lib.sh".
26

```

```

27 # shellcheck source=common_options.lib.sh
28 source "./build_and_checks_dependencies/common_options.lib.sh"
29
30 # It was rejected to enhance black with this currently.
31 # https://github.com/psf/black/issues/2370
32 # And moreover, at some point, I saw the reverse rule
33 # in black or pylint to add an empty line...
34 # This regexp may give false positives,
35 # but that's not the end of the world.
36 check_no_empty_line_after_python_function_docstring() {
37     # Options:
38     #   --verbose
39     #   --work-directory=""
40     declare -i LFBFL_i_verbose=0
41     get_verbose_option "$@"
42     local LFBFL_work_directory=""
43     get_work_directory_option "$@"
44     declare -a LFBFL_return_traps_stack
45     init_return_trap
46     pushd_to_work_directory --trap-popd
47     can_continue_after_enhanced_pushd || return 1
48
49     enhanced_set_bash_option globstar --trap-unset
50
51     [[ LFBFL_i_verbose -eq 1 ]]\
52     && printf "Checking empty lines after Python function docstrings\n"
53     pcre2grep --multiline --\
54     '$^def ([^"]|"(?!"))*""'([^^]|"(?!"))*""\n\n(?!s*def)'\
55     */*.py
56 }
57
58 python_black_complement() {
59     # Options:
60     #   --verbose
61     #   --work-directory=""
62     declare -i LFBFL_i_verbose=0
63     get_verbose_option "$@"
64     local LFBFL_work_directory=""
65     get_work_directory_option "$@"
66     declare -a LFBFL_return_traps_stack
67     init_return_trap
68     pushd_to_work_directory --trap-popd
69     can_continue_after_enhanced_pushd || return 1
70
71     declare -a LFBFL_common_options=("$@")
72     keep_options LFBFL_common_options --verbose

```

```

73
74     check_no_empty_line_after_python_function_docstring\
75         "${LFBFL_common_options[@]}"
76 }

```

3.25 build_and_checks_dependencies/python_isort_complement.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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17 #
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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "python_isort_complement.sh"
25 # to "python_isort_complement.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=lines_filters.lib.sh
31 source "${LFBFL_subdir}/lines_filters.lib.sh"
32 # shellcheck source=overwrite_if_not_equal.lib.sh
33 source "${LFBFL_subdir}/overwrite_if_not_equal.lib.sh"
34
35 check_collections_abc_place() {
36     # Options:
37     #   --verbose
38     #   --work-directory=""
39     # This function should be useless in a few months:

```



```

40 # https://github.com/PyCQA/isort/issues/2250
41 declare -i LFBFL_i_verbose=0
42 get_verbose_option "$@"
43 local LFBFL_work_directory=""
44 get_work_directory_option "$@"
45 declare -a LFBFL_return_traps_stack
46 init_return_trap
47 pushd_to_work_directory --trap-popd
48 can_continue_after_enhanced_pushd || return 1
49
50 [[ LFBFL_i_verbose -eq 1 ]]\
51   && printf\
52     "Checking that import of _collections_abc is at the right place\n"
53
54 enhanced_set_shell_option pipefail --trap-unset
55
56 declare -r LFBFL_temp=".check_collections_abc_place.temp"
57 local LFBFL_file_path
58 declare -r LFBFL_s_files_paths=$(
59   find . -type f -name "*.py" -printf '%P\n'\
60   | relevant_find
61 )
62 if [[ -z "${LFBFL_s_files_paths}" ]]; then
63   [[ LFBFL_i_verbose -eq 1 ]]\
64     && printf "check_collections_abc_place: No file found.\n"
65   return
66 fi
67 declare -a LFBFL_arr_files_paths
68 mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
69 readonly LFBFL_arr_files_paths
70 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}; do
71   sed --regext-extended --null-data\
72     --expression='s/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
73     -- "${LFBFL_file_path}"\
74     > "${LFBFL_file_path}${LFBFL_temp}"
75   overwrite_if_not_equal "${LFBFL_file_path}"\
76     "${LFBFL_file_path}${LFBFL_temp}"
77 done
78 }
79
80 python_isort_complement() {
81   # Options:
82   #   --verbose
83   #   --work-directory=""
84   declare -i LFBFL_i_verbose=0
85   get_verbose_option "$@"

```

```

86     local LFBFL_work_directory=""
87     get_work_directory_option "$@"
88     declare -a LFBFL_return_traps_stack
89     init_return_trap
90     pushd_to_work_directory --trap-popd
91     can_continue_after_enhanced_pushd || return 1
92
93     declare -a LFBFL_common_options=("$@")
94     keep_options LFBFL_common_options --verbose
95
96     check_collections_abc_place "${LFBFL_common_options[@]}"
97 }

```

3.26 build_and_checks_dependencies/repository_data.lib.sh

```

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22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24
25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.lib.sh
27 source "${LFBFL_subdir}/common_options.lib.sh"
28 # shellcheck source=lines_filters.lib.sh
29 source "${LFBFL_subdir}/lines_filters.lib.sh"
30
31 # /\ You need to activate extglob before parsing this file. /\

```

```

32 # And I'm kind enough to activate it inside functions that need it below.
33
34 get_upgrade_venvs_time_interval_in_seconds() {
35     # $1=data_file_path
36     # Options:
37     #     --verbose
38     declare -i LFBFL_i_verbose=0
39     get_verbose_option "$@"
40     declare -a LFBFL_return_traps_stack
41     init_return_trap
42
43     enhanced_set_bash_option extglob --trap-unset
44
45     grep_variable "$1"\
46         upgrade_venvs_time_interval_in_seconds\
47         --result-variable-prefix="LFBFL_"
48
49     [[ LFBFL_i_verbose -eq 1 ]]\
50     && printf "You say \"%s\".\n"\
51         "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
52
53     case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
54         +([0-9])) : ;;
55         wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
56         watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
57         *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
58     esac
59
60     [[ LFBFL_i_verbose -eq 1 ]]\
61     && printf "I say \"%s\".\n"\
62         "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
63 }

```

3.27 build_and_checks_dependencies/shell_checks_complement.lib.sh

```

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```

```

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22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "python_black_complement.sh"
25 # to "python_black_complement.lib.sh".
26
27 # shellcheck source=common_options.lib.sh
28 source "../build_and_checks_dependencies/common_options.lib.sh"
29
30 check_no_size_of_array_first_element() {
31     # Options:
32     #   --verbose
33     #   --work-directory=""
34     declare -i LFBFL_i_verbose=0
35     get_verbose_option "$@"
36     local LFBFL_work_directory=""
37     get_work_directory_option "$@"
38     declare -a LFBFL_return_traps_stack
39     init_return_trap
40     pushd_to_work_directory --trap-popd
41     can_continue_after_enhanced_pushd || return 1
42
43     enhanced_set_bash_option globstar --trap-unset
44
45     local LFBFL_s="Checking that shell scripts do not"
46     LFBFL_s+=" contain wrong array size expansion.\n"
47     # shellcheck disable=SC2059
48     [[ LFBFL_i_verbose -eq 1 ]] && printf "${LFBFL_s}"
49
50     pcre2grep --multiline -- '$\\$\\#{#(|.*_)arr_(?!.*\\[@\\])}' **/*.sh
51 }
52
53 check_no_misplaced_then() {
54     # Options:
55     #   --verbose
56     #   --work-directory=""
57     #   --max-line-length

```

```

58 declare -i LFBFL_i_verbose=0
59 get_verbose_option "$@"
60 local LFBFL_work_directory=""
61 get_work_directory_option "$@"
62 declare -a LFBFL_return_traps_stack
63 init_return_trap
64 pushd_to_work_directory --trap-popd
65 can_continue_after_enhanced_pushd || return 1
66
67 declare -i LFBFL_i_max_line_length
68 get_max_line_length_option "$@"
69 declare -i LFBFL_i_max_previous_line_length
70 LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))
71
72 enhanced_set_bash_option globstar --trap-unset
73
74 [[ LFBFL_i_verbose -eq 1 ]]\
75     && printf\
76         "Checking that shell scripts do not contain misplaced 'then'.\n"
77
78 local LFBFL_regexp
79 LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
80 LFBFL_regexp+="(?!<=\\|\\);[^\n]{0,${LFBFL_i_max_previous_line_length}})"
81 LFBFL_regexp+="(?!<!\s{0,${LFBFL_i_max_previous_line_length}}\\|\\|;)"
82 LFBFL_regexp+="\n\s*then"
83 # pcre2grep --multiline --
84 # '^[^\n]{1,70}(?!<=\\|\\);[^\n]{0,70})'
85 # '(?!<!\s{0,70}\\|\\|;)\n\s*then'
86 pcre2grep --multiline -- "${LFBFL_regexp}" **/*.sh
87 }
88
89 check_no_negation_before_bash_test() {
90     # Options:
91     #     --verbose
92     #     --work-directory=""
93     declare -i LFBFL_i_verbose=0
94     get_verbose_option "$@"
95     local LFBFL_work_directory=""
96     get_work_directory_option "$@"
97     declare -a LFBFL_return_traps_stack
98     init_return_trap
99     pushd_to_work_directory --trap-popd
100    can_continue_after_enhanced_pushd || return 1
101
102    enhanced_set_bash_option globstar --trap-unset
103

```

```

104     [[ LFBFL_i_verbose -eq 1 ]]\
105     && printf "Checking that shell scripts do not contain ! before [.\n"
106
107     pcre2grep --multiline -- '$'\!\\s*\\[\\[\' **/*.sh
108 }
109
110 shell_checks_complement() {
111     # Options:
112     #   --verbose
113     #   --work-directory=""
114     declare -i LFBFL_i_verbose=0
115     get_verbose_option "$@"
116     local LFBFL_work_directory=""
117     get_work_directory_option "$@"
118     declare -a LFBFL_return_traps_stack
119     init_return_trap
120     pushd_to_work_directory --trap-popd
121     can_continue_after_enhanced_pushd || return 1
122
123     declare -a LFBFL_common_options=("$@")
124     keep_options LFBFL_common_options --verbose --max-line-length
125
126     check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
127     check_no_misplaced_then "${LFBFL_common_options[@]}"
128     check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
129 }

```

3.28 build_and_checks_dependencies/split_score.exec.php

```

1  #!/usr/bin/php
2  <?php
3
4  /**
5   This file is part of DevOrSysAdminScripts.
6
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```

```

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20
21  You should have received a copy of
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23  along with DevOrSysAdminScripts.
24  If not, see <https://www.gnu.org/licenses/>.
25
26  @Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
27  2026-01-04T13:38+01:00: This file was renamed from
28  "split_line_at_most.exec.php"
29  to
30  "split_score.exec.php".
31  This file was renamed from
32  "call_split_score_after_before_or_before_after.exec.php" to
33  "split_line_at_most.exec.php".
34  This file was renamed from
35  "call_split_score_after_before_or_before_after.php" to
36  "call_split_score_after_before_or_before_after.exec.php".
37
38  @category Executable
39  @package DevOrSysAdminScripts
40  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
41  @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
42  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
43  */
44
45  declare(strict_types=1);
46  declare(encoding='UTF-8');
47
48  require_once './build_and_checks_dependencies/split_score.lib.php';
49
50  /*
51  php -r\
52    '$require_once(\'./build_and_checks_dependencies/\
53    $split_score.lib.php\');'\
54    '$ echo(generate_split_score\'
55    $(true, 70, [\'/\']) (\'/\', 60, true));\'
56    */
57
58  // phpcs:disable Generic.WhiteSpace.ArbitraryParenthesesSpacing
59  // The first cast "(int) $func..." is to avoid false
60  // positive errors on possible XSS if output is unescaped.
61  echo(
62    (int) generate_split_score(
63      (bool) $argv[1],

```

```

64     (int) $argv[2],
65     explode(',', $argv[3]),
66     )($argv[4], (int) $argv[5], (bool) $argv[6])
67 );
68 // phpcs:disable Generic.WhiteSpace.ArbitraryParenthesesSpacing
69 ?>

```

3.29 build_and_checks_dependencies/split_score.lib.php

```

1  <?php
2
3  /**
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25  ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
26  2026-01-04T13:38+01:00: This file was renamed from
27  "split_line_at_most.lib.php"
28  to
29  "split_score.lib.php".
30  This file was renamed from "split_line_at_most.php" to
31  "split_line_at_most.lib.php".
32
33  @category Library
34  @package DevOrSysAdminScripts
35  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
36  @copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
37  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+

```



```

38  */
39
40  declare(strict_types=1);
41  declare(encoding='UTF-8');
42
43
44
45  /**
46   Given the 3 following arguments, this function returns a closure that can
47   be used to compute the split score given the current string
48   (hopefully matching a delimiter), the current cut/split position,
49   and if that split position is after or before the current string.
50
51   @param bool          $b_larger_after
52                       Score is larger when splitting after,
53                       previous name of this parameter was $b_after_before.
54   @param int           $i_max_length
55                       The maximum length allowed (goal length).
56   @param array<string> $a_delimiter_strings_domain
57                       An array of "delimiters".
58
59   @return \Closure {
60       The "split_score" closure to use repeatedly afterward.
61
62       @param string $s_delimiter_string The string tested to be a delimiter.
63       @param int    $i_cut_position     The offset of the cut in the string.
64       @param bool   $b_is_cut_after     Is the offset taken after the string?
65
66       @return int The split score.
67   }
68  */
69  function generate_split_score(
70      bool $b_larger_after,
71      int $i_max_length,
72      array $a_delimiter_strings_domain,
73  ) : Closure {
74      return function (
75          string $s_delimiter_string,
76          int $i_cut_position,
77          bool $b_is_cut_after,
78      ) use (
79          $b_larger_after,
80          $i_max_length,
81          $a_delimiter_strings_domain,
82      ) : int {
83          if(

```

```

84         !in_array(
85             $s_delimiter_string,
86             $a_delimiter_strings_domain,
87             true,
88         )
89     ){
90         // I do not handle regexps for the moment.
91         // I'm too brainfucked. I just want to finish this feature fast
92         // for HTML and Tex/PDF listings generation with HTML and Tex on
93         // at most 70 characters.
94         return 0;
95     }
96     if($b_is_cut_after === $b_larger_after){
97         // When $b_larger_after,
98         // always way better to cut after '/' than before it.
99         // When !$b_larger_after (aka larger_before),
100        // always way better to cut before '/' than after it.
101        return 1 + $i_max_length + $i_cut_position;
102    }
103    return 1 + $i_cut_position;
104 };
105 }//end generate_split_score()
106 ?>

```

3.30 build_and_checks_dependencies/strings_functions.lib.sh

```

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```

```

21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # I think I created a file strings_functions.sh or renamed to
25 # strings_functions.sh at some point, before committing.
26 # Hence, someone must have changed or reversed the name of this file
27 # to string_functions.sh at some point.
28 # This file was renamed from "string_functions.sh" to
29 # "strings_functions.lib.sh".
30
31 LFBFL_subdir="build_and_checks_dependencies"
32 # shellcheck source=get_common_text_glob_patterns.lib.sh
33 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
34 # shellcheck source=comparisons.lib.sh
35 source "${LFBFL_subdir}/comparisons.lib.sh"
36 # shellcheck source=too_long_code_lines.lib.sh
37 source "${LFBFL_subdir}/too_long_code_lines.lib.sh"
38
39 split_line_at() {
40     # $1=line
41     # $2=position : number of characters in split_line_at_result_beginning
42     declare -g split_line_at_result_beginning="${1:0:$2}"
43     declare -g split_line_at_result_end="${1:$2}"
44     # printf "%s\n" "$split_line_at_result_beginning"
45     # printf "%s\n" "$split_line_at_result_end"
46 }
47
48 #-----
49 #-----
50 # Properties for the score for a possible split/cut in the string between
51 # two characters.
52 #-----
53 # Some preliminary remarks:
54 # a) If the character before the split position is a forbidden previous
55 #    character, the score is -1.
56 #    (Usually, either we have no or one forbidden previous character that is
57 #     a backslash.)
58 # b) If the previous character is not forbidden,
59 #     and either the character before the position is among the delimiters,
60 #     or the character after the position is among the delimiters,
61 #     then the score is computed according to the delimiter found,
62 #     the position, and the fact that the position is after or before the
63 #     delimiter.
64 # c) If two consecutive delimiters are found in the string,
65 #     the maximum score between the one after the first delimiter
66 #     and before the second delimiter is taken for the score between the

```

```

67 # two delimiters.
68 # d) Otherwise, by default,
69 # the score for cutting/splitting at some position is 0.
70 # e) A substring is given by the goal length given for the resulting
71 # strings/lines after split.
72 # f) When some scores are equal,
73 # we split at the rightmost position with maximum score in the substring.
74 # g) Remark f) implies that below we do not need to consider strict
75 # inequalities for the score.
76 # h) The score function SF(position) which will dictate our choice for
77 # split/cut position is in reality
78 # SF1a(
79 #     position, previous_character, next_character, delimiters, forbiddens
80 # )
81 # but at the cost of expliciting its internal and adding verbosity,
82 # we can see that the score really depends on the following arguments
83 # SF1b(
84 #     position, matched_delimiter_before, matched_delimiter_after,
85 #     previous_character_is_forbidden
86 # )
87 # If it is needed,
88 # then matched_delimiter_before or matched_delimiter_after is NULL.
89 # Somehow previous_character can be seen as position.previous_character,
90 # and next_character can be seen as position.next_character.
91 # We can see that the previous or next character without
92 # knowing that it matches a delimiter is irrelevant.
93 # The fact that there is 1 or more delimiters may be interesting.
94 # The fact that there is 0 or more forbidden previous characters
95 # may be interesting, but we haven't integrated it in the split_score
96 # functions.
97 # SF1a when the previous character is not forbidden
98 # needs a score function SF2a with the following parameters
99 # SF2a(position, current_character, delimiter, is_after):
100 # SF1a(
101 #     position, previous_character, next_character, delimiters, forbiddens
102 # ) = max(
103 #     For all delimiter in delimiters,
104 #     SF2a(position, previous_character, delimiter, TRUE),
105 #     SF2a(position, next_character, delimiter, FALSE),
106 # )
107 # = max(
108 #     For all delimiter in delimiters,
109 #     For all couple C in {
110 #         (position.previous_character, TRUE),
111 #         (position.next_character, FALSE)
112 #     }

```

```

113 #     SF2a(position, C.1, delimiter, C.2),
114 # )
115 # Similarly,
116 # SF1b(
117 #     position, matched_delimiter_before, matched_delimiter_after,
118 #     previous_character_is_forbidden
119 # ) = max(
120 #     For all delimiter in delimiters,
121 #     SF2b(position, matched_delimiter_before, TRUE),
122 #     SF2b(position, matched_delimiter_after, FALSE),
123 # )
124 # where the score function SF2b has the following parameters
125 # SF2b(cut_position, matched_delimiter, is_after).
126 # We have also:
127 # SF1b(
128 #     position, matched_delimiter_before, matched_delimiter_after,
129 #     previous_character_is_forbidden
130 # ) = max(
131 #     For all delimiter in delimiters,
132 #     SF2c(position-1, matched_delimiter_before, TRUE),
133 #     SF2c(position, matched_delimiter_after, FALSE),
134 # )
135 # where the score function SF2c has the following parameters
136 # SF2c(character_position, matched_delimiter, is_after).
137 # current_character is only used to check if it is equal to the
138 # delimiter, otherwise the score is zero;
139 # but when expressing properties it makes sense to compare after
140 # and before the same character position.
141 # position, delimiter, and is_after may all three impact the score
142 # once the current character matches the delimiter.
143 # Position is an integer from 1 to goal length.
144 # Cutting at position 0 would be stupid,
145 # and moreover there would be no previous character.
146 # Since we ask for a split, the current length is above goal length,
147 # and there is a next character at the last considered position.
148 # So, in search of properties that may have an impact on the algorithm,
149 # we discard the characters to only keep matched delimiters.
150 # Let us use p1 and p2 for position1 and position2,
151 # mdb1 and mdb2 for matched_delimiter_before1, etc.,
152 # mda1 and mda2 for matched_delimiter_after1, etc.,
153 # pcif for previous_character_is_forbidden, etc.,
154 # md1 and md2 for matched_delimiter1, etc.,
155 # ia1 and ia2 for is_after1, etc.
156 #
157 # What we consider as properties are of 3 forms.
158 # The first form is like:

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159 #       For all p1 in [1, goal_length], for all p2 in [1, goal_length],
160 #       for all mdb1 in delimiters union {NULL},
161 #       for all mdb2 in delimiters union {NULL},
162 #       for all mda1 in delimiters union {NULL},
163 #       for all mda2 in delimiters union {NULL},
164 #       RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
165 #       implies RelationS(
166 #           SF1b(p1, mdb1, mda1, false),
167 #           SF1b(p2, mdb2, mda2, false),
168 #       )
169 #       There are many choices for RelationP, RelationM and RelationS.
170 #       We can apply the without loss of generality argument only once when
171 #       breaking symmetry between equivalent properties.
172 #       Breaking symmetry on another relation that RelationS first
173 #       is a challenge that we do not tackle.
174 #       Indeed RelationS is the only relation where all variables appear.
175 #       And breaking symmetry on RelationP would not allow to break symmetry
176 #       on RelationM freely since some variables are correlated by the
177 #       suffixes 1 or 2.
178 #       But some choices for some relation may allow to break symmetry on
179 #       another relation.
180 #       Breaking symmetry on the result, there are only 2 choices for
181 #       RelationS:
182 #       RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
183 #       RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
184 #       But then there are 5+1 choices for RelationP:
185 #       RP1 : p1 = p2,
186 #       RP2 : p1 < p2,
187 #       RP3 : p1 <= p2,
188 #       RP4 : p1 > p2,
189 #       RP5 : p1 >= p2,
190 #       plus the choice that p1 and p2 are not necessarily related
191 #       (empty RP or RPe).
192 #       We can add RP6 : p1 = p2 - 1 and RP7 : p1 = p2 + 1, since it implies
193 #       some equalities between matched delimiters.
194 #       Note that p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
195 #       logical relations compared to p1 < p2 or p2 < p1;
196 #       it only yields subslices of other properties and open a slightly silly
197 #       door on infinity.
198 #       If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
199 #       and maybe RP6.
200 #       Note that RSe or empty RS would have no interest at all.
201 #       There are many choices for RelationM:
202 #       mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
203 #       interesting at all since both scores are 0 or -1.
204 #       mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL

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205 # implies that second score is always larger, that's not an additional
206 # property, that is a theorem.
207 # Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
208 # must (have the possibility to) be non-NULL.
209 # If we start listing the remaining relations, we have:
210 # 1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
211 #    AND mda2 is not NULL
212 # RMA1A2
213 # 2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
214 #    AND mda2 is not NULL
215 #    AND mda1 = mda2
216 # RMA1A2=
217 # 3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
218 #    AND mda2 is not NULL
219 #    AND mda1 != mda2 -> This case has no meaning:
220 #    If we only know they are different, we cannot say anything on
221 #    properties related to delimiters that are different unless the score
222 #    doesn't depend on the delimiter, and that case is already handled by
223 #    1). The same remark applies to some other choices.
224 #    The only thing that could be said would be if we had an order on
225 #    delimiters scores that are uniform for the position, or the score
226 #    before some delimiter would always be =, >=, <=, etc. the score
227 #    before or after another one. So it would only make sense with RPe,
228 #    and mda1 <= mda2 for example, but that would trivially imply RS<=.
229 #    It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
230 #    ignores NULL value. You do not deduce or specify anything new here.
231 # 4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
232 #    AND mda2 is NULL
233 # RMB1B2
234 # and so on. So if we use the short notation RM followed by A1, resp. A2,
235 # resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
236 # is not null, we have the following possibilities:
237 # RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
238 # RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
239 # with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
240 # and _4 (XX = YY = ZZ)
241 # for equalities on the 1st, 2nd or 3rd couple or between the three
242 # variables by transitivity,
243 # RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
244 # RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
245 # for one edge;
246 # RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
247 # RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
248 # RMALL_ALL square with diagonals.
249 # So we see that we listed at least 39 possibilities only for RM.
250 # We did not list yet the possibilities where the variables not present

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251 # in some equality can be NULL or not NULL, etc.
252 # We see 9 of them : RMe (40), and the following 8:
253 # RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
254 # RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
255 # RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
256 # RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
257 # RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
258 # RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
259 # RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
260 # RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
261 # We don't do complete analysis at the moment.
262 # If we wanted to filter out some possibilities,
263 # we would keep only the RM possibilities, among the 38,
264 # that are symmetric between 1 and 2:
265 # RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
266 # RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
267 # RMALL_ALL. Still 10 possibilities.
268 # We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
269 # combinations (life is too short ;) :) );
270 # if we keep 48 RM possibilities, there is 624 combinations,
271 # but we did not analyze totally which RM relations are useless.
272 #
273 # The second form is like:
274 #   For all p1 in [1, goal_length], for all p2 in [1, goal_length],
275 #   for all md1 in delimiters union {NULL},
276 #   for all md2 in delimiters union {NULL},
277 #   for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
278 #   RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
279 #   implies RelationS(
280 #       SF2b(p1, md1, ia1)
281 #       SF2b(p2, md2, ia2)
282 #   )
283 # The third form is like:
284 #   For all p1 in [1, goal_length], for all p2 in [1, goal_length],
285 #   for all md1 in delimiters union {NULL},
286 #   for all md2 in delimiters union {NULL},
287 #   for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
288 #   RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
289 #   implies RelationS(
290 #       SF2c(p1, md1, ia1)
291 #       SF2c(p2, md2, ia2)
292 #   )
293 # This time the combinatorics is much more pleasant :).
294 # As above, we have RS= and RS<=, RPe, RP1 to RP7;
295 # For RelationMM, assume that the possible delimiters correspond to
296 # integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer

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297 # in [0,i], either 0 for NULL or the delimiter index.
298 # Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
299 # information we can use is k = l or k != l.
300 # If we allow that md1 != 0 and md2 = 0,
301 # then both RS= and RS<= are false.
302 # Thus RMMe is not possible.
303 # If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
304 # whilst RS= is false.
305 # Thus properties with RS<= can trivially be extended with
306 # RMM="allow-md1-0", not useful to explicit this for the moment.
307 # If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
308 # nothing more.
309 # Thus we can focus on md1 and md2 are not NULL,
310 # and are equal RMM= or not RMM!=. It seems that a property with RMM!=
311 # could be extended to a property RMM=or!=/RMMNotNULL.
312 # and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
313 # 3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
314 # or FALSE. Wrong ! Not pleasant ! XD
315 # So for RIA, we keep either: empty RIAe (RIA8),
316 # ia1 = ia2 RIA= (RIA0 or RIA4),
317 # ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
318 # ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
319 # ia1 != ia2 = TRUE RIA!=T2 (RIA1),
320 # ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
321 # ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
322 # even less pleasant XD : 11 possibilities.
323 # In total, we have now 1 (RS=) times 5 times 2 times 11
324 # + 1 (RS<=) times 8 times 2 times 11 = 286
325 # combinations (life is even shorter ;) :).
326 # We only listed below properties of the third form.
327 # i) By the previous remark, there is a good chance that we missed some
328 # possible property.
329 # We hope we did found the properties that had a simple meaning,
330 # in particular related to the algorithm where either you
331 # loop on the string forward and keep last position with maximum score,
332 # or you loop backward and abort as soon as possible as soon you know
333 # that you found some position at the end that must have maximum score.
334 # j) These properties are "boolean" flags (integers that are powers of 2).
335 # We first put the exact integers,
336 # then wondered about switching everything to exponentiation 2**i,
337 # then we took the solution that minimizes the character length between
338 # [1-9][0-9]* and 2**i, then reverted to initial before committing.
339 # Most efficient interpreted code is what we did first,
340 # if there is no opcode cache for constants.
341 # Most safe code is when the language hides the details and handles
342 # families of flags,

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343 # but beware if some flag goes into the database.
344 # Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
345 # or $(2<<0)...
346 # What I would love to see in many languages is a flag type, as a subtype
347 # of integers, flag32, flag64 for C for example.
348 # I think there is no such thing because sooner or later you play with
349 # pseudo-flags that are combination of flags, hence integers.
350 # But just for the declaration to let the type checker check that the
351 # integer is a power of 2: flag32 MY_FLAG = 4;
352 # Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
353 # Another fact is that practice makes perfect, so working with the powers
354 # of 2 written out in base 10 gets you accustomed to them.
355 # Safe code should be favored since (pre-)compilation optimizes it,
356 # but for this project, I prefer to keep the integers in decimal and
357 # check that it doubles when reading my listing, because I'm familiar
358 # with that and the powers of 2 doesn't go beyond what I can check fast
359 # enough in my brain.
360 # k) Some of the properties below are union of other properties,
361 # and are here to simplify not only code but understanding.
362 # split_score_properties_logical_closure() function below checks that
363 # all flags that are consequences of already activated flags are set.
364 # l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
365 # Some of these remarks are repeated below in the corresponding functions,
366 # and some remarks are added just after the constants.
367 #-----
368 # Any score after a character that matches a delimiter
369 # is larger than or equal to
370 # any score before a character that matches a delimiter.
371 # RPPe RMMNotNull RIA!=T2 RS<=
372 declare -gir SSP_ALWAYS_LARGER_AFTER=1
373
374 # Any score before a character that matches a delimiter
375 # is larger than or equal to
376 # any score after a character that matches a delimiter.
377 # RPPe RMMNotNull RIA!=T1 RS<=
378 declare -gir SSP_ALWAYS_LARGER_BEFORE=2
379
380 # The score doesn't depend on the delimiter that is found.
381 # RPP1 RMMNotNull RIA= RS=
382 declare -gir SSP_DELIMITER_UNIFORM=4
383
384 # Whenever a character that matches a delimiter is found,
385 # the score after is equal to the score before.
386 # Or equivalently:
387 # For any delimiter, and for any character position,
388 # if the character at this position matches this delimiter,

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389 # the score after the character
390 # is equal to
391 # the score before the character.
392 # RPP1 RMM= RIA!= RS=
393 declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
394 # declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=
395
396 # Whenever a character that matches a delimiter is found,
397 # the score after is larger than or equal to the score before.
398 # Or equivalently:
399 # For any delimiter, and for any character position,
400 # if the character at this position matches this delimiter,
401 # the score after the character
402 # is larger than or equal to
403 # the score before the character.
404 # RPP1 RMM= RIA!=T2 RS<=
405 declare -gir SSP_LARGER_AFTER=16
406 # declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=
407
408 # Whenever a character that matches a delimiter is found,
409 # the score before is larger than or equal to the score after.
410 # Or equivalently:
411 # For any delimiter, and for any character position,
412 # if the character at this position matches this delimiter,
413 # the score before the character
414 # is larger than or equal to
415 # the score after the character.
416 # RPP1 RMM= RIA!=T1 RS<=
417 declare -gir SSP_LARGER_BEFORE=32
418 # declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=
419
420 # The score after, resp. before, a character that matches a delimiter
421 # doesn't decrease when the position in the string increases.
422 # (RPP2 or RPP3) RMM= RIA= RS<=
423 declare -gir SSP_POSITION_NOT_DECREASING=64
424
425 # The score after a character that matches a delimiter doesn't decrease
426 # when the position in the string increases.
427 # (RPP2 or RPP3) RMM= RIA=T RS<=
428 declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128
429
430 # The score before a character that matches a delimiter doesn't decrease
431 # when the position in the string increases.
432 # (RPP2 or RPP3) RMM= RIA=F RS<=
433 declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256
434

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435 # The score after, resp. before, a character that matches a delimiter
436 # doesn't depend on the position in the string.
437 # (RPP2 or RPP3) RMM= RIA= RS=
438 declare -gir SSP_POSITION_UNIFORM=512
439
440 # The score after a character that matches a delimiter doesn't depend
441 # on the position in the string.
442 # (RPP2 or RPP3) RMM= RIA=T RS=
443 declare -gir SSP_POSITION_UNIFORM_AFTER=1024
444
445 # The score before a character that matches a delimiter doesn't depend
446 # on the position in the string.
447 # (RPP2 or RPP3) RMM= RIA=F RS=
448 declare -gir SSP_POSITION_UNIFORM_BEFORE=2048
449
450 # For any delimiter,
451 # any score after a character that matches this delimiter
452 # is equal to
453 # any score before a character that matches this delimiter.
454 # RPpE RMM= RIA!= RS=
455 # <=> RPpE RMM= RIAe RS=
456 declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096
457
458 # For any delimiter,
459 # any score after a character that matches this delimiter
460 # is larger than or equal to
461 # any score before a character that matches this delimiter.
462 # RPpE RMM= RIA!=T2 RS<=
463 declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192
464
465 # For any delimiter,
466 # any score before a character that matches this delimiter
467 # is larger than or equal to
468 # any score after a character that matches this delimiter.
469 # RPpE RMM= RIA!=T1 RS<=
470 declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384
471
472 # For any character position,
473 # assuming that a character at this position matches some delimiter,
474 # the score after a character at this position that matches a delimiter
475 # is equal to
476 # the score before a character at this position that matches a delimiter.
477 # RPP1 RMMNotNull RIA!= RS=
478 declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768
479
480 # For any character position,

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481 # assuming that a character at this position matches some delimiter,
482 # the score after a character at this position that matches a delimiter
483 # is larger than or equal to
484 # the score before a character at this position that matches a delimiter.
485 # RPP1 RMMNotNull RIA!=T2 RS<=
486 declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536
487
488 # For any character position,
489 # assuming that a character at this position matches some delimiter,
490 # the score before a character at this position that matches a delimiter
491 # is larger than or equal to
492 # the score after a character at this position that matches a delimiter.
493 # RPP1 RMMNotNull RIA!=T1 RS<=
494 declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072
495
496 # For all possible split/cut positions, the score is equal as long as
497 # at least one of the previous and next characters is a delimiter.
498 # Or equivalently:
499 # Any score after a character that matches a delimiter
500 # is equal to
501 # any score before a character that matches a delimiter.
502 # is_after doesn't change anything.
503 # So we cut on the last position adjacent to a delimiter without
504 # forbidden previous character (edge case, on last position).
505 # RPpE RMMNotNull RIAe RS=
506 declare -gir SSP_UNIFORM=262144
507 # declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1
508
509 #-----
510 # m) Note that a property telling that the score is always the same
511 # even if the current character isn't among delimiters is not needed,
512 # since, in that case, a null split score command will split at the goal
513 # length.
514 #-----
515 #-----
516
517 split_score_properties_logical_closure(){
518     # $1=split_score_properties
519     # Since this code is not performance critical,
520     # we followed lexicographic order on constants,
521     # instead of an order to minimize the operations.
522     # It is also safer to avoid forgetting some relation.
523     declare -i LFBFL_i_result=$(( $1 ))
524     declare -i LFBFL_i_result_old=0
525     declare -i LFBFL_i_all_flags=$((
526         SSP_ALWAYS_LARGER_AFTER

```

```

527 | SSP_ALWAYS_LARGER_BEFORE
528 | SSP_DELIMITER_UNIFORM
529 | SSP_EQUAL_AFTER_OR_BEFORE
530 | SSP_LARGER_AFTER
531 | SSP_LARGER_BEFORE
532 | SSP_POSITION_NOT_DECREASING
533 | SSP_POSITION_NOT_DECREASING_AFTER
534 | SSP_POSITION_NOT_DECREASING_BEFORE
535 | SSP_POSITION_UNIFORM
536 | SSP_POSITION_UNIFORM_AFTER
537 | SSP_POSITION_UNIFORM_BEFORE
538 | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
539 | SSP_SAME_DELIMITER_LARGER_AFTER
540 | SSP_SAME_DELIMITER_LARGER_BEFORE
541 | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
542 | SSP_SAME_POSITION_LARGER_AFTER
543 | SSP_SAME_POSITION_LARGER_BEFORE
544 | SSP_UNIFORM
545 ))
546 until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
547     LFBFL_i_result_old=$((LFBFL_i_result))
548     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
549         LFBFL_i_result=$((
550             LFBFL_i_result
551             | SSP_LARGER_AFTER
552             | SSP_SAME_DELIMITER_LARGER_AFTER
553             | SSP_SAME_POSITION_LARGER_AFTER
554         ))
555     fi
556     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
557         LFBFL_i_result=$((
558             LFBFL_i_result
559             | SSP_LARGER_BEFORE
560             | SSP_SAME_DELIMITER_LARGER_BEFORE
561             | SSP_SAME_POSITION_LARGER_BEFORE
562         ))
563     fi
564     if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
565         LFBFL_i_result=$((
566             LFBFL_i_result
567             | SSP_POSITION_NOT_DECREASING_AFTER
568             | SSP_POSITION_NOT_DECREASING_BEFORE
569         ))
570     fi
571     if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
572         LFBFL_i_result=$((

```

```

573         LFBFL_i_result
574         | SSP_POSITION_UNIFORM_AFTER
575         | SSP_POSITION_UNIFORM_BEFORE
576         | SSP_POSITION_NOT_DECREASING
577         | SSP_POSITION_NOT_DECREASING_AFTER
578         | SSP_POSITION_NOT_DECREASING_BEFORE
579     ))
580 fi
581 if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
582     LFBFL_i_result=$((
583         LFBFL_i_result
584         | SSP_POSITION_NOT_DECREASING_AFTER
585     ))
586 fi
587 if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
588     LFBFL_i_result=$((
589         LFBFL_i_result
590         | SSP_POSITION_NOT_DECREASING_BEFORE
591     ))
592 fi
593 if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594     LFBFL_i_result=$((
595         LFBFL_i_result
596         | SSP_EQUAL_AFTER_OR_BEFORE
597         | SSP_LARGER_AFTER
598         | SSP_LARGER_BEFORE
599         | SSP_SAME_DELIMITER_LARGER_AFTER
600         | SSP_SAME_DELIMITER_LARGER_BEFORE
601     ))
602 fi
603 if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604     LFBFL_i_result=$((
605         LFBFL_i_result
606         | SSP_LARGER_AFTER
607     ))
608 fi
609 if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610     LFBFL_i_result=$((
611         LFBFL_i_result
612         | SSP_LARGER_BEFORE
613     ))
614 fi
615 if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616     LFBFL_i_result=$((
617         LFBFL_i_result
618         | SSP_DELIMITER_UNIFORM

```

```

619         | SSP_EQUAL_AFTER_OR_BEFORE
620         | SSP_LARGER_AFTER
621         | SSP_LARGER_BEFORE
622         | SSP_SAME_POSITION_LARGER_AFTER
623         | SSP_SAME_POSITION_LARGER_BEFORE
624     ))
625 fi
626 if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627     LFBFL_i_result=$((
628         LFBFL_i_result
629         | SSP_LARGER_AFTER
630     ))
631 fi
632 if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633     LFBFL_i_result=$((
634         LFBFL_i_result
635         | SSP_LARGER_BEFORE
636     ))
637 fi
638 if ((LFBFL_i_result & SSP_UNIFORM)); then
639     LFBFL_i_result=$((LFBFL_i_all_flags))
640     break
641 fi
642 if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643 && ((
644     LFBFL_i_result & (
645         SSP_ALWAYS_LARGER_BEFORE
646         | SSP_EQUAL_AFTER_OR_BEFORE
647         | SSP_LARGER_BEFORE
648         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
649         | SSP_SAME_DELIMITER_LARGER_BEFORE
650         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
651         | SSP_SAME_POSITION_LARGER_BEFORE
652     )
653 )); then
654     LFBFL_i_result=$((LFBFL_i_all_flags))
655     break
656 fi
657 if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
658 && ((
659     LFBFL_i_result & (
660         SSP_EQUAL_AFTER_OR_BEFORE
661         | SSP_LARGER_AFTER
662         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
663         | SSP_SAME_DELIMITER_LARGER_AFTER
664         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE

```



```

665         | SSP_SAME_POSITION_LARGER_AFTER
666     )
667 ); then
668     LFBFL_i_result=$((LFBFL_i_all_flags))
669     break
670 fi
671 if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
672 && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
673     LFBFL_i_result=$((
674         LFBFL_i_result
675         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
676     ))
677 fi
678 if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
679 && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
680     LFBFL_i_result=$((
681         LFBFL_i_result
682         | SSP_SAME_POSITION_LARGER_AFTER
683     ))
684 fi
685 if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
686 && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
687     LFBFL_i_result=$((
688         LFBFL_i_result
689         | SSP_SAME_POSITION_LARGER_BEFORE
690     ))
691 fi
692 if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
693 && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
694     LFBFL_i_result=$((LFBFL_i_all_flags))
695     break
696 fi
697 if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698 && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
699     LFBFL_i_result=$((
700         LFBFL_i_result
701         | SSP_ALWAYS_LARGER_AFTER
702     ))
703 fi
704 if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
705 && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
706     LFBFL_i_result=$((
707         LFBFL_i_result
708         | SSP_ALWAYS_LARGER_BEFORE
709     ))
710 fi

```

```

711 if ((LFBFL_i_result & SSP_LARGER_AFTER))\
712 && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
713     LFBFL_i_result=$((
714         LFBFL_i_result
715         | SSP_EQUAL_AFTER_OR_BEFORE
716     ))
717 fi
718 if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719 && ((
720     LFBFL_i_result & (
721         SSP_POSITION_UNIFORM
722         | SSP_POSITION_UNIFORM_AFTER
723     )
724 )); then
725     LFBFL_i_result=$((
726         LFBFL_i_result
727         | SSP_SAME_DELIMITER_LARGER_AFTER
728     ))
729 fi
730 if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731 && ((
732     LFBFL_i_result & (
733         SSP_POSITION_UNIFORM
734         | SSP_POSITION_UNIFORM_BEFORE
735     )
736 )); then
737     LFBFL_i_result=$((
738         LFBFL_i_result
739         | SSP_SAME_DELIMITER_LARGER_BEFORE
740     ))
741 fi
742 if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
743 && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
744     LFBFL_i_result=$((
745         LFBFL_i_result
746         | SSP_POSITION_NOT_DECREASING
747     ))
748 fi
749 if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
750 && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
751     LFBFL_i_result=$((
752         LFBFL_i_result
753         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
754     ))
755 fi
756 if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\

```

```

757      && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
758          LFBFL_i_result=$((LFBFL_i_all_flags))
759          break
760      fi
761      if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
762      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
763          LFBFL_i_result=$((
764              LFBFL_i_result
765              | SSP_ALWAYS_LARGER_AFTER
766          ))
767      fi
768      if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
769      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
770          LFBFL_i_result=$((
771              LFBFL_i_result
772              | SSP_ALWAYS_LARGER_BEFORE
773          ))
774      fi
775      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
776      && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
777          LFBFL_i_result=$((
778              LFBFL_i_result
779              | SSP_POSITION_UNIFORM
780          ))
781      fi
782      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
783      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
784          LFBFL_i_result=$((
785              LFBFL_i_result
786              | SSP_ALWAYS_LARGER_AFTER
787          ))
788      fi
789      if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790      && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791          LFBFL_i_result=$((
792              LFBFL_i_result
793              | SSP_ALWAYS_LARGER_BEFORE
794          ))
795      fi
796      if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797      && ((
798          LFBFL_i_result & (
799              SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800              | SSP_SAME_POSITION_LARGER_AFTER
801              | SSP_SAME_POSITION_LARGER_BEFORE
802          )

```

```

803     )); then
804         LFBFL_i_result=$((LFBFL_i_all_flags))
805         break
806     fi
807     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
808     && ((
809         LFBFL_i_result & (
810             SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811             | SSP_SAME_POSITION_LARGER_BEFORE
812         )
813     )); then
814         LFBFL_i_result=$((LFBFL_i_all_flags))
815         break
816     fi
817     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
819         LFBFL_i_result=$((
820             LFBFL_i_result
821             | SSP_ALWAYS_LARGER_AFTER
822         ))
823     fi
824     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
825     && ((
826         LFBFL_i_result & (
827             SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
828             | SSP_SAME_POSITION_LARGER_AFTER
829         )
830     )); then
831         LFBFL_i_result=$((LFBFL_i_all_flags))
832         break
833     fi
834     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
835     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
836         LFBFL_i_result=$((
837             LFBFL_i_result
838             | SSP_ALWAYS_LARGER_BEFORE
839         ))
840     fi
841     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
842     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
843         LFBFL_i_result=$((
844             LFBFL_i_result
845             | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
846         ))
847     fi
848 done

```

```

849     declare -gi i_split_score_properties_logical_closure_result=$((
850         LFBFL_i_result
851     ))
852 }
853
854 split_score_simple() {
855     # $1=larger_after (The name was "after_before" previously,
856     #                 the meaning was not modified.
857     #                 Remark not repeated below.)
858     # $2=delimiters_strings_domain concatenated characters/delimiters
859     # $3=delimiter_string a single character
860     # $4=is_cut_after
861     declare -gi i_split_score_result
862     if [[ "$2" != *$3* ]]; then
863         i_split_score_result=0
864         return
865     fi
866     if [[ "$4" == "$1" ]]; then
867         i_split_score_result=2
868         return
869     fi
870     i_split_score_result=1
871 }
872
873 get_split_score_simple() {
874     # $1=larger_after
875     # $2=max_length
876     # $3=delimiters_strings_domain
877     declare -g get_split_score_result="split_score_simple $1 '$3'"
878     declare -gi i_get_split_score_result2
879     # i_get_split_score_result2=$((11-$1*4))
880     # if [[ $1 -eq 1 ]]; then
881     #     i_get_split_score_result2=7
882     # else
883     #     i_get_split_score_result2=11
884     # fi
885     i_get_split_score_result2=$((
886         SSP_DELIMITER_UNIFORM
887         + SSP_POSITION_UNIFORM
888         + SSP_LARGER_BEFORE
889         + (
890             $1 * (
891                 SSP_LARGER_AFTER
892                 - SSP_LARGER_BEFORE
893             )
894         )

```

```

895     ))
896     # shellcheck disable=SC2248
897     split_score_properties_logical_closure ${i_get_split_score_result2}
898     i_get_split_score_result2=$((
899         i_split_score_properties_logical_closure_result
900     ))
901 }
902
903 get_split_score_exec() {
904     declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
905     SPLIT_SCORE_EXEC+="/split_score.exec.php"
906 }
907
908 split_score() {
909     # $1=larger_after
910     # $2=max_length
911     # $3=delimiters_strings_domain concatenated characters/delimiters
912     # $4=delimiter_string a single character
913     # $5=cut_position
914     # $6=is_cut_after
915     get_split_score_exec
916     declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
917     declare -gi i_split_score_result
918     i_split_score_result=$(eval -- "${LFBFL_command}")
919 }
920
921 # This function is currently not called,
922 # hence we need this shellcheck disable.
923 # shellcheck disable=SC2034
924 get_split_score() {
925     # $1=larger_after
926     # $1=max_length
927     # $2=delimiters_strings_domain
928     declare -g get_split_score_result="split_score $1 $2 '$3'"
929     declare -gi i_get_split_score_result2
930     # i_get_split_score_result2=$((9-$1*4))
931     i_get_split_score_result2=$((
932         SSP_DELIMITER_UNIFORM
933         + SSP_LARGER_BEFORE
934         + (
935             $1 * (
936                 SSP_LARGER_AFTER
937                 - SSP_LARGER_BEFORE
938             )
939         )
940     ))

```

```

941 # shellcheck disable=SC2248
942 split_score_properties_logical_closure ${i_get_split_score_result2}
943 i_get_split_score_result2=$((
944     i_split_score_properties_logical_closure_result
945 ))
946 }
947
948 split_line_at_most() {
949     # $1=line
950     # $2=max_length of beginning result string
951     # $3=split_score_command
952     # negative score means do not consider as a possible split
953     # $4=split_score_command_properties :
954     # These properties are flags.
955     # If someday some property is not a flag,
956     # then create split_score_command_properties2.
957     # Old values to get a simple idea without reading everything at the
958     # beginning of file:
959     # - null or 0 no property;
960     # - 1 split score doesn't depend on which delimiter matches;
961     # - 2 split score doesn't depend on the current position;
962     # - 4 any split score after is always larger
963     #   than any split score before;
964     # - 8 any split score before is always larger
965     #   than any split score after;
966     # $5=forbidden_previous_character
967     # $6=split_score_command_delimiter_strings_domain
968     # optimisation if small domain
969     # - this argument can be "null", then we will use $3 only instead,
970     #   in that case, only mono-character strings are considered,
971     # - this argument can be an array of characters,
972     #   optimize accordingly,
973     # - this argument can be an array of strings,
974     #   optimize accordingly,
975     # - this argument can be an array of regexps,
976     #   optimize accordingly.
977     # Not sure an array of regexps makes more sense than a single
978     # one. Maybe they could be both cases with corresponding
979     # optimizations. Same for a single character or substring.
980     #
981     # Note that we're always looking for the last position to split
982     # among the positions with maximum score;
983     # if some position gets a score from 2 delimiters,
984     # its score is always the maximum of the two corresponding scores.
985     declare -g split_line_at_most_result_start
986     declare -g split_line_at_most_result_end

```

```

987 declare -A LFBFL_positions
988 # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
989 # For my use case in Bash scripts, I will need only an array of
990 # characters. See get_split_score().
991 if [[ -n "$6" ]]; then
992     printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
993 fi
994 if [[ $2 -ge ${#1} ]]; then
995     split_line_at_most_result_start=$1
996     split_line_at_most_result_end=""
997     return
998 fi
999 if [[ $2 -lt 1 ]]; then
1000     printf "split_line_at_most: max_length should be at least 1...\n"
1001     return
1002 fi
1003 # We know that ${#1} > $2, hence ${#1} - 1 >= $2
1004 # Edge cases:
1005 # There is always at least one character that is overlength,
1006 # but we need to look only at the score before that character that is
1007 # overlength, the score after that character is invalid,
1008 # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
1009 # That first character that is overlength has index $2.
1010 # The cut position after that character has index $2 + 1.
1011 # Looping from character position going from 0 to $2, we need to drop
1012 # score before character 0 and drop score after character $2.
1013 # We do this with forbidden score -1.
1014 declare -ir LFBFL_i_max=$(( $2 ))
1015 declare -i LFBFL_i
1016 declare -i LFBFL_j
1017 local LFBFL_current_char
1018
1019 # At the beginning, each position has score 0.
1020 LFBFL_positions["0"]="-1"
1021 for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1022     LFBFL_positions["${LFBFL_i}"]="0"
1023 done
1024 LFBFL_positions["${LFBFL_i}"]="-1"
1025
1026 # If we have a forbidden previous character,
1027 # we mark the forbidden positions with -1.
1028 if [[ -n "$5" ]]; then
1029     for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1030         LFBFL_current_char="${1:${LFBFL_i}:1}"
1031         ((++LFBFL_i))
1032         if [[ "${LFBFL_current_char}" == "$5" ]]; then

```



```

1033         LFBFL_positions["${LFBFL_i}"]=" -1 "
1034     fi
1035 done
1036 fi
1037
1038 declare -r LFBFL_sort_command="sort --numeric-sort"
1039
1040 update_position_score() {
1041     # $1=position
1042     # Returns 0 if updated, 1 otherwise
1043     # printf "%s|%s\n" "${i_split_score_result}" "$1"
1044     if [[ i_split_score_result -ge 1 ]]; then
1045         # printf "valid result"
1046         if [[ ${LFBFL_positions["$1"]} != " -1" ]]; then
1047             LFBFL_positions["$1"]=$(
1048                 max "${LFBFL_sort_command}" \
1049                     "${LFBFL_positions["$1"]}" \
1050                     "${i_split_score_result}"
1051             )
1052             return 0
1053         fi
1054     fi
1055     return 1
1056 }
1057
1058 local LFBFL_split_score_command=$3
1059 declare -i LFBFL_split_score_command_properties=$4
1060
1061 normal_or_position_uniform_call() {
1062     # $1=cut_position
1063     # $2=is_cut_after
1064     if ((LFBFL_split_score_command_properties & SSP_POSITION_UNIFORM));
1065     then
1066         eval --\
1067             "${LFBFL_split_score_command} '${LFBFL_current_char}' $2"
1068     else
1069         eval --\
1070             "${LFBFL_split_score_command} '${LFBFL_current_char}' $1 $2"
1071     fi
1072 }
1073
1074 if (($4 & SSP_DELIMITER_UNIFORM)) \
1075 && (($4 & SSP_LARGER_AFTER)) \
1076 && (($4 & SSP_POSITION_NOT DECREASING_AFTER)); then
1077     for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1078         LFBFL_current_char="${1:${LFBFL_i}:1}"

```

```

1079     LFBFL_j=$((LFBFL_i + 1))
1080     # shellcheck disable=SC2248
1081     normal_or_position_uniform_call ${LFBFL_j} 1
1082     # shellcheck disable=SC2248
1083     if update_position_score ${LFBFL_j}
1084     then
1085         break
1086     fi
1087
1088     # shellcheck disable=SC2248
1089     normal_or_position_uniform_call ${LFBFL_i} 0
1090     # shellcheck disable=SC2248
1091     update_position_score ${LFBFL_i}
1092 done
1093 elif (($4 & SSP_DELIMITER_UNIFORM))\
1094 && (($4 & SSP_LARGER_BEFORE))\
1095 && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1096     for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1097         LFBFL_current_char="${1:${LFBFL_i}:1}"
1098         # We still need to break in second if, after computing score after,
1099         # because SSP_LARGER_BEFORE implies large inequality instead of
1100         # strict inequality.
1101         # And since the algorithm is to take the largest position with
1102         # maximum score, there is a possibility that we'll break on a found
1103         # score before but for splitting on a score after.
1104         LFBFL_j=$((LFBFL_i + 1))
1105         # shellcheck disable=SC2248
1106         normal_or_position_uniform_call ${LFBFL_j} 1
1107         # shellcheck disable=SC2248
1108         update_position_score ${LFBFL_j}
1109
1110         # shellcheck disable=SC2248
1111         normal_or_position_uniform_call ${LFBFL_i} 0
1112         # shellcheck disable=SC2248
1113         if update_position_score ${LFBFL_i}
1114         then
1115             break
1116         fi
1117     done
1118 else
1119     for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1120         LFBFL_current_char="${1:${LFBFL_i}:1}"
1121         eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1122         # shellcheck disable=SC2248
1123         update_position_score ${LFBFL_i}
1124         ((++LFBFL_i))

```

```

1125     eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1126     # shellcheck disable=SC2248
1127     update_position_score ${LFBFL_i}
1128 done
1129 fi
1130 declare -ir LFBFL_i_max_score=$(
1131     max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1132 )
1133 local LFBFL_value
1134 declare -i LFBFL_i_best_position
1135 for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1136     if [[ LFBFL_i -eq 0 ]]; then
1137         printf $'/!\split_line_at_most bug: no split position found./!\n'
1138     fi
1139     local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
1140     # printf "%s %s\n" "${LFBFL_i}" "${LFBFL_value}"
1141     if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
1142         LFBFL_i_best_position=$((LFBFL_i))
1143         break
1144     fi
1145 done
1146 # shellcheck disable=SC2248
1147 split_line_at "$1" ${LFBFL_i_best_position}
1148 split_line_at_most_result_start="${split_line_at_result_beginning}"
1149 split_line_at_most_result_end="${split_line_at_result_end}"
1150 # printf "%s\n" "$split_line_at_most_result_start"
1151 # printf "%s\n" "$split_line_at_most_result_end"
1152 }
1153
1154 split_last_line() {
1155     # $1=new_lines
1156     # $2=prefix
1157     # $3=max_length
1158     # $4=suffix
1159     # $5=split_score_command
1160     # $6=split_score_command_properties
1161     # $7=forbidden_previous_character ('\ ' usually)
1162     # Options:
1163     # --verbose
1164     # In general, a prefix/suffix can be a required line prefix/suffix
1165     # for all final lines like a comment prefix,
1166     # or it can be a continuation line prefix/suffix applied only
1167     # when splitting a line in 2 lines,
1168     # or it can be a prefix/suffix for the string made of all the lines.
1169     # Clearly, there is no gain to couple handling a prefix/suffix for
1170     # the string made of all the lines in this function.

```

```

1171 # Since we add a prefix/suffix when splitting,
1172 # it is easy to see that we do not need to distinguish between
1173 # required line prefix/suffix and continuation line prefix/suffix.
1174 # The $2 and $4 arguments are given as concatenation of the
1175 # required and continuation line prefixes/suffixes as needed by the
1176 # user.
1177 # Thus lines arguments should be given already prefixed
1178 # and suffixed.
1179 # This is logic, since it would be cumbersome to handle prefixing
1180 # or not already splitted lines in this function
1181 # (see repeated_split_last_line).
1182 # The given max_length is the goal.
1183 # This function adapts the effective max position for the split by
1184 # taking into account the length of the suffix.
1185 declare -i LFBFL_i_verbose=0
1186 get_verbose_option "$@"
1187 declare -a LFBFL_return_traps_stack
1188 init_return_trap
1189
1190 enhanced_set_shell_option pipefail --trap-unset
1191
1192 declare -g split_last_line_result="$1"
1193 declare -ir LFBFL_i_length2=$((3 - ${#4}))
1194 get_overlength_regexp "$3"
1195
1196 # Testing if some line is too long.
1197 printf "%s" "$1"\
1198 | grep --quiet "${get_overlength_regexp_result}"
1199 if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
1200     return
1201 fi
1202
1203 declare -r LFBFL_start=$(
1204     printf "%s" "$1"\
1205     | head --lines=-1
1206 )
1207 # printf "start: %s\n" "${LFBFL_start}"
1208 declare -r LFBFL_last_line=$(
1209     printf "%s" "$1"\
1210     | tail --lines=1
1211 )
1212 # printf "last_line: %s\n" "${LFBFL_last_line}"
1213 split_last_line_result=""
1214 if [[ -n "${LFBFL_start}" ]]; then
1215     split_last_line_result="${LFBFL_start}"
1216     # Last line-return of LFBFL_start seems dropped by $() and not ="${} ".

```

```

1217     # printf "%s %s\n" "${#split_last_line_result}"
1218     # "${split_last_line_result}"
1219     split_last_line_result+=$'\n'
1220     # something="${split_last_line_result}"
1221     # printf "%s %s\n" "${#something}" "${something}"
1222 fi
1223 if [[ -n "$5" ]]; then
1224     # shellcheck disable=SC2248
1225     split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
1226         "$5" "$6" "$7"
1227     split_last_line_result+="${split_line_at_most_result_start}"
1228     split_last_line_result+="$4"
1229     split_last_line_result+=$'\n'
1230     split_last_line_result+="$2${split_line_at_most_result_end}"
1231 else
1232     split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
1233     split_last_line_result+="$4"
1234     split_last_line_result+=$'\n'
1235     split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
1236 fi
1237 # printf "result: %s\n" "${split_last_line_result}"
1238 }
1239
1240 repeated_split_last_line() {
1241     # See split_last_line for the first arguments.
1242     # $8=final_suffix to handle when the end of the original string
1243     # is the suffix.
1244     declare -g repeated_split_last_line_result="$1"
1245     local LFBFL_result="$1x"
1246     until
1247         [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
1248     do
1249         split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
1250             "$4" "$5" "$6" "$7"
1251         LFBFL_result="${repeated_split_last_line_result}"
1252         repeated_split_last_line_result="${split_last_line_result}"
1253         # printf "%s\n" "${repeated_split_last_line_result}"
1254     done
1255     if [[ -n "$4" && -n "$8" ]]; then
1256         declare -ir LFBFL_i_offset=$((-${#4}))
1257         local LFBFL_lines_end
1258         LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
1259         if [[ "${LFBFL_lines_end}" == "$4" ]]; then
1260             repeated_split_last_line_result+="$8"
1261         fi
1262     fi

```

1263 }

3.31 build_and_checks_dependencies/too_long_code_lines.lib.sh

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
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6  # under the terms of the GNU General Public License
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9  # or (at your option) any later version.
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11 # DevOrSysAdminScripts is distributed
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13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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17 #
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19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "too_long_code_lines.sh"
25 # to "too_long_code_lines.lib.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.lib.sh
29 source "${LFBFL_subdir}/common_options.lib.sh"
30 # shellcheck source=get_common_text_glob_patterns.lib.sh
31 source "${LFBFL_subdir}/get_common_text_glob_patterns.lib.sh"
32 # shellcheck source=lines_filters.lib.sh
33 source "${LFBFL_subdir}/lines_filters.lib.sh"
34
35 get_overlength_regexp() {
36     # $1=length
37     # Option:
38     #   --length-is-overlength
39     declare -i LFBFL_i_overlength=$1
40     declare -i LFBFL_i_length_is_overlength
41     get_some_flag LFBFL_i_length_is_overlength --length-is-overlength 1 "$@"
42     if [[ LFBFL_i_length_is_overlength -eq 0 ]]; then
```

```

43     ((++LFBFL_i_overlength))
44 fi
45 readonly LFBFL_i_overlength
46 declare -g get_overlength_regexp_result=".\\${LFBFL_i_overlength}\\}"
47 # declare -g get_overlength_regexp_result="^.\\${LFBFL_i_overlength}\\}"
48 # declare -g get_overlength_regexp_result=".\\${LFBFL_i_overlength}\\}\\$"
49 # First one seems to be faster by a few percents when testing like this:
50 # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
51 # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
52 # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
53 }
54
55 too_long_code_lines(){
56     # Options:
57     #   --verbose
58     #   --work-directory=""
59     #   --max-line-length
60     declare -i LFBFL_i_verbose=0
61     get_verbose_option "$@"
62     local LFBFL_work_directory=""
63     get_work_directory_option "$@"
64     declare -a LFBFL_return_traps_stack
65     init_return_trap
66     pushd_to_work_directory --trap-popd
67     can_continue_after_enhanced_pushd || return 1
68
69     declare -i LFBFL_i_max_line_length
70     get_max_line_length_option "$@"
71
72     # shellcheck disable=SC2248
73     get_overlength_regexp ${LFBFL_i_max_line_length}
74
75     enhanced_set_shell_option pipefail --trap-unset
76
77     get_COMMON_TEXT_FILES_GLOB_PATTERNS
78     local LFBFL_pattern
79     local LFBFL_long_line
80     local LFBFL_file_path
81     local LFBFL_line
82     local LFBFL_extension
83     local LFBFL_base_name
84     local LFBFL_s_long_lines
85     declare -a LFBFL_arr_long_lines
86     for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
87         if [[ LFBFL_i_verbose -eq 1 ]]; then
88             printf "too_long_code_lines: Iterating on pattern: %s.\n\"

```

```

89     "${LFBFL_pattern}"
90 fi
91 # shellcheck disable=SC2248
92 LFBFL_s_long_lines=$(
93     find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
94     | relevant_find\
95     | not_license_find\
96     | xargs grep --line-number --with-filename\
97     --regexp="${get_overlength_regexp_result}" --
98 )
99 if [[ -z "${LFBFL_s_long_lines}" ]]; then
100     [[ LFBFL_i_verbose -eq 1 ]]\
101     && printf "too_long_code_lines: No file found with long lines.\n"
102     continue
103 fi
104 mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
105 for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
106     LFBFL_file_path=${LFBFL_long_line%:*} # Drop long ':*' suffix
107     # Remove file_path and line_number
108     LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
109     LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
110     # LFBFL_line="${LFBFL_line/%\\\/}" # Drop/Replace '\\' suffix by ''
111     LFBFL_extension=${LFBFL_file_path##*.*} # Drop long '*.' prefix
112     LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '*. ' suffix
113     if [[ "${LFBFL_extension}" == "html" ]]; then
114         if [[ -f "${LFBFL_base_name}.md" ]]; then
115             continue
116         fi
117         if grep --quiet --fixed-strings -- "${LFBFL_line}"\
118             "./build_and_checks_variables/temp/files_listing.html.sub"
119         then
120             continue
121         fi
122         printf "%s\n" "${LFBFL_long_line}"
123     elif [[ "${LFBFL_extension}" == "md" ]]; then
124         if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
125             printf "%s\n" "${LFBFL_long_line}"
126         fi
127     else
128         printf "%s\n" "${LFBFL_long_line}"
129     fi
130 done
131 done
132 }

```


3.32 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
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9  # or (at your option) any later version.
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15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "update_common_build_and_checks.sh"
25 # to "update_common_build_and_checks.exec.sh".
26
27 cd .. || exit
28 # shellcheck source=overwrite_if_not_equal.lib.sh
29 source "./build_and_checks_dependencies/overwrite_if_not_equal.lib.sh"
30 cd build_and_checks_dependencies || exit
31
32 update_common_build_and_checks() {
33     # Option:
34     #   --verbose
35     declare -i LFBFL_i_verbose=0
36     if [[ "$*" == *--verbose* ]]; then
37         printf "%s %s\n" "$0" "$*"
38         LFBFL_i_verbose=1
39     fi
40     readonly LFBFL_i_verbose
41
42     if [[ ! -o pipefail ]]; then
43         [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
44         set -o pipefail
```

```

45     trap 'set +o pipefail' RETURN
46 fi
47
48 declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
49 cp -- "${LFBFL_common_file_name}.tpl"\
50     "${LFBFL_common_file_name}.temp"
51
52 declare -ar LFBFL_files_names=(
53     "build_dependencies_notes.exec.php"
54     "build_dependencies_notes.lib.php"
55     "build_md_from_printable_md.exec.sh"
56     "check_shell_scripts_beginnings.lib.sh"
57     "check_shell_scripts_indentation.lib.sh"
58     "check_URLs.lib.sh"
59     "common_options.lib.sh"
60     "comparisons.lib.sh"
61     "create_PDF.exec.sh"
62     "generate_from_template.lib.sh"
63     "get_common_text_glob_patterns.lib.sh"
64     "grammar_and_spelling_check.lib.sh"
65     "licenses_templates/build_licenses_templates.exec.sh"
66     "licenses_templates/license_file_header_AGPLv3+.tpl"
67     "licenses_templates/license_file_header_GPLv3+.tpl"
68     "licenses_templates/license_file_header_LGPLv3+.tpl"
69     "lines_counts.lib.sh"
70     "lines_filters.lib.sh"
71     "lines_maps.lib.sh"
72     "overwrite_if_not_equal.lib.sh"
73     "python_black_complement.lib.sh"
74     "python_isort_complement.lib.sh"
75     "repository_data.lib.sh"
76     "shell_checks_complement.lib.sh"
77     "split_score.exec.php"
78     "split_score.lib.php"
79     "strings_functions.lib.sh"
80     "too_long_code_lines.lib.sh"
81     "update_or_check_files_names_listing.exec.sh"
82 )
83
84 local LFBFL_file_name
85 local LFBFL_file_sha512
86 local LFBFL_file_sha512_all
87 local LFBFL_base_file_name
88 for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
89     LFBFL_file_sha512=$(
90         sha512sum "${LFBFL_file_name}"\

```

```

91         | cut --delimiter=' ' --fields=1
92     )
93     LFBFL_file_sha512_all="LFBFL_correct_sha512="
94     LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
95     LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
96     LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
97     LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
98     LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}' "
99     LFBFL_base_file_name=$(basename -- "${LFBFL_file_name}")
100     sed --in-place\
101         "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
102         "${LFBFL_common_file_name}.temp"
103 done
104
105 overwrite_if_not_equal "${LFBFL_common_file_name}"\
106     "${LFBFL_common_file_name}.temp"
107
108 declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
109 cp "${LFBFL_main_file_name}.tpl" "${LFBFL_main_file_name}.temp"
110
111 LFBFL_file_sha512=$(
112     sha512sum "${LFBFL_common_file_name}"\
113     | cut --delimiter=' ' --fields=1
114 )
115 LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
116 LFBFL_file_sha512_all+="    LFBFL_correct_sha512="
117 LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
118 LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
119 LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
120 LFBFL_file_sha512_all+="    LFBFL_correct_sha512\+="
121 LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}' "
122 sed --in-place\
123     "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
124     "${LFBFL_main_file_name}.temp"
125
126 overwrite_if_not_equal "${LFBFL_main_file_name}"\
127     "${LFBFL_main_file_name}.temp"
128 }
129
130 update_common_build_and_checks "$@"

```

3.33 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #

```

```

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18 # You should have received a copy of
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20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 # 2025-12-30T21:33+01:00: This file was moved from
26 # "./build_and_checks_dependencies/listings"
27 # to
28 # "./build_and_checks_dependencies".
29 # This file was renamed from "update_or_check_files_names_listing.sh"
30 # to "update_or_check_files_names_listing.exec.sh".
31 # This file was renamed from "update_or_check_file_names_listing.sh"
32 # to "update_or_check_files_names_listing.sh"?
33
34 LFBFL_subdir="build_and_checks_dependencies"
35 # shellcheck source=common_options.lib.sh
36 source "${LFBFL_subdir}/common_options.lib.sh"
37 # shellcheck source=lines_filters.lib.sh
38 source "${LFBFL_subdir}/lines_filters.lib.sh"
39 # shellcheck source=strings_functions.lib.sh
40 source "${LFBFL_subdir}/strings_functions.lib.sh"
41
42 update_or_check_files_names_listing() {
43     # Options:
44     #   --verbose
45     #   --work-directory=""
46     declare -i LFBFL_i_verbose=0
47     get_verbose_option "$@"
48     local LFBFL_work_directory=""
49     get_work_directory_option "$@"

```

```

50 declare -a LFBFL_return_traps_stack
51 init_return_trap
52 pushd_to_work_directory --trap-popd
53 can_continue_after_enhanced_pushd || return 1
54
55 declare -i LFBFL_i_write
56 get_some_flag LFBFL_i_write --write 1 "$@"
57 declare -i LFBFL_i_append
58 get_some_flag LFBFL_i_append --append 1 "$@"
59
60 enhanced_set_shell_option pipefail --trap-unset
61
62 LFBFL_subdir2="build_and_checks_variables"
63 declare -r LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
64 declare -r LFBFL_data_file_path="./${LFBFL_subdir2}/repository_data.txt"
65
66 declare -i LFBFL_i_max_line_length
67 grep_variable "${LFBFL_data_file_path}" max_line_length\
68     --result-variable-prefix="LFBFL_i_"
69
70 if [[ LFBFL_i_write -eq 1 ]]; then
71     : > "${LFBFL_listing}"
72 fi
73 # Remove line returns that are here to keep lines short.
74 sed --regexp-extended --null-data --expression='s/(.)\\n/\\1/Mg'\
75     -- "${LFBFL_listing}"\
76     > "${LFBFL_listing}.temp"
77 # shopt -s dotglob was needed at some point but I don't see why now.
78 # shellcheck disable=SC2248
79 get_split_score_simple 1 ${LFBFL_i_max_line_length} /
80 declare -r LFBFL_split_score_command="${get_split_score_result}"
81 declare -i LFBFL_i_split_score_command_properties
82 LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
83 readonly LFBFL_i_split_score_command_properties
84 local LFBFL_suffix=\\ # instead of '\\' to avoid shellcheck SC1003
85 readonly LFBFL_suffix
86 # There is always a $'\n' final suffix from >>.
87 # But when origin string ends with the suffix we must add something.
88 declare -r LFBFL_final_suffix=$'\n\\'
89
90 local LFBFL_s_files_paths
91 LFBFL_s_files_paths=$(
92     find . -type f -printf '%P\n'\
93     | relevant_find\
94     | sort
95 )

```

```

96 # printf "%s\n" "${LFBFL_s_files_paths}"
97 declare -a LFBFL_arr_files_paths
98 if [[ -n "${LFBFL_s_files_paths}" ]]; then
99     if [[ LFBFL_i_verbose -eq 1 ]]; then
100         printf\
101             "update_or_check_files_names_listing: Some files to process.\n"
102         fi
103         mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
104     fi
105     # printf "%s\n" "${LFBFL_arr_files_paths[@]}"
106
107     local LFBFL_file_path
108     local LFBFL_base_file_name
109     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
110         # if [[ LFBFL_i_verbose -eq 1 ]]; then
111         #     printf\
112         #         "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
113         #         "${LFBFL_file_path}"
114         # fi
115         git check-ignore --quiet -- "${LFBFL_file_path}" && continue
116         LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
117         [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
118             || continue
119         if [[ "${LFBFL_base_file_name}" == *.md ]]; then
120             if [[ -f "${LFBFL_file_path}.tpl" ]]; then
121                 continue
122             fi
123         fi
124         if [[ LFBFL_i_verbose -eq 1 ]]; then
125             printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
126                 "${LFBFL_file_path}"
127         fi
128         # shellcheck disable=SC2248
129         repeated_split_last_line "${LFBFL_file_path}"\
130             ""\
131             "${LFBFL_i_max_line_length}"\
132             "${LFBFL_suffix}"\
133             "${LFBFL_split_score_command}"\
134             "${LFBFL_i_split_score_command_properties}"\
135             ""\
136             "${LFBFL_final_suffix}"
137         if [[ LFBFL_i_write -eq 1 ]]; then
138             # shellcheck disable=SC2001
139             printf "%s\n" "${repeated_split_last_line_result}"\
140                 >> "${LFBFL_listing}"
141             continue

```

```

142     fi
143     if ! grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
144         "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
145     then
146         printf "The file %s is not listed in %s.\n"\
147             "${LFBFL_file_path}"\
148             "${LFBFL_listing}"
149         if [[ LFBFL_i_append -eq 1 ]]; then
150             # shellcheck disable=SC2001
151             printf "%s\n" "${repeated_split_last_line_result}"\
152                 >> "${LFBFL_listing}"
153         fi
154     fi
155 done
156
157 LFBFL_s_files_paths=$(cat -- "${LFBFL_listing}.temp")
158 if [[ -n "${LFBFL_s_files_paths}" ]]; then
159     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
160 fi
161
162 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
163     [[ "${LFBFL_file_path}" != '/'/* ]] || continue
164     if [[ ! -f "${LFBFL_file_path}" ]]; then
165         printf "The non-file %s is listed in %s.\n"\
166             "${LFBFL_file_path}"\
167             "${LFBFL_listing}"
168     fi
169 done
170
171 rm -- "${LFBFL_listing}.temp"
172
173 if [[ LFBFL_i_verbose -eq 1 ]]; then
174     cat -- "${LFBFL_listing}"
175 fi
176 }
177
178 update_or_check_files_names_listing "$@"

```

3.34 build_and_checks_variables/.gitignore

```

1 temp/
2 upgrade_venvs_ts

```

3.35 build_and_checks_variables/bandit.ini

```
1 # FILE: .bandit
2 [bandit]
```

3.36 build_and_checks_variables/bandit_baseline.json

```
1 {"results":[]}
```

3.37 build_and_checks_variables/black.toml

```
1 [tool.black]
2 line-length = 75
```

3.38 build_and_checks_variables/dependencies_data.json

```
1 [
2   { "dependency_name": "Bash", "URLs": [
3     "https://www.gnu.org/software/bash/",
4     "https://tiswww.case.edu/php/chet/bash/bashtop.html",
5     "https://savannah.gnu.org/projects/bash/"
6   ] },
7   { "dependency_name": "GNU coreutils", "URLs": [
8     "https://www.gnu.org/software/coreutils/",
9     "https://savannah.gnu.org/projects/coreutils/"
10  ] },
11  { "dependency_name": "utils coreutils", "URLs": [
12    "https://utils.github.io/coreutils/",
13    "https://github.com/utils/coreutils"
14  ] },
15  { "dependency_name": "diff (GNU diffutils)", "URLs": [
16    "https://www.gnu.org/software/diffutils/",
17    "https://savannah.gnu.org/projects/diffutils/"
18  ] },
19  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
20    "https://www.gnu.org/software/findutils/",
21    "https://savannah.gnu.org/projects/findutils/"
22  ] },
23  { "dependency_name": "grep (GNU grep)", "URLs": [
24    "https://www.gnu.org/software/grep/",
25    "https://savannah.gnu.org/projects/grep/"
26  ] },
27  { "dependency_name": "sed (GNU sed)", "URLs": [
28    "https://www.gnu.org/software/sed/",
29    "https://savannah.gnu.org/projects/sed/"
30  ] },
31 ]
```



```

31 { "dependency_name": "ShellCheck", "URLs": [
32     "https://www.shellcheck.net/",
33     "https://github.com/koalaman/shellcheck"
34 ] },
35 { "dependency_name": "GNU Wget", "URLs": [
36     "https://www.gnu.org/software/wget/",
37     "https://savannah.gnu.org/projects/wget/"
38 ] },
39 { "dependency_name": "more of util-linux", "URLs": [
40     "https://www.kernel.org/pub/linux/utils/util-linux/",
41     "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42     {
43         "URL": "https://github.com/util-linux/util-linux",
44         "comment": "official?"
45     }
46 ] },
47 { "dependency_name": "pandoc", "URLs": [
48     "https://pandoc.org/",
49     "https://github.com/jgm/pandoc"
50 ] },
51 { "dependency_name": "pdftex/pdflatex", "URLs": [
52     "https://www.tug.org/applications/pdftex/",
53     "https://tug.org/texlive/svn/"
54 ] },
55 { "dependency_name": "tree", "URLs": [
56     "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57     "https://gitlab.com/OldManProgrammer/unix-tree",
58     "https://github.com/Old-Man-Programmer/tree"
59 ] },
60 { "dependency_name": "git", "URLs": [
61     "https://git-scm.com/about",
62     "https://github.com/git/git"
63 ] },
64 { "dependency_name": "pylint", "URLs": [
65     "https://www.pylint.org/",
66     "https://pylint.readthedocs.io/en/stable/",
67     "https://github.com/pylint-dev/pylint",
68     "https://pypi.org/project/pylint/"
69 ] },
70 { "dependency_name": "notify-send", "URLs": [
71     "https://gitlab.gnome.org/GNOME/libnotify/",
72     "https://gnome.pages.gitlab.gnome.org/libnotify/"
73 ] },
74 { "dependency_name": "black", "URLs": [
75     "https://github.com/psf/black",
76     "https://black.readthedocs.io/en/stable/"

```

```

77         "https://pypi.org/project/black/"
78     ] },
79     { "dependency_name": "isort", "URLs": [
80         "https://pycqa.github.io/isort/",
81         "https://github.com/pycqa/isort/",
82         "https://pypi.org/project/isort/"
83     ] },
84     { "dependency_name": "mypy", "URLs": [
85         "https://mypy-lang.org/",
86         "https://mypy.readthedocs.io/en/stable/",
87         "https://github.com/python/mypy",
88         "https://pypi.org/project/mypy/"
89     ] },
90     { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
91         "https://phpdoc.org/",
92         "https://docs.phpdoc.org/",
93         "https://github.com/phpdocumentor/phpdocumentor"
94     ] },
95     { "dependency_name": "pip", "URLs": [
96         "https://pip.pypa.io/en/stable/",
97         "https://github.com/pypa/pip",
98         {
99             "URL": "https://pypi.org/project/pip/",
100             "comment": "(egg and chicken ;) )"
101         }
102     ] },
103     { "dependency_name": "ruff", "URLs": [
104         "https://docs.astral.sh/ruff/",
105         "https://github.com/astral-sh/ruff",
106         "https://pypi.org/project/ruff/"
107     ] },
108     { "dependency_name": "xmllint (libxml2)", "URLs": [
109         "https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home",
110         "https://gitlab.gnome.org/GNOME/libxml2"
111     ] },
112     { "dependency_name": "bandit", "URLs": [
113         "https://bandit.readthedocs.io/en/latest/",
114         "https://github.com/PyCQA/bandit",
115         "https://pypi.org/project/bandit/"
116     ] },
117     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
118         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
119         "https://packagist.org/packages/squizlabs/php_codesniffer"
120     ] },
121     { "dependency_name": "phpmd", "URLs": [
122         "https://phpmd.org/",

```

```

123         "https://github.com/phpmd/phpmd",
124         "https://packagist.org/packages/phpmd/phpmd"
125     ] },
126     { "dependency_name": "php", "URLs": [
127         "https://www.php.net/",
128         "https://github.com/php/php-src"
129     ] },
130     { "dependency_name": "tex", "URLs": [
131         "https://ctan.org/tex-archive/systems/knuth/dist/tex"
132     ] },
133     { "dependency_name": "pcregrep, pcre2grep", "URLs": [
134         "https://www.pcre.org/",
135         "https://pcre2project.github.io/pcre2/",
136         "https://github.com/PCRE2Project/pcre1",
137         "https://github.com/PCRE2Project/pcre2",
138         "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
139     ] },
140     { "dependency_name": "python3", "URLs": [
141         "https://www.python.org/",
142         "https://docs.python.org/3/"
143     ] }
144 ]

```

3.39 build_and_checks_variables/dependencies_notes.txt

```

1  Bash can be found at the following URLs:
2  https://www.gnu.org/software/bash/
3  https://tiswww.case.edu/php/chet/bash/bashtop.html
4  https://savannah.gnu.org/projects/bash/
5
6  GNU coreutils can be found at the following URLs:
7  https://www.gnu.org/software/coreutils/
8  https://savannah.gnu.org/projects/coreutils/
9
10 utils coreutils can be found at the following URLs:
11 https://utils.github.io/coreutils/
12 https://github.com/utils/coreutils
13
14 diff (GNU diffutils) can be found at the following URLs:
15 https://www.gnu.org/software/diffutils/
16 https://savannah.gnu.org/projects/diffutils/
17
18 find, xargs (GNU findutils) can be found at the following URLs:
19 https://www.gnu.org/software/findutils/
20 https://savannah.gnu.org/projects/findutils/
21

```

22 grep (GNU grep) can be found at the following URLs:
23 <https://www.gnu.org/software/grep/>
24 <https://savannah.gnu.org/projects/grep/>
25
26 sed (GNU sed) can be found at the following URLs:
27 <https://www.gnu.org/software/sed/>
28 <https://savannah.gnu.org/projects/sed/>
29
30 ShellCheck can be found at the following URLs:
31 <https://www.shellcheck.net/>
32 <https://github.com/koalaman/shellcheck>
33
34 GNU Wget can be found at the following URLs:
35 <https://www.gnu.org/software/wget/>
36 <https://savannah.gnu.org/projects/wget/>
37
38 more of util-linux can be found at the following URLs:
39 <https://www.kernel.org/pub/linux/utils/util-linux/>
40 <https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/>
41 <https://github.com/util-linux/util-linux> official?
42
43 pandoc can be found at the following URLs:
44 <https://pandoc.org/>
45 <https://github.com/jgm/pandoc>
46
47 pdftex/pdflatex can be found at the following URLs:
48 <https://www.tug.org/applications/pdftex/>
49 <https://tug.org/texlive/svn/>
50
51 tree can be found at the following URLs:
52 <https://oldmanprogrammer.net/source.php?dir=projects/tree>
53 <https://gitlab.com/OldManProgrammer/unix-tree>
54 <https://github.com/Old-Man-Programmer/tree>
55
56 git can be found at the following URLs:
57 <https://git-scm.com/about>
58 <https://github.com/git/git>
59
60 pylint can be found at the following URLs:
61 <https://www.pylint.org/>
62 <https://pylint.readthedocs.io/en/stable/>
63 <https://github.com/pylint-dev/pylint>
64 <https://pypi.org/project/pylint/>
65
66 notify-send can be found at the following URLs:
67 <https://gitlab.gnome.org/GNOME/libnotify/>

68 <https://gnome.pages.gitlab.gnome.org/libnotify/>
69
70 black can be found at the following URLs:
71 <https://github.com/psf/black>
72 <https://black.readthedocs.io/en/stable/>
73 <https://pypi.org/project/black/>
74
75 isort can be found at the following URLs:
76 <https://pycqa.github.io/isort/>
77 <https://github.com/pycqa/isort/>
78 <https://pypi.org/project/isort/>
79
80 mypy can be found at the following URLs:
81 <https://mypy-lang.org/>
82 <https://mypy.readthedocs.io/en/stable/>
83 <https://github.com/python/mypy>
84 <https://pypi.org/project/mypy/>
85
86 phpdoc (phpDocumentor) can be found at the following URLs:
87 <https://phpdoc.org/>
88 <https://docs.phpdoc.org/>
89 <https://github.com/phpdocumentor/phpdocumentor>
90
91 pip can be found at the following URLs:
92 <https://pip.pypa.io/en/stable/>
93 <https://github.com/pypa/pip>
94 <https://pypi.org/project/pip/> (egg and chicken ;))
95
96 ruff can be found at the following URLs:
97 <https://docs.astral.sh/ruff/>
98 <https://github.com/astral-sh/ruff>
99 <https://pypi.org/project/ruff/>
100
101 xmllint (libxml2) can be found at the following URLs:
102 <https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home>
103 <https://gitlab.gnome.org/GNOME/libxml2>
104
105 bandit can be found at the following URLs:
106 <https://bandit.readthedocs.io/en/latest/>
107 <https://github.com/PyCQA/bandit>
108 <https://pypi.org/project/bandit/>
109
110 phpcs (PHP_CodeSniffer) can be found at the following URLs:
111 https://github.com/PHPCSStandards/PHP_CodeSniffer
112 https://packagist.org/packages/squizlabs/php_codesniffer
113

```

114  phpmd can be found at the following URLs:
115  https://phpmd.org/
116  https://github.com/phpmd/phpmd
117  https://packagist.org/packages/phpmd/phpmd
118
119  php can be found at the following URLs:
120  https://www.php.net/
121  https://github.com/php/php-src
122
123  tex can be found at the following URL:
124  https://ctan.org/tex-archive/systems/knuth/dist/tex
125
126  pcregrep, pcre2grep can be found at the following URLs:
127  https://www.pcre.org/
128  https://pcre2project.github.io/pcre2/
129  https://github.com/PCRE2Project/pcre1
130  https://github.com/PCRE2Project/pcre2
131  https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c
132
133  python3 can be found at the following URLs:
134  https://www.python.org/
135  https://docs.python.org/3/
136
137  If you see something that I missed regarding my dependencies,
138  please tell me/email me :).

```

3.40 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <!--
4  This file is part of DevOrSysAdminScripts.
5
6  DevOrSysAdminScripts is free software:
7  you can redistribute it and/or modify it
8  under the terms of the GNU Lesser General Public License
9  as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.

```

```

19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26 -->
27 <head>
28 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
29 <meta name="generator" content="DevOrSysAdminScripts">
30 <style>
31 .title{
32     margin: 2ex auto;
33     text-align: center;
34 }
35 .titlemain{
36     margin: 1ex 2ex 2ex 1ex;
37 }
38 .titlerest{
39     font-size: 1.17em;
40     margin: 0ex 2ex;
41 }
42 table{
43     border-collapse: collapse;
44 }
45 td{
46     padding: 0;
47 }
48 pre{
49     text-align: left;
50     margin-left: 0ex;
51     margin-right: auto;
52 }
53 pre.numbered_lines:before{
54     background: snow;
55     border-right: 1px black solid;
56     color: #010101;
57     content: attr(line-numbers);
58     float: left;
59     padding: 0 1ch;
60     text-align: right;
61 }
62 blockquote{
63     margin-left: 4ex;
64     margin-right: 4ex;

```

```

65     text-align: left;
66 }
67 ol.level1, ol.level2{
68     list-style-type: none;
69     /*
70     list-style-type: decimal;
71     Previous rule is shorter and more efficient if you need only level1.
72     But you don't have an accessible counter if you use decimal.
73     */
74 }
75 ol.level1{
76     counter-reset: level1;
77 }
78 ol.level2{
79     counter-reset: level2;
80 }
81 ol.level1 li:before{
82     content: counter(level1) ". ";
83     counter-increment: level1;
84 }
85 ol.level2 li:before{
86     content: counter(level1) "." counter(level2) " ";
87     counter-increment: level2;
88 }
89 </style>
90 <title>@repository_name@</title>
91 </head>
92 <body >
93 <table class="title">
94     <tr>
95         <td style="padding:1ex">
96             <h1 class="titlemain">@repository_name@</h1>
97             <h2 class="titlerest">
98                 @author_full_name@
99                 <br>
100                 <a href="@author_website@">
101                     <span style="font-family:monospace">
102                         @author_website@
103                     </span>
104                 </a>
105                 <br>
106                 <a href="mailto:@author_email@">
107                     <span style="font-family:monospace">
108                         @author_email@
109                     </span>
110                 </a>

```



```

111         </h2>
112     </td>
113 </tr>
114 </table>
115
116 <blockquote class="abstract">
117     <span style="font-weight:bold">Abstract:</span>
118     @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
123 <p>
124 Code lines: @number_of_lines@
125 </p>
126
127
128 <h2>Contents</h2>
129
130 <ol class="level1">
131     <li><a href="#section1">Files trees</a></li>
132     <li><a href="#section2">Notes on licenses and dependencies</a></li>
133     <li><a href="#section3">Listing of files</a><br>
134         <ol class="level2">
135             @files_lis@
136         </ol>
137     </li>
138 </ol>
139
140
141 <h2 id="section1">1 Files trees</h2>
142 <pre class="verbatim">
143 @current_tree_light@
144 </pre>
145 <pre class="verbatim">
146 @current_tree@
147 </pre>
148
149
150 <h2 id="section2">2 Notes on licenses and dependencies</h2>
151
152 <p>
153 This repository is dual licensed.
154 All the Bash scripts source code is covered by GPLv3+.
155 All the PHP scripts source code is covered by LGPLv3+.
156 All the Python scripts source code is covered by LGPLv3+.

```

```

157 </p>
158
159 <p>
160 The text of the licenses is available at:
161 <a href="https://www.gnu.org/licenses/"><span
162   style="font-family:monospace"
163 >https://www.gnu.org/licenses/</span></a>.
164 The git repository of this source code is also available at:
165 <a href="@repository_web_url@"><span
166   style="font-family:monospace"
167 >@repository_web_url@</span></a>.
168 Clone it with this URL:
169 <span style="font-family:monospace"
170 >@repository_git_url@</span>.
171 </p>
172
173 <div>
174 <p>
175 Here is a listing of the dependencies of the code in this repository:
176 </p>
177 <ul>
178 <li> The following dependencies are GPLv3+ licensed:<br>
179 basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
180 dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
181 sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
182 <li> The following dependencies are GPLv2+ licensed:<br>
183 more, pandoc, pdftex, pdflatex, tree;
184 <li> The following dependencies are GPLv2 licensed:<br>
185 git, pylint;
186 <li> The following dependencies are LGPLv2.1+ licensed:<br>
187 notify-send;
188 <li> The following dependencies are MIT licensed:<br>
189 black, isort, mypy, phpdoc, pip, ruff, xmllint;
190 <li> The following dependencies are Apache-2.0 licensed:<br>
191 bandit;
192 <li> The following dependencies are BSD-3-Clause licensed:<br>
193 phpcs, phpmd;
194 <li> php is licensed under PHP License v3.01;
195 <li> tex is licensed under Knuth license;
196 <li> pcregrep, pcre2grep has a custom license from Cambridge
197 University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
198 <li> python3 is licensed under PSF License Version 2 and the Zero-Clause
199 BSD license.
200 </ul>
201 </div>
202

```

```

203 <p>
204 Remark: Some commands listed under GPLv3+ are POSIX
205 and can have different implementations under another license;
206 for example, utils coreutils is MIT licensed,
207 whilst GNU coreutils is GPLv3+ licensed.
208 </p>
209
210 @dependencies_notes@
211
212
213 <h2 id="section3">3 Listing of files</h2>
214
215 @files_listing_HTMLPreInput@
216
217 <p>
218 @acknowledgments@
219 </p>
220
221 <script>
222 "use strict";
223 const pres = Array.from(
224   document.getElementsByClassName('numbered_lines')
225 );
226 pres.forEach(
227   (pre) => {
228     const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
229     let numbers = "";
230     for(let i = 1; i <= nb; ++i){
231       numbers += i.toString();
232       if(i < nb){ numbers += "\n "; }
233     }
234     pre.setAttribute('line-numbers', numbers);
235   }
236 );
237 </script>
238 </html>

```

3.41 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```

1 % This file is part of DevOrSysAdminScripts.
2 %
3 % DevOrSysAdminScripts is free software:
4 % you can redistribute it and/or modify it
5 % under the terms of the GNU Lesser General Public License
6 % as published by the Free Software Foundation,
7 % either version 3 of the License,

```

```

8 % or (at your option) any later version.
9 %
10 % DevOrSysAdminScripts is distributed
11 % in the hope that it will be useful,
12 % but WITHOUT ANY WARRANTY;
13 % without even the implied warranty of
14 % MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 % See the GNU Lesser General Public License for more details.
16 %
17 % You should have received a copy of
18 % the GNU Lesser General Public License
19 % along with DevOrSysAdminScripts.
20 % If not, see <https://www.gnu.org/licenses/>.
21 %
22 % ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
23
24 % 2025-12-29T17:08+01:00: This file was moved from
25 % "./latex"
26 % to
27 % "./build_and_checks_variables".
28
29 \documentclass{article}
30
31 \usepackage[utf8]{inputenc}
32 \usepackage[T1]{fontenc}
33 \usepackage{subfigure}
34 \usepackage{amsmath}
35 \usepackage{amssymb}
36 \usepackage{amsthm}
37 \usepackage[pdftex]{hyperref}
38 \usepackage{tikz}
39 \usepackage{caption}
40 \usepackage[round]{natbib}
41 \usepackage{fancyhdr}
42 \usepackage{amsfonts}
43 \usepackage{times}
44 \usepackage{ifpdf}
45 \usepackage{latexsym}
46 \usepackage{graphicx}
47 \usepackage{enumerate}
48 \usepackage{pmbboxdraw}
49 \usepackage{fancyvrb}
50 \usepackage[french,english]{babel}
51
52 % *** les environnements ***
53 %\theoremstyle{break}

```

```

54 \newtheorem{definition}{Definition}[section]
55 \newtheorem{proposition}[definition]{Proposition}
56 \newtheorem{theorem}[definition]{Theorem}
57 \newtheorem{lemma}[definition]{Lemma}
58 \newtheorem{corollary}[definition]{Corollary}
59 \newtheorem{remark}[definition]{Remark}
60 \newtheorem{openproblem}[definition]{Open problem}
61
62 \begin{document}
63
64 \author{
65     @author_full_name@\\
66     \url{@author_website@}\\
67     \href{mailto:@author_email@}{@author_email@}
68 }
69 \title{@repository_name@}
70
71 \maketitle
72 \begin{abstract}
73 @abstract@
74 \end{abstract}
75
76 Current version: @current_date@
77
78 Current number of commits: @number_of_commits@
79
80 Current git SHA1: @current_git_SHA1@
81
82 Code lines: @number_of_lines@
83
84 \tableofcontents
85 \newpage
86
87 \section{Files trees}
88 \label{section:trees}
89
90 \begin{verbatim}
91 @current_tree_light@
92 \end{verbatim}
93
94 \begin{verbatim}
95 @current_tree@
96 \end{verbatim}
97
98 \section{Notes on licenses and dependencies}
99 \label{section:notes_on}

```

```

100
101 This repository is dual licensed.
102 All the Bash scripts source code is covered by GPLv3+.
103 All the PHP scripts source code is covered by LGPLv3+.
104 All the Python scripts source code is covered by LGPLv3+.
105
106 The text of the licenses is available at:
107 \url{https://www.gnu.org/licenses/}.
108 The git repository of this source code is also available at:
109 \url{@repository_web_url@}.
110 Clone it with this URL:
111 \url{@repository_git_url@}.
112
113 Here is a listing of the dependencies of the code in this repository:
114 \begin{itemize}
115 \item The following dependencies are GPLv3+ licensed:\newline
116 basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
117 dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
118 sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
119 \item The following dependencies are GPLv2+ licensed:\newline
120 more, pandoc, pdftex, pdflatex, tree;
121 \item The following dependencies are GPLv2 licensed:\newline
122 git, pylint;
123 \item The following dependencies are LGPLv2.1+ licensed:\newline
124 notify-send;
125 \item The following dependencies are MIT licensed:\newline
126 black, isort, mypy, phpdoc, pip, ruff, xmllint;
127 \item The following dependencies are Apache-2.0 licensed:\newline
128 bandit;
129 \item The following dependencies are BSD-3-Clause licensed:\newline
130 phpcs, phpmd;
131 \item php is licensed under PHP License v3.01;
132 \item tex is licensed under Knuth license;
133 \item pcregrep, pcre2grep has a custom license from Cambridge
134 University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
135 \item python3 is licensed under PSF License Version 2 and the Zero-Clause
136 BSD license.
137 \end{itemize}
138
139 Remark: Some commands listed under GPLv3+ are POSIX
140 and can have different implementations under another license;
141 for example, utils coreutils is MIT licensed,
142 whilst GNU coreutils is GPLv3+ licensed.
143
144 @dependencies_notes@
145

```

```

146 \section{Listing of files}
147 \label{section:listing}
148
149 @files_listing_VerbatimInput@
150
151 @acknowledgments@
152
153 \end{document}

```

3.42 build_and_checks_variables/files_names_listing.txt

```

1 // This file is part of DevOrSysAdminScripts.
2 //
3 // DevOrSysAdminScripts is free software:
4 // you can redistribute it and/or modify it under the terms
5 // of the GNU Lesser General Public License
6 // as published by the Free Software Foundation,
7 // either version 3 of the License,
8 // or (at your option) any later version.
9 //
10 // DevOrSysAdminScripts is distributed in the hope
11 // that it will be useful,
12 // but WITHOUT ANY WARRANTY;
13 // without even the implied warranty of
14 // MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
15 // See the GNU Lesser General Public License for more details.
16 //
17 // You should have received a copy of
18 // the GNU Lesser General Public License
19 // along with DevOrSysAdminScripts.
20 // If not, see <https://www.gnu.org/licenses/>.
21 //
22 // ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
23 // 2024-10-31T21:37+01:00: This file was moved from
24 // "../build_and_checks_dependencies/listings"
25 // to
26 // "../build_and_checks_variables".
27 // Blasé20240517Résultat.txt
28 // Blasé20240517.sh
29 // Blasé20240518_2.txt
30 // Blasé20240518_3.txt
31 build_and_checks_dependencies/build_and_checks.exec.sh.tpl
32 build_and_checks_dependencies/build_dependencies_notes.exec.php
33 build_and_checks_dependencies/build_dependencies_notes.lib.php
34 build_and_checks_dependencies/build_md_from_printable_md.exec.sh
35 build_and_checks_dependencies/check_shell_scripts_beginnings.lib.sh

```

```

36 build_and_checks_dependencies/check_shell_scripts_indentation.lib.sh
37 build_and_checks_dependencies/check_URLs.lib.sh
38 build_and_checks_dependencies/common_build_and_checks.exec.sh
39 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
40 build_and_checks_dependencies/common_options.lib.sh
41 build_and_checks_dependencies/comparisons.lib.sh
42 build_and_checks_dependencies/create_PDF.exec.sh
43 build_and_checks_dependencies/generate_from_template.lib.sh
44 build_and_checks_dependencies/get_common_text_glob_patterns.lib.sh
45 build_and_checks_dependencies/grammar_and_spelling_check.lib.sh
46 build_and_checks_dependencies/licenses_templates/\
47 build_licenses_templates.exec.sh
48 build_and_checks_dependencies/licenses_templates/\
49 license_file_header_AGPLv3+.tpl
50 build_and_checks_dependencies/licenses_templates/\
51 license_file_header_GPLv3+.tpl
52 build_and_checks_dependencies/licenses_templates/\
53 license_file_header_LGPLv3+.tpl
54 build_and_checks_dependencies/lines_counts.lib.sh
55 build_and_checks_dependencies/lines_filters.lib.sh
56 build_and_checks_dependencies/lines_maps.lib.sh
57 // build_and_checks_dependencies/listings/files_listing.tex.tpl
58 build_and_checks_dependencies/overwrite_if_not_equal.lib.sh
59 build_and_checks_dependencies/python_black_complement.lib.sh
60 build_and_checks_dependencies/python_isort_complement.lib.sh
61 build_and_checks_dependencies/repository_data.lib.sh
62 build_and_checks_dependencies/shell_checks_complement.lib.sh
63 build_and_checks_dependencies/split_score.exec.php
64 build_and_checks_dependencies/split_score.lib.php
65 build_and_checks_dependencies/strings_functions.lib.sh
66 build_and_checks_dependencies/too_long_code_lines.lib.sh
67 build_and_checks_dependencies/update_common_build_and_checks.exec.sh
68 build_and_checks_dependencies/\
69 update_or_check_files_names_listing.exec.sh
70 build_and_checks_variables/.gitignore
71 build_and_checks_variables/bandit.ini
72 build_and_checks_variables/bandit_baseline.json
73 build_and_checks_variables/black.toml
74 build_and_checks_variables/dependencies_data.json
75 build_and_checks_variables/dependencies_notes.txt
76 build_and_checks_variables/DevOrSysAdminScripts.html.tpl
77 // build_and_checks_variables/DevOrSysAdminScripts.tex
78 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
79 build_and_checks_variables/files_names_listing.txt
80 build_and_checks_variables/licenses_notes.txt
81 build_and_checks_variables/phpcs_ruleset.xml

```



```

82 build_and_checks_variables/phpmd_baseline.xml
83 build_and_checks_variables/pylintrc.toml
84 build_and_checks_variables/repository_data.txt
85 build_and_checks_variables/repository_data.txt.tpl
86 build_and_checks_variables/ruff.toml
87 build_and_checks_variables/shellcheck.ini
88 build_and_checks_variables/shellcheck_notes.txt
89 // build_and_checks_variables/upgrade_venvs_ts
90 build_and_checks.exec.sh
91 COPYING
92 COPYING.LESSER
93 // DevOrSysAdminScripts.html
94 // DevOrSysAdminScripts.pdf
95 // EncoreBlasé20240517-20240518.txt
96 git_helpers/git_checks.lib.sh
97 git_helpers/post-commit
98 git_helpers/post_commit_renaming_comment.lib.sh
99 git_helpers/pre-commit
100 IMAP_transfert.exec.py
101 LiensUtiles.txt
102 // Uncomment some files in miscellaneous to do some testing.
103 // miscellaneous/\
104 12345678901234567890123456789012345678901234567890123456789012345
105 // miscellaneous/\
106 1234567890123456789012345678901234567890123456789012345678901234\
107 56
108 // miscellaneous/\
109 1234567890123456789012345678901234567890123456789012345678901234\
110 56\
111 \
112 // miscellaneous/\
113 1234567890123456789012345678901234567890123456789012345678901234\
114 5\
115 \
116 // miscellaneous/\
117 1234567890123456789012345678901234567890123456789012345678901234\
118 \
119 // miscellaneous/\
120 123456789012345678901234567890123456789012345678901234567890123\
121 \
122 // miscellaneous/\
123 a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
124 e_that_it_works.txt
125 miscellaneous/funny.sh
126 miscellaneous/spamty_data.txt
127 miscellaneous/string_escaping.html

```

```

128 miscellaneous/string_escaping.xhtml
129 // miscellaneous/\
130 yet_another_long_long_long_long_long_long_long_filename_so_that_\
131 you_can_see_that_it_works.txt
132 // README.html
133 // README.md
134 README.md.tpl
135 string_escaping.lib.php
136 unicode.lib.php
137 upgrade_build_and_checks.exec.sh
138 wget_sha512.lib.sh

```

3.43 build_and_checks_variables/licenses_notes.txt

```

1 This repository is dual licensed.
2 All the Bash scripts source code is covered by GPLv3+.
3 All the PHP scripts source code is covered by LGPLv3+.
4 All the Python scripts source code is covered by LGPLv3+.
5
6 Here is a listing of the dependencies of the code in this repository:
7 - The following dependencies are GPLv3+ licensed:
8 basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
9 dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
10 sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
11 - The following dependencies are GPLv2+ licensed:
12 more, pandoc, pdftex, pdflatex, tree;
13 - The following dependencies are GPLv2 licensed:
14 git, pylint;
15 - The following dependencies are LGPLv2.1+ licensed:
16 notify-send;
17 - The following dependencies are MIT licensed:
18 black, isort, mypy, phpdoc, pip, ruff, xmllint;
19 - The following dependencies are Apache-2.0 licensed:
20 bandit;
21 - The following dependencies are BSD-3-Clause licensed:
22 phpcs, phpmd;
23 - php is licensed under PHP License v3.01;
24 - tex is licensed under Knuth license;
25 - pcregrep, pcre2grep has a custom license from Cambridge University,
26 but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
27 - python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
28 license.
29
30 Remark: Some commands listed under GPLv3+ are POSIX
31 and can have different implementations under another license;
32 for example, utils coreutils is MIT licensed,

```

33 whilst GNU coreutils is GPLv3+ licensed.

3.44 build_and_checks_variables/phpcs_ruleset.xml

```
1 <?xml version="1.0"?>
2 <ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
3   <description>A custom coding standard</description>
4   <config name="tab_width" value="2"/>
5
6   <file>../</file>
7
8   <arg name="extensions" value="php"/>
9   <arg name="report" value="summary"/>
10  <arg name="colors"/>
11  <arg value="sp"/>
12
13  <!--
14  I added all the standards, then excluded all the rules that gave me
15  false positives or that seemed inappropriate when looking at the
16  documentation.
17  Some are bugs, some are conflicting with my own coding style.
18  You may delete an exclude and see what pops up to maybe understand
19  my choices.
20  -->
21
22  <rule ref="Generic">
23    <!-- You have to allow long or short or both syntaxes. -->
24    <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
25    <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
26    <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
27    <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
28    <!-- Back to my habits at Teliae with always a closing tag. -->
29    <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
30    <!--
31    Not standard in Unix/Linux world to keep files not ending with \n.
32    This is always an option of the code editors.
33    Since PHP doesn't output the final \n after ?>, it is ok.
34    -->
35    <exclude name="Generic.Files.EndFileNoNewline.Found"/>
36    <exclude name="Generic.Files.ExecutableFile.Executable"/>
37    <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
38    <exclude
39      name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
40    />
41    <exclude
42      name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
```

```

43     />
44     <!-- Applies only to true, false and null. -->
45     <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
46     <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
47     <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
48     <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
49 </rule>
50 <rule ref="PEAR">
51     <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
52     <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
53     <!--
54     Look at the comment before generate_split_score.
55     PEAR and Squiz require to align the parameter names parts in the same
56     comment, and also the parameter comments parts in the same comment,
57     but with a long parameter name like $a_delimiter_strings_domain
58     and a line length limit at 75 characters, parameter comments parts
59     have only 24 characters left per line, which is ridiculous.
60     Our choice to put the parameter comments parts starting on another line
61     with same "indent" as parameter names parts is way better in those
62     cases. Alignment has always been a trap with short lines.
63     Since phpDocumentor handles the parameter comments parts correctly,
64     it is enough for me.
65     Worse is the handling of "nested doc block comment" for the Closure
66     returned by generate_split_score().
67     Our solution makes at least the source code readable.
68     Maybe one day the dev tools will handle it correctly.
69     -->
70     <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
71     <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
72     <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
73     <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
74     <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
75     <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
76     <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
77     <exclude
78         name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
79     />
80     <exclude
81         name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
82     />
83     <exclude
84         name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
85     />
86     <!--
87     Don't laugh when looking at the false positives of the two exclude
88     below ;) XD.

```

```

89     They don't check for multiline function declarations apparently.
90     -->
91     <exclude
92         name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
93     />
94     <exclude
95         name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
96     />
97     <exclude
98         name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
99     />
100    <exclude
101        name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
102    />
103    </rule>
104    <rule ref="PSR1">
105    </rule>
106    <rule ref="PSR2">
107        <!-- Back to my habits at Teliae with always a closing tag. -->
108        <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
109        <exclude
110            name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
111        />
112        </rule>
113        <rule ref="PSR12">
114            <exclude
115                name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
116            />
117            <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
118            <!--
119            3 false positives, see:
120            https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
121            -->
122            <exclude
123                name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
124            />
125            <exclude
126                name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
127            />
128            <exclude
129                name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
130            />
131            <!-- and even a fourth and fifth false positives -->
132            <exclude
133                name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
134            />

```

```

135     <exclude
136         name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
137     />
138     <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
139     <exclude
140         name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
141     />
142     <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
143     <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
144     <!-- I agree except for string concatenation. -->
145     <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
146     <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
147 </rule>
148 <rule ref="Squiz">
149     <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
150     <!--
151     It doesn't know the //*/ trick...
152     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
153     -->
154     <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
155     <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
156     <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
157     <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
158     <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
159     <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
160     <!-- Problem for doc block comment before namespace -->
161     <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
162     <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
163     <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
164     <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
165     <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
166     <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
167     <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
168     <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
169     <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
170     <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
171     <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
172     <!-- asks for boolean instead of bool which is incorrect -->
173     <exclude
174         name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
175     />
176     <!-- asks for boolean instead of bool which is incorrect -->
177     <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
178     <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
179     <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
180     <exclude

```

```

181     name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
182   />
183   <exclude
184     name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
185   />
186   <!-- Problem for doc block comment before namespace -->
187   <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
188   <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
189   <!-- It doesn't know the /*/ trick... -->
190   <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
191   <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
192   <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
193   <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
194   <exclude
195     name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
196   />
197   <exclude
198 name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
199   />
200   <exclude
201     name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
202   />
203   <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
204   <exclude name="Squiz.Functions.GlobalFunction.Found"/>
205   <exclude
206     name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
207   />
208   <exclude
209 name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
210   />
211   <exclude
212     name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
213   />
214   <exclude
215     name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
216   />
217   <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
218   <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
219   <exclude name="Squiz.PHP.CommentedOutCode"/>
220   <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
221   <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
222   <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
223   <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
224   <exclude
225     name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
226   />

```

```

227 </rule>
228 <rule ref="Zend">
229     <!-- Back to my habits at Teliae with always a closing tag. -->
230     <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
231     <exclude
232         name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
233     />
234 </rule>
235
236 <rule ref="Generic.Arrays.ArrayIndent">
237     <properties>
238         <property name="indent" value="2" />
239     </properties>
240 </rule>
241 <rule ref="Generic.Files.LineLength">
242     <properties>
243         <property name="lineLimit" value="75"/>
244         <property name="absoluteLineLimit" value="75"/>
245     </properties>
246 </rule>
247 <rule ref="Generic.Formatting.SpaceAfterNot">
248     <properties>
249         <property name="spacing" value="0" />
250     </properties>
251 </rule>
252 <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
253     <properties>
254         <property name="checkFunctions" value="true"/>
255         <property name="checkClosures" value="true"/>
256     </properties>
257 </rule>
258 <rule ref="Generic.Metrics.CyclomaticComplexity">
259     <properties>
260         <property name="complexity" value="40" />
261         <property name="absoluteComplexity" value="80" />
262     </properties>
263 </rule>
264 <rule ref="Generic.Strings.UnnecessaryStringConcat">
265     <properties>
266         <property name="allowMultiline" value="true" />
267     </properties>
268 </rule>
269 <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
270     <properties>
271         <property name="ignoreNewlines" value="true" />
272     </properties>

```



```

273 </rule>
274 <rule ref="Generic.WhiteSpace.ScopeIndent">
275     <properties>
276         <property name="ignoreIndentationTokens" type="array">
277             <element value="T_COMMENT"/>
278             <element value="T_DOC_COMMENT_OPEN_TAG"/>
279         </property>
280         <property name="indent" value="2"/>
281     </properties>
282 </rule>
283
284 <rule ref="PEAR.Functions.FunctionCallSignature">
285     <properties>
286         <property name="indent" value="2"/>
287     </properties>
288 </rule>
289 <rule ref="PEAR.Functions.FunctionDeclaration">
290     <properties>
291         <property name="indent" value="2"/>
292     </properties>
293 </rule>
294 <rule ref="PEAR.WhiteSpace.ScopeIndent">
295     <properties>
296         <property name="ignoreIndentationTokens" type="array">
297             <element value="T_COMMENT"/>
298             <element value="T_DOC_COMMENT_OPEN_TAG"/>
299         </property>
300         <property name="indent" value="2"/>
301     </properties>
302 </rule>
303
304 <rule ref="PSR2.Methods.FunctionCallSignature">
305     <properties>
306         <property name="indent" value="2"/>
307     </properties>
308 </rule>
309
310 <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
311     <properties>
312         <property name="indent" value="2"/>
313     </properties>
314 </rule>
315
316 <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
317     <properties>
318         <property name="indent" value="2"/>

```

```

319     </properties>
320 </rule>
321 <rule ref="Squiz.Strings.ConcatenationSpacing">
322     <properties>
323         <property name="ignoreNewlines" value="true" />
324     </properties>
325 </rule>
326 <rule ref="Squiz.WhiteSpace.FunctionSpacing">
327     <properties>
328         <property name="spacing" value="3" />
329         <property name="spacingBeforeFirst" value="0" />
330         <property name="spacingAfterLast" value="0" />
331     </properties>
332 </rule>
333 </ruleset>

```

3.45 build_and_checks_variables/phpmd_baseline.xml

```

1  <?xml version="1.0"?>
2  <phpmd-baseline>
3      <violation rule="PHPMD\Rule\CyclomaticComplexity"
4          file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
5      />
6      <violation rule="PHPMD\Rule\Design\NpathComplexity"
7          file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
8      />
9      <violation rule="PHPMD\Rule\Design\LongMethod"
10         file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
11     />
12     <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
13         file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
14     />
15     <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
16         file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
17     />
18     <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
19         file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
20     />
21     <violation rule="PHPMD\Rule\Naming\LongVariable"
22         file="build_and_checks_dependencies/build_dependencies_notes.lib.php"
23     />
24     <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
25         file="build_and_checks_dependencies/split_score.lib.php"
26     />
27     <violation rule="PHPMD\Rule\Naming\LongVariable"
28         file="build_and_checks_dependencies/split_score.lib.php"

```

```

29     />
30     <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
31         file="build_and_checks_dependencies/split_score.lib.php"
32     />
33     <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
34         file="string_escaping.lib.php"
35     />
36     <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
37         file="string_escaping.lib.php"
38     />
39     <violation rule="PHPMD\Rule\Design\LongMethod"
40         file="unicode.lib.php"
41     />
42     <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
43         file="unicode.lib.php"
44     />
45     <violation rule="PHPMD\Rule\Naming\LongVariable"
46         file="unicode.lib.php"
47     />
48     <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
49         file="unicode.lib.php"
50     />
51     <!-- https://github.com/phpmd/phpmd/issues/910 -->
52     <violation rule="PHPMD\Rule\CleanCode\MissingImport"
53         file="unicode.lib.php"
54     />
55     <violation rule="PHPMD\Rule\CyclomaticComplexity"
56         file="unicode.lib.php"
57     />
58     <violation rule="PHPMD\Rule\Design\NpathComplexity"
59         file="unicode.lib.php"
60     />
61 </phpmd-baseline>

```

3.46 build_and_checks_variables/pylintrc.toml

```

1 [tool.pylint.main]
2 exit-zero = true
3 ignore = ["node_modules"]
4 jobs = 0
5 [tool.pylint.basic]
6 # We use uppercase acronyms like HTML, IMAP, etc.
7 module-rgx = "^[_a-zA-Z0-9+]+$"
8 function-rgx = "^[_a-zA-Z0-9+]+$"
9 [tool.pylint."messages control"]
10 # Code like R0912 is better to order disables by category.

```

```

11  disable = [
12      "C1804", # use-implicit-booleaness-not-comparison-to-string
13      "I0011", # locally-disabled
14      "I0020", # suppressed-message
15      "R0912", # too-many-branches
16      "R0913", # too-many-arguments
17      "R0914", # too-many-locals
18      "R0915", # too-many-statements
19      "R0917", # too-many-positional-arguments
20  ]
21  enable = "all"
22  [tool.pylint.reports]
23  output-format = "colorized"

```

3.47 build_and_checks_variables/repository_data.txt

```

1  Look at grep_variable to understand the structure of this file.
2  Basically, it has almost no structure apart from some lines
3  starting with a "variable_name=".
4  Anything else is comments.
5
6  repository_name=DevOrSysAdminScripts
7  repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
8  repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git
9
10 abstract=Some "small" useful scripts
11 abstract=for enhancing development or system-administrators tasks
12
13 license=GPLv3+
14 license2=LGPLv3+
15
16 author_email=Laurent.Lyaudet@gmail.com
17 author_full_name=Laurent Frédéric Bernard François Lyaudet
18 author_last_name=Lyaudet
19 author_name=Laurent Lyaudet
20 author_website=https://lyaudet.eu/laurent/
21
22 acknowledgments=Merci Dieu !
23 acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !
24
25 Great day! Now I allow myself lines of 75 characters instead of 70 :).
26 I checked how many characters fit in the PDF file generated by tex with
27 default paper size,
28 when using verbatim environment with line numbers for source code listing.
29 With a screen capture in GIMP,
30 I mesured 197px between two-digits line numbers and the left border of

```

```

31 the page.
32 (It is less than 197px for three-digits or more line numbers.)
33 The same distance from the right border yields 75 characters allowed.
34 Since when printing the HTML file, the number of allowed characters in
35 <pre> tags is way larger, 75 is the right limit with the tools I use.
36
37 max_line_length=75
38
39 latex_command=pdflatex
40
41 grammar_and_spelling_checker_command=
42
43
44 Why one venv for each Python dependency ?
45 Because no language of my knowledge did it right to handle all
46 possible dependency conflicts.
47 So in case of such a conflict, one specialized venv can get you out
48 of any problem.
49 The best solution would have been to allow install of different
50 versions of the same package, and have a mechanism:
51 - first intent, use the latest available version of a package among
52 those that satisfy the declared dependency constraints of the package
53 that has this dependency;
54 - second intent, check a dependency overload configuration file
55 with tree structure: if calling python package is A version 1.1.1
56 with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
57 you could have along that another rule saying that when called
58 directly B 2.2.2 should use C 4.4.4;
59 and even more complicated:
60 if calling python package is A version 1.1.1 with dependency B and D
61 both themselves with dependency C, use B 2.2.2, D 0.1.0,
62 and C 3.3.3 for B, and C 4.4.4 for D;
63 once you've seen that it can never be more complicated than that once
64 the same version number guarantees that the code is the same,
65 you create a config language "à la CSS" for cascading dependencies ;)
66 XD fit into JSON the dumbest possible way and you stop bothering
67 about dependency conflicts for the rest of history of informatics.
68 {
69   "A":{
70     "1.1.1": {
71       "B": {
72         "2.2.2": {
73           "C": {
74             // Dumb, you did not found it by yourself? Failed exam.
75             "3.3.3" : {}
76           }
77         }
78       }
79     }
80   }
81 }

```

```

77         }
78     },
79     "D": {
80         "0.1.0": {
81             "C": {
82                 // Dumb, you did not found it by yourself? Failed exam.
83                 "4.4.4" : {}
84             }
85         }
86     }
87 },
88 "B": {
89     "2.2.2": {
90         "C": {
91             // Dumb, you did not found it by yourself? Failed exam.
92             "4.4.4" : {}
93         }
94     }
95 }
96 }
97 }
98 Wonderful someone rediscovered nested structures XD
99 Seriously so many engineers or computer scientists are dumb to cry.
100
101 bandit_venv=../venv_bandit
102 black_venv=../venv_black
103 isort_venv=../venv_isort
104 mypy_venv=../venv_mypy
105 pylint_venv=../venv_pylint
106 ruff_venv=../venv_ruff
107
108
109 upgrade_venvs can be auto or ask, any other value is interpreted
110 as ask.
111
112 upgrade_venvs=ask
113
114
115 time interval can be 0 to always upgrade or ask to upgrade,
116 or a number at most bash max int
117 https://superuser.com/questions/1030122/
118 what-is-the-maximum-value-of-a-numeric-bash-shell-variable
119 it should be enough to simulate never "upgrade or ask for upgrade"
120 unless you claim to live many billions years ;)
121 If it is wat, this is a random int XD (see RANDOM).
122 Note that the probability that it upgrades your dependencies is not

```

```

123 1/2, it increases over time, and is 1 after slightly
124 less than one day.
125 If it is watyouwant?, see SRANDOM XD.
126
127 upgrade_venvs_time_interval_in_seconds=wat
128
129
130 For the Nu HTML, CSS, and SVG checker.
131
132 vnu_jar_path=../jar_venv/vnu.jar
133
134 Not fully implemented yet, it will only warn that vnu jar was not
135 updated since some time.
136 "wat" works, but "watyouwant?" joke was not repeated ;).
137
138 upgrade_vnu_jar_time_interval_in_seconds=604800
139
140 use_vnu_for_xhtml=false
141
142 xmllint_files_without_DTD=./build_and_checks_variables/phpcs_ruleset.xml
143 xmllint_files_without_DTD=./build_and_checks_variables/phpmd_baseline.xml

```

3.48 build_and_checks_variables/repository_data.txt.tpl

```

1 Look at grep_variable to understand the structure of this file.
2 Basically, it has almost no structure apart from some lines
3 starting with a "variable_name=".
4 Anything else is comments.
5 The goal of this template is to list all possible variables.
6
7 repository_name=
8 repository_web_url=
9 repository_git_url=
10
11 abstract=
12 abstract=
13
14 license=
15 license2=
16
17 author_email=
18 author_full_name=
19 author_last_name=
20 author_name=
21 author_website=
22

```

```

23 acknowledgments=
24 acknowledgments=
25
26 max_line_length=
27
28 latex_command=
29
30 grammar_and_spelling_checker_command=
31
32 bandit_venv=
33 black_venv=
34 isort_venv=
35 mypy_venv=
36 pylint_venv=
37 ruff_venv=
38
39 upgrade_venvs=
40
41 upgrade_venvs_time_interval_in_seconds=
42
43 vnu_jar_path=
44
45 upgrade_vnu_jar_time_interval_in_seconds=
46
47 use_vnu_for_xhtml=
48
49 npm_lint_directories=
50 npm_lint_directories=
51
52 xmllint_files_without_DTD=
53 xmllint_files_without_DTD=

```

3.49 build_and_checks_variables/ruff.toml

```

1 [lint]
2 select = [
3     "ALL",
4     "D211", # blank-line-before-class no...before-class-docstring!
5 ]
6 # ignore = ["D203", "D212", "D404"]
7 ignore = [
8     "C901", # complex-structure AKA give-me-toy-code
9     "D203", # (want-)?incorrect-blank-line-before-class
10    "D205", # missing-blank-line-after-summary So so cute|true ;) XD.
11    "D212", # multi-line-summary-first-line
12    "D404", # docstring-starts-with-this Shame! XD

```



```

13
14 # raw-string-in-exception may be useful for a real project
15 # with "dictionnaires" of errors made of constants.
16 "EM101",
17
18 # commented-out-code Of course git log is always faster...
19 # Note that if you're tired or someone else in your team is,
20 # it may catch some commit that lacked some review,
21 # or automated tests: "Last manual test succeeded, ah but when I
22 # debugged at some point I commented a part that was needed in
23 # another case."...
24 "ERA001",
25
26 "N802", # invalid-function-name IMAP, HTML, etc.
27 "PLR0912", # too-many-branches AKA ...
28 "PLR0913", # too-many-arguments AKA ...
29 "PLR0915", # too-many-statements AKA ...
30 # Not enough reading when young makes brain slow to read and
31 # understand lengthy functions.
32 # If the project is finished, you may play with AKA rules.
33 ]
34 [lint.per-file-ignores]
35 "build_and_checks_dependencies/licenses_templates/*.py" = [
36     "INP001" # implicit-namespace-package
37 ]
38 "IMAP_transfert.exec.py" = [
39     "T201", # print Use log instead in full applications.
40     "TRY003", # raise-vanilla-args
41 ]

```

3.50 build_and_checks_variables/shellcheck.ini

```

1 enable=all
2 check-sourced=true
3 external-sources=true
4 source-path="."
5 source-path="SCRIPTDIR"
6 source-path="./build_and_checks_dependencies"
7 source-path="./git_helpers"

```

3.51 build_and_checks_variables/shellcheck_notes.txt

```

1 A listing of the current suppressed warnings/errors from shellcheck,
2 and also some of the warnings that were suppressed in some previous
3 state of the git repository (in that case, we use X suffix, see SC2162X):
4 - SC1090 : Can't follow non-constant source.

```

```

5         Use a directive to specify location
6         # shellcheck source=bla/bli.sh
7         Only remaining cases are in common_build_and_checks
8         when sourcing some Python venv
9         source venv/bin/active
10    - SC1091 : Not following (file not found most of the time).
11             Only remaining cases are in common_build_and_checks
12             when sourcing some Python venv
13             source venv/bin/active
14    - SC1111 : Use of slanted double quotes (valid inside single quotes)
15             False positive, but may catch some problem.
16    - SC2001 : Replace some sed with bash builtin ${a//b/c}
17             False positive when c contains \n.
18    - SC2016 : Expressions don't expand in single quotes,
19             use double quotes for that.
20    - SC2034 : Unused variable
21             False positive for globals that are declared in functions
22             that are not currently called.
23             Somehow it can be used to detect dead code.
24    - SC2059 : Don't use variables in the printf format string.
25             Use printf '..%s..' "$foo".
26             False positive when using a variable to build the format string.
27    - SC2086 : Double quote to prevent globbing and word splitting.
28             False positive when returning integer variable
29             via an indirection like return ${!foo}.
30             See SC2248 below and the corresponding issue on GitHub.
31    - SC2119 : Use foo "$@" if function's $1 should mean script's $1.
32             False positive when passing arguments would yield
33             a wrong result.
34    - SC2120 : foo references arguments, but none are ever passed.
35             False positives when some arguments are meant for debug in CLI.
36    - SC2125 : Brace expansions and globs are literal.
37             False positive when assigning a variable declared with -i.
38             https://github.com/koalaman/shellcheck/issues/914
39    - SC2145 : Argument mixes string and array.
40             False positive when using "$@" line return and pipes.
41             https://github.com/koalaman/shellcheck/issues/3192
42             Example:
43             # shellcheck disable=SC2145
44             all_self_code_lines "$@"\
45             | empty_lines_after_file_name
46    - SC2154 : var is referenced but not assigned.
47             Very useful but false positive when I play with
48             variable variable names.
49    - SC2162X: read without -r will mangle backslashes.
50             False positive when read is used without variable:

```

```

51         https://github.com/koalaman/shellcheck/issues/623
52 - SC2248 : Prefer double quoting
53         False positive on returning integer values.
54         https://github.com/koalaman/shellcheck/issues/3405
55 - SC2309 : -eq treats this as a variable.
56         Use = to compare as string (or expand explicitly with $var).
57         Very useful but false positive when I play with
58         variable variable names.
59 - SC2312 : Some command is not invoked separately and thus its return
60         value is masked. Frequent with pipes.
61 - SCNNNNX: Use printf instead of echo.
62         https://github.com/koalaman/shellcheck/issues/3419
63
64 The command to check if there is anything else:
65 grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$\\"
66     -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$\\"
67     -e "=SC2016$" -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$\\"
68     -e "=SC2120$" -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$\\"
69     -e "=SC2309$" -e "=SC2312$" -e "=SC2154,SC2309$"

```

3.52 build_and_checks.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # The file "build_and_checks.exec.sh.tpl" was renamed from

```

```

25 # "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
26 # The file "build_and_checks.exec.sh" generated from the file
27 # "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
28 # was renamed from
29 # "build_and_checks.sh" to "build_and_checks.exec.sh".
30
31 build_and_checks() {
32     # $1=work_directory
33     # Options:
34     #   --check-download=y|keep (=keep to keep the downloaded files)
35     #   --verbose
36     local LFBFL_work_directory="."
37     if [[ -n "$1" ]]; then
38         LFBFL_work_directory="$1"
39     fi
40     readonly LFBFL_work_directory
41
42     local LFBFL_verbose=""
43     if [[ "$*" == *--verbose* ]]; then
44         printf "%s %s\n" "$0" "$*"
45         LFBFL_verbose="--verbose"
46     fi
47     readonly LFBFL_verbose
48
49     local LFBFL_check_download=""
50     if [[ "$*" == *--check-download=keep* ]]; then
51         LFBFL_check_download="--check-download=keep"
52     elif [[ "$*" == *--check-download=y* ]]; then
53         LFBFL_check_download="--check-download=y"
54     fi
55     readonly LFBFL_check_download
56
57     source ./wget_sha512.lib.sh
58
59     LFBFL_subdir="build_and_checks_dependencies"
60     mkdir --parents "${LFBFL_subdir}/licenses_templates"
61     local LFBFL_variables_directory
62     LFBFL_variables_directory="${LFBFL_work_directory}/"
63     LFBFL_variables_directory+="build_and_checks_variables"
64     readonly LFBFL_variables_directory
65     mkdir --parents -- "${LFBFL_variables_directory}/temp"
66     if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
67         printf "temp/\nupgrade_venvs_ts\n" \
68             > "${LFBFL_variables_directory}/.gitignore"
69     fi
70

```

```

71 local LFBFL_dependencies_URL
72 LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73 LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74 readonly LFBFL_dependencies_URL
75
76 declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77 local LFBFL_script_download_URL
78 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79 LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80 readonly LFBFL_script_download_URL
81 declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
82 local LFBFL_correct_sha512
83 LFBFL_correct_sha512='f03ab16e274619d5b4514ee6fa7016411bcaa916db31b'
84 LFBFL_correct_sha512+='1d8b408862a4ac15a3b57b3d777e1a509bdc4dbdc694'
85 LFBFL_correct_sha512+='4a48c7646c3264763f0578606d9ff5e26b3e716'
86 wget_sha512 "${LFBFL_file_path}" \
87     "${LFBFL_script_download_URL}" \
88     "${LFBFL_correct_sha512}" \
89     "${LFBFL_verbose}" \
90     "${LFBFL_check_download}"
91 chmod +x "${LFBFL_file_path}"
92
93 "${LFBFL_file_path}" "${LFBFL_work_directory}" \
94     "${LFBFL_dependencies_URL}" \
95     "${LFBFL_verbose}" \
96     "${LFBFL_check_download}"
97 }
98
99 if [[ "$*" == *--fixed-point-build* ]]; then
100     printf -- "--fixed-point-build\n"
101     source "./build_and_checks_dependencies/lines_filters.lib.sh"
102     LFBFL_work_directory="."
103     if [[ -n "$1" ]]; then
104         LFBFL_work_directory="$1"
105     fi
106     LFBFL_variables_directory="${LFBFL_work_directory}/"
107     LFBFL_variables_directory+="build_and_checks_variables"
108     LFBFL_data_file_name="${LFBFL_variables_directory}/"
109     LFBFL_data_file_name+="repository_data.txt"
110     LFBFL_repository_name=""
111     grep_variable "${LFBFL_data_file_name}" repository_name \
112         --result-variable-prefix="LFBFL_" \
113         --replace-line-returns-by=""
114     # Touching the 3 following files first let us iterate
115     # upgrade_build_and_checks 2 times instead of 3 to reach
116     # a fixed point.

```

```

117 touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
118 touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
119 touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
120 # The two iterations of build_and_checks needs to complete
121 # during the same minute to have a fixed point.
122 build_and_checks "$@"
123 # build_and_checks "$@" second iteration below
124 # Then calling this script without --fixed-point-build should not
125 # yield any new superficial modifications.
126 fi
127
128 build_and_checks "$@"

```

3.53 COPYING

```

1          GNU GENERAL PUBLIC LICENSE
2          Version 3, 29 June 2007
3
4  Copyright (C) 2007 Free Software Foundation, Inc. <https://fsf.org/>
5  Everyone is permitted to copy and distribute verbatim copies
6  of this license document, but changing it is not allowed.
7
8          Preamble
9
10 The GNU General Public License is a free, copyleft license for
11 software and other kinds of works.
12
13 The licenses for most software and other practical works are designed
14 to take away your freedom to share and change the works. By contrast,
15 the GNU General Public License is intended to guarantee your freedom to
16 share and change all versions of a program--to make sure it remains free
17 software for all its users. We, the Free Software Foundation, use the
18 GNU General Public License for most of our software; it applies also to
19 any other work released this way by its authors. You can apply it to
20 your programs, too.
21
22 When we speak of free software, we are referring to freedom, not
23 price. Our General Public Licenses are designed to make sure that you
24 have the freedom to distribute copies of free software (and charge for
25 them if you wish), that you receive source code or can get it if you
26 want it, that you can change the software or use pieces of it in new
27 free programs, and that you know you can do these things.
28
29 To protect your rights, we need to prevent others from denying you
30 these rights or asking you to surrender the rights. Therefore, you have
31 certain responsibilities if you distribute copies of the software, or if

```

32 you modify it: responsibilities to respect the freedom of others.

33
34 For example, if you distribute copies of such a program, whether
35 gratis or for a fee, you must pass on to the recipients the same
36 freedoms that you received. You must make sure that they, too, receive
37 or can get the source code. And you must show them these terms so they
38 know their rights.

39
40 Developers that use the GNU GPL protect your rights with two steps:
41 (1) assert copyright on the software, and (2) offer you this License
42 giving you legal permission to copy, distribute and/or modify it.

43
44 For the developers' and authors' protection, the GPL clearly explains
45 that there is no warranty for this free software. For both users' and
46 authors' sake, the GPL requires that modified versions be marked as
47 changed, so that their problems will not be attributed erroneously to
48 authors of previous versions.

49
50 Some devices are designed to deny users access to install or run
51 modified versions of the software inside them, although the manufacturer
52 can do so. This is fundamentally incompatible with the aim of
53 protecting users' freedom to change the software. The systematic
54 pattern of such abuse occurs in the area of products for individuals to
55 use, which is precisely where it is most unacceptable. Therefore, we
56 have designed this version of the GPL to prohibit the practice for those
57 products. If such problems arise substantially in other domains, we
58 stand ready to extend this provision to those domains in future versions
59 of the GPL, as needed to protect the freedom of users.

60
61 Finally, every program is threatened constantly by software patents.
62 States should not allow patents to restrict development and use of
63 software on general-purpose computers, but in those that do, we wish to
64 avoid the special danger that patents applied to a free program could
65 make it effectively proprietary. To prevent this, the GPL assures that
66 patents cannot be used to render the program non-free.

67
68 The precise terms and conditions for copying, distribution and
69 modification follow.

70 71 TERMS AND CONDITIONS

72 73 0. Definitions.

74
75 "This License" refers to version 3 of the GNU General Public License.

76
77 "Copyright" also means copyright-like laws that apply to other kinds of

78 works, such as semiconductor masks.

79
80 "The Program" refers to any copyrightable work licensed under this
81 License. Each licensee is addressed as "you". "Licensees" and
82 "recipients" may be individuals or organizations.

83
84 To "modify" a work means to copy from or adapt all or part of the work
85 in a fashion requiring copyright permission, other than the making of an
86 exact copy. The resulting work is called a "modified version" of the
87 earlier work or a work "based on" the earlier work.

88
89 A "covered work" means either the unmodified Program or a work based
90 on the Program.

91
92 To "propagate" a work means to do anything with it that, without
93 permission, would make you directly or secondarily liable for
94 infringement under applicable copyright law, except executing it on a
95 computer or modifying a private copy. Propagation includes copying,
96 distribution (with or without modification), making available to the
97 public, and in some countries other activities as well.

98
99 To "convey" a work means any kind of propagation that enables other
100 parties to make or receive copies. Mere interaction with a user through
101 a computer network, with no transfer of a copy, is not conveying.

102
103 An interactive user interface displays "Appropriate Legal Notices"
104 to the extent that it includes a convenient and prominently visible
105 feature that (1) displays an appropriate copyright notice, and (2)
106 tells the user that there is no warranty for the work (except to the
107 extent that warranties are provided), that licensees may convey the
108 work under this License, and how to view a copy of this License. If
109 the interface presents a list of user commands or options, such as a
110 menu, a prominent item in the list meets this criterion.

111 112 1. Source Code.

113
114 The "source code" for a work means the preferred form of the work
115 for making modifications to it. "Object code" means any non-source
116 form of a work.

117
118 A "Standard Interface" means an interface that either is an official
119 standard defined by a recognized standards body, or, in the case of
120 interfaces specified for a particular programming language, one that
121 is widely used among developers working in that language.

122
123 The "System Libraries" of an executable work include anything, other

124 than the work as a whole, that (a) is included in the normal form of
125 packaging a Major Component, but which is not part of that Major
126 Component, and (b) serves only to enable use of the work with that
127 Major Component, or to implement a Standard Interface for which an
128 implementation is available to the public in source code form. A
129 "Major Component", in this context, means a major essential component
130 (kernel, window system, and so on) of the specific operating system
131 (if any) on which the executable work runs, or a compiler used to
132 produce the work, or an object code interpreter used to run it.

133
134 The "Corresponding Source" for a work in object code form means all
135 the source code needed to generate, install, and (for an executable
136 work) run the object code and to modify the work, including scripts to
137 control those activities. However, it does not include the work's
138 System Libraries, or general-purpose tools or generally available free
139 programs which are used unmodified in performing those activities but
140 which are not part of the work. For example, Corresponding Source
141 includes interface definition files associated with source files for
142 the work, and the source code for shared libraries and dynamically
143 linked subprograms that the work is specifically designed to require,
144 such as by intimate data communication or control flow between those
145 subprograms and other parts of the work.

146
147 The Corresponding Source need not include anything that users
148 can regenerate automatically from other parts of the Corresponding
149 Source.

150
151 The Corresponding Source for a work in source code form is that
152 same work.

153 154 2. Basic Permissions.

155
156 All rights granted under this License are granted for the term of
157 copyright on the Program, and are irrevocable provided the stated
158 conditions are met. This License explicitly affirms your unlimited
159 permission to run the unmodified Program. The output from running a
160 covered work is covered by this License only if the output, given its
161 content, constitutes a covered work. This License acknowledges your
162 rights of fair use or other equivalent, as provided by copyright law.

163
164 You may make, run and propagate covered works that you do not
165 convey, without conditions so long as your license otherwise remains
166 in force. You may convey covered works to others for the sole purpose
167 of having them make modifications exclusively for you, or provide you
168 with facilities for running those works, provided that you comply with
169 the terms of this License in conveying all material for which you do

170 not control copyright. Those thus making or running the covered works
171 for you must do so exclusively on your behalf, under your direction
172 and control, on terms that prohibit them from making any copies of
173 your copyrighted material outside their relationship with you.

174
175 Conveying under any other circumstances is permitted solely under
176 the conditions stated below. Sublicensing is not allowed; section 10
177 makes it unnecessary.

178 179 3. Protecting Users' Legal Rights From Anti-Circumvention Law.

180
181 No covered work shall be deemed part of an effective technological
182 measure under any applicable law fulfilling obligations under article
183 11 of the WIPO copyright treaty adopted on 20 December 1996, or
184 similar laws prohibiting or restricting circumvention of such
185 measures.

186
187 When you convey a covered work, you waive any legal power to forbid
188 circumvention of technological measures to the extent such circumvention
189 is effected by exercising rights under this License with respect to
190 the covered work, and you disclaim any intention to limit operation or
191 modification of the work as a means of enforcing, against the work's
192 users, your or third parties' legal rights to forbid circumvention of
193 technological measures.

194 195 4. Conveying Verbatim Copies.

196
197 You may convey verbatim copies of the Program's source code as you
198 receive it, in any medium, provided that you conspicuously and
199 appropriately publish on each copy an appropriate copyright notice;
200 keep intact all notices stating that this License and any
201 non-permissive terms added in accord with section 7 apply to the code;
202 keep intact all notices of the absence of any warranty; and give all
203 recipients a copy of this License along with the Program.

204
205 You may charge any price or no price for each copy that you convey,
206 and you may offer support or warranty protection for a fee.

207 208 5. Conveying Modified Source Versions.

209
210 You may convey a work based on the Program, or the modifications to
211 produce it from the Program, in the form of source code under the
212 terms of section 4, provided that you also meet all of these conditions:

- 213
214 a) The work must carry prominent notices stating that you modified
215 it, and giving a relevant date.

216
217 b) The work must carry prominent notices stating that it is
218 released under this License and any conditions added under section
219 7. This requirement modifies the requirement in section 4 to
220 "keep intact all notices".
221
222 c) You must license the entire work, as a whole, under this
223 License to anyone who comes into possession of a copy. This
224 License will therefore apply, along with any applicable section 7
225 additional terms, to the whole of the work, and all its parts,
226 regardless of how they are packaged. This License gives no
227 permission to license the work in any other way, but it does not
228 invalidate such permission if you have separately received it.
229
230 d) If the work has interactive user interfaces, each must display
231 Appropriate Legal Notices; however, if the Program has interactive
232 interfaces that do not display Appropriate Legal Notices, your
233 work need not make them do so.
234
235 A compilation of a covered work with other separate and independent
236 works, which are not by their nature extensions of the covered work,
237 and which are not combined with it such as to form a larger program,
238 in or on a volume of a storage or distribution medium, is called an
239 "aggregate" if the compilation and its resulting copyright are not
240 used to limit the access or legal rights of the compilation's users
241 beyond what the individual works permit. Inclusion of a covered work
242 in an aggregate does not cause this License to apply to the other
243 parts of the aggregate.
244
245 6. Conveying Non-Source Forms.
246
247 You may convey a covered work in object code form under the terms
248 of sections 4 and 5, provided that you also convey the
249 machine-readable Corresponding Source under the terms of this License,
250 in one of these ways:
251
252 a) Convey the object code in, or embodied in, a physical product
253 (including a physical distribution medium), accompanied by the
254 Corresponding Source fixed on a durable physical medium
255 customarily used for software interchange.
256
257 b) Convey the object code in, or embodied in, a physical product
258 (including a physical distribution medium), accompanied by a
259 written offer, valid for at least three years and valid for as
260 long as you offer spare parts or customer support for that product
261 model, to give anyone who possesses the object code either (1) a

262 copy of the Corresponding Source for all the software in the
263 product that is covered by this License, on a durable physical
264 medium customarily used for software interchange, for a price no
265 more than your reasonable cost of physically performing this
266 conveying of source, or (2) access to copy the
267 Corresponding Source from a network server at no charge.

268
269 c) Convey individual copies of the object code with a copy of the
270 written offer to provide the Corresponding Source. This
271 alternative is allowed only occasionally and noncommercially, and
272 only if you received the object code with such an offer, in accord
273 with subsection 6b.

274
275 d) Convey the object code by offering access from a designated
276 place (gratis or for a charge), and offer equivalent access to the
277 Corresponding Source in the same way through the same place at no
278 further charge. You need not require recipients to copy the
279 Corresponding Source along with the object code. If the place to
280 copy the object code is a network server, the Corresponding Source
281 may be on a different server (operated by you or a third party)
282 that supports equivalent copying facilities, provided you maintain
283 clear directions next to the object code saying where to find the
284 Corresponding Source. Regardless of what server hosts the
285 Corresponding Source, you remain obligated to ensure that it is
286 available for as long as needed to satisfy these requirements.

287
288 e) Convey the object code using peer-to-peer transmission, provided
289 you inform other peers where the object code and Corresponding
290 Source of the work are being offered to the general public at no
291 charge under subsection 6d.

292
293 A separable portion of the object code, whose source code is excluded
294 from the Corresponding Source as a System Library, need not be
295 included in conveying the object code work.

296
297 A "User Product" is either (1) a "consumer product", which means any
298 tangible personal property which is normally used for personal, family,
299 or household purposes, or (2) anything designed or sold for incorporation
300 into a dwelling. In determining whether a product is a consumer product,
301 doubtful cases shall be resolved in favor of coverage. For a particular
302 product received by a particular user, "normally used" refers to a
303 typical or common use of that class of product, regardless of the status
304 of the particular user or of the way in which the particular user
305 actually uses, or expects or is expected to use, the product. A product
306 is a consumer product regardless of whether the product has substantial
307 commercial, industrial or non-consumer uses, unless such uses represent

308 the only significant mode of use of the product.

309
310 "Installation Information" for a User Product means any methods,
311 procedures, authorization keys, or other information required to install
312 and execute modified versions of a covered work in that User Product from
313 a modified version of its Corresponding Source. The information must
314 suffice to ensure that the continued functioning of the modified object
315 code is in no case prevented or interfered with solely because
316 modification has been made.

317
318 If you convey an object code work under this section in, or with, or
319 specifically for use in, a User Product, and the conveying occurs as
320 part of a transaction in which the right of possession and use of the
321 User Product is transferred to the recipient in perpetuity or for a
322 fixed term (regardless of how the transaction is characterized), the
323 Corresponding Source conveyed under this section must be accompanied
324 by the Installation Information. But this requirement does not apply
325 if neither you nor any third party retains the ability to install
326 modified object code on the User Product (for example, the work has
327 been installed in ROM).

328
329 The requirement to provide Installation Information does not include a
330 requirement to continue to provide support service, warranty, or updates
331 for a work that has been modified or installed by the recipient, or for
332 the User Product in which it has been modified or installed. Access to a
333 network may be denied when the modification itself materially and
334 adversely affects the operation of the network or violates the rules and
335 protocols for communication across the network.

336
337 Corresponding Source conveyed, and Installation Information provided,
338 in accord with this section must be in a format that is publicly
339 documented (and with an implementation available to the public in
340 source code form), and must require no special password or key for
341 unpacking, reading or copying.

342 343 7. Additional Terms.

344
345 "Additional permissions" are terms that supplement the terms of this
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348 be treated as though they were included in this License, to the extent
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370 Notices displayed by works containing it; or
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373 requiring that modified versions of such material be marked in
374 reasonable ways as different from the original version; or
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394 License, you may add to a covered work material governed by the terms
395 of that license document, provided that the further restriction does
396 not survive such relicensing or conveying.

397
398 If you add terms to a covered work in accord with this section, you
399 must place, in the relevant source files, a statement of the

400 additional terms that apply to those files, or a notice indicating
401 where to find the applicable terms.

402
403 Additional terms, permissive or non-permissive, may be stated in the
404 form of a separately written license, or stated as exceptions;
405 the above requirements apply either way.

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445

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451 for enforcing compliance by third parties with this License.

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518 you grant is automatically extended to all recipients of the covered
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562

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611
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622
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624
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628
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632 the "copyright" line and a pointer to where the full notice is found.

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651
652 If the program does terminal interaction, make it output a short
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654
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656 This program comes with ABSOLUTELY NO WARRANTY; for details type `show w'.
657 This is free software, and you are welcome to redistribute it
658 under certain conditions; type `show c' for details.

659
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23 by the Library, but which is not otherwise based on the Library.
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25 of using an interface provided by the Library.
26
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28 Application with the Library. The particular version of the Library
29 with which the Combined Work was made is also called the "Linked
30 Version".
31
32  The "Minimal Corresponding Source" for a Combined Work means the
33 Corresponding Source for the Combined Work, excluding any source code
34 for portions of the Combined Work that, considered in isolation, are
35 based on the Application, and not on the Linked Version.
36
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39 and utility programs needed for reproducing the Combined Work from the
40 Application, but excluding the System Libraries of the Combined Work.
41
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43
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54
55 a) under this License, provided that you make a good faith effort to
56 ensure that, in the event an Application does not supply the
57 function or data, the facility still operates, and performs
58 whatever part of its purpose remains meaningful, or

59
60 b) under the GNU GPL, with none of the additional permissions of
61 this License applicable to that copy.

63 3. Object Code Incorporating Material from Library Header Files.

64
65 The object code form of an Application may incorporate material from
66 a header file that is part of the Library. You may convey such object
67 code under terms of your choice, provided that, if the incorporated
68 material is not limited to numerical parameters, data structure
69 layouts and accessors, or small macros, inline functions and templates
70 (ten or fewer lines in length), you do both of the following:

71
72 a) Give prominent notice with each copy of the object code that the
73 Library is used in it and that the Library and its use are
74 covered by this License.

75
76 b) Accompany the object code with a copy of the GNU GPL and this license
77 document.

79 4. Combined Works.

80
81 You may convey a Combined Work under terms of your choice that,
82 taken together, effectively do not restrict modification of the
83 portions of the Library contained in the Combined Work and reverse
84 engineering for debugging such modifications, if you also do each of
85 the following:

86
87 a) Give prominent notice with each copy of the Combined Work that
88 the Library is used in it and that the Library and its use are
89 covered by this License.

90

91 b) Accompany the Combined Work with a copy of the GNU GPL and this license
92 document.

93

94 c) For a Combined Work that displays copyright notices during
95 execution, include the copyright notice for the Library among
96 these notices, as well as a reference directing the user to the
97 copies of the GNU GPL and this license document.

98

99 d) Do one of the following:

100

101 0) Convey the Minimal Corresponding Source under the terms of this
102 License, and the Corresponding Application Code in a form
103 suitable for, and under terms that permit, the user to
104 recombine or relink the Application with a modified version of
105 the Linked Version to produce a modified Combined Work, in the
106 manner specified by section 6 of the GNU GPL for conveying
107 Corresponding Source.

108

109 1) Use a suitable shared library mechanism for linking with the
110 Library. A suitable mechanism is one that (a) uses at run time
111 a copy of the Library already present on the user's computer
112 system, and (b) will operate properly with a modified version
113 of the Library that is interface-compatible with the Linked
114 Version.

115

116 e) Provide Installation Information, but only if you would otherwise
117 be required to provide such information under section 6 of the
118 GNU GPL, and only to the extent that such information is
119 necessary to install and execute a modified version of the
120 Combined Work produced by recombining or relinking the
121 Application with a modified version of the Linked Version. (If
122 you use option 4d0, the Installation Information must accompany
123 the Minimal Corresponding Source and Corresponding Application
124 Code. If you use option 4d1, you must provide the Installation
125 Information in the manner specified by section 6 of the GNU GPL
126 for conveying Corresponding Source.)

127

128 5. Combined Libraries.

129

130 You may place library facilities that are a work based on the
131 Library side by side in a single library together with other library
132 facilities that are not Applications and are not covered by this
133 License, and convey such a combined library under terms of your
134 choice, if you do both of the following:

135

136 a) Accompany the combined library with a copy of the same work based

137 on the Library, uncombined with any other library facilities,
 138 conveyed under the terms of this License.

139

140 b) Give prominent notice with the combined library that part of it
 141 is a work based on the Library, and explaining where to find the
 142 accompanying uncombined form of the same work.

143

144 6. Revised Versions of the GNU Lesser General Public License.

145

146 The Free Software Foundation may publish revised and/or new versions
 147 of the GNU Lesser General Public License from time to time. Such new
 148 versions will be similar in spirit to the present version, but may
 149 differ in detail to address new problems or concerns.

150

151 Each version is given a distinguishing version number. If the
 152 Library as you received it specifies that a certain numbered version
 153 of the GNU Lesser General Public License "or any later version"
 154 applies to it, you have the option of following the terms and
 155 conditions either of that published version or of any later version
 156 published by the Free Software Foundation. If the Library as you
 157 received it does not specify a version number of the GNU Lesser
 158 General Public License, you may choose any version of the GNU Lesser
 159 General Public License ever published by the Free Software Foundation.

160

161 If the Library as you received it specifies that a proxy can decide
 162 whether future versions of the GNU Lesser General Public License shall
 163 apply, that proxy's public statement of acceptance of any version is
 164 permanent authorization for you to choose that version for the
 165 Library.

3.55 git_helpers/git_checks.lib.sh

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.
3 #
4 # DevOrSysAdminScripts is free software:
5 # you can redistribute it and/or modify it
6 # under the terms of the GNU General Public License
7 # as published by the Free Software Foundation,
8 # either version 3 of the License,
9 # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of

```



```

15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:57+02:00: This file was renamed from
25 # "git_checks.lib.sh"
26 # to
27 # "git_checks.lib.sh".
28 # 2024-05-31T01:57+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32 # 2024-05-31T01:49+02:00: This file was renamed from
33 # "git_checks.lib.sh"
34 # to
35 # "git_checks.lib.sh".
36 # 2024-05-31T01:49+02:00: This file was moved from
37 # "."
38 # to
39 # "./git_helpers".
40 # This file was renamed from "git_checks.sh" to "git_checks.lib.sh".
41
42 check_files() {
43     # Checks used on Django projects.
44     # Use `git commit --no-verify` after check.
45     declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
46     declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
47     local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
48     LFBFL_send_body_2+=" dès la création du modèle"
49     readonly LFBFL_send_body_2
50     declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
51     declare -r LFBFL_send_body_3=\
52 "Les nouveaux preloadings sont interdits -> #Prefetch()"
53     declare -i LFBFL_i_error=0
54     git diff --cached --name-only\
55     > /tmp/DOSAS_django_git_check1.temp
56     if grep models /tmp/DOSAS_django_git_check1.temp;
57     then
58         notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
59         notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
60         LFBFL_i_error=1

```

```

61     fi
62     if grep serializer /tmp/DOSAS_django_git_check1.temp;
63     then
64         notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65         LFBFL_i_error=1
66     fi
67     rm /tmp/DOSAS_django_git_check1.temp
68     # shellcheck disable=SC2248
69     return ${LFBFL_i_error}
70 }
71
72
73 check_no_abusive_trailing_comma(){
74     # Useful for Python code, although it regularly yields
75     # false positives in Python code
76     # (but it is easy to distinguish by hand).
77     # Use `git commit --no-verify` after check.
78     declare -r LFBFL_send_summary_1="ATTENTION"
79     local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80     LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81     readonly LFBFL_send_body_1
82     declare -i LFBFL_i_error=0
83     git diff --cached\
84     > /tmp/DOSAS_django_git_check2.temp
85     if grep --perl-regexp ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86     then
87         notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88         LFBFL_i_error=1
89     fi
90     rm /tmp/DOSAS_django_git_check2.temp
91     # shellcheck disable=SC2248
92     return ${LFBFL_i_error}
93 }
94
95
96 check_black_code_formatting(){
97     # attention ça ne marche que sur les fichiers "stagés" avec git add
98     declare -r LFBFL_s_files_paths=$(
99         git diff --cached --name-only\
100         | grep '\.py'
101     )
102     printf "%s\n" "${LFBFL_s_files_paths}"
103     declare -a LFBFL_arr_files_paths
104     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
105     readonly LFBFL_arr_files_paths
106     declare -i LFBFL_i_error=0

```

```

107 local LFBFL_file_path
108 for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
109     printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
110     if ! black --check --diff -- "${LFBFL_file_path}";
111     then
112         LFBFL_i_error=1
113     fi
114 done
115 # shellcheck disable=SC2248
116 return ${LFBFL_i_error}
117 }

```

3.56 git_helpers/post-commit

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it under the terms
6  # of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed in the hope
12 # that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2024 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:49+02:00: This file was renamed from
25 # "post-commit"
26 # to
27 # "post-commit".
28 # 2024-05-31T01:49+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32

```

```

33 LFBFL_some_directory="../DevOrSysAdminScripts"
34 declare -i LFBFL_i_pushd_result3
35 pushd "${LFBFL_some_directory}" || {
36     LFBFL_i_pushd_result3=$?
37     printf "post-commit pushd %s no such directory\n"\
38         "${LFBFL_some_directory}"
39     # shellcheck disable=SC2248
40     exit ${LFBFL_i_pushd_result3}
41 }
42 # shellcheck source=post_commit_renaming_comment.lib.sh
43 source "../git_helpers/post_commit_renaming_comment.lib.sh"
44 # shellcheck source=lines_filters.lib.sh
45 source "../build_and_checks_dependencies/lines_filters.lib.sh"
46 declare -i LFBFL_i_popd_result3
47 popd || {
48     LFBFL_i_popd_result3=$?
49     printf "post-commit no popd\n"
50     # shellcheck disable=SC2248
51     exit ${LFBFL_i_popd_result3}
52 }
53
54 LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
55 LFBFL_max_comment_line_length=""
56 if [[ -f "${LFBFL_data_file_name}" ]]; then
57     LFBFL_max_line_length=""
58     grep_variable "${LFBFL_data_file_name}" "max_line_length"\
59         --result-variable-prefix="LFBFL_"
60     LFBFL_max_comment_line_length="--max-comment-line-length="
61     LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
62 fi
63
64 commit_a_file_renamed_comment --verbose --log-directory-change\
65     "${LFBFL_max_comment_line_length}"
66 exit 0

```

3.57 git_helpers/post_commit_renaming_comment.lib.sh

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.
3 #
4 # DevOrSysAdminScripts is free software:
5 # you can redistribute it and/or modify it
6 # under the terms of the GNU General Public License
7 # as published by the Free Software Foundation,
8 # either version 3 of the License,
9 # or (at your option) any later version.

```

```

10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:49+02:00: This file was renamed from
25 # "post_commit_renaming_comment.lib.sh"
26 # to
27 # "post_commit_renaming_comment.lib.sh".
28 # 2024-05-31T01:49+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32 # This file was renamed from "post_commit_renaming_comment"
33 # to "post_commit_renaming_comment.lib.sh".
34 # (Before first commit it seems. But I had the old name in listing.)
35
36 LFBFL_this_file_path=$(realpath -- "${BASH_SOURCE[0]}")
37 LFBFL_this_file_directory=$(dirname -- "${LFBFL_this_file_path}")
38 LFBFL_some_directory="${LFBFL_this_file_directory}/../"
39 declare -i LFBFL_i_pushd_result2
40 pushd -- "${LFBFL_some_directory}" || {
41     LFBFL_i_pushd_result2=$?
42     printf\
43         "post_commit_renaming_comment.lib.sh pushd %s no such directory\n"\
44         "${LFBFL_some_directory}"
45     # shellcheck disable=SC2248
46     exit ${LFBFL_i_pushd_result2}
47 }
48 # shellcheck source=common_options.lib.sh
49 source "./build_and_checks_dependencies/common_options.lib.sh"
50 # shellcheck source=strings_functions.lib.sh
51 source "./build_and_checks_dependencies/strings_functions.lib.sh"
52 declare -i LFBFL_i_popd_result2
53 popd || {
54     LFBFL_i_popd_result2=$?
55     printf "post_commit_renaming_comment.lib.sh no popd\n"

```

```

56     # shellcheck disable=SC2248
57     exit ${LFBFL_i_popd_result2}
58 }
59
60 # Never forget that the worse security is the illusion of security
61 # (Kevin Mitnick). So, do not expect to have an accurate trackability
62 # of code history just because you use this post-commit hook or
63 # something else more evolved. This is useful but not a guarantee.
64 # For this reason, this function will only handle when all files
65 # impacted in the preceding commit are only renamed.
66 # It assumes that 'diff' is the current default difference utility
67 # parametered with 'git'. This point could be enhanced.
68 commit_a_file_renamed_comment() {
69     # Options:
70     #   --verbose
71     #   --log-directory-change
72     #   --max-comment-line-length
73     declare -i LFBFL_i_verbose=0
74     get_verbose_option "$@"
75     declare -i LFBFL_i_log_directory_change
76     get_some_flag LFBFL_i_log_directory_change --log-directory-change 1 "$@"
77     declare -i LFBFL_i_max_comment_line_length
78     get_some_option LFBFL_i_max_comment_line_length\
79         --max-comment-line-length 70 70 1 "$@"
80     if [[ LFBFL_i_verbose -eq 1 ]]; then
81         LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
82         LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
83         # shellcheck disable=SC2059,SC2248
84         printf "${LFBFL_s_format}"\
85             "${LFBFL_i_log_directory_change}"\
86             "${LFBFL_i_max_comment_line_length}"
87     fi
88     if [[ LFBFL_i_max_comment_line_length -lt 10 ]]; then
89         declare -r LFBFL_send_summary="Max comment line length below 10."
90         notify-send "${LFBFL_send_summary}" "${LFBFL_send_summary}"
91         if [[ LFBFL_i_verbose -eq 1 ]]; then
92             printf "Max comment line length below 10, aborting.\n"
93         fi
94         return
95     fi
96     local LFBFL_s_format
97     #-----
98
99     enhanced_set_shell_option pipefail\
100     && trap 'enhanced_unset_shell_option pipefail' RETURN
101

```

```

102 declare -r LFBFL_renaming_lines=$(
103     git log --max-count=1 --patch\
104     | grep --after-context=1 '^diff --git'
105 )
106 if [[ LFBFL_i_verbose -eq 1 ]]; then
107     printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
108 fi
109
110 declare -a LFBFL_renaming_lines_array
111 mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
112 readonly LFBFL_renaming_lines_array
113
114 declare -r LFBFL_timestamp=$(date --iso-8601="minute")
115 if [[ LFBFL_i_verbose -eq 1 ]]; then
116     printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
117 fi
118 local LFBFL_split_score_command1
119 declare -i LFBFL_i_split_score_command_properties1
120 # shellcheck disable=SC2248
121 get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
122 LFBFL_split_score_command1="${get_split_score_result}"
123 LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
124 readonly LFBFL_split_score_command1
125 readonly LFBFL_i_split_score_command_properties1
126 local LFBFL_split_score_command2
127 declare -i LFBFL_i_split_score_command_properties2
128 # shellcheck disable=SC2248
129 get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
130 LFBFL_split_score_command2="${get_split_score_result}"
131 LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
132 readonly LFBFL_split_score_command2
133 readonly LFBFL_i_split_score_command_properties2
134 declare -i LFBFL_i_renaming_happened=0
135 declare -i LFBFL_i
136 declare -i LFBFL_j
137 declare -r LFBFL_special_file_name="files_names_listing.txt"
138 local LFBFL_diff_line
139 local LFBFL_similarity_line
140 local LFBFL_old_file_path
141 local LFBFL_old_file_name
142 local LFBFL_old_file_directory
143 local LFBFL_new_file_path
144 local LFBFL_new_file_name
145 local LFBFL_new_file_directory
146 local LFBFL_useful_file_name
147 local LFBFL_extension

```

```

148 local LFBFL_comment_prefix
149 local LFBFL_comment_prefix2
150 local LFBFL_old_file_name2
151 local LFBFL_old_file_directory2
152 local LFBFL_new_file_name2
153 local LFBFL_new_file_directory2
154 local LFBFL_new_comment
155 declare -i LFBFL_i_copyright_line_number
156 declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
157 declare -i LFBFL_i_line_count
158 declare -i LFBFL_i_lines_after
159 local LFBFL_temp_file_path
160 for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
161     LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
162     LFBFL_i=$((LFBFL_i + 1))
163     LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
164     LFBFL_i=$((LFBFL_i + 1))
165     # We need to increment 3 times since grep adds "--" lines between.
166     LFBFL_i=$((LFBFL_i + 1))
167     if [[ LFBFL_i_verbose -eq 1 ]]; then
168         printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n" \
169             "${LFBFL_diff_line}" \
170             "${LFBFL_similarity_line}"
171     fi
172     if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
173         continue
174     fi
175
176     # Extract old file name from diff line. -----
177     LFBFL_old_file_path=$(
178         printf "%s" "${LFBFL_diff_line}" \
179         | cut --delimiter=' ' --fields=3
180     )
181     # It starts by "a/".
182     LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
183     LFBFL_old_file_name=$(basename -- "${LFBFL_old_file_path}")
184     LFBFL_old_file_directory=$(dirname -- "${LFBFL_old_file_path}")
185     # Extract new file name from diff line. -----
186     LFBFL_new_file_path=$(
187         printf "%s" "${LFBFL_diff_line}" \
188         | cut --delimiter=' ' --fields=4
189     )
190     # It starts by "b/".
191     LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
192     LFBFL_new_file_name=$(basename -- "${LFBFL_new_file_path}")
193     LFBFL_new_file_directory=$(dirname -- "${LFBFL_new_file_path}")

```



```

194 if [[ LFBFL_i_verbose -eq 1 ]]; then
195     LFBFL_s_format="LFBFL_old_file_path: %s\n"
196     LFBFL_s_format+="LFBFL_old_file_name: %s\n"
197     LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
198     LFBFL_s_format+="LFBFL_new_file_path: %s\n"
199     LFBFL_s_format+="LFBFL_new_file_name: %s\n"
200     LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
201     # shellcheck disable=SC2059
202     printf "${LFBFL_s_format}"\
203         "${LFBFL_old_file_path}"\
204         "${LFBFL_old_file_name}"\
205         "${LFBFL_old_file_directory}"\
206         "${LFBFL_new_file_path}"\
207         "${LFBFL_new_file_name}"\
208         "${LFBFL_new_file_directory}"
209 fi
210 # Extract "LFBFL_extension" with smart guessing for .tpl -----
211 # and set prefix for line with ©Copyright. -----
212 LFBFL_useful_file_name="${LFBFL_new_file_name}"
213 LFBFL_extension="${LFBFL_useful_file_name##*.}"
214 if [[ LFBFL_i_verbose -eq 1 ]]; then
215     printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
216         "${LFBFL_useful_file_name}"\
217         "${LFBFL_extension}"
218 fi
219 for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
220     if [[ "${LFBFL_extension}" != "tpl" ]]; then
221         break
222     fi
223     LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
224     LFBFL_extension="${LFBFL_useful_file_name##*.}"
225     if [[ LFBFL_i_verbose -eq 1 ]]; then
226         printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
227             "${LFBFL_useful_file_name}"\
228             "${LFBFL_extension}"
229     fi
230 done
231 LFBFL_comment_prefix=""
232 if [[ "${LFBFL_extension}" == "sh" ]]; then
233     LFBFL_comment_prefix="#"
234 fi
235 if [[ "${LFBFL_extension}" == "sql" ]]; then
236     LFBFL_comment_prefix="-- "
237 fi
238 if [[ "${LFBFL_extension}" == "tex" ]]; then
239     LFBFL_comment_prefix="% "

```

```

240     fi
241     if [[ "${LFBFL_useful_file_name}" == "${LFBFL_special_file_name}" ]];
242     then
243         LFBFL_comment_prefix="// "
244     fi
245     if [[ LFBFL_i_verbose -eq 1 ]]; then
246         printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
247     fi
248     #-----
249     # Split if necessary old_file_name, new_file_name, etc. to have
250     # comment lines below max_comment_line_length.
251     LFBFL_comment_prefix2="${LFBFL_comment_prefix}"' "'
252     # -----
253     LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
254     LFBFL_old_file_name2+="${LFBFL_old_file_name}"
255     # shellcheck disable=SC2248
256     repeated_split_last_line "${LFBFL_old_file_name2}"\
257         "${LFBFL_comment_prefix2}"\
258         "${LFBFL_i_max_comment_line_length}"\
259         ' "'\
260         "${LFBFL_split_score_command1}"\
261         "${LFBFL_i_split_score_command_properties1}"
262     LFBFL_old_file_name2="${repeated_split_last_line_result}"
263     # -----
264     LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
265     LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
266     # shellcheck disable=SC2248
267     repeated_split_last_line "${LFBFL_old_file_directory2}"\
268         "${LFBFL_comment_prefix2}"\
269         "${LFBFL_i_max_comment_line_length}"\
270         ' "'\
271         "${LFBFL_split_score_command2}"\
272         "${LFBFL_i_split_score_command_properties2}"
273     LFBFL_old_file_directory2="${repeated_split_last_line_result}"
274     # -----
275     LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
276     LFBFL_new_file_name2+="${LFBFL_new_file_name}"
277     # shellcheck disable=SC2248
278     repeated_split_last_line "${LFBFL_new_file_name2}"\
279         "${LFBFL_comment_prefix2}"\
280         "${LFBFL_i_max_comment_line_length}"\
281         ' "'\
282         "${LFBFL_split_score_command1}"\
283         "${LFBFL_i_split_score_command_properties1}"
284     LFBFL_new_file_name2="${repeated_split_last_line_result}"
285     # -----

```

```

286 LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
287 LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
288 # shellcheck disable=SC2248
289 repeated_split_last_line "${LFBFL_new_file_directory2}"\
290     "${LFBFL_comment_prefix2}"\
291     "${LFBFL_i_max_comment_line_length}"\
292     '\ '\
293     "${LFBFL_split_score_command2}"\
294     "${LFBFL_i_split_score_command_properties2}"
295 LFBFL_new_file_directory2="${repeated_split_last_line_result}"
296 # Create new comments strings. -----
297 LFBFL_new_comment=""
298 if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
299     LFBFL_new_comment="${LFBFL_comment_prefix}"
300     LFBFL_new_comment+="${LFBFL_timestamp}"
301     LFBFL_new_comment+=": This file was renamed from"
302     LFBFL_new_comment+=$'\n'
303     LFBFL_new_comment+="${LFBFL_old_file_name2}"' "'
304     LFBFL_new_comment+=$'\n'
305     LFBFL_new_comment+="${LFBFL_comment_prefix}to"
306     LFBFL_new_comment+=$'\n'
307     LFBFL_new_comment+="${LFBFL_new_file_name2}"' ". '
308 fi
309 if [[
310     LFBFL_i_log_directory_change -eq 1
311     && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
312 ]];
313 then
314     if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
315         LFBFL_new_comment+=$'\n'
316     fi
317     LFBFL_new_comment+="${LFBFL_comment_prefix}"
318     LFBFL_new_comment+="${LFBFL_timestamp}"
319     LFBFL_new_comment+=": This file was moved from"
320     LFBFL_new_comment+=$'\n'
321     LFBFL_new_comment+="${LFBFL_old_file_directory2}"' "'
322     LFBFL_new_comment+=$'\n'
323     LFBFL_new_comment+="${LFBFL_comment_prefix}to"
324     LFBFL_new_comment+=$'\n'
325     LFBFL_new_comment+="${LFBFL_new_file_directory2}"' ". '
326 fi
327 if [[ LFBFL_i_verbose -eq 1 ]]; then
328     printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
329 fi
330 # Find line with ©Copyright. -----
331 LFBFL_i_copyright_line_number=$(

```

```

332     grep --line-number -- '@Copyright' "${LFBFL_new_file_path}"\
333     | head --lines=1\
334     | cut --delimiter=':' --fields=1
335 )
336 if [[ LFBFL_i_verbose -eq 1 ]]; then
337     printf "LFBFL_i_copyright_line_number: %s\n"\
338     "${LFBFL_i_copyright_line_number}"
339 fi
340 if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
341     notify-send "${LFBFL_send_summary_2}"\
342     "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
343     if [[ LFBFL_i_verbose -eq 1 ]]; then
344         printf "No copyright line. Skipping file.\n"
345     fi
346     continue
347 fi
348 # Get total number of lines. -----
349 LFBFL_i_line_count=$(
350     wc --lines\
351     < "${LFBFL_new_file_path}"
352 )
353 if [[ LFBFL_i_verbose -eq 1 ]]; then
354     printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
355 fi
356 # Compute number of lines after. -----
357 LFBFL_i_lines_after=$((
358     LFBFL_i_line_count - LFBFL_i_copyright_line_number
359 ))
360 if [[ LFBFL_i_verbose -eq 1 ]]; then
361     printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
362 fi
363 # Update file. -----
364 LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
365 {
366     head --lines="${LFBFL_i_copyright_line_number}"\
367     -- "${LFBFL_new_file_path}"
368     printf "%s\n" "${LFBFL_new_comment}"
369     tail --lines="${LFBFL_i_lines_after}" -- "${LFBFL_new_file_path}"
370 } >> "${LFBFL_temp_file_path}"
371 mv -- "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
372 # Add file to stage. -----
373 git add -- "${LFBFL_new_file_path}"
374 LFBFL_i_renaming_happened=1
375 done
376 readonly LFBFL_i_renaming_happened
377 if [[ LFBFL_i_renaming_happened -eq 1 ]]; then

```

```

378     # Commit -----
379     local LFBFL_message="commit_a_file_renamed_comment()"
380     LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
381     readonly LFBFL_message
382     printf "Commit message:\n%s\n" "${LFBFL_message}"
383     git commit --no-verify --message="${LFBFL_message}"
384 fi
385 }

```

3.58 git_helpers/pre-commit

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it under the terms
6  # of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed in the hope
12 # that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
24 # 2024-05-31T01:57+02:00: This file was renamed from
25 # "pre-commit"
26 # to
27 # "pre-commit".
28 # 2024-05-31T01:57+02:00: This file was moved from
29 # "."
30 # to
31 # "./git_helpers".
32
33 LFBFL_some_directory="../DevOrSysAdminScripts"
34 declare -i LFBFL_i_pushd_result3
35 pushd "${LFBFL_some_directory}" || {

```

```

36     LFBFL_i_pushd_result3=$?
37     printf "pre-commit pushd %s no such directory\n"\
38         "${LFBFL_some_directory}"
39     # shellcheck disable=SC2248
40     exit ${LFBFL_i_pushd_result3}
41 }
42 # shellcheck source=git_checks.lib.sh
43 source "./git_helpers/git_checks.lib.sh"
44 declare -i LFBFL_i_popd_result3
45 popd || {
46     LFBFL_i_popd_result3=$?
47     printf "pre-commit no popd\n"
48     # shellcheck disable=SC2248
49     exit ${LFBFL_i_popd_result3}
50 }
51
52 declare -gi LFBFL_i_git_hook_error=0
53 if ! check_files; then
54     LFBFL_i_git_hook_error=1
55 fi
56 if ! check_no_abusive_trailing_comma; then
57     LFBFL_i_git_hook_error=1
58 fi
59 if ! check_black_code_formatting; then
60     LFBFL_i_git_hook_error=1
61 fi
62 readonly LFBFL_i_git_hook_error
63 # shellcheck disable=SC2248
64 exit ${LFBFL_i_git_hook_error}

```

3.59 IMAP_transfert.exec.py

```

1  """
2  This file is part of DevOrSysAdminScripts.
3
4  DevOrSysAdminScripts is free software:
5  you can redistribute it and/or modify it
6  under the terms of the GNU Lesser General Public License
7  as published by the Free Software Foundation,
8  either version 3 of the License,
9  or (at your option) any later version.
10
11  DevOrSysAdminScripts is distributed
12  in the hope that it will be useful,
13  but WITHOUT ANY WARRANTY;
14  without even the implied warranty of

```

```

15  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16  See the GNU Lesser General Public License for more details.
17
18  You should have received a copy of
19  the GNU Lesser General Public License
20  along with DevOrSysAdminScripts.
21  If not, see <https://www.gnu.org/licenses/>.
22
23  ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24  2026-05-12T12:06+02:00: This file was renamed from
25  "IMAP_transfert.py"
26  to
27  "IMAP_transfert.exec.py".
28
29  Pour transférer d'un compte IMAP à un autre
30  Penser à autoriser les apps moins sûres sur Gmail si nécessaire
31  """
32
33  import argparse
34  import email
35  import imaplib
36
37
38  class IMAPTransfertBadArgumentError(Exception):
39      """
40      A class to avoid a linter warning,
41      that needs an Error suffix to avoid another linter warning,
42      that needs a docstring to avoid another linter warning.
43      If you don't see its intent, try badminton.
44      """
45
46
47  class SurpriseError(Exception):
48      """
49      A class to avoid a linter warning,
50      that needs an Error suffix to avoid another linter warning,
51      that needs a docstring to avoid another linter warning.
52      If you don't see its intent after seeing its use, try volleyball.
53      But "Cheer up!", now you don't need "pass" XD.
54      """
55
56
57  def IMAP_transfert(
58      from_login: str,
59      from_password: str,
60      from_host: str,

```

```

61     from_port: int,
62     to_login: str,
63     to_password: str,
64     to_host: str,
65     to_port: int,
66     *,
67     to_new_folder: str = "",
68     verbose: bool = False,
69 ) -> None:
70     """
71     Transfert all emails from some IMAP account
72     to another IMAP account.
73     """
74     empty = ""
75     if from_login == "":
76         raise IMAPTransfertBadArgumentError("From login is empty.")
77     # if from_password == "":
78     if from_password == empty:
79         # https://github.com/PyCQA/bandit/issues/714
80         # Dumb
81         raise IMAPTransfertBadArgumentError("From password is empty.")
82     if from_host == "":
83         raise IMAPTransfertBadArgumentError("From host is empty.")
84     if from_port <= 0:
85         raise IMAPTransfertBadArgumentError(
86             "From port is not positive.",
87         )
88     if to_login == "":
89         raise IMAPTransfertBadArgumentError("To login is empty.")
90     if to_password == empty:
91         # And Dumber The infinite collection XD
92         raise IMAPTransfertBadArgumentError("To password is empty.")
93     if to_host == "":
94         raise IMAPTransfertBadArgumentError("To host is empty.")
95     if to_port <= 0:
96         raise IMAPTransfertBadArgumentError(
97             "To port is not positive.",
98         )
99     # pylint: disable=pointless-string-statement
100     """
101     Maybe one day Python will ship something like that in its standard
102     library:
103     https://www.php.net/manual/fr/function.filter-var.php
104     But now urlparse doesn't do any validation.
105     You may want to check that the hosts or other fields are not
106     "anything".

```



```

107     """
108
109     # pylint: disable=unreachable
110     raise SurpriseError(
111         "I haven't tested this version of my script... Ooooooh!\n"
112         "But I linted it without this exception ;). Aaaaaaaah!\n"
113         "XD XD XD\n"
114         "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
115     )
116     # """
117     # I haven't tested any version with an empty to_new_folder...
118
119     if verbose:
120         print('Connecting to "from account".')
121     from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
122     from_account.login(from_login, from_password)
123
124     if verbose:
125         print('Connecting to "to account".')
126     to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
127     to_account.login(to_login, to_password)
128
129     if to_new_folder != "":
130         if verbose:
131             print('Creating new folder in "to account".')
132             to_account.create(to_new_folder)
133
134     if verbose:
135         print('Fetching all emails ids in "from account".')
136     from_account.select()
137     result, ids = from_account.uid("search", "all")
138     if verbose:
139         print(f'from_account.uid("search") result:{result} ids:{ids}')
140     ids = ids[0].decode().split()
141     if verbose:
142         print('Fetching and transferring each email in "from account".')
143     for my_id in ids:
144         result, raw_message = from_account.uid(
145             "fetch",
146             my_id,
147             "(RFC822)",
148         )
149         if verbose:
150             print(f'from_account.uid("fetch", {my_id}) result:{result}')
151         message = raw_message[0][1]
152         my_email = email.message_from_string(message.decode("utf-8"))

```

```

153         flags = imaplib.ParseFlags(raw_message[0][0])
154         # flags = " ".join(flags) mypy is unhappy but IMAP flags are
155         # ASCII strings, we give mypy what it wants.
156         flags_as_strs = [flag.decode("ascii") for flag in flags]
157         flags_str = " ".join(flags_as_strs)
158         # date = imaplib.Internaldate2tuple(raw_message[0][0])
159         date = ''' + my_email["date"] + '''
160         if verbose:
161             print(
162                 f'from_account.uid("fetch", {my_id})'
163                 f" flags:{flags} date:{date}",
164             )
165         to_account.append(to_new_folder, flags_str, date, message)
166
167
168 def main() -> None:
169     """
170     Parse command-line arguments and
171     transfert all emails from some IMAP account
172     to another IMAP account.
173     """
174     parser = argparse.ArgumentParser(
175         description=(
176             "Transfert all emails from some IMAP account"
177             " to another IMAP account"
178         ),
179     )
180     parser.add_argument(
181         "from_login",
182         type=str,
183         help="Email account login of user one.",
184     )
185     parser.add_argument(
186         "from_password",
187         type=str,
188         help="Email account password of user one.",
189     )
190     parser.add_argument(
191         "to_login",
192         type=str,
193         help="Email account login of user two.",
194     )
195     parser.add_argument(
196         "to_password",
197         type=str,
198         help="Email account password of user two.",

```

```

199     )
200     parser.add_argument (
201         "--to_new_folder",
202         type=str,
203         default="",
204         help="Transfert emails into a new folder of this name.",
205     )
206     parser.add_argument (
207         "--from_host",
208         type=str,
209         default="imap.gmail.com",
210         help="Host of first email account",
211     )
212     parser.add_argument (
213         "--from_port",
214         type=int,
215         default=993,
216         help="Port on host of first email account",
217     )
218     parser.add_argument (
219         "--to_host",
220         type=str,
221         default="imap.gmail.com",
222         help="Host of second email account",
223     )
224     parser.add_argument (
225         "--to_port",
226         type=int,
227         default=993,
228         help="Port on host of second email account",
229     )
230     parser.add_argument (
231         "-v",
232         "--verbose",
233         action="store_true",
234         help="Print details of operations",
235     )
236
237     args = parser.parse_args()
238
239     if args.verbose:
240         print (
241             "-----\n"
242             "IMAP transfert:\n"
243             "-----\n"
244             "From account:\n"

```

```

245         f" - Login: {args.from_login}\n"
246         f" - Password: {args.from_password}\n"
247         f" - Host: {args.from_host}\n"
248         f" - Port: {args.from_port}\n"
249         "To account:\n"
250         f" - Login: {args.to_login}\n"
251         f" - Password: {args.to_password}\n"
252         f" - Host: {args.to_host}\n"
253         f" - Port: {args.to_port}\n"
254         "-----\n",
255     )
256     if args.to_new_folder is not None:
257         print(
258             f"Emails will go in folder {args.to_new_folder}"
259             " after it is created.",
260         )
261     try:
262         IMAP_transfert(
263             args.from_login,
264             args.from_password,
265             args.from_host,
266             args.from_port,
267             args.to_login,
268             args.to_password,
269             args.to_host,
270             args.to_port,
271             to_new_folder=args.to_new_folder,
272             verbose=args.verbose,
273         )
274     except (IMAPTransfertBadArgumentError, SurpriseError) as e:
275         print(e)
276
277
278 if __name__ == "__main__":
279     main()

```

3.60 LiensUtiles.txt

```

1  https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
2  -configurez-le-service-de-migration-de-donn%C3%A9es
3  -----
4  https://stackoverflow.com/a/78413334/5796086
5  https://stackoverflow.com/questions/21612980/\
6  why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
7  https://web.archive.org/web/20240501105801/https://stackoverflow.com\
8  /questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/

```

```

9 78413334#78413334
10 -----
11 https://news.ycombinator.com/item?id=22214690
12 https://github.com/koalaman/shellcheck/issues/2005
13 https://opensource.stackexchange.com/questions/6271/\
14 calling-gpld-programs-from-a-shell-script
15 -----
16 https://pubs.opengroup.org/onlinepubs/9699919799/
17 When looking 'man sed' GNU sed says to use POSIX -E instead of
18 -r or --regexp-extended for portability.
19 But the link above says that still now only -e, -f, -n are POSIX options
20 for sed.
21 For this reason we replaced all short option names by long option names
22 when possible, even for sed.

```

3.61 miscellaneous/funny.sh

```

1 #!/usr/bin/env bash
2 # This file is part of DevOrSysAdminScripts.
3 #
4 # DevOrSysAdminScripts is free software:
5 # you can redistribute it and/or modify it
6 # under the terms of the GNU General Public License
7 # as published by the Free Software Foundation,
8 # either version 3 of the License,
9 # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24
25 unprotected_funny(){
26     # Works in a script but not in shell if job control is active.
27     shopt -s lastpipe
28     declare -i LFBFL_i_result=0
29     local LFBFL_line

```

```

30 printf "hahaha\n" | while read -r LFBFL_line; do
31     if [[ "${LFBFL_line}" == "hahaha" ]]; then
32         LFBFL_i_result=1
33     fi
34 done
35 [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
36 }
37
38 protected_funny() {
39     # Always works regarding the echoes.
40     printf "hahaha\n" | while read -r LFBFL_line; do
41         unprotected_funny
42     done
43 }

```

3.62 miscellaneous/spamty_data.txt

```

1
2
3
4 a
5 a
6 empty.txt:
7 one_space.txt:
8 all_space.txt:
9 normal.txt:a
10 normal_again.txt: a

```

3.63 miscellaneous/string_escaping.html

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="utf-8">
5     <title>Truc</title>
6 </head>
7 <body>
8 <header id="title-block-header">
9 <h1 class="title">Truc</h1>
10 </header>
11 &lt;abc> &amp; &lt;de>
12 <p title="<&>"
13   abcd`\' machin">
14 This is some text
15 </p>
16

```

```

17 <pre>
18 "&lt;&amp;>'&quot;;
19 abcd\\ machin"
20 This is some text
21 </pre>
22
23 <script>
24 let a = 'abcd & < > \\ \n  \"\'';
25 alert(a);
26 let b = "abcd & < > \\ \n  \"\'";
27 alert(b);
28 </script>
29
30 </body>
31 </html>

```

3.64 miscellaneous/string_escaping.xhtml

```

1 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
2   "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
3 <html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
4 <head>
5 <meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
6   <title>Truc</title>
7 </head>
8 <body>
9 <h1 class="title">Truc</h1>
10 &lt;abc> &amp; &lt;de>
11
12 <p title="&lt;&amp;>'&#10;
13   abcd\\ machin">
14   This is some text
15 </p>
16
17 <pre>
18 "&lt;&amp;>'&quot;;
19 abcd\\ machin"
20 This is some text
21 </pre>
22
23 <script type="text/javascript">
24 let a = 'abcd &amp; &lt; > \\ \n  \"\'';
25 alert(a);
26 let b = "abcd &amp; &lt; > \\ \n  \"\'";
27 alert(b);
28 </script>

```

```

29
30 </body>
31 </html>

```

3.65 README.md.tpl

```

1  # DevOrSysAdminScripts
2
3  [[CodeFactor-badge]][CodeFactor-package-page]
4  [[Codacy-badge]][Codacy-package-page]
5  [[GitHub-top-language-badge]]
6  [[GitHub-code-size-in-bytes-badge]]
7  [[Security:bandit:badge]][Bandit-GitHub.com]
8  [[Code-style:black:badge]][Black-GitHub.com]
9  [[Imports:isort:badge]][Isort-GitHub.io]
10 [[Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
11 [[Linting:pylint:badge]][Pylint-GitHub.com]
12 [[Linting:ruff:badge]][Ruff-GitHub.com]
13
14 |                               **Some "small" useful scripts**                               |
15 | :-----:
16 | **for enhancing development or system-administrators tasks** |
17
18 Currently, the biggest part of this repository is in
19 a suite of shell scripts about build and checks that
20 I use in some of my other repositories.
21
22 I tried to follow some conventions for my bash code:
23
24 - I tried to encapsulate most of the code in functions.
25 - Unless used as return values, variables in functions are defined
26   as local variables (`local` keyword or `declare`)
27   with prefix "LFBFL_".
28 - When used as return values, variables in functions are defined
29   as global variables (`declare -g`) with prefix
30   "@function_name@_result_".
31 - Whenever a variable is no more modified after some point,
32   add the keyword `readonly` or use `declare -r`.
33 - Whenever a variable will only contain integer values (or boolean
34   values as 0 or 1), use `declare -i`.
35 - I named bash files with suffixes ".exec.sh" when the script can
36   be runned
37   (executable, abbreviation stops before a vowel).
38 - I named bash files with suffixes ".lib.sh" when the script contains
39   only functions definitions
40   (library, abbreviation stops before a vowel).

```



```

41 - But the files "pre-commit" and "post-commit" were not renamed,
42     since it is not possible to give them other names to use them as
43     pre-commit/post-commit hooks in 'git'.
44 - When some ".exec.sh" code is encapsulated into functions,
45     either these functions can be reused and go in some ".lib.sh" file,
46     either they are truly specific to this script and they are kept
47     in ".exec.sh" file.
48 - When using 'source' builtin, although any path/filename with a slash
49     will only be searched for from the current directory (tested),
50     we add the prefix "./", to make it explicit.
51 - We use mapfile instead of read -r most of the time, but with the
52     following exceptions: real use of read -r to read the command line of the
53     shell, comments (of course), and a file funny.sh where it is kind of
54     required to use read -r.
55 - We replaced all echo calls by printf calls to follow
56     "The Open Group Base Specifications" since at least 2 decades:
57     "New applications are encouraged to use printf instead of echo.",
58     <https://pubs.opengroup.org/onlinepubs/000095399/>.
59 - We used '_?i_' or '_?j_', etc. in integer variables names in Bash
60     scripts.
61
62 [CodeFactor-badge]: https://www.codefactor.io/repository/github/\
63 llyaudet/DevOrSysAdminScripts/badge
64
65 [CodeFactor-package-page]: https://www.codefactor.io/repository/\
66 github/llyaudet/DevOrSysAdminScripts
67
68 [Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
69 3dd6ef85e8db4460991d1ebc404b68dd
70
71 [Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
72 DevOrSysAdminScripts/dashboard?utm\_source=gh&utm\_medium=referral\
73 &utm\_content=&utm\_campaign=Badge\_grade
74
75 [GitHub-top-language-badge]: https://img.shields.io/github/\
76 languages/top/llyaudet/DevOrSysAdminScripts
77
78 [GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
79 languages/code-size/llyaudet/DevOrSysAdminScripts
80
81 [Security:bandit:badge]: https://img.shields.io/badge/\
82 security-bandit-yellow.svg
83
84 [Bandit-GitHub.com]: https://github.com/PyCQA/bandit
85
86 [Code-style:black:badge]: https://img.shields.io/badge/\

```

```

87 code%20style-black-000000.svg
88
89 [Black-GitHub.com]: https://github.com/psf/black
90
91 [Imports:isort:badge]: https://img.shields.io/badge/\
92 %20imports-isort-%231674b1?style=flat&labelColor=ef8336
93
94 [Isort-GitHub.io]: https://pycqa.github.io/isort/
95
96 [Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
97 mypy_badge.svg
98
99 [Typecheck-mypy-lang.org]: https://mypy-lang.org/
100
101 [Linting:pylint:badge]: https://img.shields.io/badge/\
102 linting-pylint-yellowgreen
103
104 [Pylint-GitHub.com]: https://github.com/pylint-dev/pylint
105
106 [Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
107 https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
108 badge/v2.json
109
110 [Ruff-GitHub.com]: https://github.com/astral-sh/ruff

```

3.66 string_escaping.lib.php

```

1 <?php
2
3 /**
4  This file is part of DevOrSysAdminScripts.
5
6  DevOrSysAdminScripts is free software:
7  you can redistribute it and/or modify it
8  under the terms of the GNU Lesser General Public License
9  as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19

```

```

20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26
27 @category Library
28 @package DevOrSysAdminScripts
29 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
30 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
31 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
32 */
33
34 declare(strict_types=1);
35 declare(encoding='UTF-8');
36
37
38
39 /**
40 The namespace string_escaping is splitted into subnamespaces for each
41 language.
42 This subnamespace is for HTML (surprised?)...
43
44 @package DevOrSysAdminScripts
45 @subpackage string_escaping\HTML
46 */
47 namespace string_escaping\HTML {
48     /**
49      When you want that the string appears as is in HTML text block.
50
51      @param string $s_string The string you want to escape.
52
53      @return string The escaped string.
54     */
55     function escape_text(string $s_string) : string {
56         return str_replace(
57             [
58                 '&',
59                 '<',
60             ],
61             [
62                 '&amp;',
63                 '&lt;',
64             ],
65             $s_string,

```

```

66     );
67 }//end escape_text()
68
69
70
71 /**
72  When you want that the string appears as is in an HTML tag attribute
73  that starts and ends with double quotes: my_attribute="something".
74  Example: a title attribute.
75  /\ in double quotes /\ my_attribute='something' is not supported.
76
77  @param string $s_string The string you want to escape.
78
79  @return string The escaped string.
80  */
81 function escape_attribute(string $s_string) : string {
82     return str_replace(
83         ['"',
84         ['&quot;'],
85         $s_string,
86     );
87 }//end escape_attribute()
88
89
90
91 /**
92  When you want that the string appears verbatim inside HTML <pre> tags.
93
94  @param string $s_string The string you want to escape.
95
96  @return string The escaped string.
97  */
98 function escape_pre(string $s_string) : string {
99     return str_replace(
100         [
101             '&',
102             '<',
103         ],
104         [
105             '&amp;',
106             '&lt;',
107         ],
108         $s_string,
109     );
110 }//end escape_pre()
111 }//end namespace string_escaping\HTML

```

```

112
113
114
115
116 /**
117 The namespace string_escaping is splitted into subnamespaces for each
118 language.
119 This subnamespace is for XHTML (still surprised?)...
120
121 @package DevOrSysAdminScripts
122 @subpackage string_escaping\XHTML
123 */
124 namespace string_escaping\XHTML {
125     /**
126     When you want that the string appears as is in XHTML text block.
127
128     @param string $s_string The string you want to escape.
129
130     @return string The escaped string.
131     */
132     function escape_text(string $s_string) : string {
133         return str_replace(
134             [
135                 '&',
136                 '<',
137             ],
138             [
139                 '&',
140                 '<',
141             ],
142             $s_string,
143         );
144     } //end escape_text()
145
146
147
148 /**
149 When you want that the string appears as is in an XHTML tag attribute
150 that starts and ends with double quotes: my_attribute="something".
151 Example: a title attribute.
152 /!\ in double quotes /!\ my_attribute='something' is not supported.
153
154 @param string $s_string The string you want to escape.
155
156 @return string The escaped string.
157 */

```

```

158     function escape_attribute(string $s_string) : string {
159         return str_replace(
160             [
161                 '&',
162                 '<',
163                 '"',
164                 "\n",
165             ],
166             [
167                 '&amp;',
168                 '&lt;',
169                 '&quot;',
170                 '&#10;',
171             ],
172             $s_string,
173         );
174     } //end escape_attribute()
175
176
177
178     /**
179     When you want that the string appears verbatim inside XHTML <pre> tags.
180
181     @param string $s_string The string you want to escape.
182
183     @return string The escaped string.
184     */
185     function escape_pre(string $s_string) : string {
186         return str_replace(
187             [
188                 '&',
189                 '<',
190             ],
191             [
192                 '&amp;',
193                 '&lt;',
194             ],
195             $s_string,
196         );
197     } //end escape_pre()
198 } //end namespace string_escaping\XHTML
199
200
201
202
203     /**

```

```

204 The namespace string_escaping is splitted into subnamespaces for each
205 language.
206 This subnamespace is for JS (you're hopeless?)...
207
208 @package DevOrSysAdminScripts
209 @subpackage string_escaping\JS
210 */
211 namespace string_escaping\JS {
212
213     use function string_escaping\XHTML\escape_text as XHTML_escape_text;
214
215
216
217     /**
218     When you want that the string can be put in a JS string definition.
219
220     @param string $s_string The string you want to escape.
221
222     @return string The escaped string.
223     */
224     function escape(string $s_string) : string {
225         return str_replace(
226             [
227                 '\\',
228                 '\n',
229                 '\"',
230                 '\'',
231             ],
232             [
233                 '\\\\',
234                 '\\n',
235                 '\\\\\'',
236                 '\\\\\"',
237             ],
238             $s_string,
239         );
240     } //end escape()
241
242
243
244     /**
245     When you want that the string can be put in a JS string definition
246     to be included inside an XHTML file <script> tag.
247
248     @param string $s_string The string you want to escape.
249

```

```

250 @return string The escaped string.
251 */
252 function escape_for_string_definition_inside_XHTML(string $s_string)
253 : string {
254     return XHTML_escape_text(escape($s_string));
255 } //end escape_for_string_definition_inside_XHTML()
256 } //end namespace string_escaping\JS
257
258
259
260
261 /**
262 The namespace string_escaping is splitted into subnamespaces for each
263 language.
264 This subnamespace is for TeX.
265
266 @package DevOrSysAdminScripts
267 @subpackage string_escaping\TeX
268 */
269 namespace string_escaping\TeX {
270     /**
271     When you want that the string can be written to a TeX file inside a
272     part where TeX is in text mode.
273
274     @param string $s_string The string you want to escape.
275
276     @return string The escaped string.
277     */
278     function escape_text(string $s_string) : string {
279         // https://tex.stackexchange.com/a/34586/270015
280         return str_replace(
281             [
282                 '\\',
283                 '&',
284                 '%',
285                 '$',
286                 '#',
287                 '_',
288                 '{',
289                 '}',
290                 '~',
291                 '^',
292             ],
293             [
294                 '\\textbackslash',
295                 '\\&',

```



```

296         '\\%',
297         '\\$',
298         '\\#',
299         '\\_',
300         '\\{',
301         '\\}',
302         '\\textasciitilde',
303         '\\textasciicircum',
304     ],
305     $s_string,
306 );
307 } //end escape_text()
308 } //end namespace string_escaping\\TeX
309
310 /*
311 $ php
312 <?php
313 require_once('string_escaping.lib.php');
314 var_dump(string_escaping\\HTML\\escape_text("<&\\\"\\n'truc\\>"));
315 var_dump(string_escaping\\HTML\\escape_attribute("<&\\\"\\n'truc\\>"));
316 var_dump(string_escaping\\HTML\\escape_pre("<&\\\"\\n'truc\\>"));
317 var_dump(string_escaping\\XHTML\\escape_text("<&\\\"\\n'truc\\>"));
318 var_dump(string_escaping\\XHTML\\escape_attribute("<&\\\"\\n'truc\\>"));
319 var_dump(string_escaping\\XHTML\\escape_pre("<&\\\"\\n'truc\\>"));
320 var_dump(string_escaping\\JS\\escape("abcd & < > \\ \\n \\'"));
321 var_dump(string_escaping\\JS\\escape_for_string_definition_inside_XHTML(
322     "abcd & < > \\ \\n \\'"));
323 );
324 */
325 ?>

```

3.67 unicode.lib.php

```

1  <?php
2
3  /**
4   This file is part of DevOrSysAdminScripts.
5
6   DevOrSysAdminScripts is free software:
7   you can redistribute it and/or modify it
8   under the terms of the GNU Lesser General Public License
9   as published by the Free Software Foundation,
10  either version 3 of the License,
11  or (at your option) any later version.
12
13  DevOrSysAdminScripts is distributed

```

```

14  in the hope that it will be useful,
15  but WITHOUT ANY WARRANTY;
16  without even the implied warranty of
17  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18  See the GNU Lesser General Public License for more details.
19
20  You should have received a copy of
21  the GNU Lesser General Public License
22  along with DevOrSysAdminScripts.
23  If not, see <https://www.gnu.org/licenses/>.
24
25  ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26  This file was renamed from "unicode.php" to "unicode.lib.php".
27
28  @category Library
29  @package DevOrSysAdminScripts
30  @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31  @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32  @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33  */
34
35  declare(strict_types=1);
36  declare(encoding='UTF-8');
37
38
39
40  /**
41   Converts an int to the byte-string corresponding to the UTF-8 encoding
42   of the Unicode code point of same integer value.
43   decimal_code_point is a slight abuse of language to say that you write the
44   code point in decimal notation in PHP code and you get the UTF-8.
45   If you provide the integer of the code point without using decimal
46   notation, it will also work.
47
48   @param int $i_code_point_in_decimal_notation The code point.
49
50   @throws Exception When the code point is outside of Unicode range.
51
52   @return string The corresponding UTF-8 character.
53   */
54   function decimal_code_point_to_UTF8(
55       int $i_code_point_in_decimal_notation
56   ) : string {
57       // var_dump($i_code_point_in_decimal_notation);
58       // 0xxxxxxx ASCII
59       if($i_code_point_in_decimal_notation < 128){

```

```

60     return chr($i_code_point_in_decimal_notation);
61 }
62 $i_continuation_base_value = 128;
63 // 110xxxxx 10xxxxxx
64 if($i_code_point_in_decimal_notation < 2048){ // 2**11
65     $i_first_byte_base_value = 192;
66     /*
67     $i_last_byte_significant_bits = (
68         $i_code_point_in_decimal_notation % 64 // 2**6
69     );
70     $i_first_byte_significant_bits = intdiv(
71         (
72             $i_code_point_in_decimal_notation
73             - $i_last_byte_significant_bits
74         ),
75         64, // 2**6
76     );
77     $s_result = chr(
78         $i_first_byte_base_value
79         + $i_first_byte_significant_bits
80     );
81     $s_result .= chr(
82         $i_continuation_base_value
83         + $i_last_byte_significant_bits
84     );
85     return $s_result;
86     */
87     return (
88         chr(
89             $i_first_byte_base_value
90             + ($i_code_point_in_decimal_notation >> 6)
91         )
92         .chr(
93             $i_continuation_base_value
94             + ($i_code_point_in_decimal_notation & 63)
95         )
96     );
97 }//end if($i_code_point_in_decimal_notation < 2048)
98 // 1110xxxx 10xxxxxx 10xxxxxx
99 if($i_code_point_in_decimal_notation < 65536){ // 2**16
100     $i_first_byte_base_value = 224;
101     /*
102     $arr_arr_data_per_byte_reverse = [
103         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
104         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
105         ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],

```

```

106     ];
107     $i_rest = $i_code_point_in_decimal_notation;
108     $s_result = '';
109     foreach($arr_arr_data_per_byte_reverse as $arr_data){
110         // _Pragma unroll
111         // J'aurais pu "roller" les 4 cas des ifs.
112         // Mais pour les perfs, il faudrait tout unroller vu le nombre
113         // d'itérations. Et j'aimerais bien comparer du code unrollé
114         // aux 2 niveaux à la main avec celui unrollé par le compilateur
115         // sur les 2 niveaux. Je préjuge/présume qu'il unroll
116         // l'intérieur d'abord (pour ne pas refaire le même boulot)
117         // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
118         // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
119         $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
120         $i_rest = intdiv(
121             $i_rest - $i_significant_bits,
122             2**$arr_data['i_bits'],
123         );
124         // .= or =.
125         // Below could have been a case of "=".
126         $s_result = (
127             chr($i_significant_bits + $arr_data['i_base_value'])
128             .$s_result
129         );
130     }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
131     return $s_result;
132 */
133     return (
134         chr(
135             $i_first_byte_base_value
136             + ($i_code_point_in_decimal_notation >> 12)
137         )
138         .chr(
139             $i_continuation_base_value
140             + (($i_code_point_in_decimal_notation >> 6) & 63)
141         )
142         .chr(
143             $i_continuation_base_value
144             + ($i_code_point_in_decimal_notation & 63)
145         )
146     );
147 }//end if($i_code_point_in_decimal_notation < 65536)
148 // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
149 if($i_code_point_in_decimal_notation < 2097152){ // 2**21
150     $i_first_byte_base_value = 240;
151     /*

```

```

152     $arr_arr_data_per_byte_reverse = [
153         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
154         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
155         ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
156         ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
157     ];
158     $i_rest = $i_code_point_in_decimal_notation;
159     $s_result = '';
160     foreach($arr_arr_data_per_byte_reverse as $arr_data){
161         // _Pragma unroll
162         $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
163         $i_rest = intdiv(
164             $i_rest - $i_significant_bits,
165             2**$arr_data['i_bits'],
166         );
167         // .= or =.
168         // Below could have been a case of "=".
169         $s_result = (
170             chr($i_significant_bits + $arr_data['i_base_value'])
171             . $s_result
172         );
173     }
174     return $s_result;
175 */
176     return (
177         chr(
178             $i_first_byte_base_value
179             + ($i_code_point_in_decimal_notation >> 18)
180         )
181         .chr(
182             $i_continuation_base_value
183             + (($i_code_point_in_decimal_notation >> 12) & 63)
184         )
185         .chr(
186             $i_continuation_base_value
187             + (($i_code_point_in_decimal_notation >> 6) & 63)
188         )
189         .chr(
190             $i_continuation_base_value
191             + ($i_code_point_in_decimal_notation & 63)
192         )
193     );
194 }//end if($i_code_point_in_decimal_notation < 2097152)
195 // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
196 throw new Exception(
197     "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."

```

```

198     ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
199     ." 3 octets au lieu de 4 :face-palm:."
200     ." Du coup, ça peut donner quelques \"surprises\"."
201     ." Sinon, y'a aussi ASCII dans MySQL."
202     ." Et comme ça joue sur la mémoire et les performances,"
203     ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
204     ." :) "
205 );
206 // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
207 }//end decimal_code_point_to_UTF8()
208
209
210
211 /**
212  Converts a string of hexadecimal digits to the byte-string corresponding
213  to the UTF-8 encoding of the Unicode code point of same integer value
214  that is denoted by the hexadecimal digits.
215
216  @param string $s_code_point_in_hexadecimal_notation The code point.
217
218  @throws Exception When the code point is outside of Unicode range.
219
220  @return string The corresponding UTF-8 character.
221  */
222 function hexa_code_point_to_UTF8(
223     string $s_code_point_in_hexadecimal_notation
224 ) : string {
225     $s_code_point_in_hexadecimal_notation = strtoupper(
226         $s_code_point_in_hexadecimal_notation
227     );
228     $i_code_point_in_decimal_notation = 0;
229     $i_ord_0 = ord('0');
230     $i_ord_A = ord('A');
231     $i_digit_offset = -$i_ord_0;
232     $i_letter_offset = 10 - $i_ord_A;
233     for(
234         $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
235         $i < $i_max;
236         ++$i
237     ) {
238         $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
239         $i_code_point_in_decimal_notation *= 16;
240         if ctype_digit($s_hexa_digit) {
241             $i_code_point_in_decimal_notation += (
242                 ord($s_hexa_digit) + $i_digit_offset
243             );

```

```

244         continue;
245     }
246     if ctype_xdigit($s_hexa_digit){
247         $i_code_point_in_decimal_notation += (
248             ord($s_hexa_digit) + $i_letter_offset
249         );
250         continue;
251     }
252     throw new Exception('Not an hexadecimal digit.');
```

```

253 }
254 return decimal_code_point_to_UTF8(
255     $i_code_point_in_decimal_notation
256 );
257 }//end hexa_code_point_to_UTF8()
258
259
260
261 /**
262  This function throws an exception with extended debug informations
263  if the input string is not valid ASCII.
264  Otherwise, it returns true.
265
266  @param string $s_string The input string.
267
268  @throws Exception When the input string is not valid ASCII.
269
270  @return bool
271  */
272 function check_string_is_valid_ASCII(string $s_string) : bool {
273     $i_offset_in_octets_from_string_start = 0;
274     $i_current_line_number = 1;
275     $i_offset_in_octets_from_line_start = -1;
276     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
277         $i_current_octet = ord($s_string[$i]);
278         $i_offset_in_octets_from_string_start = $i;
279         ++$i_offset_in_octets_from_line_start;
280         if($i_current_octet > 127){
281             throw new Exception(
282                 'Non-ASCII character found on line '
283                 . $i_current_line_number
284                 . ' ; the octet/character '
285                 . ($i_offset_in_octets_from_line_start + 1)
286                 . ' has value '
287                 . $i_current_octet
288                 . ' which is not ASCII.'
289                 . ' (Sequential position without line splitting:'
```

```

290         .' This is at octet/character '
291         .($i_offset_in_octets_from_string_start + 1)
292         .'.)',
293     );
294 }
295 // if($s_string[$i] === "\n"){
296 if($i_current_octet === 10){
297     ++$i_current_line_number;
298     $i_offset_in_octets_from_line_start = -1;
299 }
300 }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
301 return true;
302 }//end check_string_is_valid_ASCII()
303
304
305
306 /**
307 This function throws an exception with extended debug informations
308 if the content of the file with given file_path is not valid ASCII.
309 Otherwise, it returns true.
310
311 @param string $s_file_path The input file_path.
312
313 @throws Exception When the input string is not valid ASCII.
314
315 @return bool
316 */
317 function check_file_is_valid_ASCII(string $s_file_path) : bool {
318     $s_string = file_get_contents($s_file_path);
319     if($s_string === false){
320         throw new Exception('File '.$s_file_path.' not found.');
```



```

336 @return bool
337 */
338 function check_string_is_valid_UTF8(string $s_string) : bool {
339     $i_continuation_octet_needed = 0;
340     $i_offset_in_octets_from_string_start = 0;
341     $i_offset_in_characters_from_string_start = -1;
342     $i_character_start_position_from_string_start = 0;
343     $i_current_line_number = 1;
344     $i_offset_in_octets_from_line_start = -1;
345     $i_offset_in_characters_from_line_start = -1;
346     $i_character_start_position_from_line_start = -1;
347     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
348         $i_current_octet = ord($s_string[$i]);
349         $i_offset_in_octets_from_string_start = $i;
350         ++$i_offset_in_octets_from_line_start;
351         if($i_continuation_octet_needed > 0){
352             if($i_current_octet < 128 || $i_current_octet >= 192){
353                 throw new Exception(
354                     'Non-UTF8 character found on line '
355                     . $i_current_line_number
356                     . ' ; the octet '
357                     . ($i_offset_in_octets_from_line_start + 1)
358                     . ' , part of the character '
359                     . ($i_offset_in_characters_from_line_start + 1)
360                     . ' , has value '
361                     . $i_current_octet
362                     . ' which is not a continuation octet.'
363                     . ' This character starts at octet '
364                     . ($i_character_start_position_from_line_start + 1)
365                     . ' of the current line.'
366                     . ' (Sequential positions without line splitting:'
367                     . ' This is at character '
368                     . ($i_offset_in_characters_from_string_start + 1)
369                     . ' and octet '
370                     . ($i_offset_in_octets_from_string_start + 1)
371                     . ' .'
372                     . ' This character starts at octet '
373                     . ($i_character_start_position_from_string_start + 1)
374                     . ' .)' ,
375                 );
376             } //end if($i_current_octet < 128 || $i_current_octet >= 192)
377             --$i_continuation_octet_needed;
378             continue;
379         } //end if($i_continuation_octet_needed > 0)
380
381         ++$i_offset_in_characters_from_string_start;

```

```

382     ++$i_offset_in_characters_from_line_start;
383     $i_character_start_position_from_string_start = (
384         $i_offset_in_octets_from_string_start
385     );
386     $i_character_start_position_from_line_start = (
387         $i_offset_in_octets_from_line_start
388     );
389     if($i_current_octet < 128){ // 0xxxxxxx ASCII
390         // if($s_string[$i] === "\n"){
391             if($i_current_octet === 10){
392                 ++$i_current_line_number;
393                 $i_offset_in_octets_from_line_start = -1;
394                 $i_offset_in_characters_from_line_start = -1;
395             }
396             continue;
397         }
398         if($i_current_octet >= 128 && $i_current_octet < 192){
399             throw new Exception(
400                 'Non-UTF8 character found on line '
401                 . $i_current_line_number
402                 . ' ; the octet '
403                 . ($i_offset_in_octets_from_line_start + 1)
404                 . ' , part of the character '
405                 . ($i_offset_in_characters_from_line_start + 1)
406                 . ' , has value '
407                 . $i_current_octet
408                 . ' which is a continuation octet.'
409                 . ' This character starts at octet '
410                 . ($i_character_start_position_from_line_start + 1)
411                 . ' of the current line.'
412                 . ' (Sequential positions without line splitting:'
413                 . ' This is at character '
414                 . ($i_offset_in_characters_from_string_start + 1)
415                 . ' and octet '
416                 . ($i_offset_in_octets_from_string_start + 1)
417                 . ' .'
418                 . ' This character starts at octet '
419                 . ($i_character_start_position_from_string_start + 1)
420                 . ' .)' ,
421             );
422         }
423         if($i_current_octet >= 192 && $i_current_octet < 224){
424             // 110xxxxx 10xxxxxx
425             $i_continuation_octet_needed = 1;
426             continue;
427         }

```

```

428     if($i_current_octet >= 224 && $i_current_octet < 240){
429         // 1110xxxx 10xxxxxx 10xxxxxx
430         $i_continuation_octet_needed = 2;
431         continue;
432     }
433     if($i_current_octet >= 240 && $i_current_octet < 248){
434         // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
435         $i_continuation_octet_needed = 3;
436         continue;
437     }
438     throw new Exception(
439         'Non-UTF8 character found on line '
440         . $i_current_line_number
441         . ' ; the octet '
442         . ($i_offset_in_octets_from_line_start + 1)
443         . ' , part of the character '
444         . ($i_offset_in_characters_from_line_start + 1)
445         . ' , has value '
446         . $i_current_octet
447         . ' which is invalid.'
448         . ' This character starts at octet '
449         . ($i_character_start_position_from_line_start + 1)
450         . ' of the current line.'
451         . ' (Sequential positions without line splitting:'
452         . ' This is at character '
453         . ($i_offset_in_characters_from_string_start + 1)
454         . ' and octet '
455         . ($i_offset_in_octets_from_string_start + 1)
456         . ' .'
457         . ' This character starts at octet '
458         . ($i_character_start_position_from_string_start + 1)
459         . ' .)' ,
460     );
461 } //end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
462 return true;
463 } //end check_string_is_valid_UTF8()
464
465
466
467 / **
468 This function throws an exception with extended debug informations
469 if the content of the file with given file_path is not valid ASCII.
470 Otherwise, it returns true.
471
472 @param string $s_file_path The input file_path.
473

```

```

474 @throws Exception When the input string is not valid UTF-8.
475
476 @return bool
477 */
478 function check_file_is_valid_UTF8(string $s_file_path) : bool {
479     $s_string = file_get_contents($s_file_path);
480     if($s_string === false){
481         throw new Exception('File '.$s_file_path.' not found.');
```

```

482     }
483     return check_string_is_valid_UTF8($s_string);
484 }//end check_file_is_valid_UTF8()
485
486
487
488 /*
489 ?>
490 <?php
491 require_once("./unicode.lib.php");
492 var_dump(hexa_code_point_to_UTF8("002B"));
493 var_dump(hexa_code_point_to_UTF8("00E6"));
494 var_dump(hexa_code_point_to_UTF8("1400"));
495 var_dump(hexa_code_point_to_UTF8("10111"));
496 // https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
497 // Some invisible unicode characters that are a problem:
498 // 2000-200F 2028-202F 205F 2060-206F
499 $arr = [
500     '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
501     '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
502     '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
503     '205F',
504     '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
505     '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
506 ];
507 $arr_s = [];
508 $arr_s2 = [];
509 $arr_s3 = [];
510 echo("sed\\n");
511 foreach($arr as $s_hexa){
512     $s = hexa_code_point_to_UTF8($s_hexa);
513     $arr_s []= $s;
514     if(strlen($s) === 2){
515         $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
516     }
517     if(strlen($s) === 3){
518         $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
519             .'\x'.bin2hex($s[2]);

```

```

520     }
521     $arr_s2 []= $s2;
522     $s3 = " --expression='s/\".$s2.\"/ /g'\\\"";
523     $arr_s3 []= $s3;
524     echo($s3.\"\\n\");
525 }
526 //*/
527 ?>

```

3.68 upgrade_build_and_checks.exec.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023–2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "upgrade_build_and_checks.sh"
25 # to "upgrade_build_and_checks.exec.sh".
26
27 upgrade_build_and_checks() {
28     # Options:
29     # --add-test-files
30     # --check-download=y|keep (=keep to keep the downloaded files)
31     # --fixed-point-build
32     # --verbose
33     local LFBFL_verbose=""
34     if [[ "$*" == *--verbose* ]]; then
35         printf "%s %s\\n" "$0" "$*"

```

```

36     LFBFL_verbose="--verbose"
37 fi
38 readonly LFBFL_verbose
39 local LFBFL_fixed_point_build=""
40 if [[ "$*" == *--fixed-point-build* ]]; then
41     LFBFL_fixed_point_build="--fixed-point-build"
42 fi
43 readonly LFBFL_fixed_point_build
44 local LFBFL_check_download=""
45 if [[ "$*" == *--check-download=keep* ]]; then
46     LFBFL_check_download="--check-download=keep"
47 elif [[ "$*" == *--check-download=y* ]]; then
48     LFBFL_check_download="--check-download=y"
49 fi
50 readonly LFBFL_check_download
51 declare -i LFBFL_add_test_files=0
52 if [[ "$*" == *--add-test-files* ]]; then
53     LFBFL_add_test_files=1
54 fi
55 readonly LFBFL_add_test_files
56 ( \
57     cd build_and_checks_dependencies/\
58     && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}" \
59 )
60 if [[ LFBFL_add_test_files -eq 1 ]]; then
61     mkdir --parents -- --a/
62     {
63         printf "<?php\nfunction my_function(;;){\n\n}\n"
64         # shellcheck disable=SC2016
65         printf '$my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
66         printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
67     } > --a.php
68     cp -- --a.php --a/
69     {
70         printf "from typing import Iterable\n"
71         printf "from _collections_abc import dict_items\n"
72         printf "\n"
73         printf "from python_none_objects import NoneIterable\n"
74         printf "\n"
75         printf "def my_function(my_argument : int) -> None:\n"
76         printf "    print(my_argument)\n"
77         printf "\n"
78         printf "my_function(\"abcd\")\n"
79         printf "\n"
80         printf "def my_function2(my_argument : int) -> None:\n"
81         printf "    \"\"\"Some comment.\"\"\"\n"

```

```

82     printf "    print(my_argument)\n"
83     printf "\n"
84     printf "def my_function3(my_argument : int) -> None:\n"
85     printf "    \"\"\"Some comment.\"\"\"\n"
86     printf "\n"
87     printf "    print(my_argument)\n"
88     printf "\n"
89     printf "def my_function4(my_argument : int):\n"
90     printf "    \"\"\"Some comment.\"\"\"\n"
91     printf "    def my_sub_function():\n"
92     printf "        print(my_argument)\n"
93     printf "    return my_sub_function\n"
94     printf "\n"
95     printf "def my_function5(my_argument : int):\n"
96     printf "    \"\"\"Some comment.\"\"\"\n"
97     printf "\n"
98     printf "    def my_sub_function():\n"
99     printf "        print(my_argument)\n"
100    printf "    return my_sub_function\n"
101    printf "\n"
102 } > --a.py
103 cat -- --a.py
104 cp -- --a.py --a/
105 {
106     printf "def my_function(;;):\n\n\n"
107     printf 'my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'\n"
108     printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
109 } > --a2.py
110 cat -- --a2.py
111 cp -- --a2.py --a/
112 {
113     # Error 0 is missing shell header.
114     printf "my_function(){\n"
115     printf "    # Error 1: Indentation is not correct.\n"
116     printf "    sed -e 's/a/b/g' \\\n"
117     printf "        -e 's/a/b/g' \"\$1\" # Indentation is correct.\n"
118     printf "    my_variable=\"%s\\n\"\\n"
119     # shellcheck disable=SC2016
120     printf '    printf "${my_variable}" # Error 2: shellcheck warning\n'
121     local LFBFL_ttp="ttp"
122     local LFBFL_URL="h${LFBFL_ttp}://example.com"
123     printf '    printf "%s" # Error 3: http -> https\n' "${LFBFL_URL}"
124     # shellcheck disable=SC2016
125     printf '    printf "${#some_'\n"
126     printf 'arr_1}" # Error 4: no_size_of_array_first_element\n'
127     # shellcheck disable=SC2016

```

```

128     printf '    printf "${#'\n'
129     printf 'arr_1}" # Error 5: no_size_of_array_first_element\n'
130     printf '    if !'\n'
131     # shellcheck disable=SC2016
132     printf '    [[ $1 -eq 1 ]]; # Error 6: no_negation_before_bash_test\n'
133     printf '    then # Error 7: misplaced then\n'
134     printf '        my_variable="a"\n'
135     printf '    fi\n'
136     printf "\n"
137     # shellcheck disable=SC2016
138     printf '    if [[ $1 -eq 1 ]];\n'
139     printf '    then # Error 8: another case of misplaced then\n'
140     printf '        my_variable="a"\n'
141     printf '    fi\n'
142     printf '    my_variable="aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'\n'
143     printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa' # Error 9 line too long.\n'
144     printf '}\n'
145 } > --a.sh
146 cp -- --a.sh --a/
147 fi
148 ./build_and_checks.exec.sh ". "\n'
149 "${LFBFL_verbose}"\n'
150 "${LFBFL_fixed_point_build}"\n'
151 "${LFBFL_check_download}"\n'
152 if [[ LFBFL_add_test_files -eq 1 ]]; then
153     cat -- --a.py
154     cat -- --a2.py
155     rm --force --recursive -- --a/
156     rm --force -- --a.php --a.py --a2.py --a.sh
157 fi
158 }
159
160 upgrade_build_and_checks "$@"

```

3.69 wget_sha512.lib.sh

```

1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
5  # you can redistribute it and/or modify it
6  # under the terms of the GNU General Public License
7  # as published by the Free Software Foundation,
8  # either version 3 of the License,
9  # or (at your option) any later version.
10 #

```



```

11 # DevOrSysAdminScripts is distributed
12 # in the hope that it will be useful,
13 # but WITHOUT ANY WARRANTY;
14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "wget_sha512.sh"
25 # to "wget_sha512.lib.sh".
26
27 wget_sha512() {
28     # $1=file_path
29     # $2=download_URL
30     # $3=correct_sha512
31     # Options:
32     #   --check-download=y|keep (=keep to keep the downloaded file)
33     #   --verbose
34     local LFBFL_verbose=""
35     if [[ "$*" == *--verbose* ]]; then
36         printf "%s wget_sha512 %s\n" "$0" "$*"
37         LFBFL_verbose="--verbose"
38     fi
39     readonly LFBFL_verbose
40     local LFBFL_check_download=""
41     if [[ "$*" == *--check-download=keep* ]]; then
42         LFBFL_check_download="keep"
43     elif [[ "$*" == *--check-download=y* ]]; then
44         LFBFL_check_download="y"
45     fi
46     readonly LFBFL_check_download
47     declare -i LFBFL_i_error=0
48
49     if [[ ! -f "$1" ]]; then
50         wget ${LFBFL_verbose:+"${LFBFL_verbose}"} --output-document="$1"\
51             -- "$2"
52     fi
53     declare -r LFBFL_present_sha512=$(
54         sha512sum -- "$1"\
55         | cut --fields=1 --delimiter=' '
56     )

```

```

57 if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
58     printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
59         "$1" "$3" "${LFBFL_present_sha512}"
60     LFBFL_i_error=1
61 fi
62 if [[ -n "${LFBFL_check_download}" ]]; then
63     wget ${LFBFL_verbos:+"${LFBFL_verbos}"}\
64         --output-document="$1.wget_sha512.temp"\
65         -- "$2"
66     declare -r LFBFL_present_sha512_2=$(
67         sha512sum -- "$1.wget_sha512.temp"\
68         | cut --fields=1 --delimiter=' '
69     )
70     if [[ "${LFBFL_check_download}" != "keep" ]]; then
71         rm -- "$1.wget_sha512.temp"
72     fi
73     if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
74         printf\
75             "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
76             "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
77         LFBFL_i_error=1
78     fi
79 fi
80 # shellcheck disable=SC2248
81 return ${LFBFL_i_error}
82 }

```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !